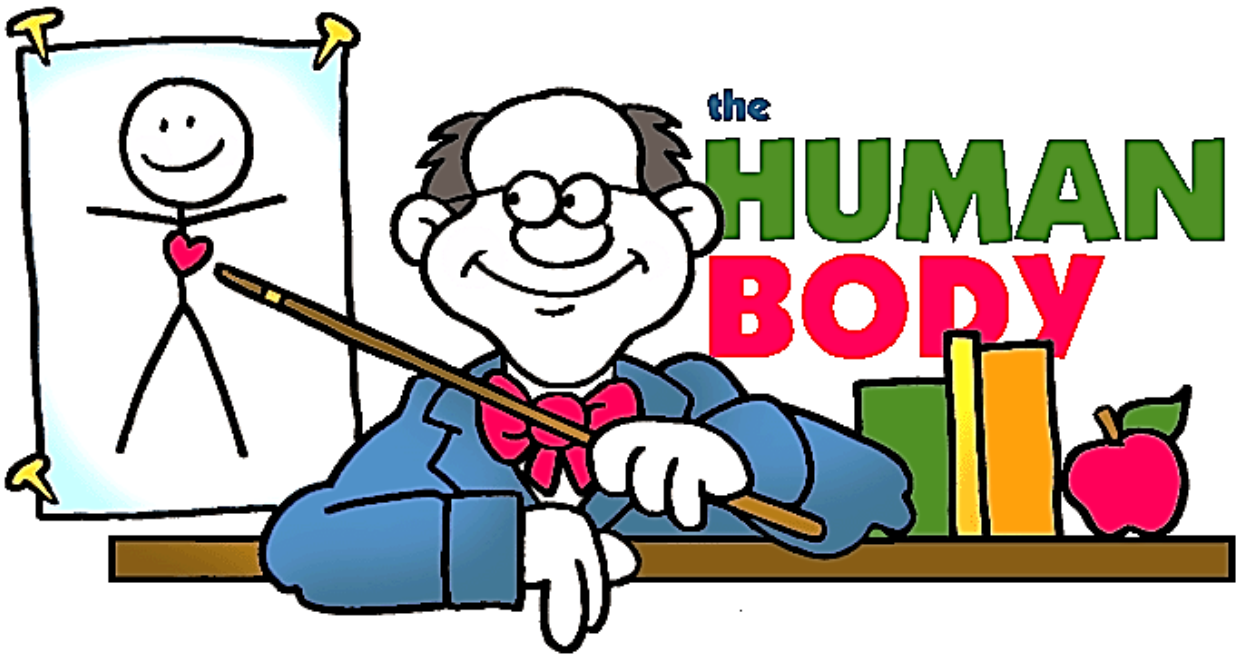
HUMAN ANATOMY

AND PHYSIOLOGY

(II)



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Dedication

This work is lovingly dedicated to My Family

For their abundant support, for their patience and understanding, and for their love.

I would also like to dedicate this work to the immaculate spirit of my parents , my daughter Lara, my sister and my brother

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CHAPTER EIGHT

CARDIOVASCULAR SYSTEM

CHAPTER EIGHT CARDIOVASCULAR SYSTEM

The cardiovascular system is the transport system of the body by which food, oxygen, water and all other essentials are carried to the tissue cells and their waste products are carried away. It :consists of three parts

1. **The blood**, Which is the fluid in which materials are carried to and from the tissue

2. **The heart**, which is the driving force which propels the blood

3. **The blood vessels**, the routes by which the blood travels to and through the tissues and back to the heart

The Blood

Blood is classified as a connective tissue, since nearly half of it is made up of cells. However, it differ from other connective tissues in that its cells are not fixed in position, instead they move .freely in the liquid portion of the blood, the plasma

Blood is a viscous (thick) fluid that varies in color from bright to dark red, depending on how much oxygen it is carrying. Its quantity differs with the size of the person; the average adult male, weighing 70 kg has about 5-6 liters of blood. This volume accounts for about 8% of the total body weight. It is carried through a closed system of vessels pumped by the heart. The circulating blood is of fundamental importance in maintaining the internal environment in a constant state (homeostasis)

Functions of the Blood

1. Transportation

A. Oxygen- from inhaled air diffuses into the blood through the thin lung membranes and is carried to all the tissue of the body. Carbon dioxide, a waste product of cell metabolism, is carried from the .tissues to the lungs, where it is breathed out

B. The blood transports foods and other needed substances such as minerals and vitamins, to the cells. These materials may inter the blood from the digestive system or may be released into the .blood from body stores

C. The blood transports waste products from the cells to the sites from which they are released. The kidney removes excess water, minerals, and urea from protein metabolism and maintains the .acid-base balance of the blood. The liver removes bile pigments and drugs

.**D. The blood carries hormones** from their sites of origin to the organs they affect

2. Regulation

A. Buffers in the blood help keep the PH of body fluids at about 7.4

B. The blood serves to regulate the amount of fluid in the tissues by means of substances (mainly proteins) that maintain the proper osmotic pressure.

C. The blood transports heat that is generated in the muscles to other parts of the body thus aiding in the regulation of body temperature by the blood, thus aiding in the regulation of body .temperature

3. Protection

A. The blood carries the cells that are among the body's defenders against pathogens. It also .contains substances (antibodies) that are concerned with immunity to disease

B. The blood contains factors that protect against blood loss.

Composition of Blood

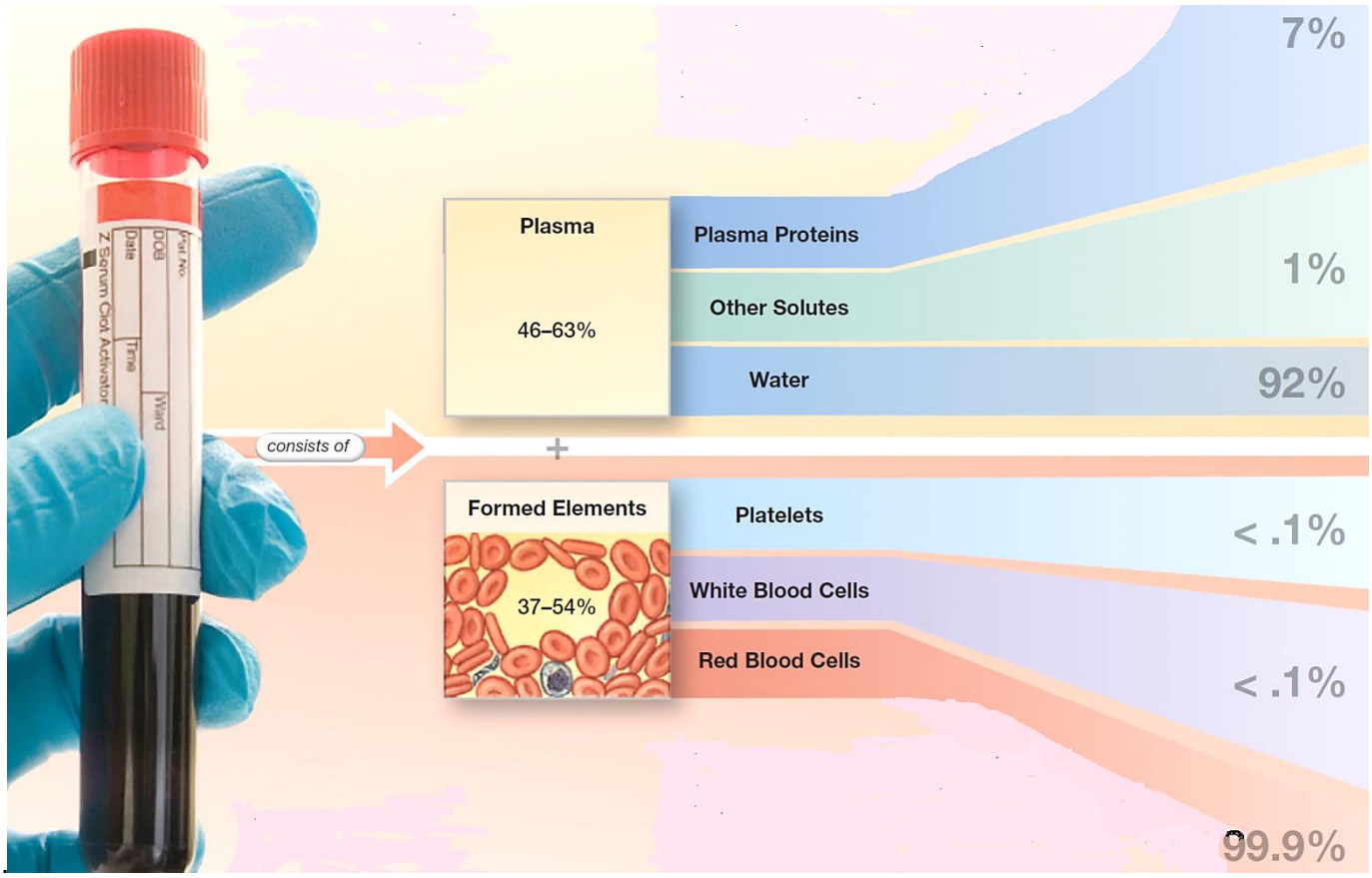
The blood is composed of two prime elements: as already mentioned, the liquid element is called plasma; the cells and fragments of cells are called formed elements or corpuscles.

:The formed elements are classified as follows :

.**1. Erythrocytes,** from erythro, meaning "red," are the red blood cells, which transport oxygen

2. Leukocytes, from leuko, meaning "white," are the several types of white blood cells, which .protect against infection

3. Platelets, also called thrombocytes, are cell fragments that participate in blood clotting



Blood Plasma

Over half of the total volume of blood is plasma. The plasma itself is 90% water. Many different substances dissolved or suspended in the water, make up the other 10%. The plasma content varies somewhat, since the substances carried by the blood to and from the organs get used and

added to. However the body tends to maintain a fairly constant level of these substances. For example, the level of glucose, a simple sugars maintained at a remarkably constant level of about .on tenth of a 1% solution

After water, the next largest percentage of material in the plasma is protein. Proteins are the principal constituents of cytoplasm and are essential to the growth and the rebuilding of body tissues. The plasma proteins include the following:

1. Albumin, the most abundant protein in plasma, is important for maintaining the osmotic .pressure of the blood. This protein is manufactured in the liver

2. The antibodies combat infection.

3. The blood clotting factors are also manufactured in the liver.

4. A system of enzymes made of several proteins, collectively known as complement, helps **.**antibodies in their fight **against pathogens**

Nutrients are also found in the plasma. One group of nutrients is the **carbohydrates**. The principal form of carbohydrate found in the plasma is glucose, which is absorbed by the capillaries of the intestine following digestion. Glucose is stored mainly in the liver as glycogen and released as needed to supply energy.

Amino acids, the products of protein digestion, are also found in the plasma. These are also absorbed into the blood through the intestinal capillaries. Lipids constitute a small percentage of blood plasma. Lipids include fats. They may be stored as fat for reserve energy or carried to the cells as a source of energy. The mineral salts in the plasma appear primarily as chloride،carbonate, or phosphate salts of sodium, potassium, and magnesium.

These salts have a variety of functions, including :

- The formation of bone (calcium and phosphorus),
- The production of hormones by certain glands (iodine for the production of thyroid hormone),
- The transportation of the gases oxygen and carbon dioxide (iron), and
- The maintenance of the acid base balance (sodium and potassium carbonates and phosphates).

Small amounts of other elements also help maintain homeostasis.

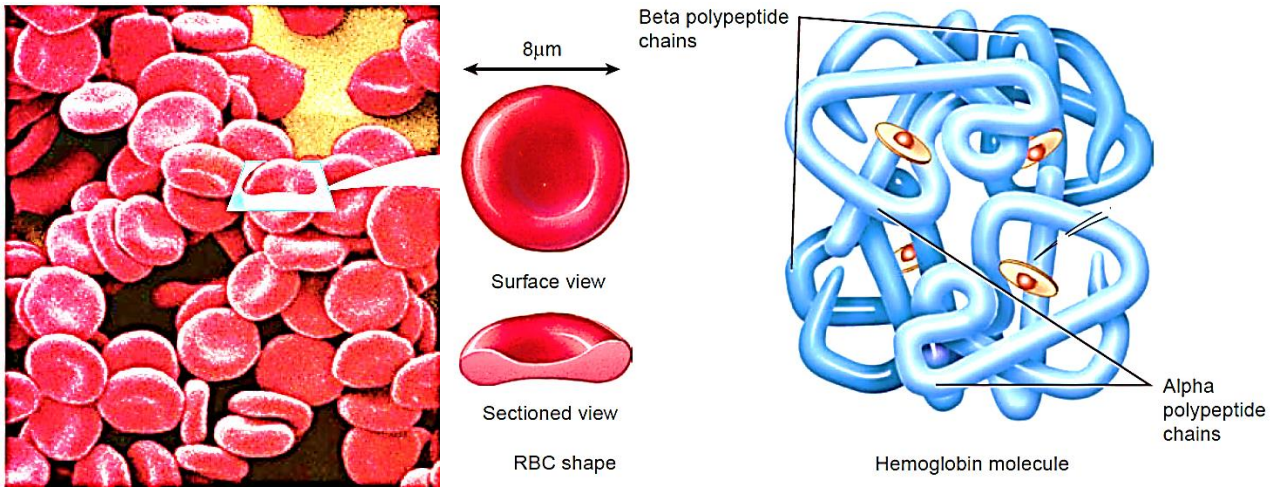
Many other materials, such as waste products and hormones, are also transported in the plasma.

The Formed Elements

❖ Erythrocytes

Erythrocytes, the red cells, are tiny, disk-shaped bodies with a central area that is thinner than the edges. They are different from other cells in that the mature form found in the circulating blood does not have a nucleus. These cells, like almost all the blood cells, live a much shorter time (120 days) than most other cells in the body, some of which last a lifetime. One purpose of the red cells is to carry oxygen from the lungs to the tissues. The oxygen is bound in the red cells to

hemoglobin, a protein that contains iron. Hemoglobin combined with oxygen gives the blood its characteristic red color. The more oxygen carried by the hemoglobin, the brighter is the red color of the blood.



Therefore, the blood that goes from the lungs to the tissues is a bright red because it carries a great supply of oxygen; in contrast, the blood that returns to the lings is a much darker red, since it has given up much of its oxygen to the tissues.

Hemoglobin that has given up its oxygen is able to carry hydrogen ions; in this way, hemoglobin acts as a buffer and plays an important role in acid-base balance. The red cells also

carry a small amount of carbon dioxide from the tissues to the lungs for elimination in exhalation. Carbon monoxide is a harmful gas that combines with hemoglobin to form a stable compound.

It displaces the oxygen that is normally carried by the hemoglobin and reduces the oxygen-carrying ability of the blood. Carbon monoxide maybe produced by the incomplete burning of various fuels, such as gasoline, coal, wood, and other carbon containing materials. It also occurs in automobile exhaust fumes and in cigarette smoke.

The erythrocytes are by far the most numerous of the corpuscles, averaging from 4.5 to 5 million per microliter (L) of blood,.

- **Leukocytes**

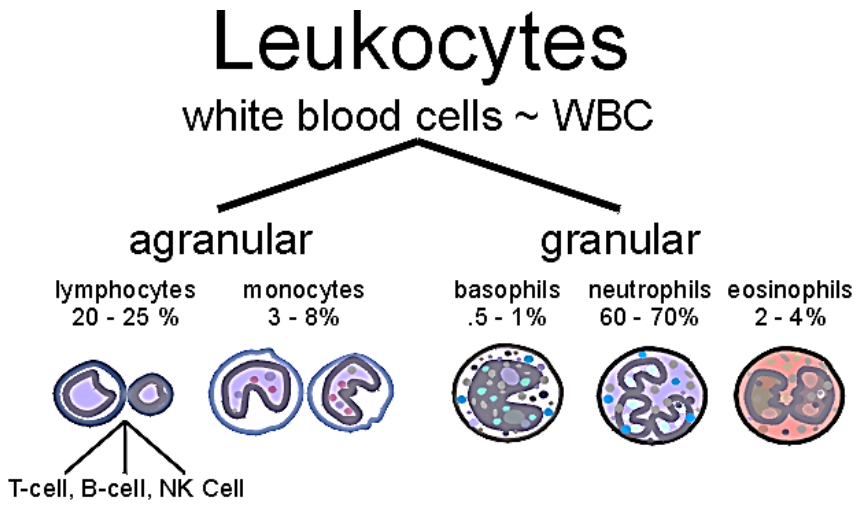
The leukocytes, or white blood cells, are very different from the erythrocytes in appearance, quantity, and function.

They contain nuclei of varying shapes and sizes; the cells themselves are round. Leukocytes are outnumbered by red cells by 700 to 1, numbering 5,000 to 10,000 per cubic millimeter of blood.

Whereas the red cells have a definite color, the leukocytes tend to be colorless.

The different types of white blood cells are identified by their size, the shape of the nucleus, and the appearance of granules in the cytoplasm when the cells are stained, usually with Wright's blood stain. Granulocytes include neutrophils, which show lavender granules; eosinophil's, which have beadlike, bright pink granules; and basophils, which have large, dark blue granules that often obscure the nucleus. The neutrophils are the most numerous of the white cells, constituting up to 60%of all leukocytes. Because the nuclei of the nuclei of the neutrophils are of various

shapes, they are also called polymorphs (meaning "many forms") or simply polys. The granulocytes, so named because they lack easily visible granules, are the lymphocytes and monocytes. The ratio of the different types of leukocytes is often a valuable clue in arriving at a diagnosis .

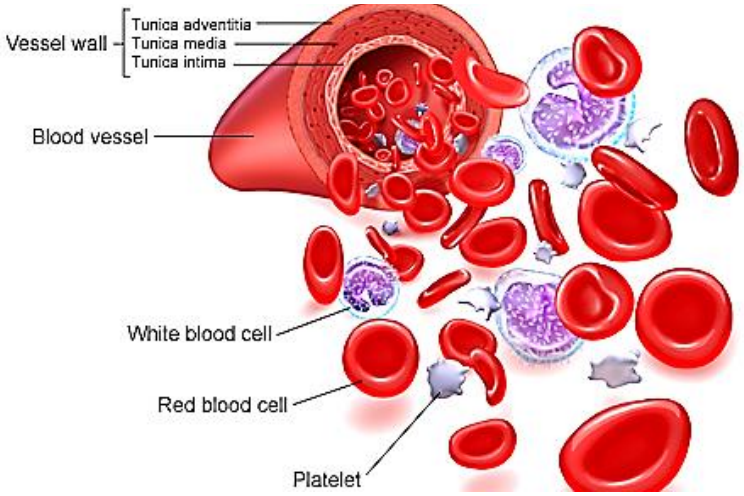


The most important function of the leukocytes is to destroy pathogens. Whenever pathogens enter the tissues, as through a wound, certain white blood cells (neutrophils and monocytes are attracted to that area. They leave the blood vessels and proceed by amoeboid or ameba-like motion to the area of infection. There they engulf the invaders by a process called phagocytosis. If the pathogens are extremely strong or numerous, they may destroy the leukocytes.

A collection of dead and living leukocytes, forms pus. A collection of pus localized in one area is known as abscess. The Lymphocytes destroy foreign invaders by attacking the cells directly or by producing antibodies that circulate in the blood and help destroy the cells

- **Platelets.**

Of all the formed elements, the blood platelets (thrombocytes)are the smallest. These tiny structures are not cells in themselves, but fragments of cells. The number of platelets in the circulating blood has to 400,000 per cubic millimeter. Platelets are essential to blood coagulation ···been estimated at 200 (clotting). When, as a result of injury, blood comes in contact with any tissue other than the lining of the blood vessels, the platelets stick together and form a plug that seals the wound. They then release chemicals that take partin a series of reactions that eventually results in the formation of a clot.



The last step in these reactions is the conversion of a plasma protein called fibrinogen into solid threads of fibrin, which form the clot.

Blood Clotting

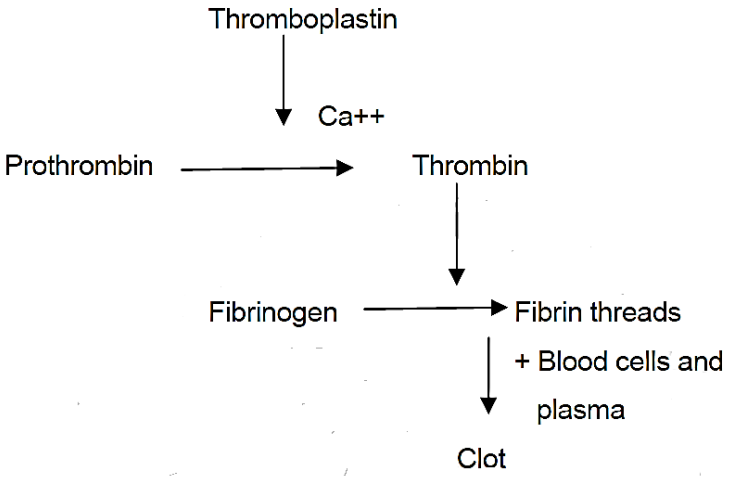
Blood clotting, or coagulation, is a protective device that prevents blood loss when a blood vessel is ruptured by an injury. The many substances necessary for clotting are normally inactive in the blood stream. A balance is maintained between compounds that promote clotting, known as pro coagulants, and those that prevent clotting known as anticoagulants. In addition, there are also chemicals in the circulation that act to dissolve clots. Under normal conditions the substances that prevent clotting prevail. However, when an injury occurs, the procoagulants are activated and a clot is formed.

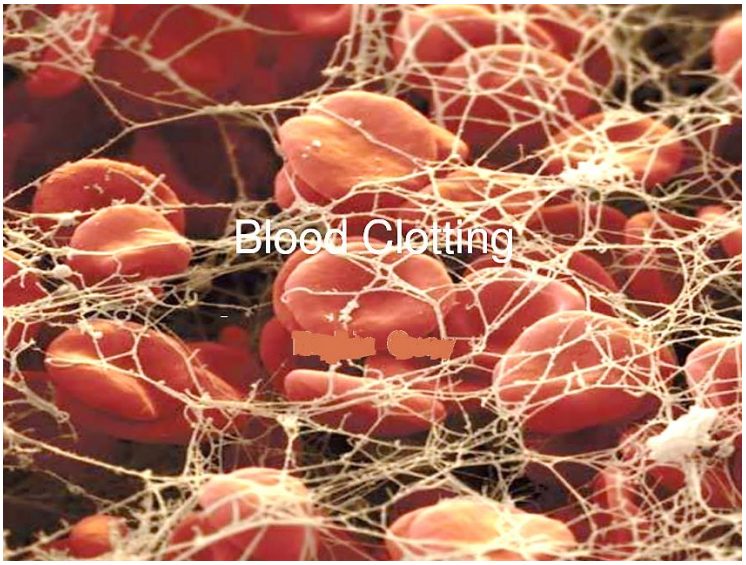
Basically, the clotting process consists of the following essential steps :

1. The injured tissues release thromboplastin, a substance that triggers the clotting mechanism.

2. Thromboplastin reacts with certain protein factors and calcium ions to form prothrombin activator, which in turner acts with calcium ions to convert the prothrombin to thrombin.

3. Thrombin, in turn, converts soluble fibrinogen into insoluble fibrin. Fibrin forms a network of threads that entraps red blood cells and platelets to form clot.





Blood Groups

If for some reason the amount of blood in the body is severely reduced, through hemorrhage or disease, the body cells suffer from lack of oxygen and food. The obvious measure to take in such an emergency is to inject blood from another person into the veins of the patient, a procedure **called transfusion.**

The patient's plasma may contain substances called antibodies that can cause the red cells of the donor's blood to become clumped, a process **called agglutination**. Alternatively, the donor's red blood cells may rupture and release their hemoglobin; such cells are said to be hemolysed, and the .resulting condition can be very dangerous

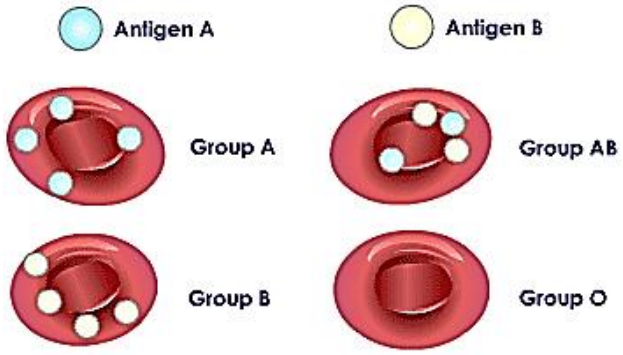
These reactions are determined largely by certain proteins, called antigens, on the surface .membrane of the red blood cells

There are many types of these proteins but only two groups are particularly likely to cause a transfusion reaction, the so-**called A and B antigens and the Rh factor.**

- **Four blood types involving the A and B antigens have been recognized: A, B, AB, and O.** these letters indicate the types of antigen present on the red cells, with O indicating that neither A nor B antigen is present. It is these antigens ton the donor's red fells that react with the antibodies in the patient's plasma and cause a transfusion reaction.
- **Blood serum containing antibodies** that can agglutinate and destroy red cells that have A antigens on the surface is called anti-A serum; blood serum containing antibodies that can destroy red cells with B antigen on the surface is called anti-B serum.

These sera are used to test for blood type Persons with type O blood are said to be universal donors because they lack the AB red cell antigens and in an emergency their blood can be given to anyone. Type AB individuals are called universal recipients, since their blood contains no antibodies to agglutinate red cells and they can therefor receive blood from most donors .

Usually a person can safely give blood to any person with the same blood type. However, because of other factors that may be present in the blood, determination of blood type must be accompanied by additional tests (cross matching) for compatibility before a transfusion is given.



The Rh factor

Rh factor is another red cell antigen that determines the blood group.

Those individuals who possess this antigen in their red cell surface are said to be **Rh positive.**

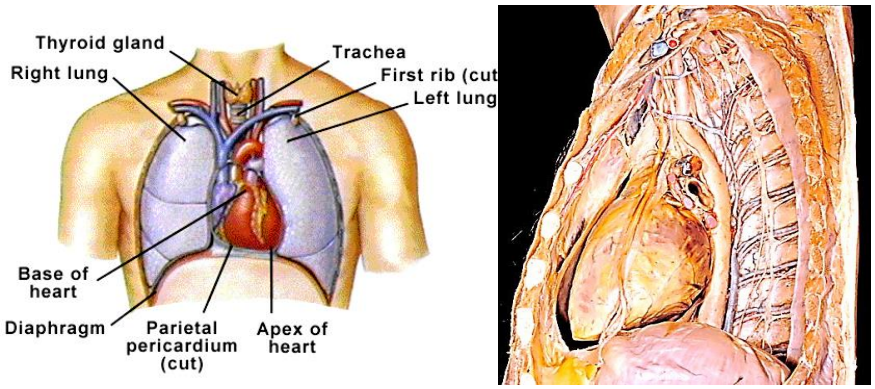
Those who lack this antigen are said to be **Rh negative.**

If Rh positive blood is given to an Rh negative person, he or she may become sensitized to the protein in the Rh positive blood. The sensitized person's blood cells may then produce antibodies to the "foreign" Rh antigens and destroy the transfused red cells.

The Heart

The heart is a muscular pump that drives the blood through the blood vessels. Slightly bigger than a fist, this organ is located between the lungs in the center and a bit to the left on the midline of the body. The importance of the heart has been recognized for centuries. The fact that its rate of beating is affected by the emotions may be responsible for the very frequent references to the heart in song

and poetry. However , the vital functions of the heart and its disorders are of more practical importance to us.



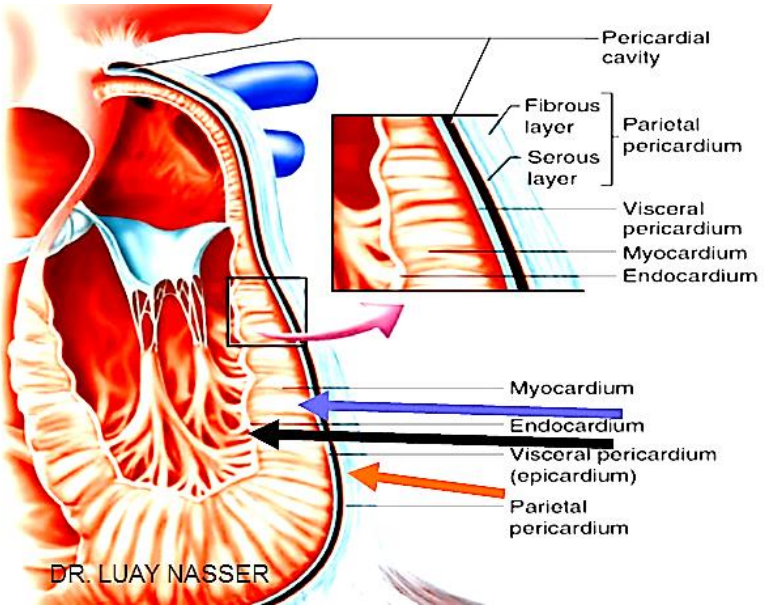
Structure of the Heart

The heart is a hollow organ the walls of which are formed of three different layers. The heart wall has three tissue layers :

1. The **endocardium** is a very thin smooth layer of cells that resembles squamous epithelium. This membrane lines the interior of the heart. The valves of the heart are formed by reinforced folds of this material.

2. The **myocardium**, the muscle of the heart, is the thickest layer.

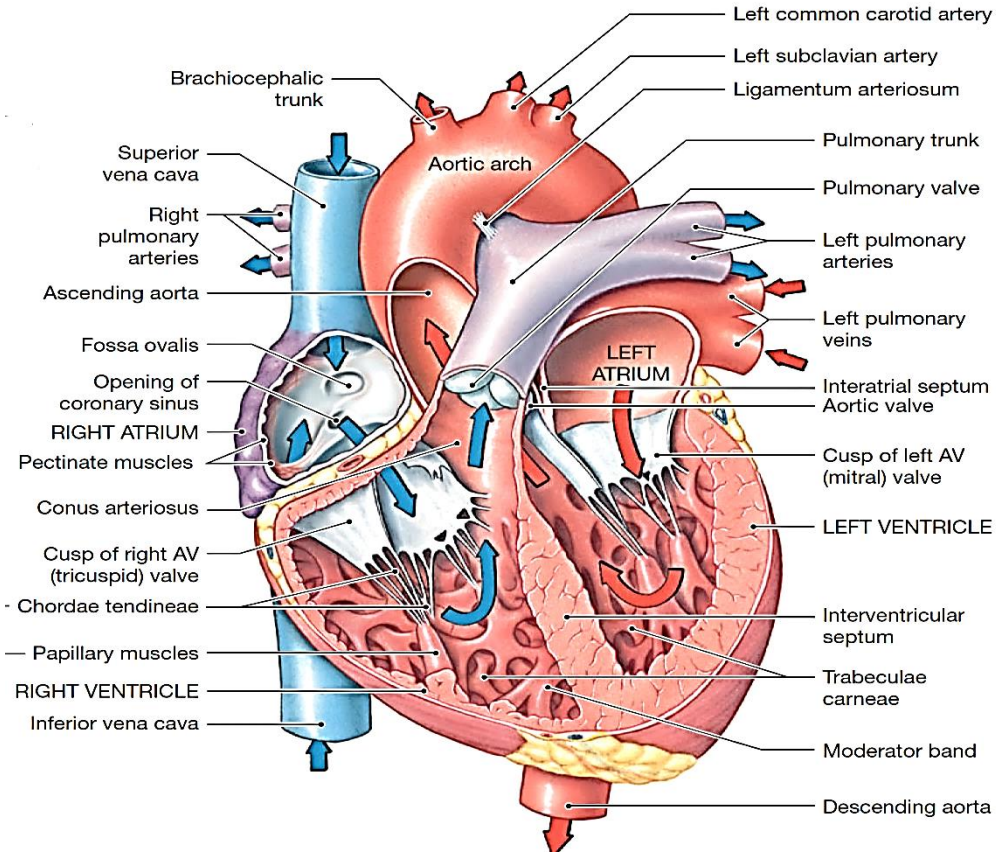
3. The **epicardium** forms the thin outermost layer of the heart wall and is continuous with the serous lining of the fibrous sac that encloses the heart. These membranes together make up the pericardium. The serous lining of the pericardial sac is separated from the epicardium on the heart surface by a thin fluid- filled space.



- **Two Hearts and a Partition**

Physicians often refer to the right heart and the left heart. This is because the human heart is really a double pump. The two sides are completely separated from each other by a partition called the septum. The upper part of this partition is called interatrial septum; while the larger the lower

portion is called interventricular septum. The septum, like the heart wall, consists largely of myocardium.



- **Four Chambers**

On either side of the heart are two chambers, one a receiving chamber (atrium) and the other a pumping chamber (ventricle):

1.The right atrium is a thin-walled chamber that receives the blood retuning from the body tissues. This blood, which is low in oxygen, is carried in the veins, the blood vessels leading to the heart from the body tissues.

2. The right ventricle pumps the venous blood received from the right atrium and sends it to the lungs.

3. The **left atrium** receives blood high in oxygen content as it returns from the lungs.

4. The **left ventricle**, which has the thickest walls of all, pumps, oxygenated blood to all parts of the body. This blood goes through the arteries, the vessels that take blood from the heart to the tissues.

- **Four Valves**

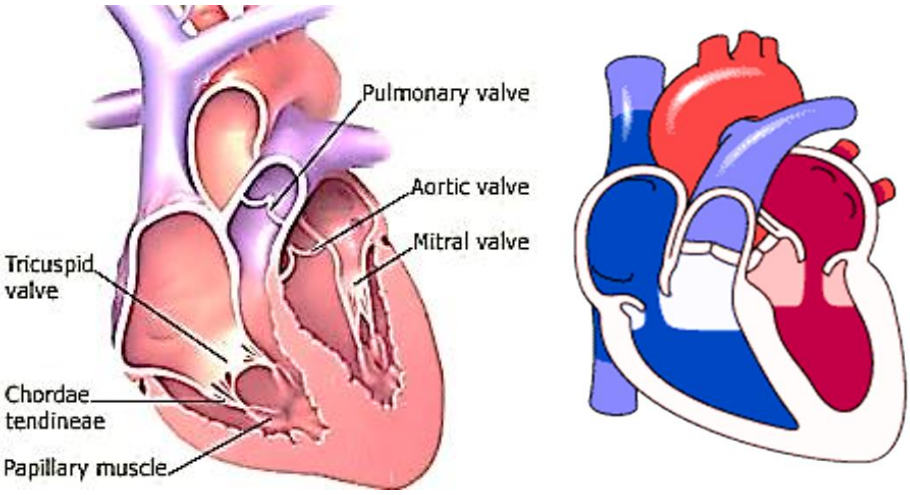
Since the ventricles are the pumping chambers, the valves, which are all one way, are located at the entrance and the exit of each ventricle. The entrances valves are the **atrioventricular valves,** while the exit valves are the **semilunar valves.**

- **Semilunar means "resembling a half moon." Each valve has a specific name, as follows:**

1. The right atrioventricular valve also is known as the **tricuspid valve**, since it has three cusps, or flaps, that open and closes. When this valve is open, blood flows freely from the right atrium into

the right ventricle. However, when the right ventricle begins to contract, the valve closes so that blood cannot return to the right atrium; this ensures forward flow into the pulmonary artery.

2. The left atrioventricular valve is the bicuspid valve, but it is usually referred to as the mitral valve. It has two rather heavy cusps that permit blood to flow freely from the left atrium into the left ventricle. However, the cusps close when the left ventricle begins to contract; this prevents blood from returning to the left atrium and ensures the forward flow of blood into the aorta. Both the tricuspid and mitral valves are attached by means of thin fibrous threads to the wall of the ventricles. The function of these threads, called the chordae tendineae, is to keep the valve flaps from flipping up into the atria when the ventricles contract and thus causing a backflow of blood.

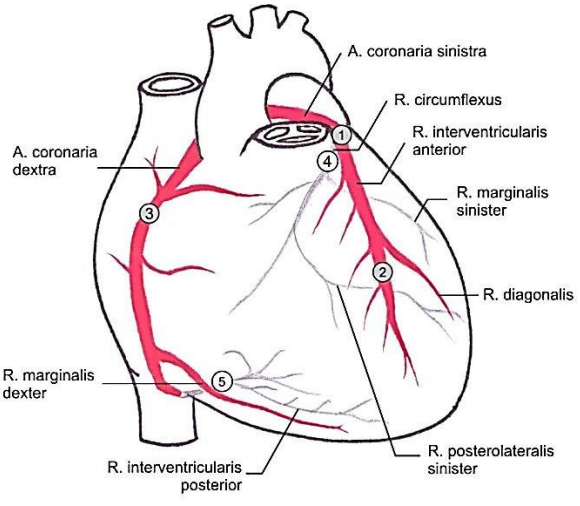


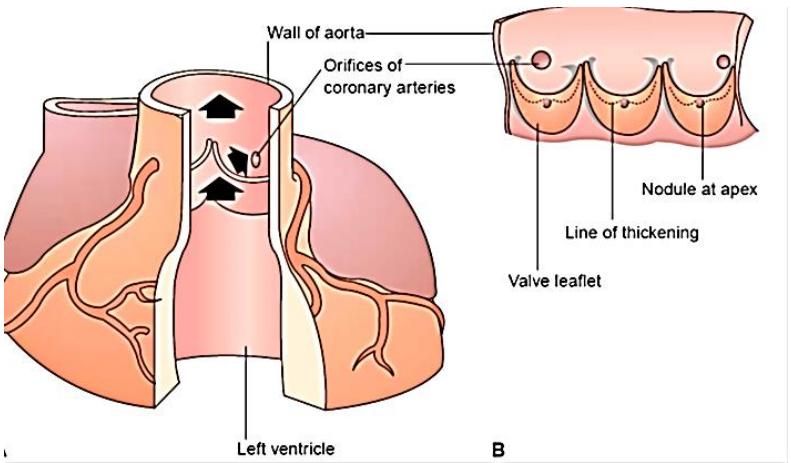
3. The pulmonic (semilunar)valve is located between the right ventricle and the pulmonary artery that leads to the lungs. As soon as the right ventricle has finished emptying itself, the valve closes .in order to prevent blood on its way to the lungs from returning to the ventricle

4. The aortic (semilunar) valve is located between the left ventricle and the aorta. Following contraction of the left ventricle, the aortic valve closes to prevent the flow of blood back from the .aorta to the ventricle

Blood Supply to the Myocardium

Although blood flows through the heart chambers, only the endocardium comes into contact with it. Therefore, the myocardium must have its own blood vessels to provide oxygen and nourishment and to remove waste products. The arteries that supply blood to the muscle of the heart are called the **right and left coronary arteries**.





These arteries, which are the first branches of the aorta, arise just above the aortic semilunar valve. They receive blood when the heart relaxes. After passing through capillaries in the myocardium,

blood drains into the cardiac veins and finally into the coronary (venous) sinus for return to the right atrium.

Physiology of the Heart

Physiology of Cardiac Muscle

The heart is composed of three major types of cardiac muscle:

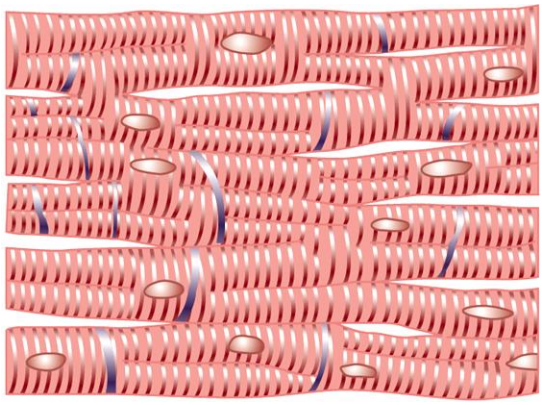
1. *Atrial muscle,*
2. *Ventricular muscle, and*
3. *Specialized excitatory and conductive muscle fibers.*

The atrial and ventricular types of muscle contract in much the same way as skeletal muscle, except that the duration of contraction is much longer

Cardiac Muscle as a Syncytium.

The dark areas crossing the cardiac muscle fibers in are called intercalated discs; they are actually cell membranes that separate individual cardiac muscle cells from one another.

That is, cardiac muscle fibers are made up of many individual cells connected in series and in parallel with one another.



❖ The Work of the Heart

Although the right and left side of the heart are separated from each other, they work together. The blood is squeezed through the chambers by a contraction of heart muscle beginning in the thin-walled upper chambers, the atria, followed by a contraction of the thick muscle of the lower chambers, the ventricles. This active phase is **called systole**, and in each case it is followed by a resting period **known as diastole**. The contraction of the walls of the atria is completed at the time the contraction of the ventricles begins. Thus, the resting phase (diastole) begins in the atria at the same time as the contraction (systole) begins in the ventricles. After the ventricles have emptied, both chambers are relaxed for a short period of time as they fill with blood. Then another beat begins with contraction of the ventricles.

This sequence of heart relaxation and contraction **is called the cardiac cycle.**

Each cycle takes an average of 0.8 seconds. Cardiac muscle tissue has several unique properties.

One of these is the interconnection of the muscle fibers. The fibers are interwoven so the stimulation that causes the contraction of one fiber results in the contraction of the whole group. This plays an important role in the process of conduction and the working of the heart muscle.

Another property of heart muscle is its ability to adjust contraction strength to the amount of blood received. When the heart chamber is filled and the wall stretched (within limits), the contraction is strong. As less blood enters the heart, the contraction becomes weaker. As more blood enters the heart, as occurs during exercise, the muscle contracts, with greater strength so push the larger volume of blood out into the blood vessels.

The volume of blood pumped by each ventricle in 1 minute is **termed the cardiac output.**

- **It is determined by the volume of blood ejected from the ventricle with each beat–the stroke volume**–and the number of beats of the heart per minute–the heart rate. The cardiac output averages 5 litres/minute for an adult at rest.

❖ The Conduction System of the Heart

The cardiac cycle is regulated by specialized areas in the heart wall that forms the conduction system of the heart. Two of these areas are tissue mass called nodes; the third is a group of fibers called the atrioventricular bundle. The sinoatrial node, which is located I the upper wall of the right atrium an initiates the heart beat, is called the pacemaker. The second node, located in the interatrial septum at the bottom of the right atrium, is called the atrioventricular node. The atrioventricular bundle ,also known as the bundle of His, is located at the top of the interventricular septum; it has branches that extend to all parts of the ventricle walls. Fibers travel first down both sides of the interventricular septum in groups called the right and left bundle branches. Smaller Purkinje fibers then travel in a branching network throughout the myocardium of the ventricles .

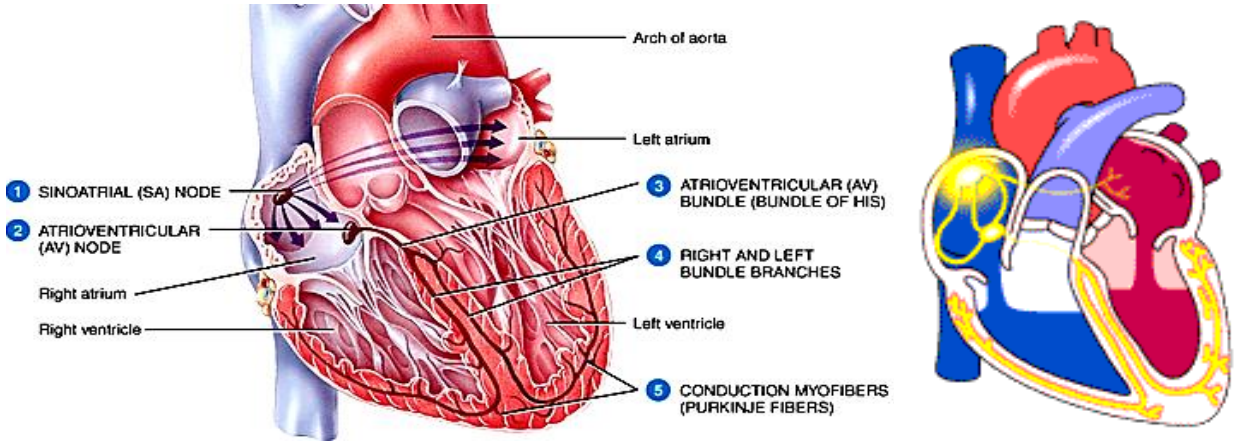
The order in which the impulses travel is as follows:

1. The sinoatrial node generates the electric impulse that begins the heart beat.

2. the excitation wave travels throughout the muscle of each atrium, causing it to contract.

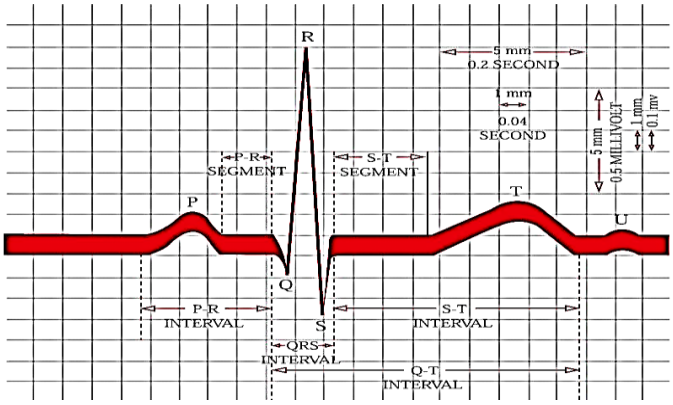
3. the atrioventricular node is stimulated. The relatively slower conduction through this node allows time for the atria to contract and complete the filling of the ventricles.

4. The excitation wave travels rapidly through the bundle of His and then throughout the ventricular walls by means of the bundle branches and Purkinje fibers. The entire musculature of the ventricles contracts practically at once.

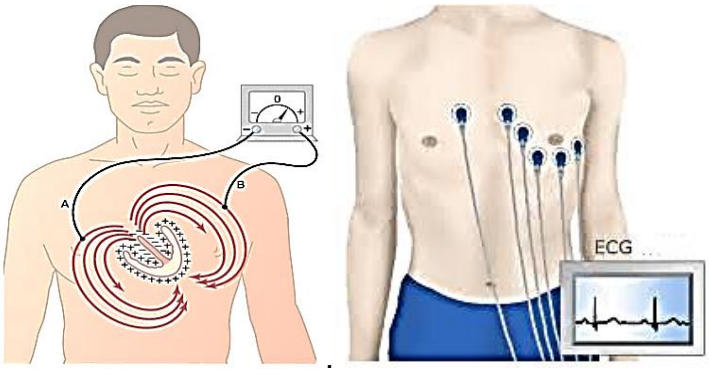


As a safety measure, a region of the conduction system other than the sinoatrial node fails, but it does so at a slower rate.

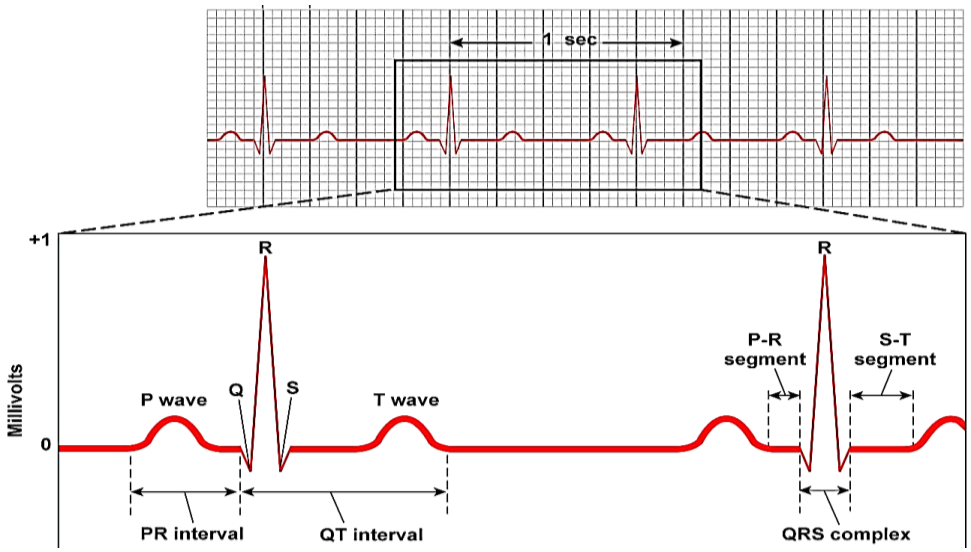
Electrocardiogram(ECG)



When the cardiac impulse passes through the heart, electrical current also spreads from the heart into the adjacent tissues surrounding the heart. A small portion of the current spreads all the way to the surface of the body. If electrodes are placed on the skin on opposite sides of the heart, electrical potentials generated by the current can be recorded; the recording is known as an electrocardiogram



The normal electrocardiogram is composed of a P wave, a QRS complex, and a T wave. The QRS complex is often, but not always, three separate waves: the Q wave, the R wave, and the S wave.



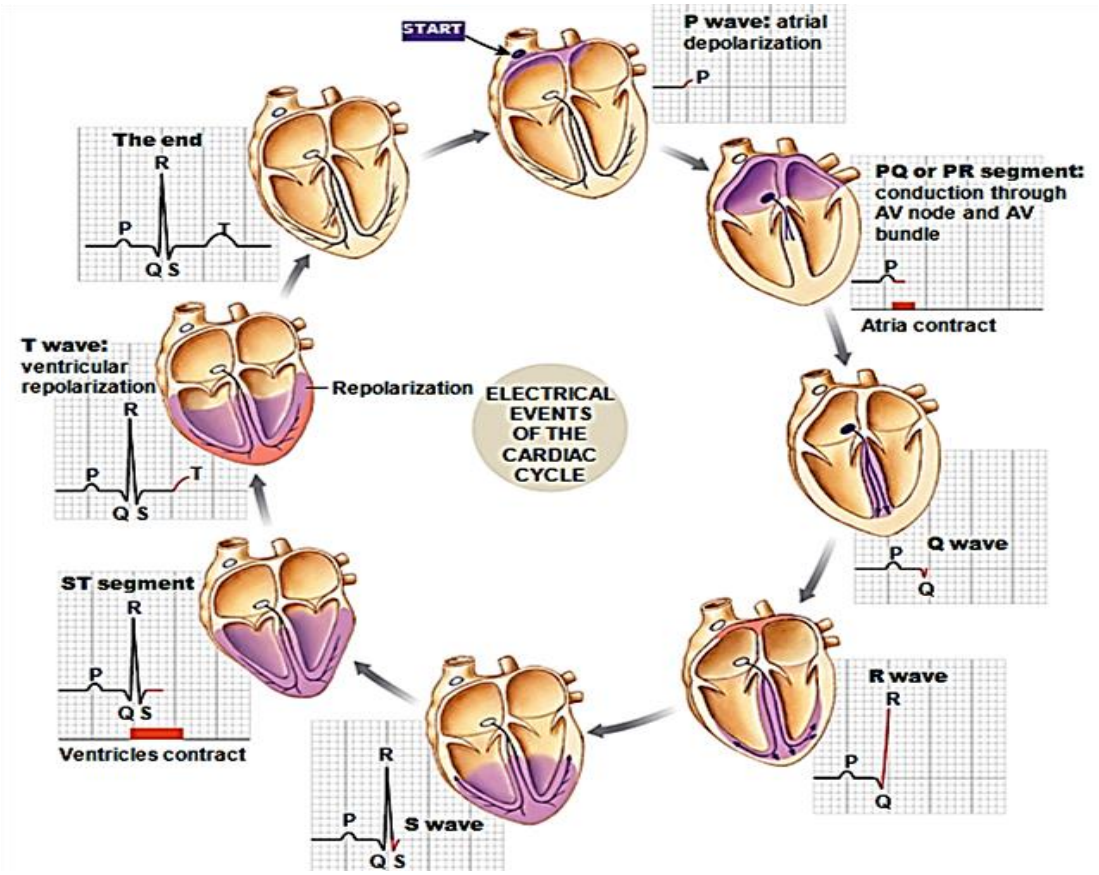
The P wave is caused by electrical potentials generated when the atria depolarize before atrial contraction begins.

The QRS complex is caused by potentials generated when the ventricles depolarize before contraction, that is, as the depolarization wave spreads through the ventricles.

Therefore, both the P wave and the components of the QRS complex are depolarization waves.

The T wave is caused by potentials generated as the ventricles recover from the state of depolarization. This process normally occurs in ventricular muscle 0.25to 0.35 second after depolarization, and the T wave is known as a repolarization wave.

Thus, the electrocardiogram is composed of both depolarization and repolarization waves.



- **The Cardiac Cycle**

The cardiac events that occur from the beginning of one heartbeat to the beginning of the next are called the cardiac cycle.

Each cycle is initiated by spontaneous generation of an action potential in the sinus node

The cardiac cycle consists of a period of relaxation called diastole, during which the heart fills with blood, followed by a period of contraction called systole. During each cardiac cycle two chief sounds can be heard by (the aid of a stethoscope in man).

The 1st called systolic sound, is due to the contraction of the ventricular muscles and the closure of the auriculoventricular valves.

The second sound, called diastolic sound, is due to the closure of the semilunar valves during the ventricular diastole.

Cardiac output (CO) is the volume of blood pumped by each ventricle per minute.

Cardiac output depends on the heart rate ((beats per minute)the stroke volume (volume of blood pumped per beat or stroke).

- **Stroke volume:** Amount of blood pumped by one ventricle during a contraction EDV – ESV = stroke volume
- **Cardiac output Volume** of blood pumped by one ventricle in a given period of time
  Cardiac output = heart rate × stroke volume = 70 beats/min × 70 ml/beat =4900 ml/min ≅ 5 L/min

The cardiac output is the amount of blood pumped to the peripheral circulation per minute. It depends upon:

End Diastolic Volume(EDV):the volume of blood that enter the ventricle during diastole =110 -120 ml

When heart rate increase ,the duration of diastole is shortened and EDV is reduced

End Systolic Volume (ESV): : The amount of blood left in the ventricle at the end of systole when ejection is complete is the end systolic Volume (ESV) =40 -50 ml

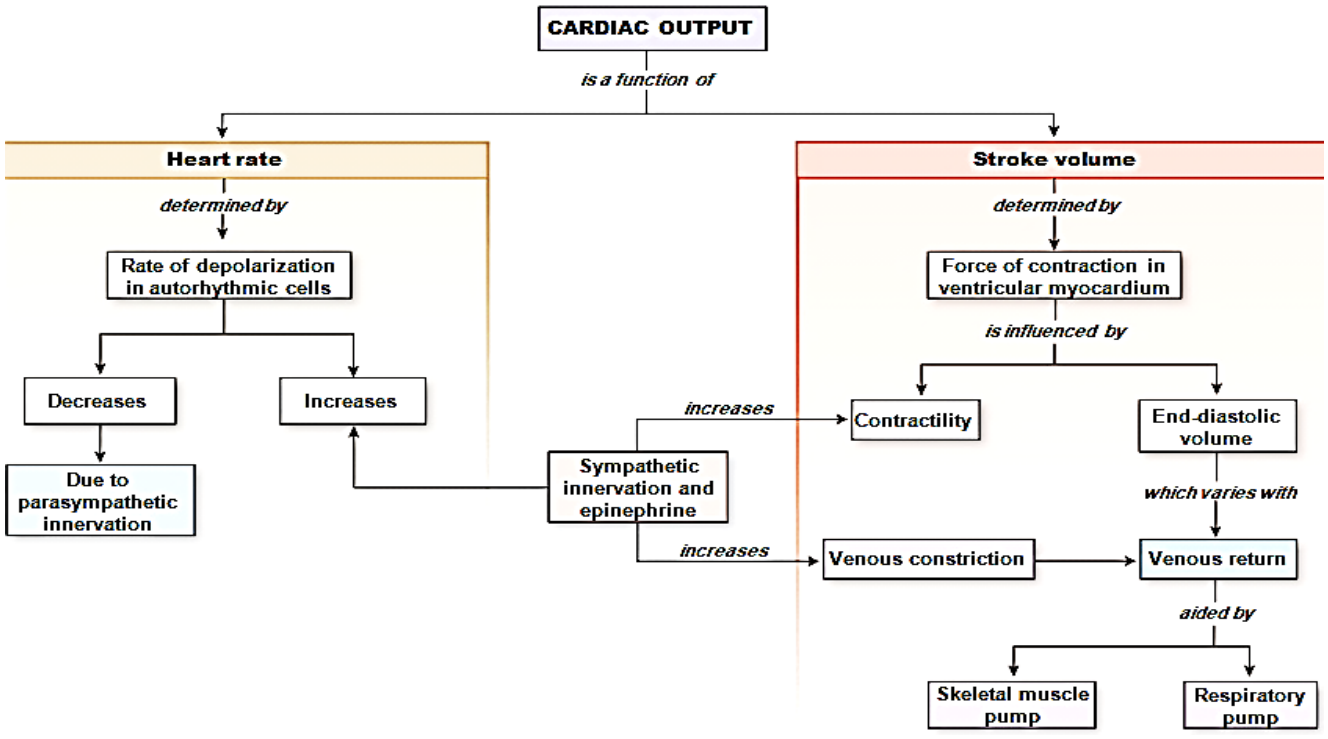
Diastole: period of relaxation Systole :period of contraction **SV=EDV-ESV= 70ml**

The contractility of the myocardium determines the ejection fraction of the heart, which is the ratio of the volume of blood ejected from the left ventricle per beat (stroke volume) to the volume of blood in the left ventricle at the end of diastole (end diastolic volume): Ejection fraction = 60%:

During exercise when sympathetic stimulation to the heart is increased, the ejection fraction may increase to more than 80%resulting in greater stroke volume and cardiac output.

The degree of tension on the muscle when it begins to contract, which is called the preload,

The tension needed by the muscle to exerts its contractile force, which is called the afterload.



❖ **Phases of Cardiac Cycle**

☺ **1. Isometric (isovolumetric contraction)**

All valves closed, Ventricle contract isometrically, Ventricular volume is constant, Ventricular pressure increase.

☺ **2 Ventricular systole :**

A-Maximal (rapid) ejection (70% of blood pumped) :Semilunar valves open (when Lt. ventricular pressure reach 80cmHg aortic valve open) blood pumped through aorta ,ventricular pressure increase.

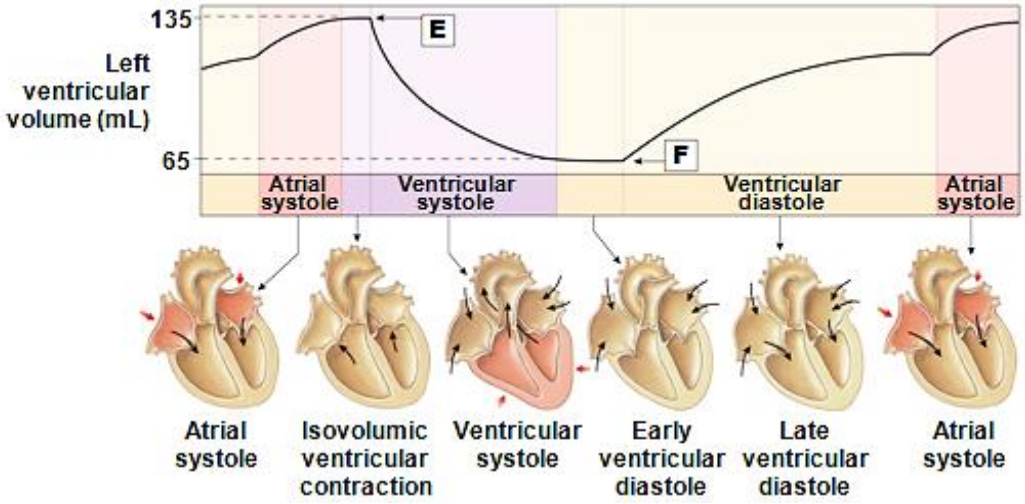
B-Reduced ejection (30% of blood pumped):Blood pumped through aorta more slowly ,ventricular pressure decrease.

☺ **3.Isometric Relaxation:**

All valves Closed ventricle relax isometrically, ventricular volume is constant ,ventricular pressure decrease. Lt. ventricular pressure drops to zero

☺ **4. Ventricular Diastole**

A-V valves open, Volume increase in the ventricles, Pressure increase slightly in ventricles



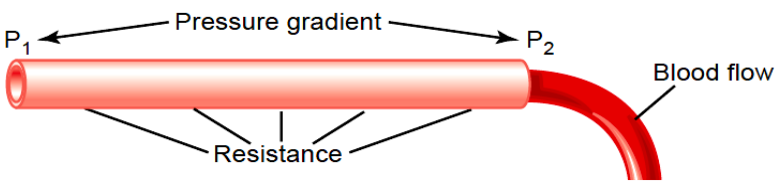
❖ **Frank starling mechanism** :The heart has an inherent, self-regulating mechanism by which it can alter its force of contraction based upon the volume of blood that flows into it.

This intrinsic mechanism , the Frank–Starling law of the heart , states that when ventricular filling is increased, this increased volume of blood stretches the walls of these chambers and, as a result, the ventricles contract more forcefully.

Stroke volume increase as EDV (ending diastolic volume) increases – stretch -> more force

❖ **Blood Flow:** Means simply the quantity of blood that passes a given point in the circulation in a given period of time .

Determined by : pressure difference and vascular resistance Flow =delta pressure /resistance



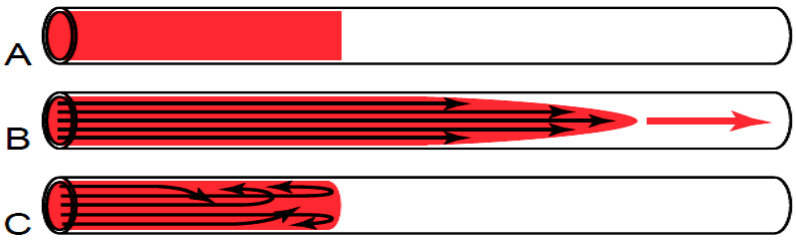
❖ **Types of blood flow:** When blood flows at a steady rate through a long, smooth blood vessel, it flows in streamlines, with each layer of blood remaining the same distance from the vessel wall.
Also, the central most portion of the blood stays in the center of the vessel.

This type of flow is called laminar flow or streamline flow, and it is the opposite of turbulent flow, which is blood flowing in all directions in the vessel and continually mixing within the vessel

Turbulent flow means that the blood flows cross wise in the vessel as well as along the vessel, usually forming whorls in the blood called eddy currents.

A, Two fluids (one dyed red, and the other clear) before flow begins;

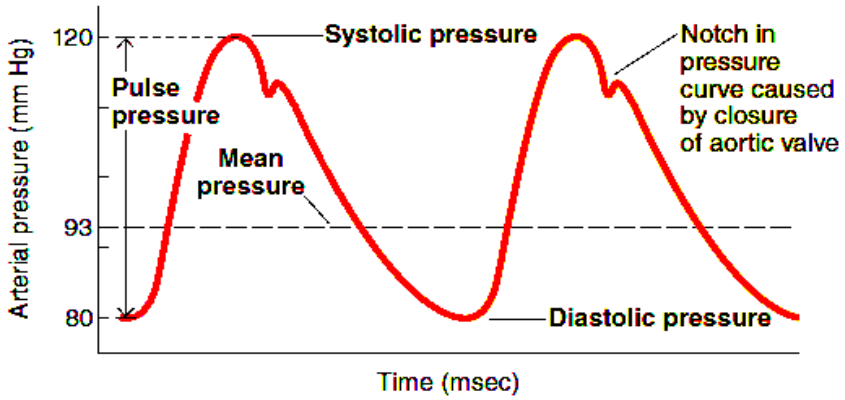
B, the same fluids 1 second after flowbegins; C, turbulent flow, with elements of the fluid moving in a disorderly pattern.



The tendency for turbulent flow increases in direct proportion to the velocity of blood flow, the diameter of the blood vessel, and the density of the blood, and is inversely proportional to the viscosity of the blood,

❖ **Blood Pressure**: Actually, blood pressure means the force exerted by the blood against any unit area of the vessel wall. When one says that the pressure in a vessel is 50 mm Hg, one means that the force exerted is sufficient to push a column of mercury against gravity up to a level 50 mm high. If the pressure is 100 mm Hg, it will push the column of mercury up to 100 millimeters. Occasionally, pressure is measured in centimeters of water (cmH2O).Pulse pressure=systolic – diastolic pressure.

Mean Arterial Pressure: The average of arterial pressure measured millisecond by millisecond over a period of time Arterial Pressure = diastolic pressure +1/3Pulse pressure 40%SP =+ 60% DP



❖ **Vascular Distensibility: All blood vessels are distensible**
The ability of blood vessels to accommodate more volume without increasing of pressure

Arteries are less distensible

When the pressure in blood vessels is increased, this dilates the blood vessels and therefore decreases their resistance

Local blood flow control

1- Acute control Rapid changes in local vasodilatation or vasoconstriction of vessels within seconds to minutes

2- Long term regulation: Increase or decrease in physical size and number of vessels supplying tissues within days, weeks or months.

- **Acute control**

Metabolic rate and the availability of O2 to tissue determine the amount of blood that flows to certain tissue

Tow basic theories can explain the mechanism

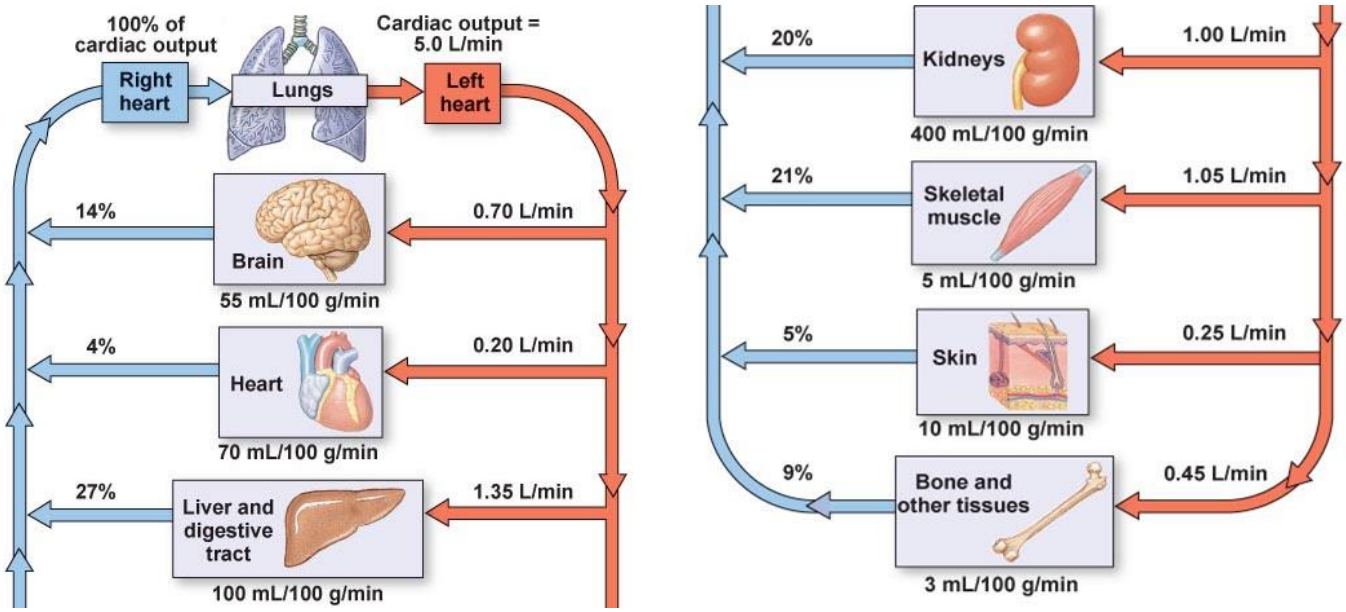
1-Vasodilatory theory: increasing the metabolic rate in the tissue cause vasodilatation of the vessels that supply this tissue

2-O2 lack theory O2 needs increases blood flow to the tissues

- **Long term regulation :**

Gives more complete regulation It is used when long term metabolic demands of tissue change

The arterioles and capillaries increase both in number and size within a few weeks to match the tissue needs



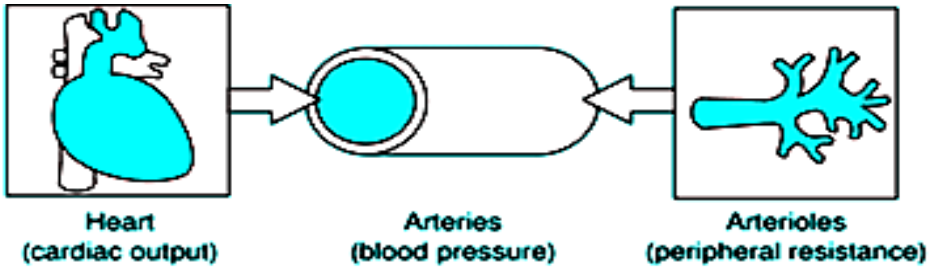
Distribution of cardiac output at rest. The lungs receive all the blood pumped out by the right side of the heart, whereas the systemic organs each receive some of the blood pumped out by the left side of the heart.

The arterial blood pressure reflects the rhythmic ejection of blood from the left ventricle into the aorta.

It rises as the left ventricle contracts and falls as it relaxes.

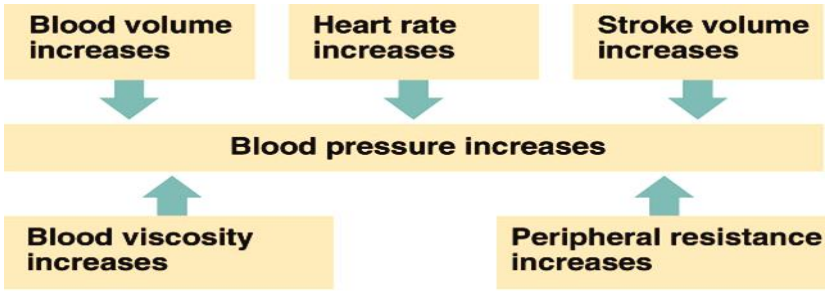
The systolic and diastolic components of blood pressure are determined by the cardiac output and the peripheral vascular resistance and can be expressed as a product of the two (blood pressure = cardiac output × peripheral vascular resistance)

The cardiac output is the product of the stroke volume (amount of blood ejected from the heart with each beat) and the heart rate.
The peripheral vascular resistance reflects changes in the radius of the arterioles as well as the viscosity or thickness of the blood.
The arterioles often are referred to as the resistance vessels
because they can selectively constrict or relax to control the resistance to outflow of blood into the capillaries.

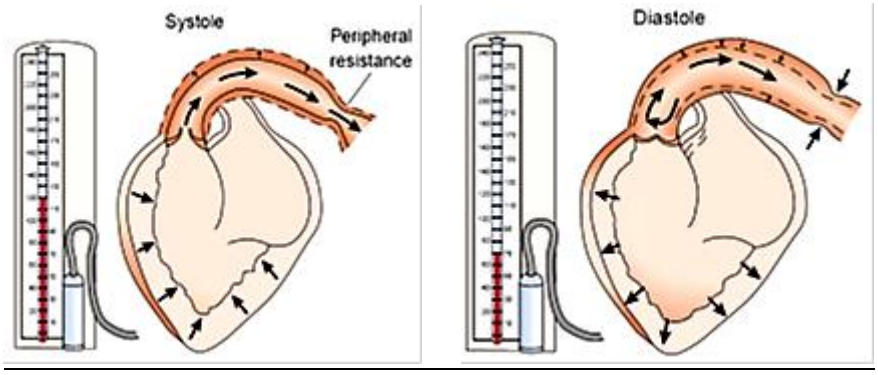


The body maintains its blood pressure by :
1.adjusting the cardiac output to compensate for changes in peripheral vascular resistance,
2.and it changes the peripheral vascular resistance to compensate for changes in cardiac output.
In hypertension and disease conditions that affect blood pressure, changes in blood pressure are influenced by :

1. the stroke volume,
2. the rapidity with which blood is ejected from the heart,
3. the elastic properties of the aorta and large arteries and their ability to accept various amounts of blood as it is ejected from the heart,
4. the properties of the resistance blood vessels that control the runoff of blood into the smaller vessels and capillaries that connect the arterial and venous circulations.



■ **The systolic blood pressure** is largely determined by the characteristics of the stroke volume being ejected from the heart and the ability of the aorta to stretch and accommodate the stroke volume



■ **The diastolic pressure** is largely determined by the energy that is stored in the aorta as its elastic fibers are stretched during systole and by the resistance to the runoff of blood from the peripheral blood vessels.

The pulse pressure is the difference between these two pressures, and the mean arterial pressure reflects the average pressure throughout the cardiac cycle.

Top number (systolic) 120/80 Bottom number (diastolic)

Mechanisms of Blood Pressure Regulation:

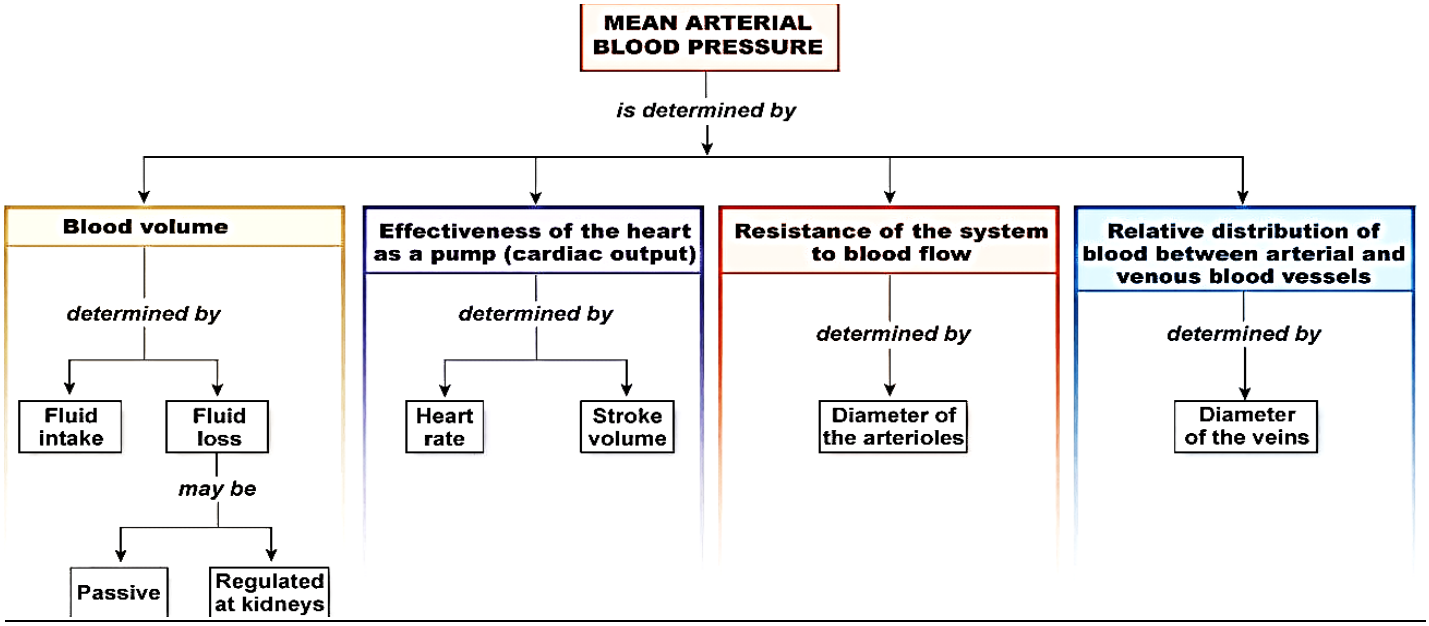
- **Short-Term Regulation**

The mechanisms for short-term regulation of blood pressure, those occurring over minutes or hours, are intended to correct temporary imbalances in blood pressure, such as occur during physical exercise and changes in body position.

These mechanisms also are responsible for maintenance of blood pressure at survival levels during life-threatening situations.

The short-term regulation of blood pressure relies mainly on neural and hormonal mechanisms, the most rapid of which are the neural mechanisms.

► **Factors that Influence Mean Arterial Pressure**



- **Neural Mechanisms:**

The neural control center for the regulation of blood pressure is located in the reticular formation of the lower pons and medulla of the brain where integration and modulation of autonomic nervous system (ANS) responses occur.

o intrinsic circulatory reflexes, extrinsic reflexes, and higher neural control centers.

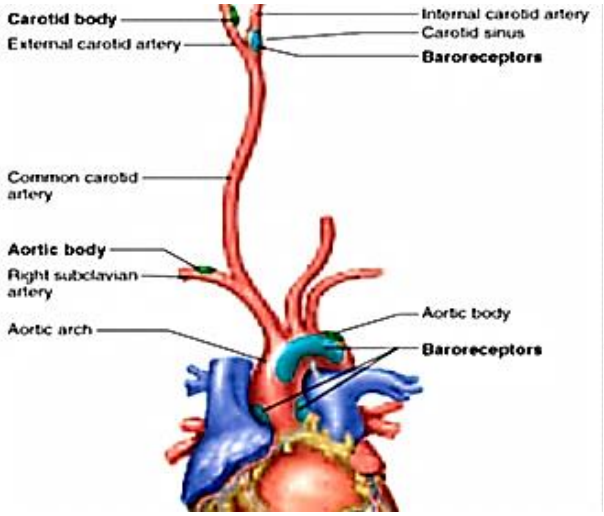
The intrinsic reflexes, including the baroreflex and chemoreceptor-mediated reflex, are located in the circulatory system and are essential for rapid and short-term regulation of blood pressure.

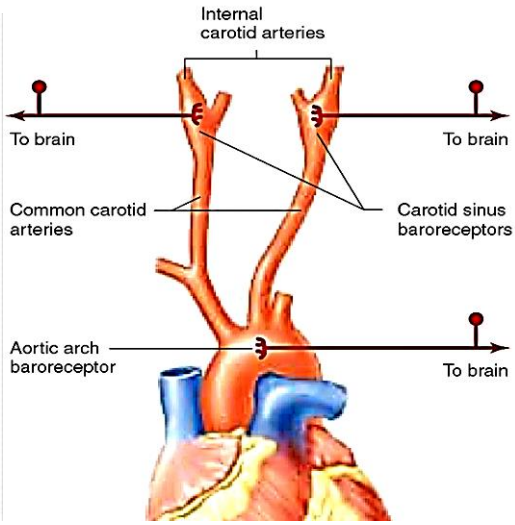
The sensors for extrinsic reflexes are found outside the circulation.

They include blood pressure responses associated with factors such as pain and cold.

Chemoreceptors are located in the carotid bodies and aortic bodies. Baroreceptors are located in the ascending aorta, aortic arch, and carotid sinus.

The structures shown here in the right carotid arteries are repeated in the left carotids.

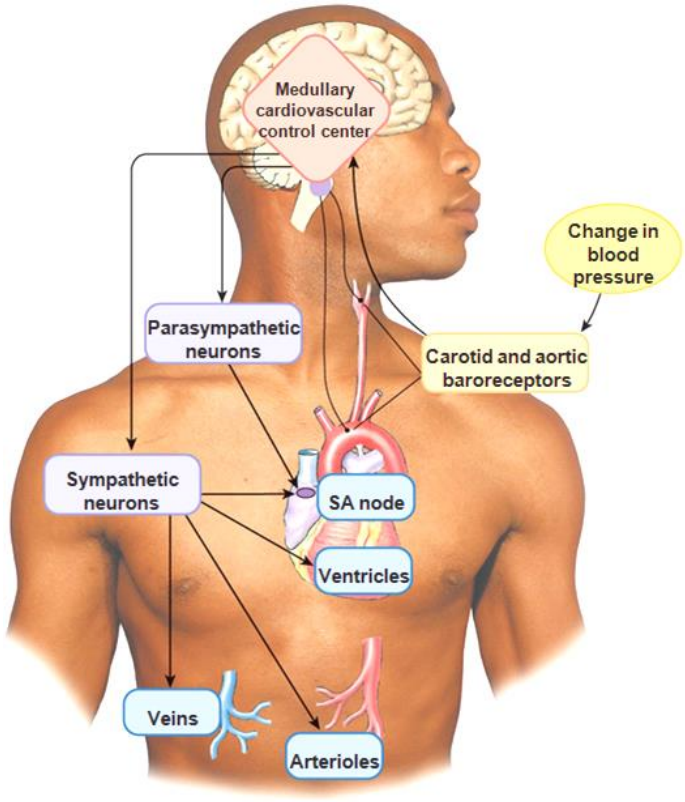


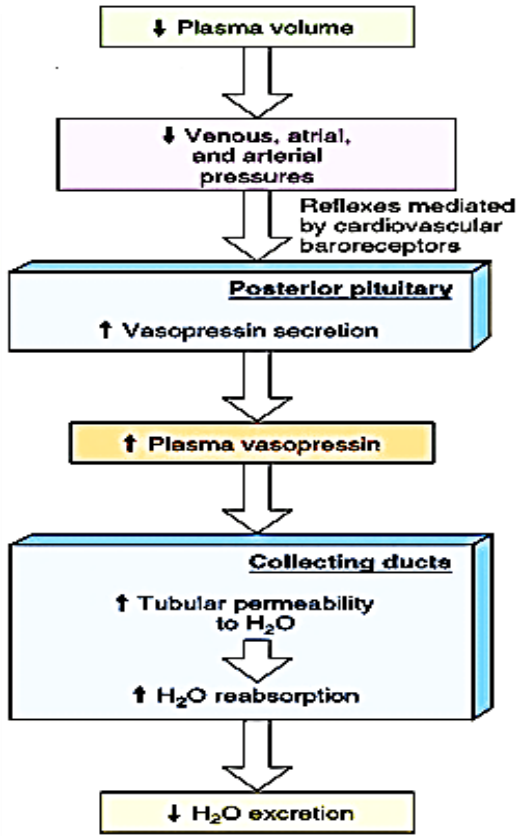


The baroreceptors are pressure sensitive receptors located in the walls of blood vessels and the heart.

The carotid and aortic baroreceptors are located in strategic positions between the heart and the brain .

They respond to changes in the stretch of the vessel wall by sending impulses to cardiovascular centers in the brain stem to effect appropriate changes in heart rate and vascular smooth muscle tone





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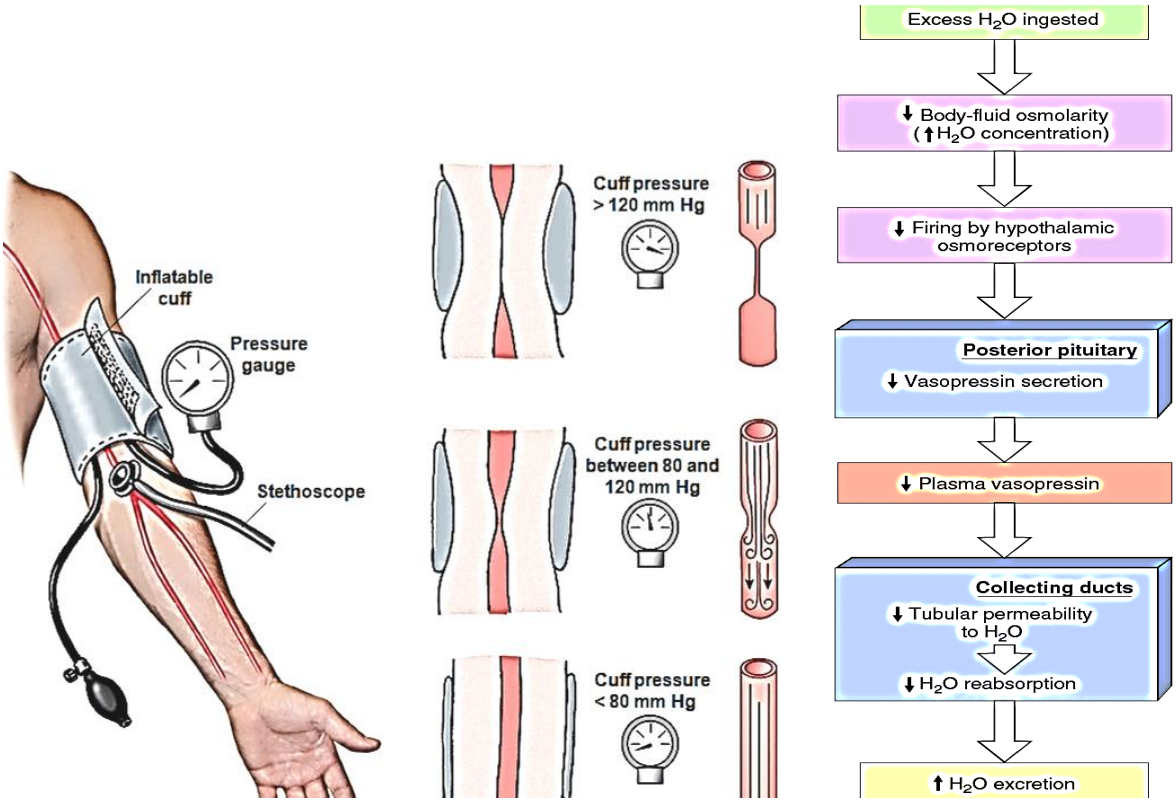
Baroreceptor pathway by which vasopressin secretion is increased when plasma volume is decreased.

The opposite events (culminating in a decrease in vasopressin secretion) occur when plasma volume increases.

Osmoreceptor pathway by which vasopressin secretion is lowered and water excretion raised when excess water is ingested.

The opposite events (an increase in vasopressin secretion) occur when osmolarity increases, as during water deprivation.

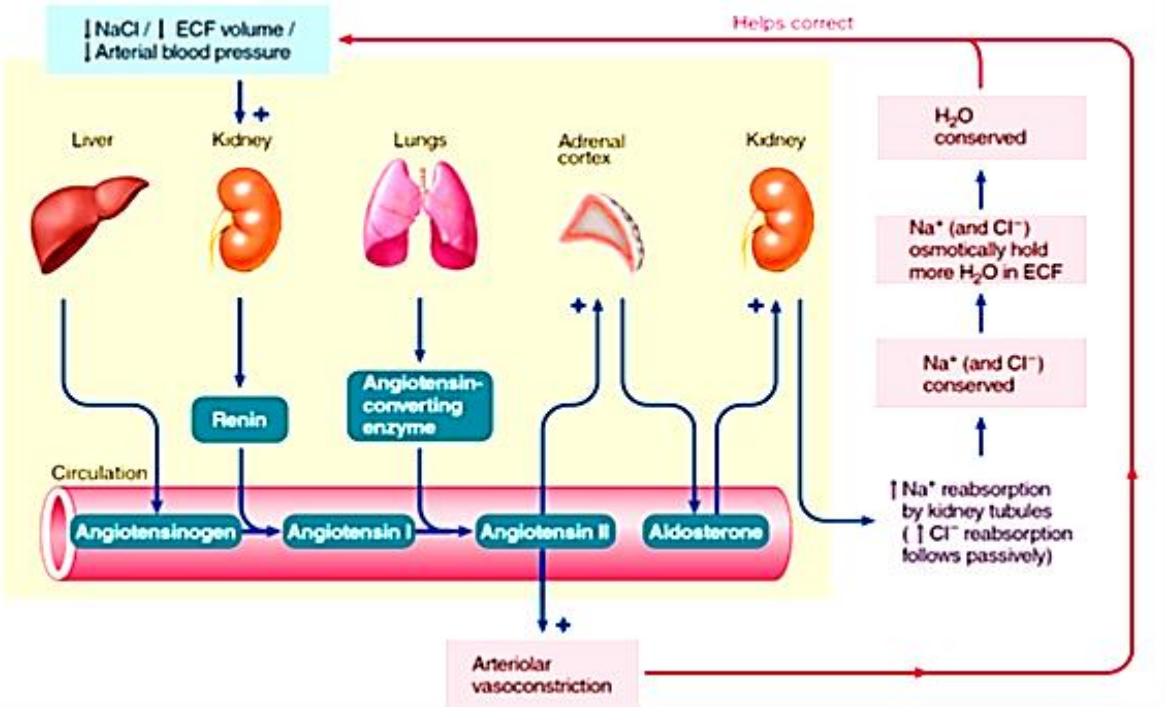


The arterial chemoreceptors are sensitive to changes in the oxygen, carbon dioxide, and hydrogen ion content of the blood.

They are located in the carotid bodies, which lie in the bifurcation of the two common carotids, and in the aortic bodies of the aorta .

Humoral Mechanisms: A number of hormones and humoral mechanisms contribute to blood pressure regulation, including the renin-angiotensin-aldosterone mechanism and vasopressin. Other humoral substances such as epinephrine, a sympathetic neurotransmitter released from the adrenal gland, has the effect of directly stimulating an increase in heart rate, cardiac contractility, and vascular tone.

Summary of the renin-angiotensin system and the stimulation of aldosterone secretion by angiotensin II. Converting enzyme is located on the surface of capillary endothelial cells, particularly in the lungs



Control of the Heart Rate

Although the fundamental beat of the heart originates within the heart itself, the heart rate can be influenced by the nervous system and by other factors in the internal environment.

Stimulation from the sympathetic nervous system increases the heart rate and the stimulation from the parasympathetic nervous system decreases the heart rate. These influences allow the heart to meet changing need rapidly. The heart rate is also affected by such factors as hormones, ions, and drugs in the blood.

- **Heart Rates**

1. Bradycardia is a relatively slow heart rate of less than 60beats/minute. During rest and sleep, the heart may beat less than 60 beats/minute but usually does not fall below50 beats/minute.

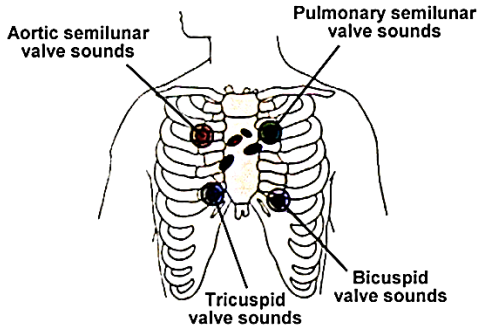
2. Tachycardia refers to a heart rate over 100beats/minute.

3. Sinus arrhythmia is a regular variation in heart rate due to changes in the rate and depth of breathing. It is normal phenomenon.

4. Premature beats, **also called extra systoles** are beats that come in before the expected normal beats. They may occur in normal persons initiated by caffeine, nicotine, or psychological stresses. They are also common in persons with heart disease.

- **Heart Sounds and Murmurs**

The normal heart sounds are usually described by the syllables "lubb" and "dupp." The first is a longer, lower pitched sound that occurs at the start of ventricular systole. It is probably caused by a combination of things, including closure of the atrioventricular valves. The second, or "dupp," sound is shorter and sharper. It occurs at the beginning of ventricular relaxation and is due in large part to sudden closure of the semilunar valves. Some abnormal sounds **called murmurs** are usually due to faulty action of the valves.



For example, if the valves fail to close tightly and blood leaks back, a murmur is heard. Another condition giving rise to an abnormal sound is the narrowing (stenosis) of a valve opening. The many

conditions that can cause abnormal heart sounds include congenital defects, disease, and physiological variations. A murmur due to rapid filling of the ventricles is called a **functional (flow) murmur;** such a murmur is not abnormal.

An abnormal sound caused by any structural change in the heart or the vessels connected with the heart is called an **organic murmur**.

Blood Vessels

❖ Functional classification

The blood vessels, together with the four chambers of the heart, from a closed system for the flow of blood; only if there is an injury to some part of the wall of this system does any blood escape. On the basis of function, blood vessels may be **classified into three groups**

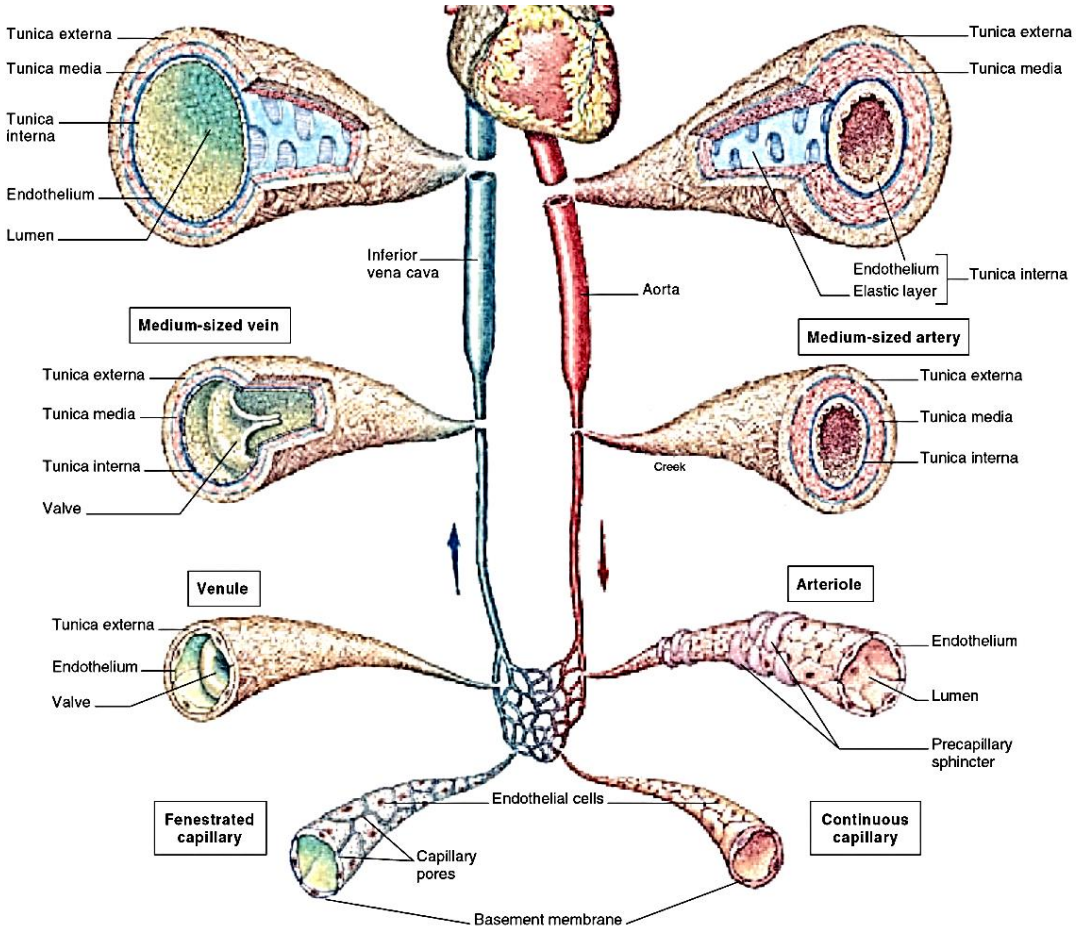
1. **Arteries** carry blood from the ventricles (pumping chambers) of the heart out to the capillaries .in organs and tissue. The smallest arteries are called arterioles

2. Veins drain capillaries in the tissues and organs and return the blood to the heart. The smallest veins are the **venules**.

3. Capillaries allow for exchanges between the blood and body cells, or between the blood and air .in the lung tissues. The capillaries connect the arterioles and venules

Structure of blood vessels

Arteries, veins and capillaries differ in structure. Three coats or layers are found in both arteries and veins



The outer most layer is called the tunica externa. Note smooth muscle is found in the middle layer or tunica media of arteries and veins. However, the muscle layer is much thicker in arteries than in veins. Why is this important? Because the thicker muscle layer in the artery wall is able to resist great pressures generated by ventricular systole**. In arteries, the tunica medial** plays a

critical role in maintaining blood pressure and controlling blood distribution in the body. This isa smooth muscle, so it is controlled by the autonomic nervous system

A thin layer of elastic and white fibrous tissue covers an inner layer of endothelial cells called the tunica interna in arteries and veins. **The tunica interna** is actually a single layer of squamous .epithelial cells called endothelium that lines the inner surface of the entire circulatory system

Veins have a unique structural feature not present in arteries. They are equipped with one way valves that prevent the backflow of blood. When a surgeon cuts into the body, only arteries, arterioles, veins, and venules can be seen. Capillaries cannot be seen because they are microscopic. **The most important structural feature of capillaries is their extreme thinness−only** one layer of flat،endothelial cells composes the capillary membrane. Instead of three layers or coats, the capillary wall is composed of only one−the tunica interna.

Substances such as glucose, oxygen and wastes can quickly pass through it on their way to or from the cells. Smooth muscle cells that are called precapillary sphincters guard the entrance to the capillary and determine into which capillary blood will flow.,

Location of common arteries and veins

Names of systemic arteries

- **The Aorta and Its Parts**

The aorta is the largest artery in the body. It extends upward and to the right from the left ventricle. Then it curves backward and to the left. It continues down behind the heart just in front of the vertebral column, through the diaphragm, and into the abdomen .

The aorta is one continuous artery, but it may be divided into sections:

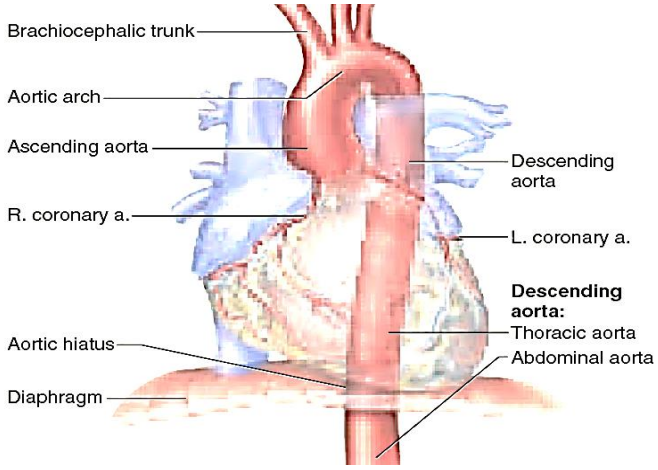
1. The ascending aorta is near the heart and inside pericardial sac.

2. The aortic arch curves from the right to the left and also extends backward.

3. The thoracic aorta lies just in front of the vertebral column behind the heart and in the space behind the pleura.

4. The abdominal aorta is the longest section of the aorta, spanning the abdominal cavity.

The thoracic and abdominal aorta together makes up the descending aorta



Branches of the Ascending Aorta

The first, or ascending, part of the aorta has two branches near the heart, called the left and right coronary arteries that supply the heart muscle. These from a crown around the base of the heart and give off branches to all parts of the myocardium.

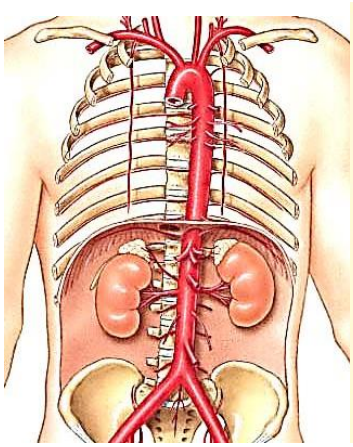
Branches of the Aortic Arch

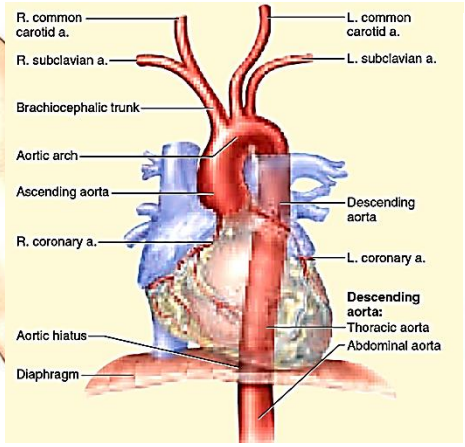
The arch of aorta, located immediately beyond the ascending aorta, gives off three large branches.

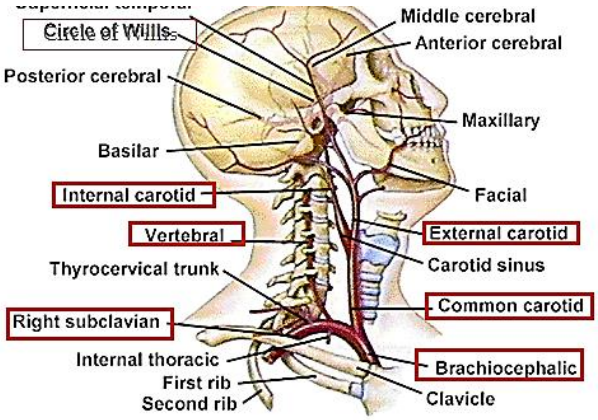
1. The brachiocephalic trunk is a short artery formerly called the innominate. Its name means that is supplies the head and the arm. After extending upward somewhat less than 5cm (2 inches), it divides into the right subclavian artery, which supplies the right side of the head and the neck.

2. The left common carotid artery extends upward from the highest part of the aortic arch. It supplies the left side of the neck and the head.

3. The left subclavian artery extends under the left collar bone(clavicle) and supplies the left upper extremity. This is the last branch of the aortic arch.

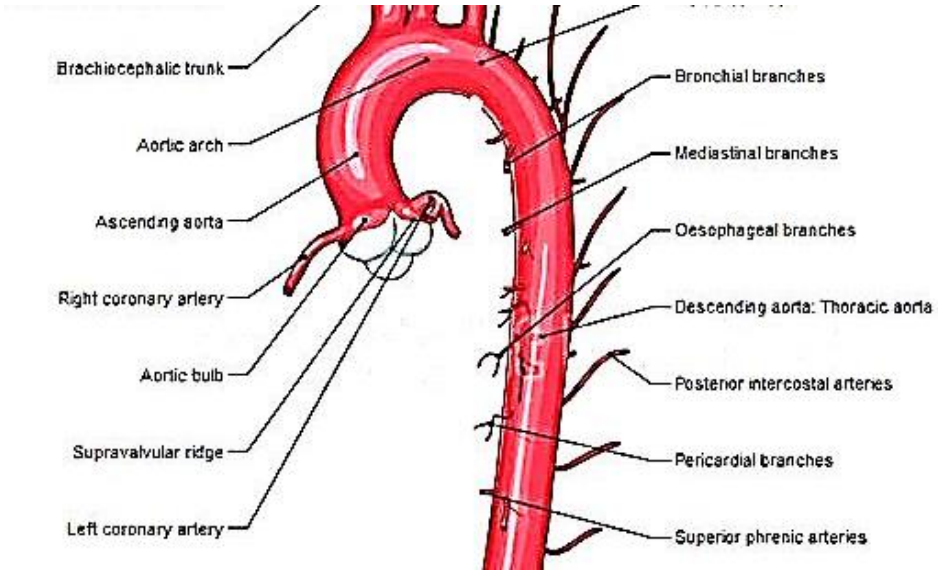






Branches of the Thoracic Aorta

The third part of the aorta supplies branches to the chest wall, to the esophagus, and to the bronchi and their treelike subdivisions in the lungs. There are usually nine to ten pairs of intercostal arteries that extend between the ribs, sending branches to the muscles and other structures of the chest wall.



Branches of the Abdominal Aorta

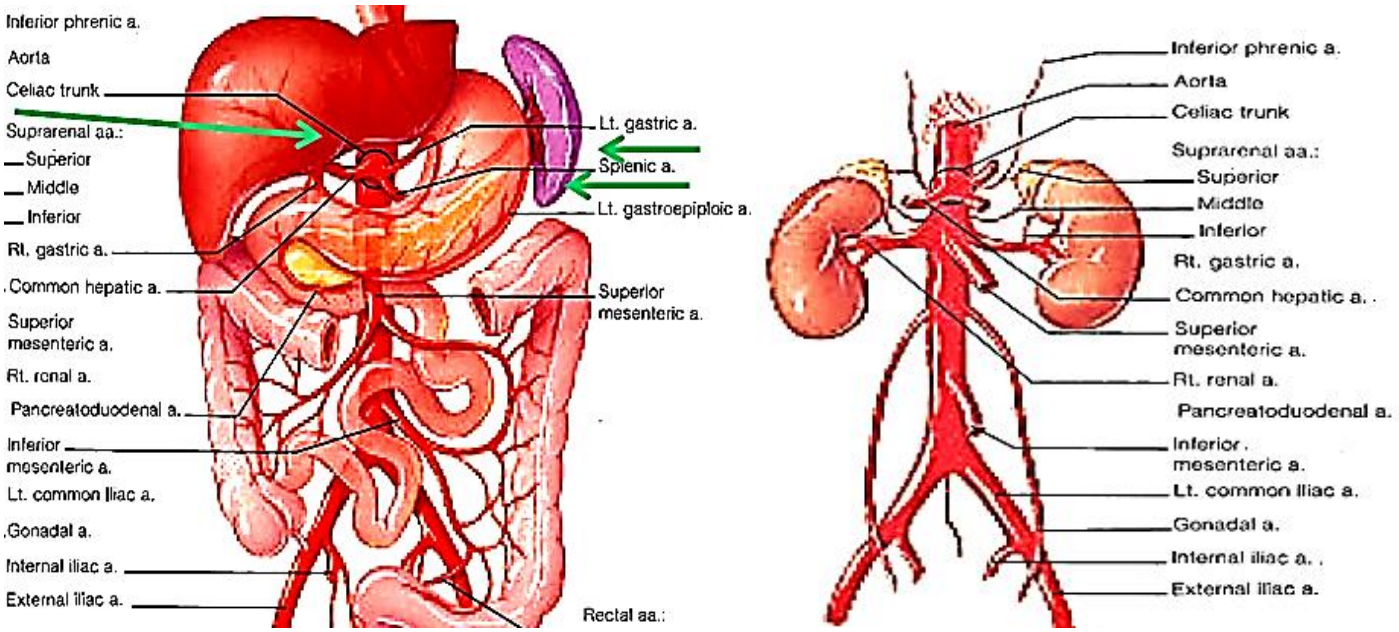
As in the case of the thoracic aorta, there are unpaired branches extending forward and paired arteries extending toward the side.

The unpaired vessels are large arteries that supply the abdominal viscera. The most important of these visceral branches are listed below:

1. The celiac trunk is a short artery about 1.25 cm long that subdivides into three branches: the left gastric artery goes to the stomach, the splenic artery goes to the spleen, and the hepatic artery carries oxygenated blood to the liver.

2. The superior mesenteric artery, the largest of these branches, carries blood to most of the small intestine as well as to the first half of the large intestine.

3. The much smaller inferior mesenteric artery, located below the superior mesenteric and near the end of the abdominal aorta, supplies the second one half of the large intestine.



The lateral (paired) branches of the abdominal aorta include the following right and left divisions:

1. The phrenic arteries supply the diaphragm.

2. The suprarenal arteries supply the adrenal (suprarenal) glands.

3. The renal arteries, the largest in this group, carry blood to the kidneys.

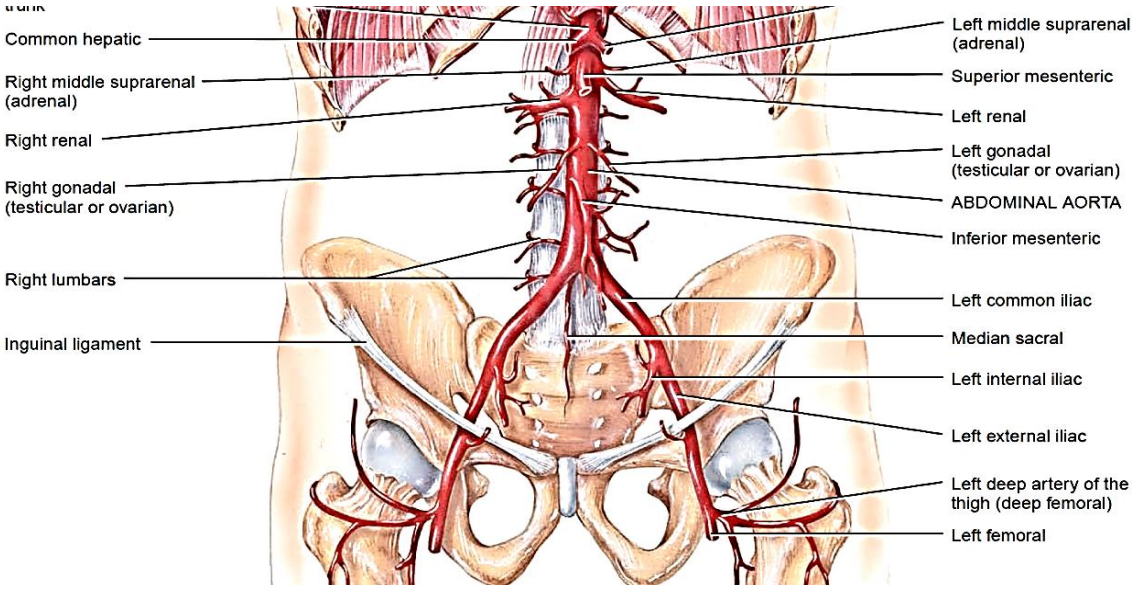
4. The ovarian arteries in the female and testicular arteries in the male (formerly called the spermatic arteries), supply the sex glands.

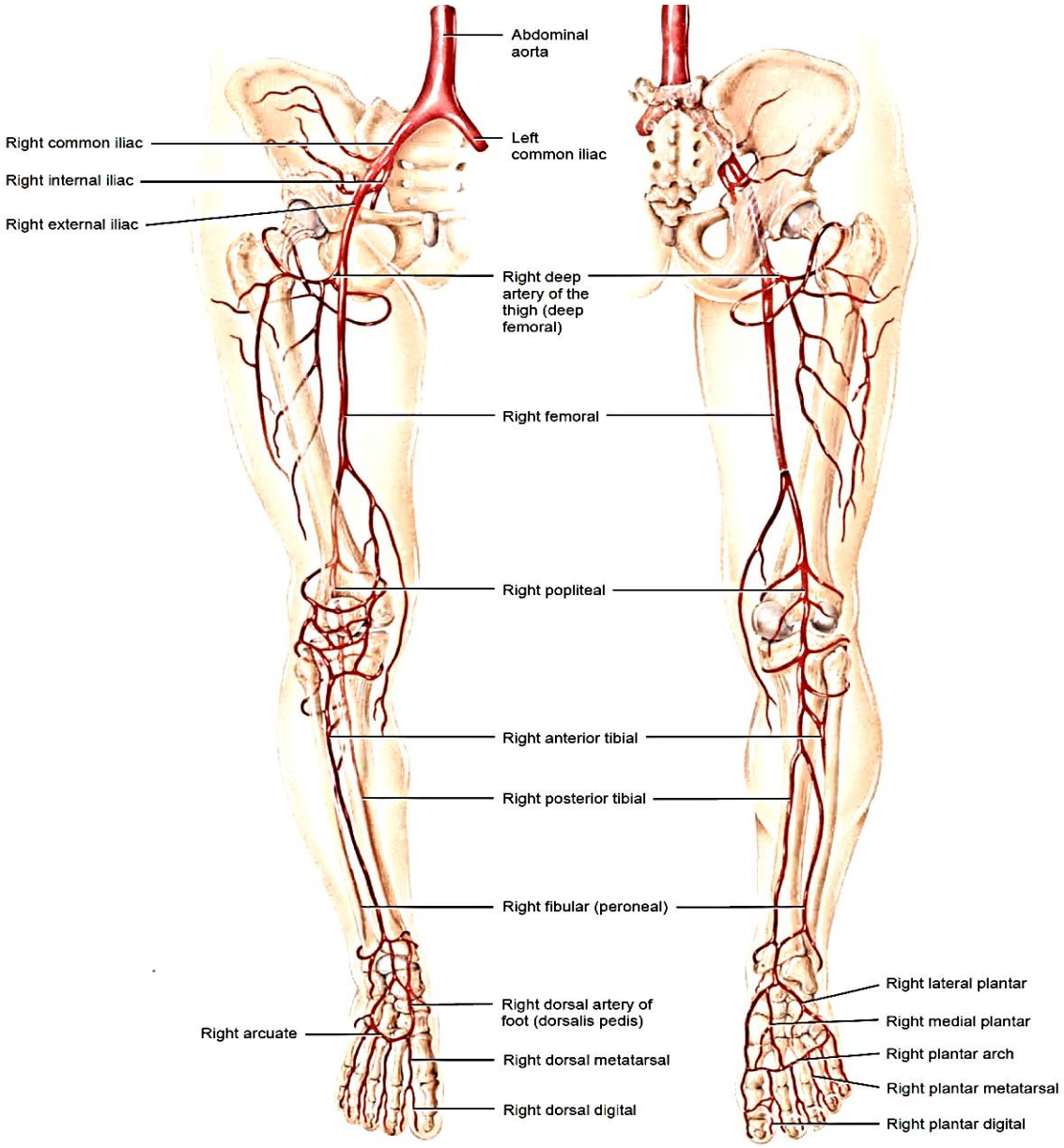
5. Four pairs of lumbar arteries extend into the musculature of the abdominal wall.

Iliac Arteries and Their Subdivisions

The abdominal aorta finally **divides into two common iliac arteries.** Both of these vessels, about 5 cm long, extend into the pelvis, where each one subdivides into an internal and an external iliac artery. The internal iliac vessels then send branches to the pelvic organs, including the urinary

bladder, the rectum, and some of the reproductive organs. The external iliac arteries continue into the thigh as the femoral arteries. These vessels give off branches in the thigh and then become the popliteal arteries, which subdivide below the knee. The subdivisions include the tibial arteries and dorsalis pedis, which supply the leg and the foot.

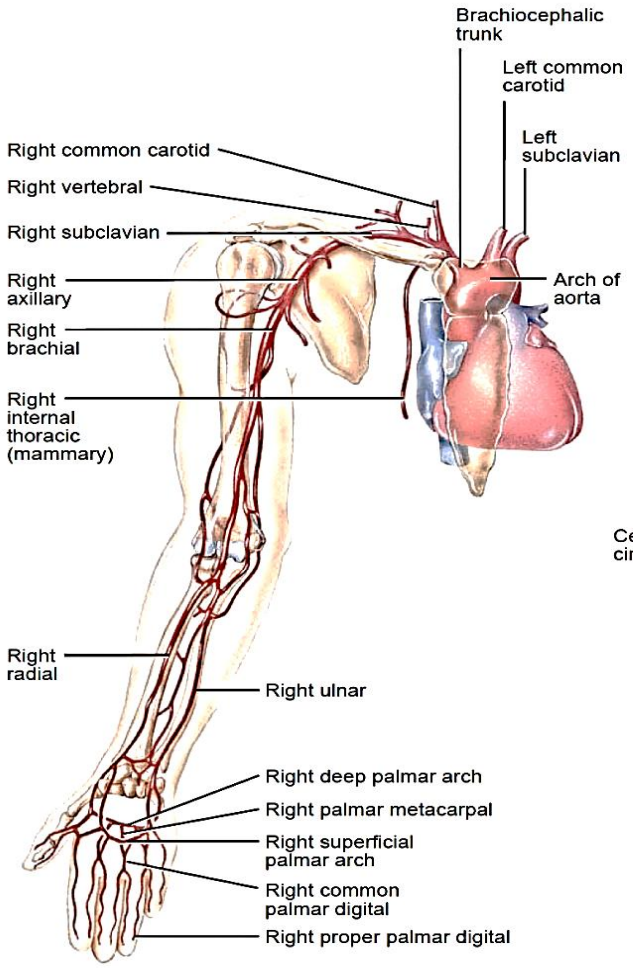


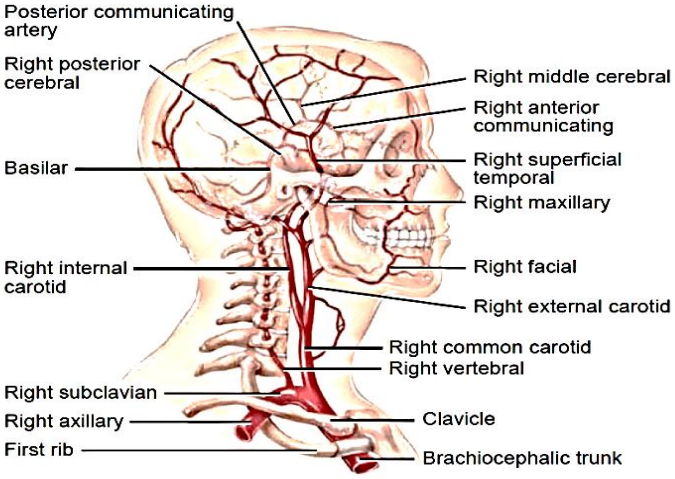


Other Subdivisions of Systemic Arteries

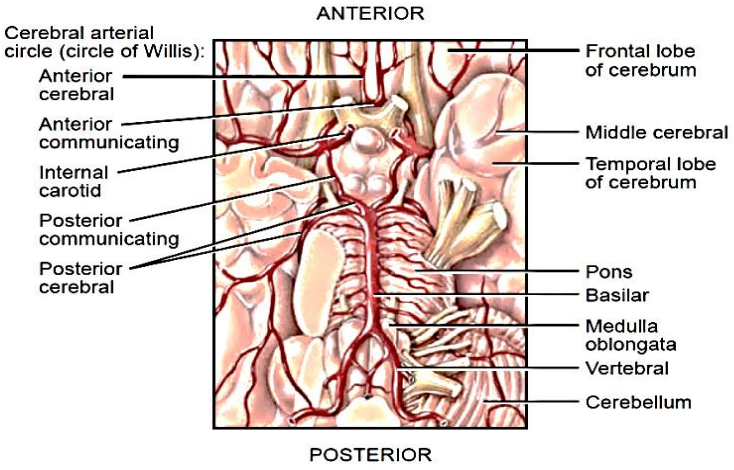
Just as the larger branches of a tree give off limbs of varying sizes, so the arterial tree has a multitude of subdivisions.

Hundreds of names might be included, but we shall mention only a few. For example, each common carotid artery gives off branches to the thyroid gland and other structures in the neck before dividing into the external and internal carotid artery, which supplies parts of the head. The hand receives blood from the subclavian artery, which becomes the axillary in the axilla (armpit). The longest part of this vessel, the brachial artery, is in the arm proper. It subdivides into two branches near the elbow :the radial artery, which continues down the thumb side of the forearm and wrist, and the ulnar artery, which extends along the medial or little finger side into the hand.





(b) Right lateral view of branches of brachiocephalic trunk in neck and head

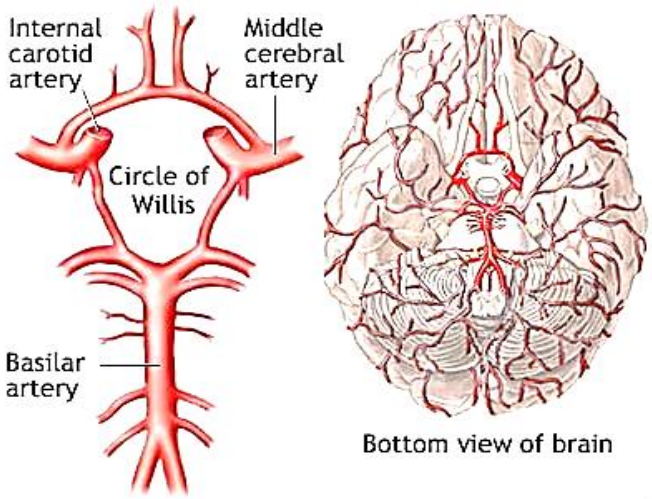


Anastomosis

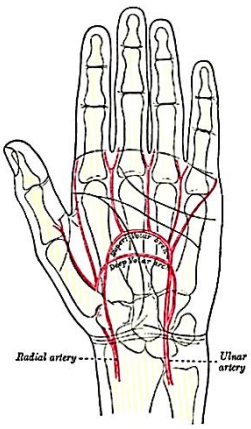
A communication between two arteries is called an anastomosis. By this means, blood reaches vital organs by more than one route.

Some examples of such unions of end arteries are described below:

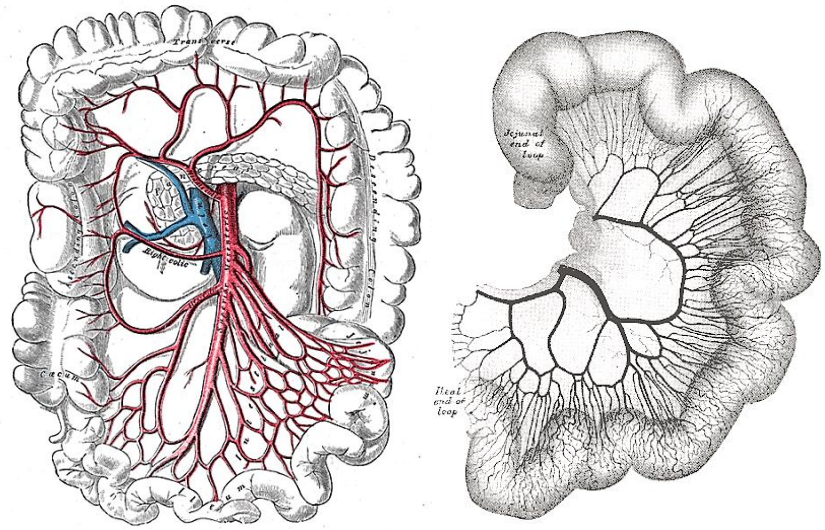
1. The circle of Willis receives blood from the two internal carotid arteries as well as from the basilar artery, which is formed by the union of two vertebral arteries. This arterial circle lies just under the center of the brain and sends branches to the cerebrum and other parts of the brain.



2. The volar arch is formed by the union of the radial and ulnar arteries in the hand. It sends branches to the hand and fingers.



3. The mesenteric arches are made of communications between branches of the vessels that supply blood to the intestinal.



4. Arterial arches are formed by the union of branches of the tibial arteries in the foot, and similar anastomoses are found in various parts of the body. Arteiovenous anastomoses are found in a few parts of the body ,including the external ears, the hands, and the feet. Vessels that have muscular walls connect arteries directly with veins and thus bypass the capillaries. This provides a more rapid flow and a greater volume of blood to these areas the elsewhere, thus protecting these exposed parts from freezing in cold weather.

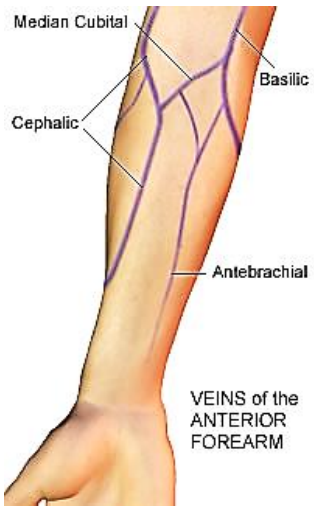
Names of Systemic Veins

- **Superficial Veins**

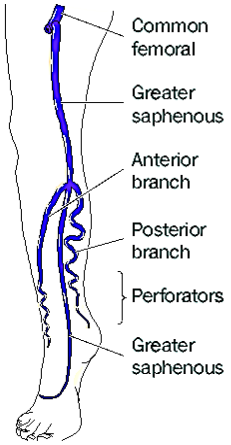
Whereas most arteries are located in protected and rather deep areas of the body, many veins are found near the surface. The most important of these superficial veins are in the extremities.

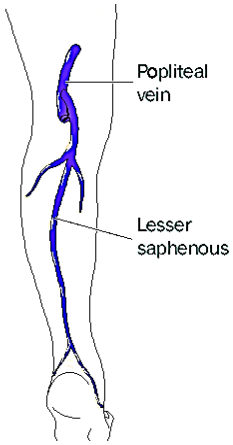
These include the following:

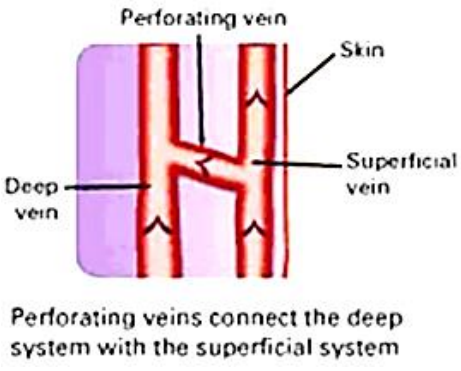
1. The veins on the back of the hand and at the front of the elbow. Those at the elbow are often used for removing blood samples for test purposes, as well as for intravenous injections. The largest of this group of veins are the cephalic, the basilic, and the median cubital veins.



2. The saphenous veins of the lower extremities, which are the longest veins of the body. The great saphenous vein begins in the foot and extends up the medial side of the leg, the knee, and the thigh. It finally empties into the femoral vein near the groin.

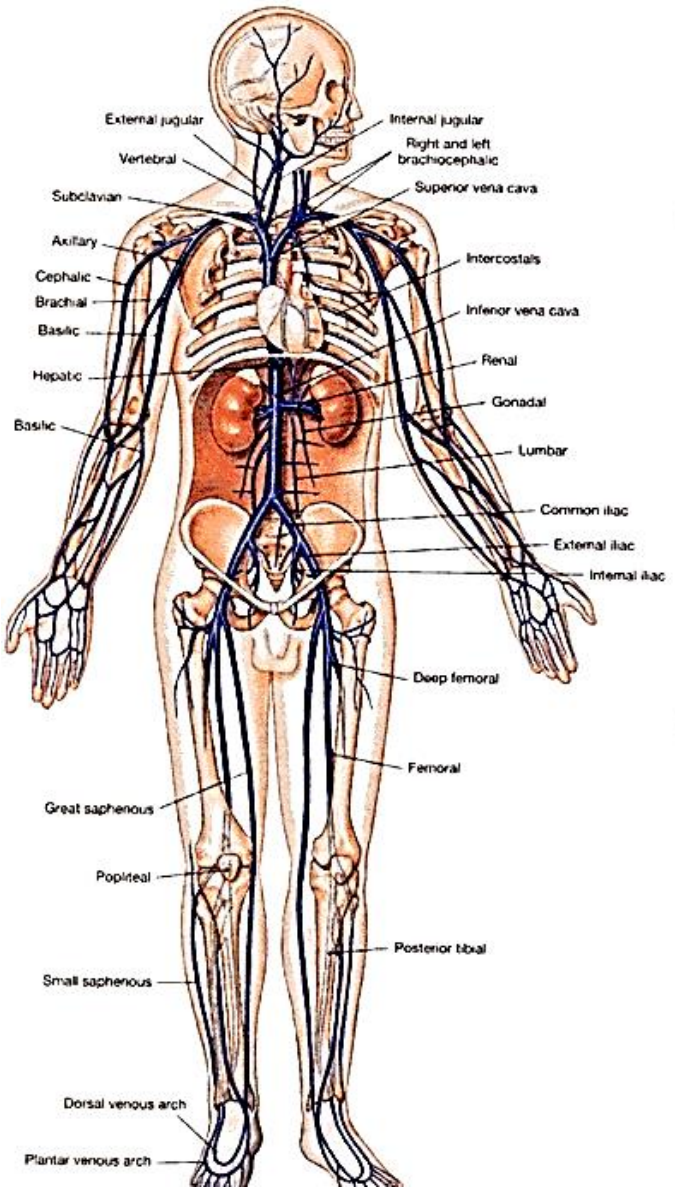


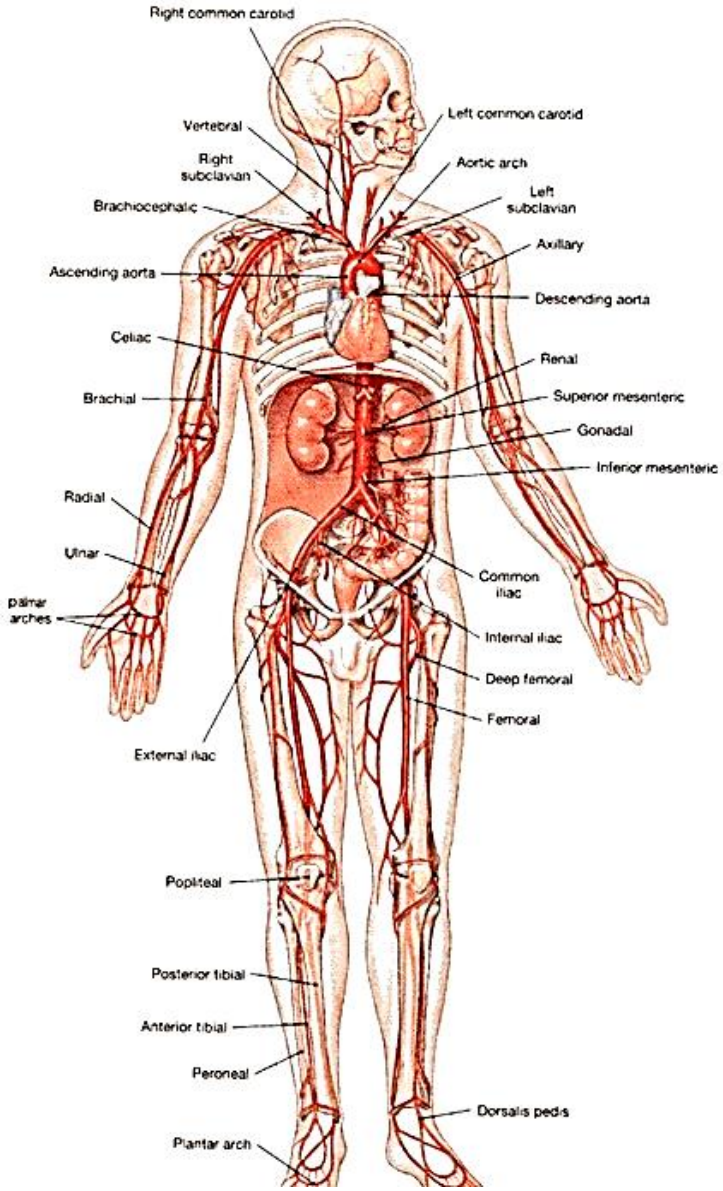




- **Deep Veins**

The deep veins tend to parallel arteries and usually have the same names as the corresponding arteries. Examples of these include the femoral and the iliac vessels of the lower part of the body and the brachial, axillary, and subclavian vessels of the upper extremities. However, exceptions are found in the veins of the head and the neck. The jugular veins drain the areas supplied by the carotid arteries. Two brachiocephalic (innominate) veins are formed, one on each side, by the union of the subclavian and the jugular veins.





Superior Vena Cava

The veins of the head, neck, upper extremities, and chest all drain into the superior vena cava, which goes to the heart. It is formed by the union of the right and left brachiocephalic veins, which drain the head, neck, and upper extremities. The azygos vein drains the veins of the chest wall and empties into the superior vena cava just before the latter empties into the heart

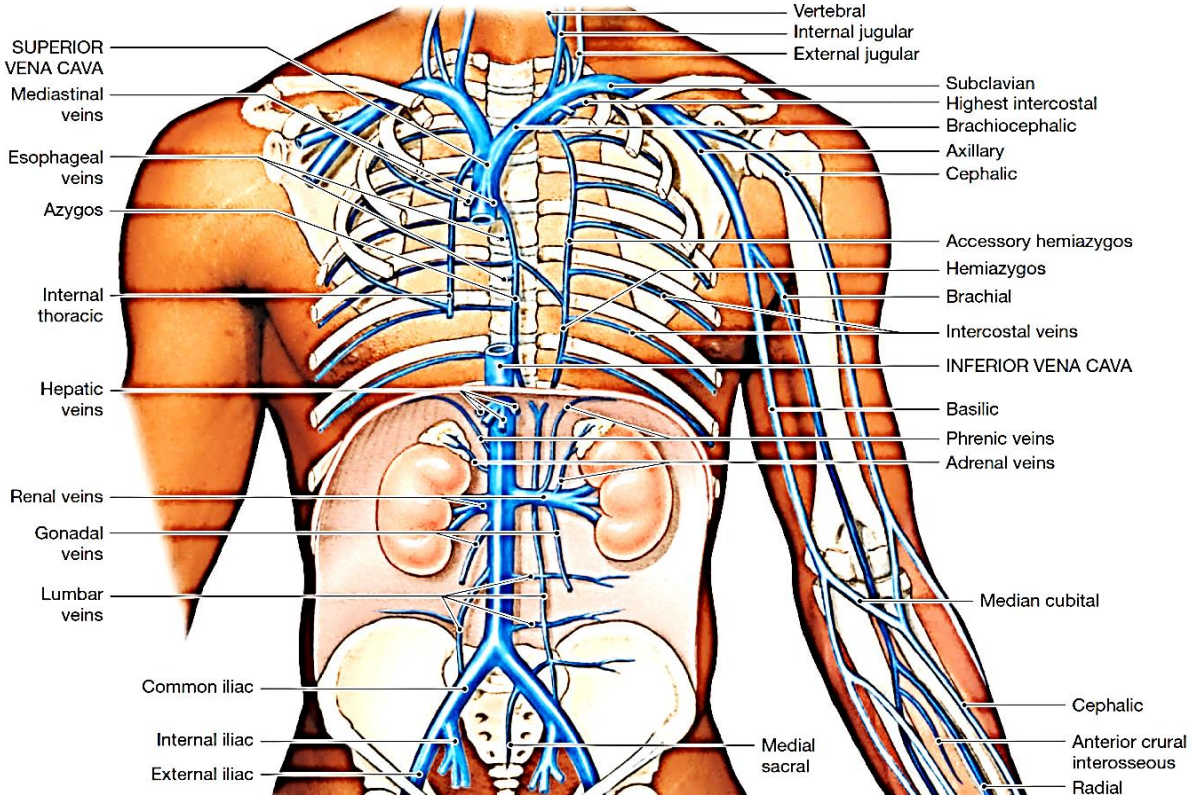
Inferior Vena Cava

The inferior vena cava, which is much longer than the superior vena cava, returns the blood from the parts of the body below the diaphragm. It begins in the lower abdomen with the union of the two common iliac veins. It then ascends along the back wall of the abdomen, through a groove in the posterior part of the liver, through the diaphragm, and finally through the lower thorax to empty into the right atrium of the heart. Drainage into the inferior vena cava is more complicated than drainage into the superior vena cava. The large veins below the diaphragm may be divided into two groups:

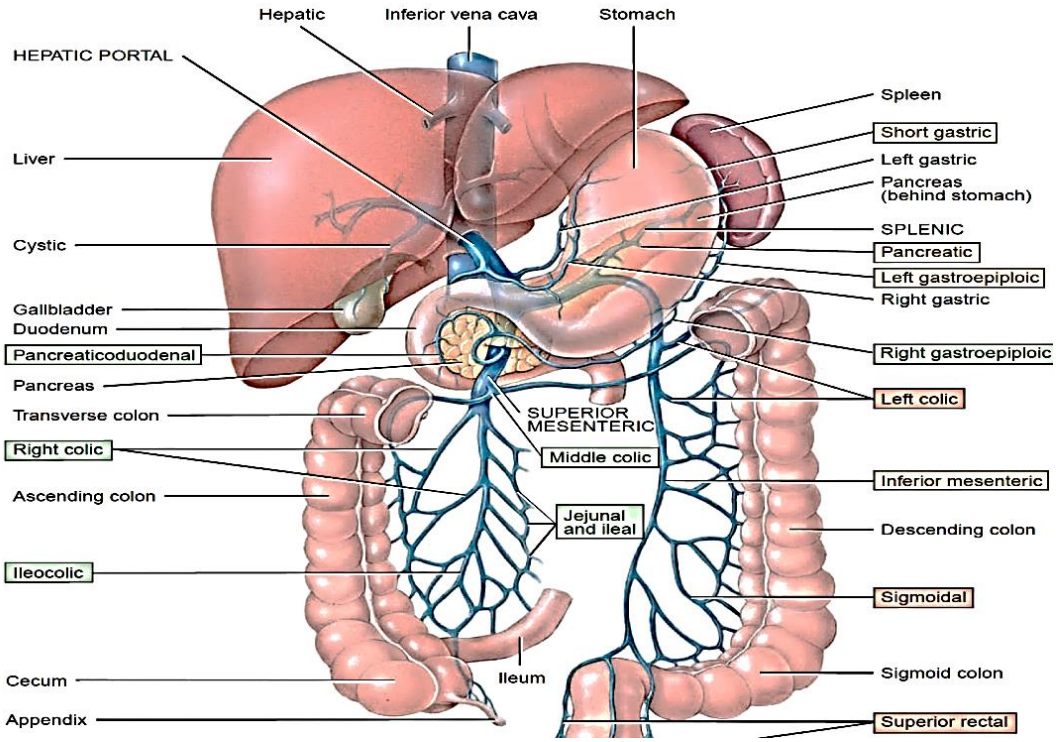
1. The right and left veins that drain paired parts and organs.

They include the iliac veins from near the groin, four pairs of lumbar veins from the dorsal part of the trunk and from the spinal cord, the testicular veins from the testes of the male and the ovarian

veins from the ovaries of the female, the renal and suprarenal veins from the kidneys and adrenal glands near the kidneys, and finally the large hepatic veins from the liver. For the most part, these vessels empty directly into the inferior vena cava. The left testicular in the male and the left ovarian in the female empty into the left renal vein, which then take this blood to the inferior vena cava; these veins thus constitute exceptions to the rule that the paired veins empty directly into vena cava.



2. Unpaired veins that come from the spleen and from parts of the digestive tract (stomach and intestine) and empty into a vein called the portal vein. Unlike other veins, which empty into the inferior vena cava, the hepatic portal vein is part of a special system that enables blood to circulate through the liver before returning to the heart.



Circulatory Routes or Circuits

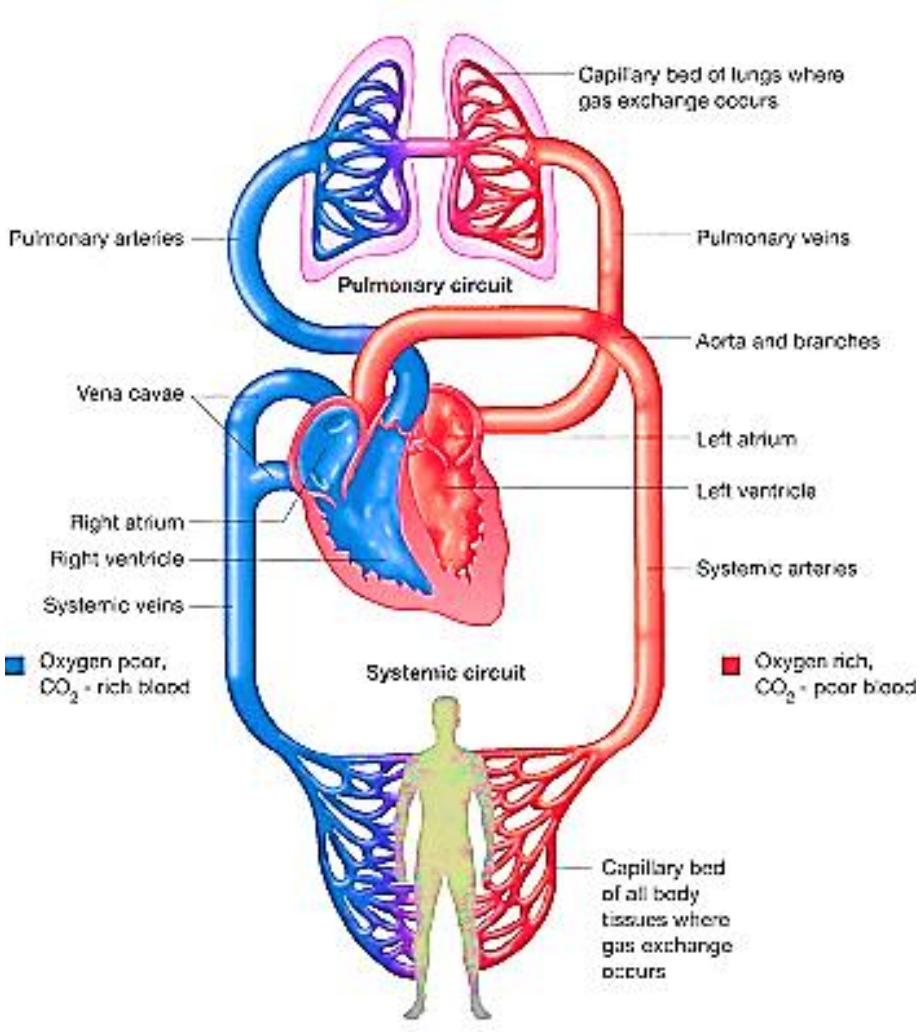
All the vessels together may be subdivided into two groups or circuits: pulmonary and systemic.

1. Pulmonary circulation: carry blood to and from the lungs. They in include the pulmonary artery and its branches to the capillaries in the lungs, as well as the veins that drain those capillaries. The pulmonary arteries carry blood low in oxygen from the right ventricle, while the pulmonary veins carry blood high in oxygen from the lungs into the left atrium. This circuit functions to eliminate carbon dioxide from the blood and replenish its supply of oxygen.

2. Systemic circulation: it is the largest circulatory route. It takes oxygenated blood from the left ventricle through the aorta to all parts of the body, including some lung tissue (not air sac or alveolus) and returns the deoxygenated blood to the right atrium, through the systemic veins; the superior vena cava, the inferior vena cava, and the coronary sinus. It has several subdivisions.

Two of the several subdivisions are :

- the coronary circulation and the hepatic portal system or circulation.



Hepatic Portal System

Almost always, when blood leaves a capillary bed it returns directly to the heart. In a portal system, however, blood circulates through a second capillary bed, usually in a second organ, before returning to the heart. Thus, a portal system is a kind of detour in the pathway of venous return that can transport materials directly from one organ to another. The portal system between

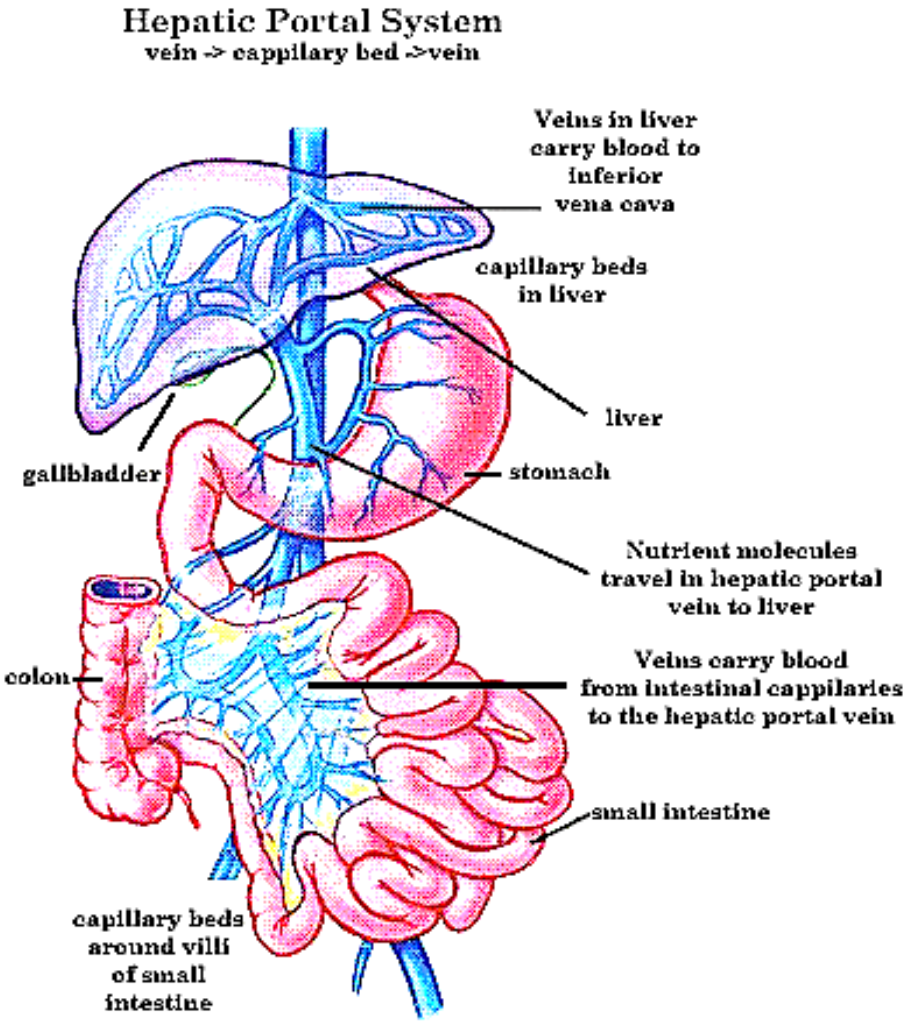
the hypothalamus and the anterior pituitary has already been described. The largest portal system in the body is the hepatic portal system, which carries blood from the abdominal organs to the liver.

The hepatic portal system includes the veins drains blood from capillaries in the spleen, stomach, pancreas, and intestine. Instead of emptying their blood directly into the inferior vena cava, they deliver it by way of the hepatic portal vein to the liver. The largest tributary of the portal vein is the superior mesenteric vein it is joined by the splenic vein just under the liver. Other tributaries of the portal circulation are the gastric, pancreatic, and inferior mesenteric veins.

Upon entering the liver, the portal vein divides and subdivides into ever smaller branches. Eventually, the portal blood flows into a vast network of sinus like vessels called sinusoids. These are enlarged capillaries that serve as blood channels within the tissues of the liver, spleen, thyroid gland, and other structures.

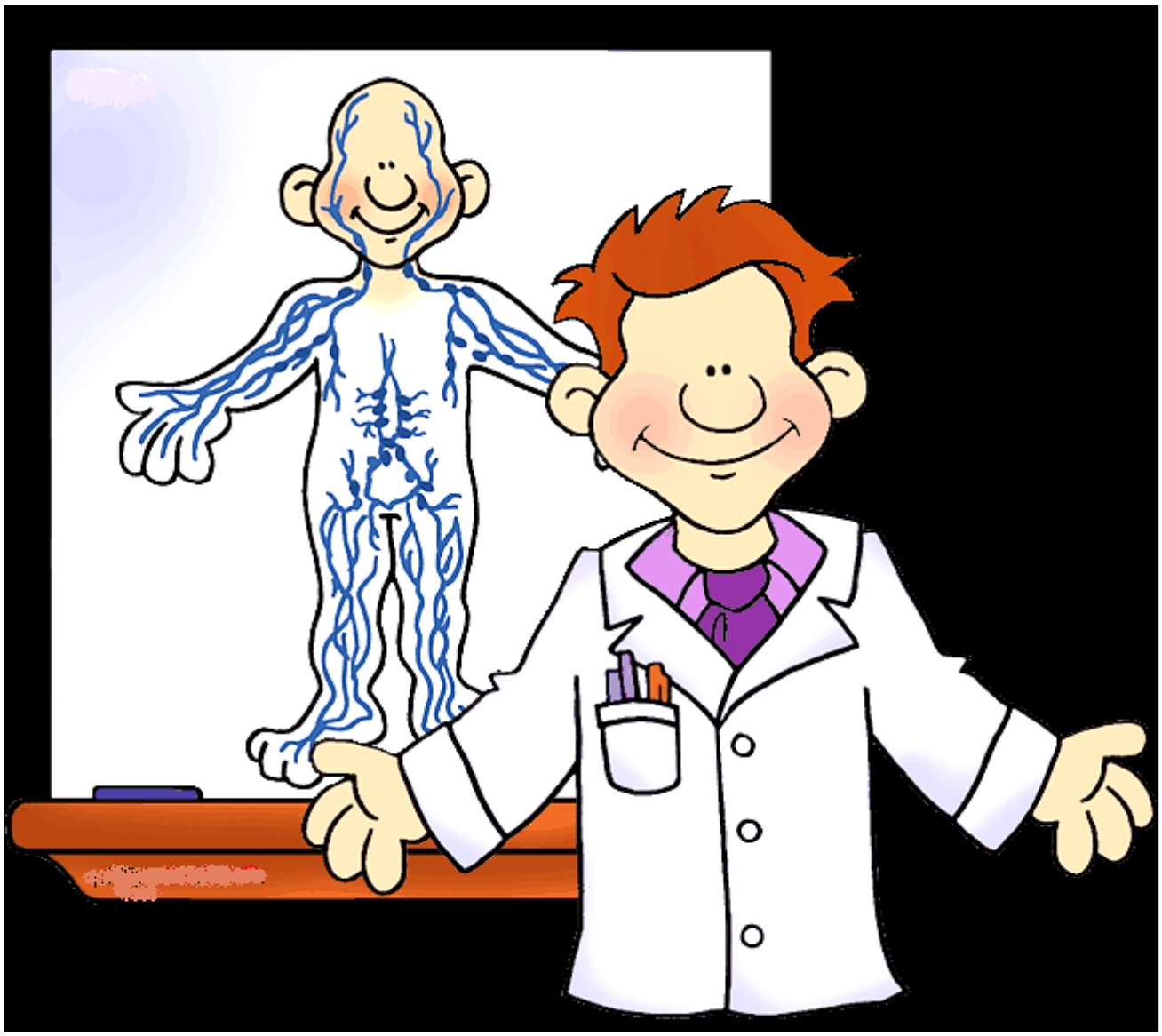
After leaving the sinusoids, blood is finally collected by the hepatic veins, which empty into the inferior vena cava.

The purpose of the portal system of veins is to the liver sinusoids so the liver cells can carry out their functions. For example, when food is digested, most of the end products are absorbed from the small intestine into the blood stream and transported to the liver by the portal system. In the liver, these nutrients are processed, stored, and released as needed into the general circulation.



CHAPTER NINE

The Lymphatic System



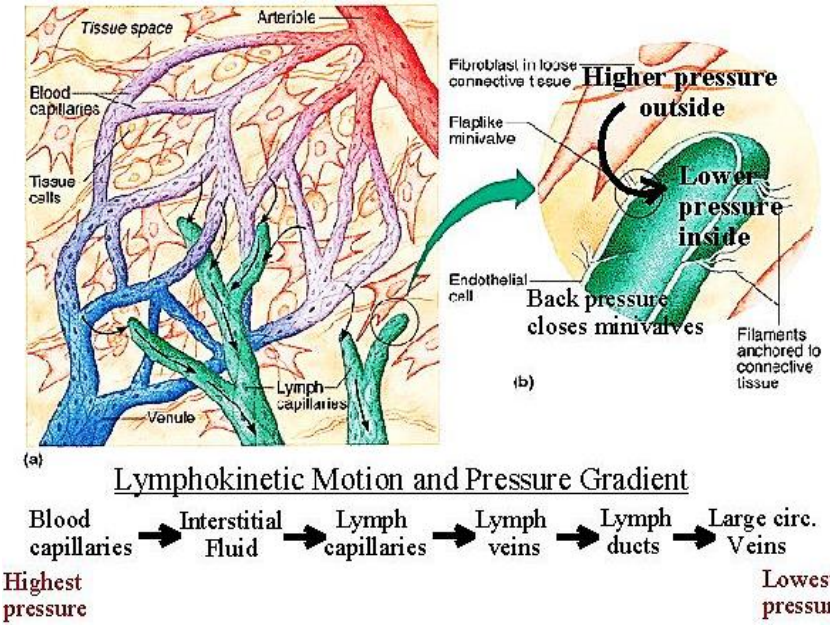
CHAPTER NINE The Lymphatic System

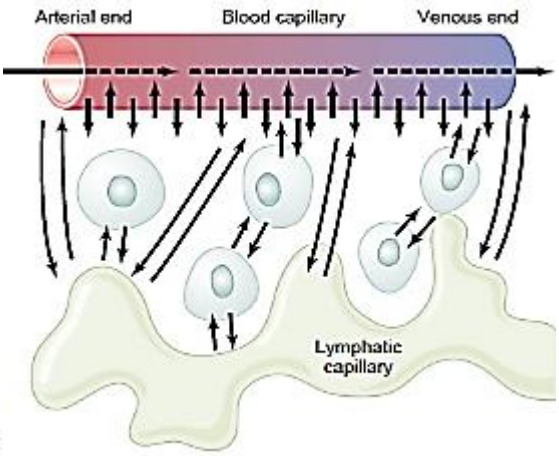
The lymphatic system communicates with the blood circulatory system and is closely associated with it.

Lymph Channels of the Body

Almost all tissues of the body have special lymph channels that drain excess fluid directly from the interstitial spaces. The exceptions include the superficial portions of the skin, the central nervous system, the endomysium of muscles, and the bones. But, even these tissues have minute interstitial channels called prelymphatics through which interstitial fluid can flow; this fluid eventually empties either into lymphatic vessels or, in the case of the brain, into the cerebrospinal fluid and then directly back into the blood.

As blood passes through the capillaries, some of the fluid diffuses into the surrounding tissues. One function of the lymphatic system is to collect and recycle this fluid (called lymph). Lymph passes from capillaries to lymph vessels and flows through lymph nodes that are located along the course of these vessels.



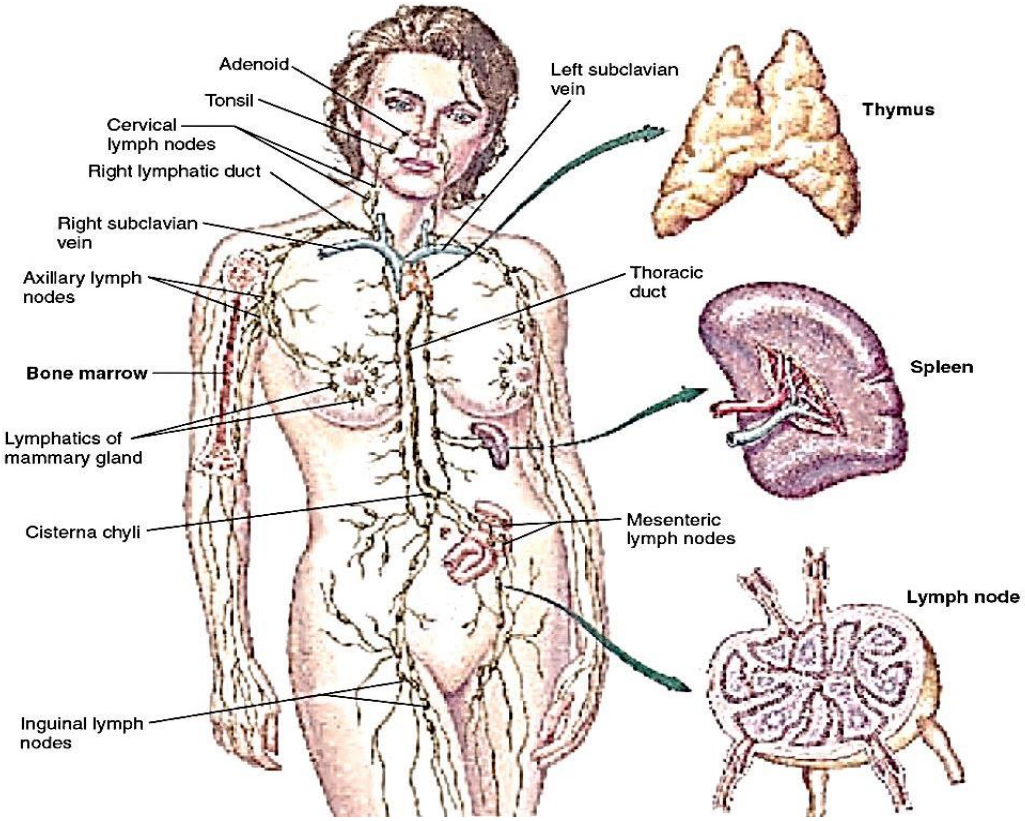


The lymphoid system is the part of the immune system comprising a network of conduits called lymphatic vessels that carry a clear fluid called lymph (from Latin lympha "water") unidirectionally toward the heart.

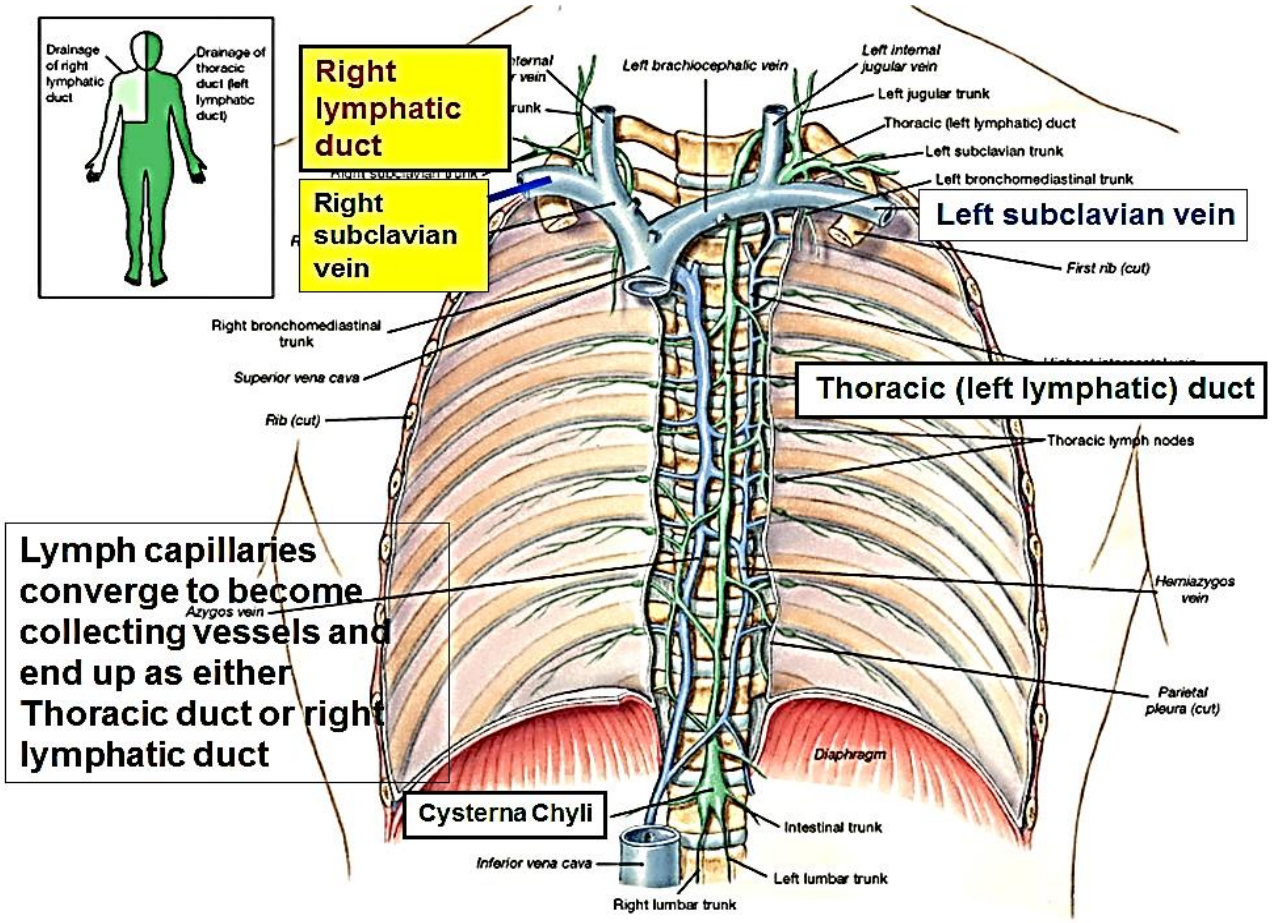
Lymphoid tissue is found in many organs, particularly the lymph nodes, and in the lymphoid follicles associated with the digestive system such as the tonsils.

The system also includes all the structures dedicated to the circulation and production of lymphocytes, which includes :

- The spleen,
- Thymus,
- Bone marrow
- The lymphoid tissue associated with the digestive system.

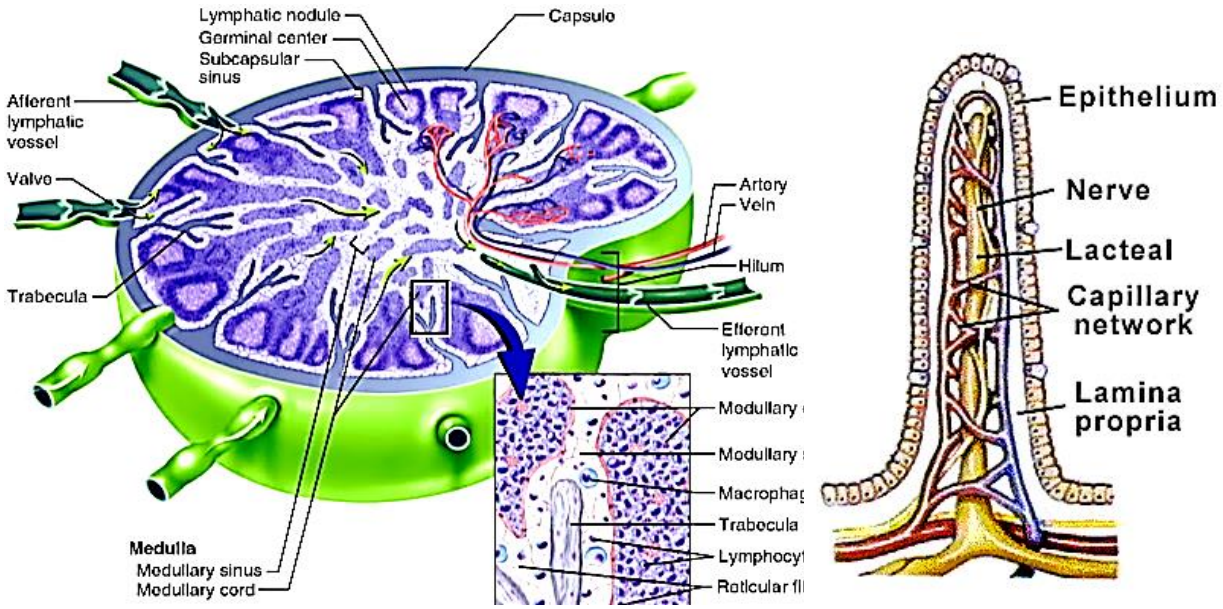


Lymph is the fluid that is formed when interstitial fluid enters the initial lymphatic vessels of the lymphatic system. The lymph is then moved along the lymphatic vessel network by either intrinsic contractions of the lymphatic passages or by extrinsic compression of the lymphatic vessels via external tissue forces (e.g. the contractions of skeletal muscles). Eventually, the lymph vessels empty into the lymphatic ducts, which drain into one of the two subclavian veins (near the junctions of the subclavian veins with the internal jugular veins).



Main functions of the lymphatic system

1. To collect and return interstitial fluid, including plasma protein to the blood, and thus help maintain fluid balance,
2. To defend the body against disease by producing lymphocytes,
3. To absorb lipids from the intestine and transport them to the blood."

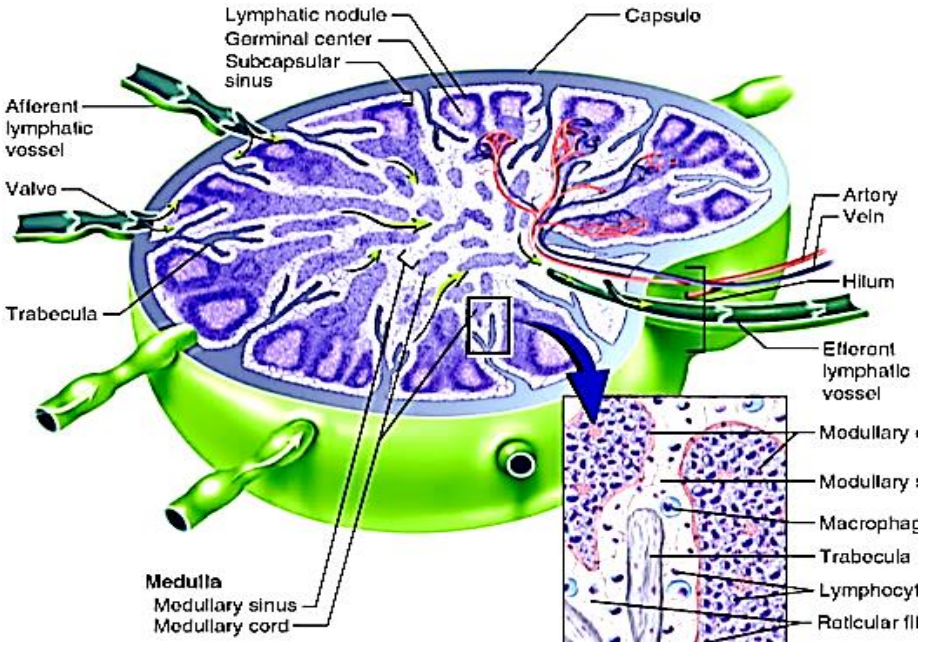


Other lymphoid tissue:

1. Lymph nodes:

"Human lymph nodes are bean-shaped and range in size from a few millimeters to about 1-2 cm in their normal state. They may become enlarged due to a tumor or infection. White blood cells are located within honeycomb structures of the lymph nodes. Lymph nodes are enlarged when the body is infected due to enhanced production of some cells and division of activated T and B cells.

In some cases they may feel enlarged due to past infections; although one may be healthy, one may still feel them residually enlarged."

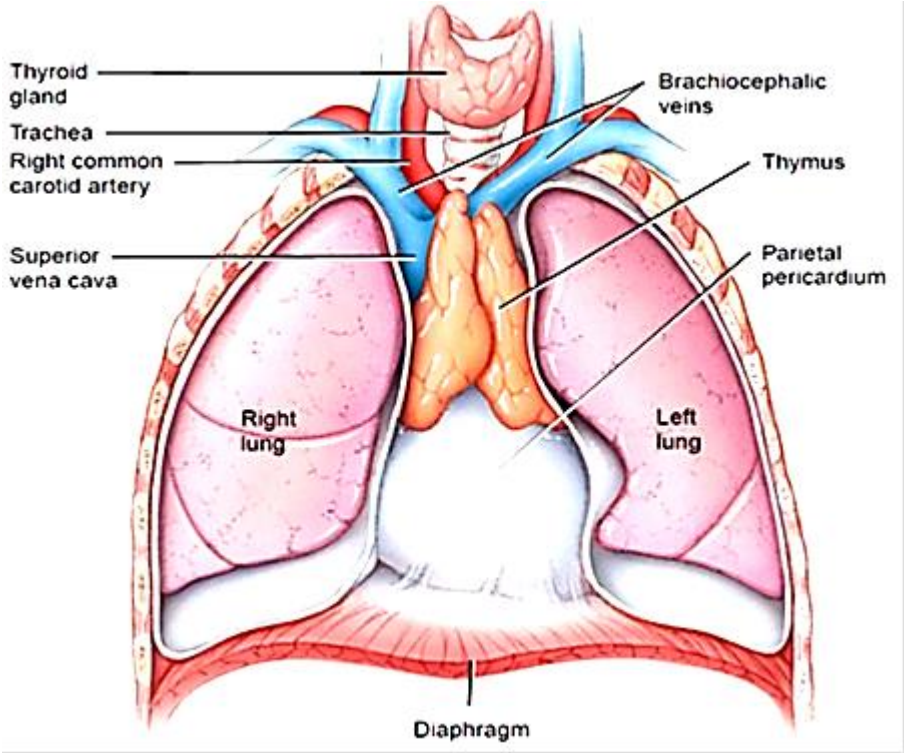


2. **Diffuse Lymphatic Tissue and Lymphatic nodules:**

The alimentary canal, respiratory passages, and genitourinary tract are guarded by accumulations of lymphatic tissue that are not enclosed by a capsule (i.e. they are diffuse) and are found in connective tissue beneath the epithelial mucosa. In general these are single and random but are more concentrated in the GI tract in the ileum, appendix, cecum, and tonsils. These are collectively called the Gut Associated Lymphatic Tissue (GALT).Mucosa Associated Lymphatic Tissue (MALT) includes these plus the diffuse lymph tissue in the respiratory tract.

3. **The thymus:**

The thymus is where immature lymphocytes differentiate into T-lymphocytes. The thymus is fully formed and functional at birth. Characteristic features of thymic structure persist until about puberty, when lymphocyte processing and proliferation are dramatically reduced and eventually eliminated and the thymic tissue is largely replaced by adipose tissue.

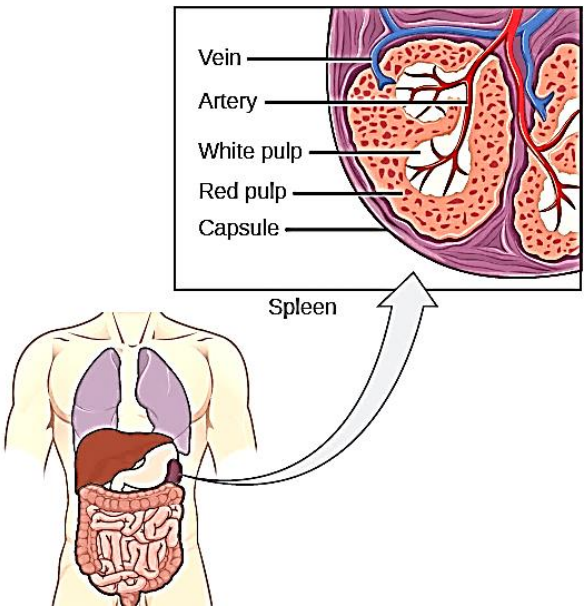


The lymphocytes released by the thymus are carried to lymph nodes, spleen, and other lymphatic tissue where they form colonies. These colonies form the basis of T-lymphocyte proliferation in the specific immune response. T-lymphocytes survive for long periods and recirculate through lymphatic tissues. The transformation of primitive or immature lymphocytes into T-lymphocytes and their proliferation in the lymph nodes is promoted by a thymic hormone called thymosin.

4. **The spleen:**

The spleen filters the blood and reacts immunologically to blood-borne antigens. The human spleen holds relatively little blood compared to other mammals, but it has the capacity for contraction to release this blood into the circulation during anoxic stress.

White pulp in the spleen contains lymphocytes and is equivalent to other lymph tissue, while red pulp contains large numbers of red blood cells that it filters and degrades.



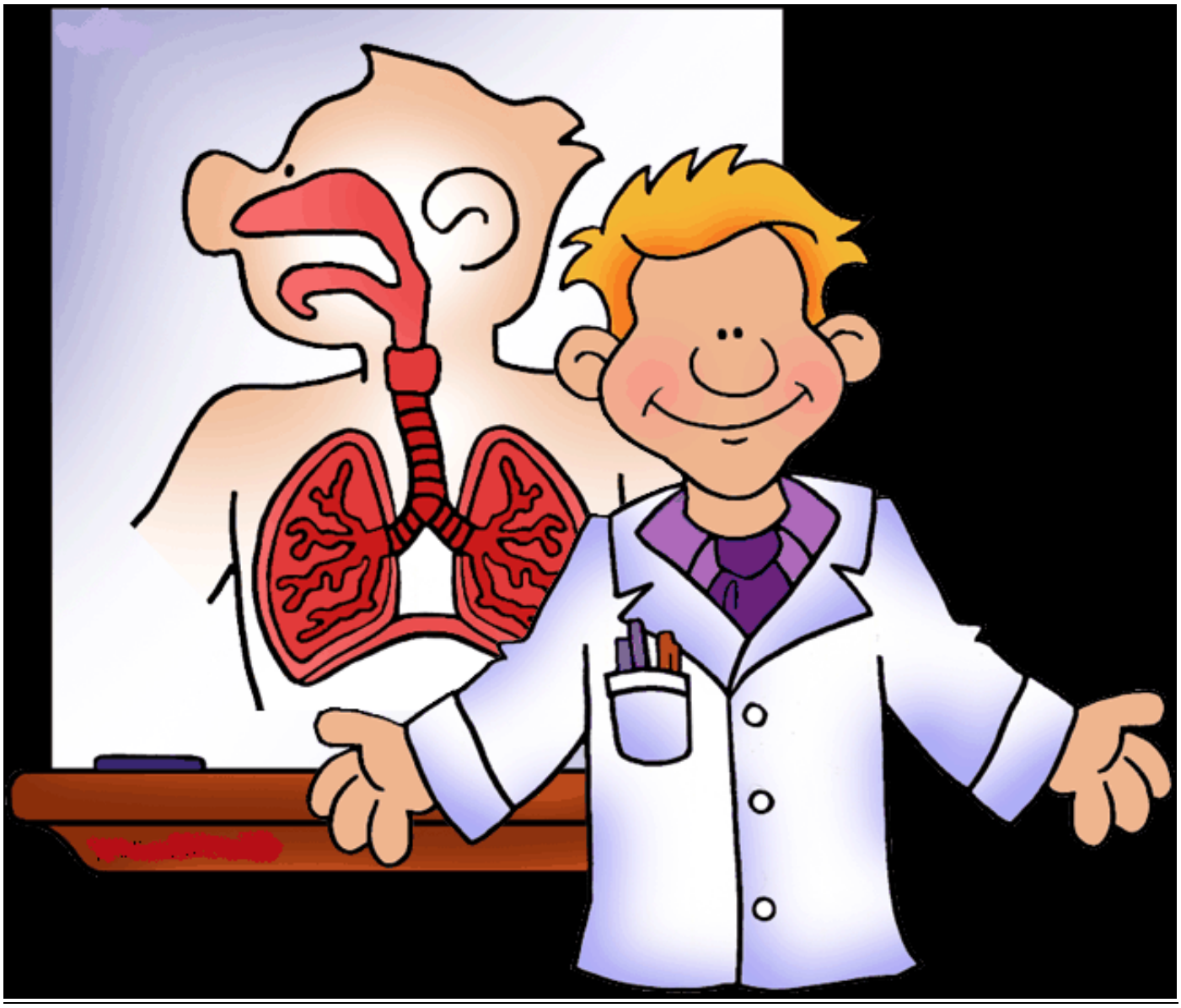
The spleen functions in both immune and hematopoietic systems. Immune functions include:

- Proliferation of lymphocytes,
- Production of antibodies,
- Removal of antigens from the blood.

Hematopoietic functions include: formation of blood cells during fetal life, removal and destruction of aged, damaged and abnormal red cells and platelets, retrieval of iron from hemoglobin degradation, storage of red blood cells.

CHAPTER TEN

RESPIRATORY SYSTEM

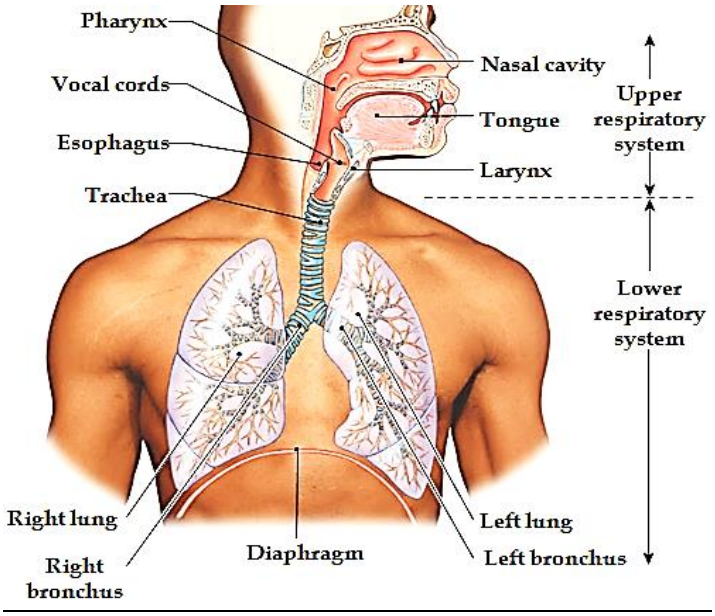


CHAPTER TEN RESPIRATORY SYSTEM

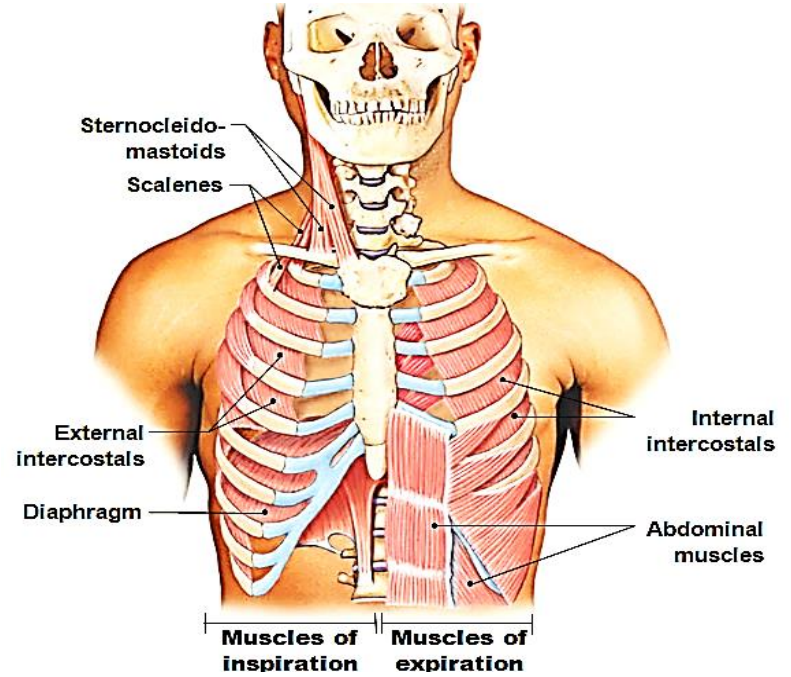
Structure and Function of Respiratory Pathways

Respiratory System Components

- **Conducting system**
- **Alveoli**
- **Bones and muscle of thorax**



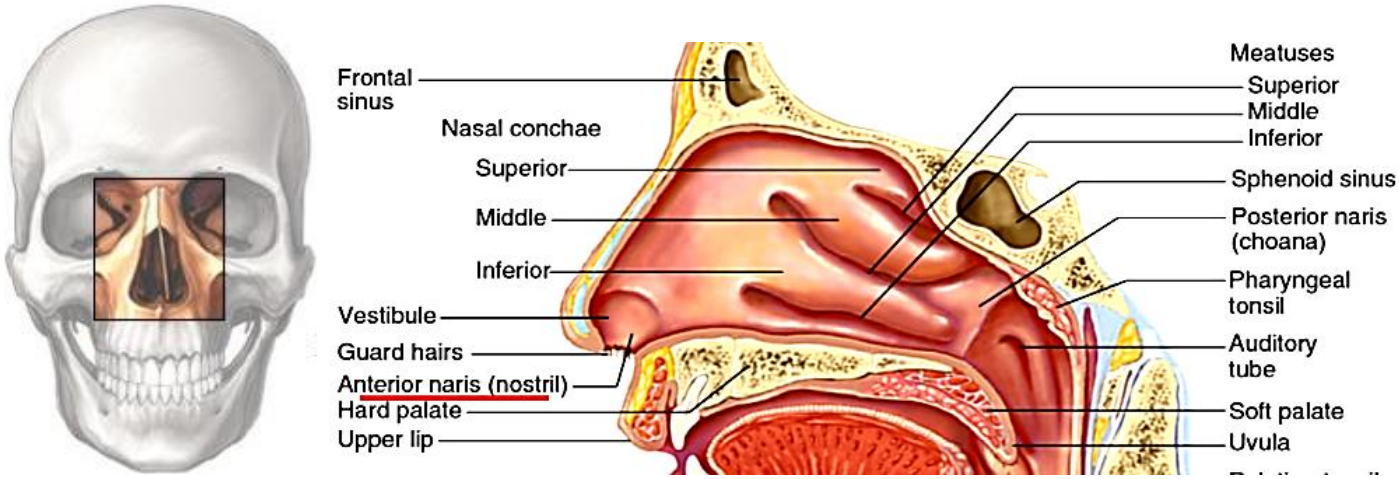
Muscles Used for Ventilation



The Nasal Cavities

Air makes its initial entrance into the body through the openings in the nose called the nostrils. Immediately inside the nostrils, located between the roof of the mouth and the cranium, are the two spaces known as the nasal cavities.

These two spaces are separated from each other by a partition, the nasal septum. The septum and the walls of the nasal cavities are constructed of bone covered with mucous membrane. From the lateral (side) walls of each nasal cavityare three projections called the conchae. The conchae greatly increase the surface over wich air must travel on its way through the nasal cavities.



The lining of the nasal cavities is a mucous membrane, which contains many blood vessels that bring heat and moisture to it. The cells of this membrane secrete a large amount of fluid.

It is better to breath through the nose than through the mouth because of changes produced in the air as it comes in contact with the lining of the nose:

1. Foreign bodies, such as dust particles and pathogens, are filtered out by the hairs of the nostrils or caught in the surface mucus.

2. Air is warned by the blood in the vascular membrane.

3. Air is moistened by the liquid secretion

Paranasal Sinuses

The sinuses are small cavities lined with mucous membrane in the bones of the skull. The sinuses communicate with the nasal cavities, and they are highly susceptible to infection.

Four bones of the skull contain paired air spaces called the paranasal sinuses.

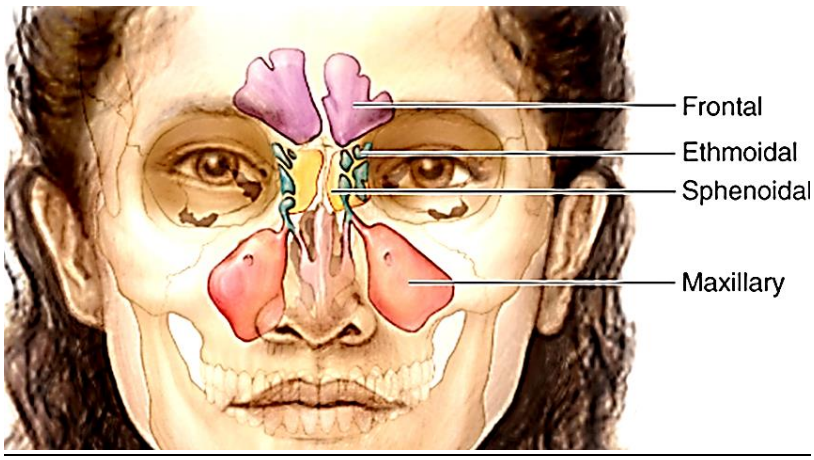
- decrease skull bone weight

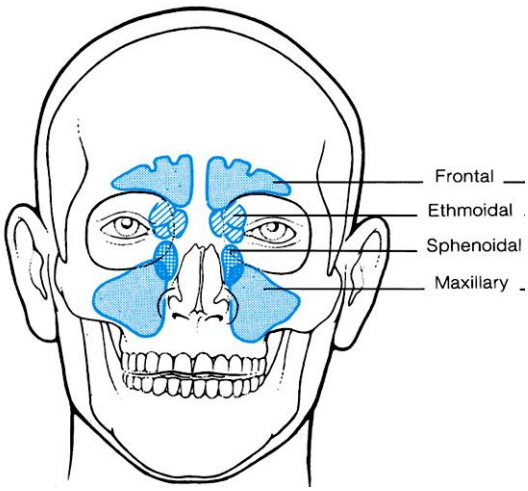
Named for the bones in which they are housed.

  - frontal
  - ethmoidal
  - sphenoidal
  - maxillary

Communicate with the nasal cavity by ducts.

Covered with the same pseudostratified ciliated columnar epithelium as the nasal cavity.

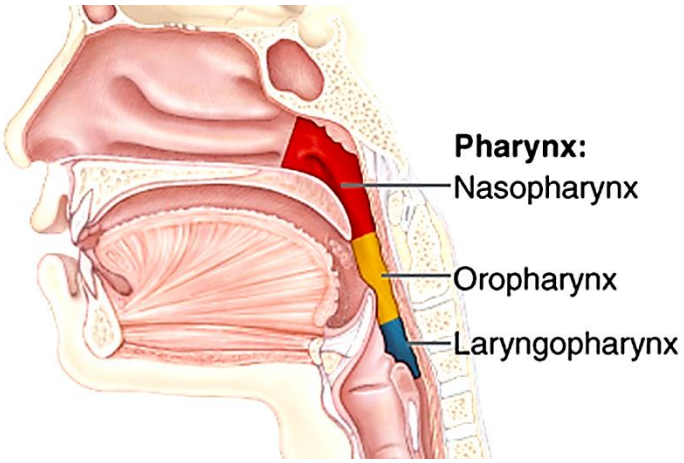




The Pharynx

Is a tube about 12,5 cm long that extend from the base of the skull to the esophagus .

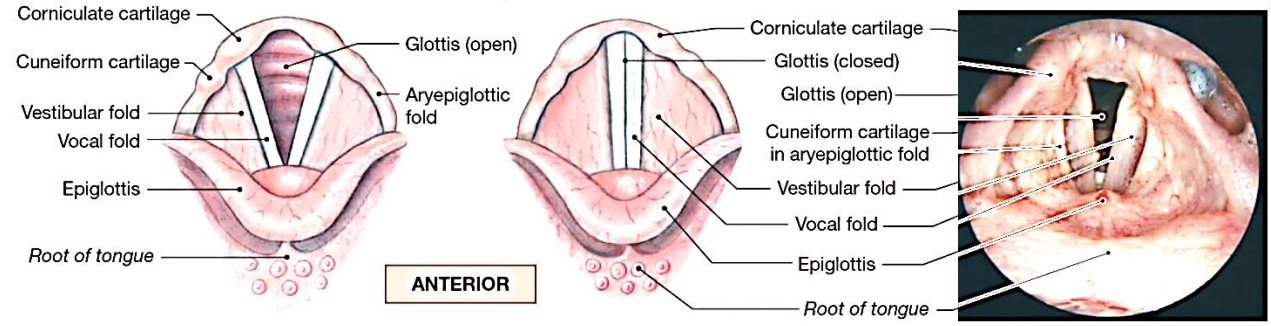
The muscular pharynx (throat) carries air into the respiratory tract and foods and liquids into the digestive system. The upper portion located immediately behind the nasal cavity is called the nasopharynx , the middle section located behind the mouth is called the oropharynx, and the lowest portion is called the laryngeal pharynx. This last section opens into the larynx toward the front and into the esophagus toward the back.



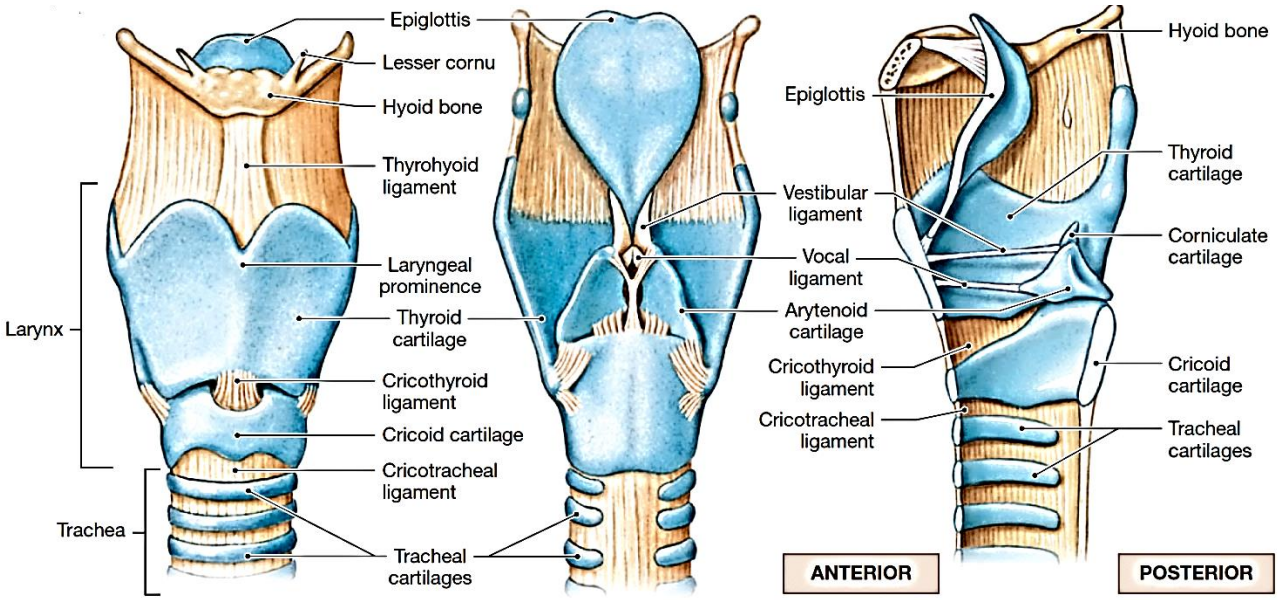
The Larynx

The larynx (voice box) is located between the pharynx and the trachea. It has a framework of cartilage that protrudes in the front of the neck and some times is referred to as the **Adam's apple.** The larynx is considerably larger in the male than in the female; hence, the Adam's apple is much more prominent in the male. At the upper end of the larynx are the vocal cords, which serve in the production of speech. They are set into vibration by the flow of air from the lungs. A difference in the size of the larynx is what accounts for the difference between the male and female voices; because a man's larynx is larger than a woman's, his voice is lower in pitch.

The nasal cavities, the sinuses, and the pharynx all serve as resonating chambers for speech, just as the cabinet does for a stereo speaker.



The space between these two vocal cords is called the glottis, and the little leaf-shaped cartilage that covers the larynx during swallowing is called the epiglottis. The epiglottis helps keep food out of the remainder of the respiratory tract.



As the larynx moves upward and forward during swallowing, the epiglottis moves downward, covering the opening into the larynx. You can feel the larynx move upward toward the epiglottis during this process by placing the flat ends of your fingers on your larynx as you swallow.

The larynx is lined with ciliated mucous membrane. The cilia trap dust and other particles, moving them upward to the pharynx to be expelled by coughing, sneezing, or blowing the nose.

The Trachea (Windpipe)

Is a tube about 11cm long and about 2,5 in diameter It's upper half is situated in the midline of the neck ,

It's lower half in the superior mediastinum of the thorax ,in front of the esophagus .It divided into the two main bronchi at 4-5-th thoracic vertebra.

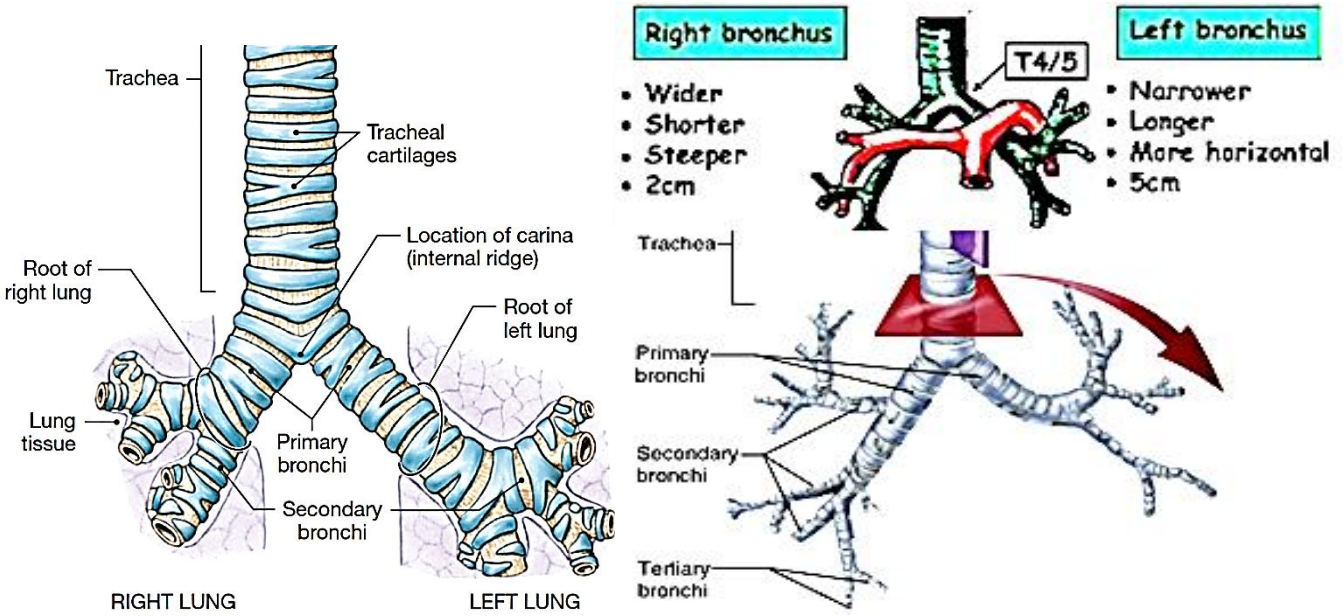
The trachea is a tube that extends from the lower edge of the larynx to the upper part of the chest above the heart. It has a frame work of cartilages to keep it open. These cartilages, shaped somewhat like a tiny horseshoe or the letter C, are found along the entire length of the trachea.

All the open sections of these cartilages are at the back so that the esophagus can bulge into this section during swallowing. The purpose of the trachea is to conduct air between the larynx and the lungs.

The Bronchi and Bronchioles

The trachea divides into two bronchi which enter the lungs.

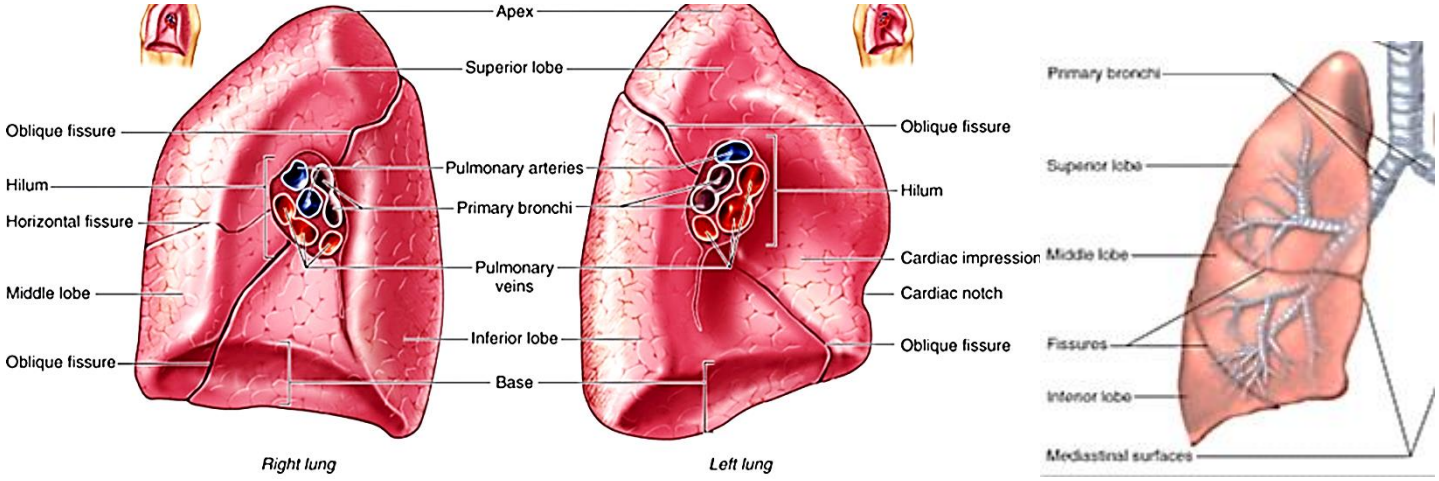
The right bronchus is considerably larger in diameter than the left and extends downward in a more vertical direction.



Therefore, if a foreign body is inhaled, it is likely to enter the right lung. Each bronchus enters the lung at a notch or depression called the hilus or hilum. The blood vessels and nerves also connect with the lung in this region.

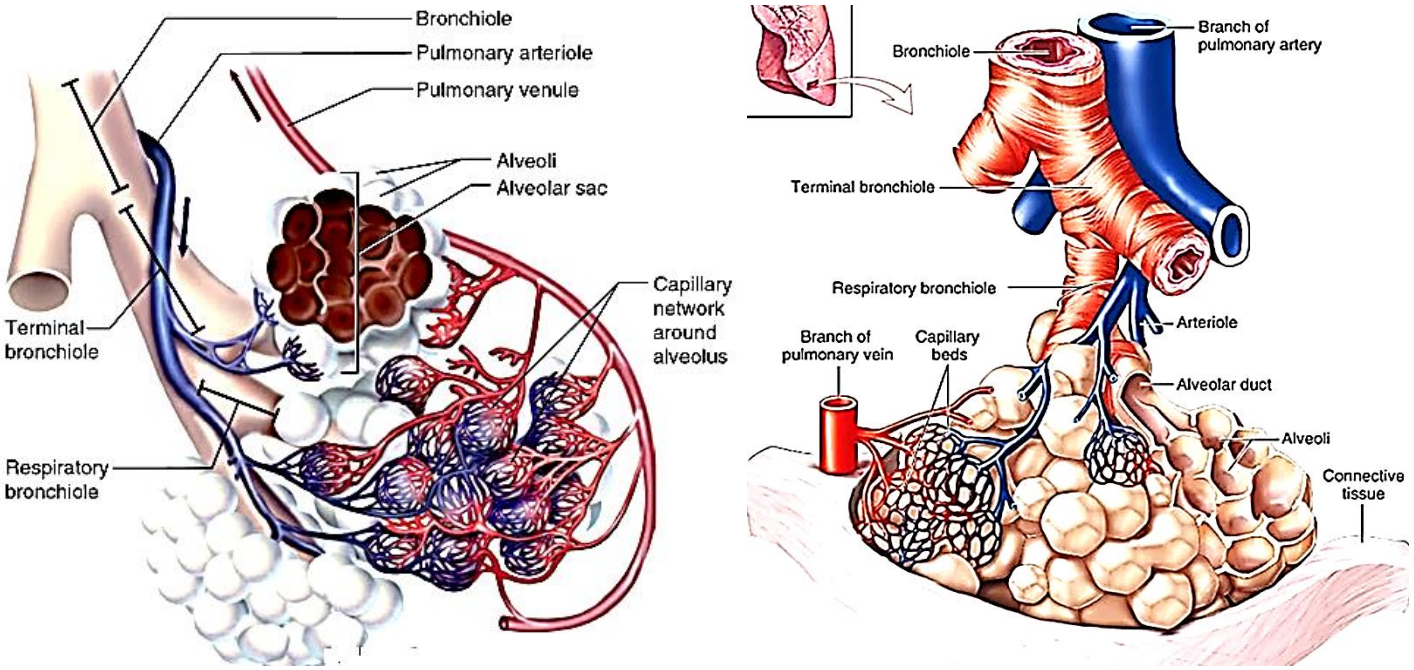
The Lungs

The two lungs, set side by side in the thoracic cavity, are constructed in the following manner: Each bronchus enters the lung at the hilus and immediately subdivides. Because the subdivision of the bronchi resembles the branches of a tree, they have been given the common name bronchial tree. The bronchi subdivide again and again, forming progressively smaller divisions, the smallest of which are called bronchioles. The bronchi contain small bits of cartilage, which give firmness to the walls and serve to hold the passageways open so that air can pass in and out easily.



However, as the bronchi become smaller, the cartilage decreases in amount. In the bronchioles there is no cartilage at all; what remains is mostly smoothly muscle, which is under the control of the autonomic nervous system.

At the end of each of the smallest subdivisions of the bronchial tree, called terminal bronchioles, is a cluster of air sacs, resembling a bunch of grapes. These sacs are known as alveoli. Each alveolus is a single-cell layer of squamous (flat) epithelium. This very thin wall provides easy passage for the gases entering and leaving the blood as it circulates through millions of tiny capillaries of the alveoli. Certain cells in the alveolar wall produce surfactant, a substance that prevents the alveoli from collapsing by reducing the surface tension ("pull") of the fluids that line them. There are millions of alveoli in the human lung.

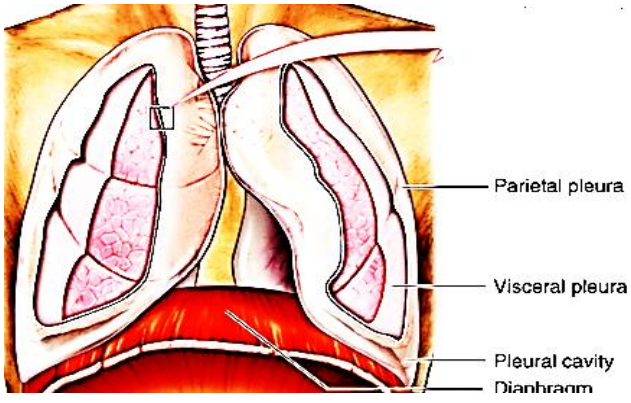


In the lungs blood passes through the capillaries around the alveoli, where the gas exchange takes place.

The Lung Cavities

The lungs occupy a considerable portion of the thorax cavity, which is separated from the abdominal cavity by the muscular partition known as the diaphragm. Each lung is enveloped in a

double sac of serous membrane called the pleura. The portion of the pleura that is attached to the chest wall is called parietal pleura, while the portion that is reflected onto the surface of the lung is called visceral pleura.



The pleural cavity around the lungs is an air-tight space with a partial vacuum, which causes the pressure in this space to be less than atmospheric pressure. Because the pressure inside the lungs is higher than that in the surrounding pleural cavity, the lungs tend to remain inflated. The entire thoracic cavity is flexible, capable of expanding and contracting along with the lungs. The region between the lungs, the mediastinum, contains the heart, great blood vessels, esophagus, trachea ,and lymph nodes.

The apex extend into the neck 2-3 cm above the clavicle and the base is concave and situated above the diaphragm

Physiology of Respiration

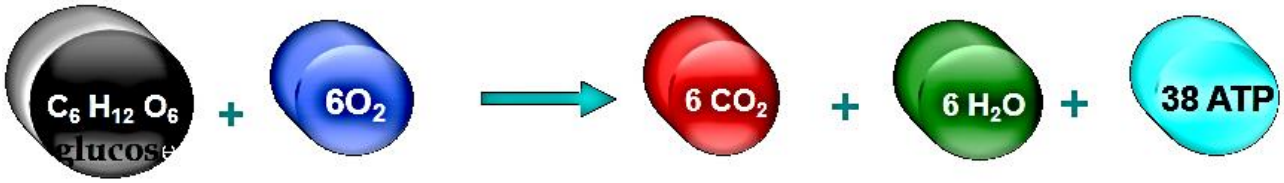
Pulmonary Ventilation : Ventilation is the movement of air into and out of the lungs, asin breathing.

There are two phases of ventilation :

1. **Inhalation** is the drawing of air into the lungs.

2. **Exhalation** is the expulsion of air from the lungs

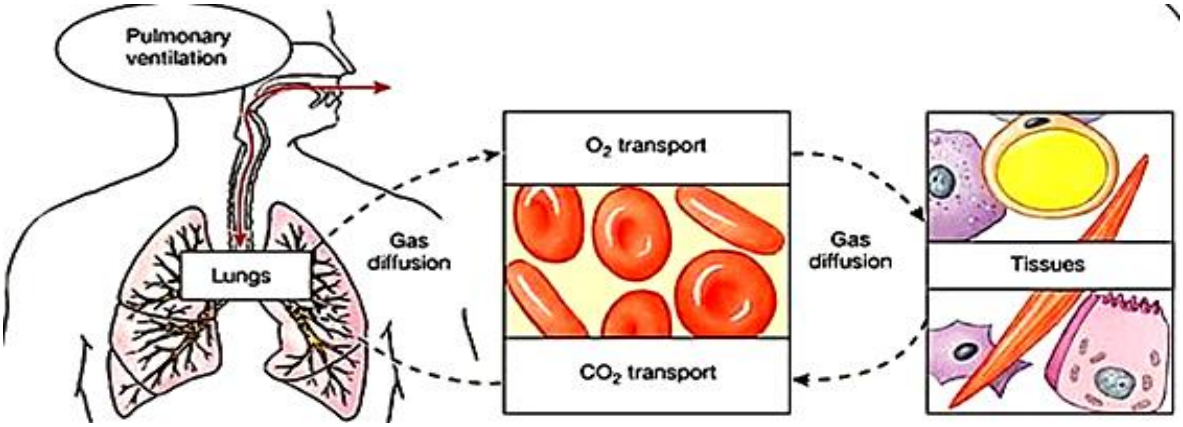
Respiration is the process by means of which O2 is used by the living cells for oxidation of the organic substances.



The process of Respiration is divided into:
I. External Respiration:
II. Transport of O2 & CO2 by Blood
III. Internal Respiration

- **I . External Respiration: It occurs at the lungs and consists of:**

1.Pulmonary Ventilation: Gas exchange between atmospheric air and air present inside the lungs.
2) Pulmonary Respiration: Gas exchange between alveolar air and venous blood. Venous blood gets rid of CO2 and becomes **loaded with O2 i.e. becomes arterial blood.**
3) Pulmonary perfusion: The alveoli are adequately perfused with blood which is essential for gas exchange between the **alveolar air and the venous blood.**



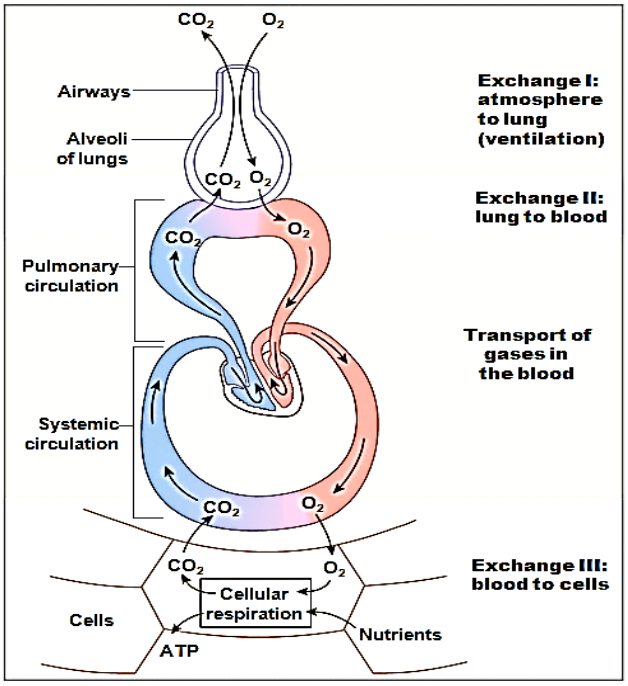
► **II. Transport of O2 & CO2 by Blood:**

O2 is carried from the lungs to the tissues to be utilized, and CO2 is carried from the the tissues to the lungs to be expired.

► **III. Internal Respiration:**

It occurs at tissues and consists of:

1. Exchange of gases between the arterial blood and the tissues; the arterial blood gives O2 to the tissues and becomes loaded with CO2 i.e. becomes venous blood.
2. Utilization of O2 by the tissues by the help of oxidative enzymes.



Other Functions of the Respiratory System

➢ **Lung Defense Mechanisms**

□ They humidify and cool or warm the inspired air so that even very hot or very cold air is at or near body temperature by the time it reaches the alveoli.

□ Bronchial secretions contain secretory immunoglobulins (IgA) and other substances that help resist infections and maintain the integrity of the mucosa.

□ The pulmonary alveolar macrophages (PAMs, "dust cells") are another important component of the pulmonary defense mechanisms.

Various mechanisms operate to prevent foreign matter from reaching the alveoli.

The hairs in the nostrils strain out many particles larger than 10 μm in diameter. Most of the remaining particles of this size settle on mucous membranes in the nose and pharynx; because of their momentum, they do not follow the airstream as it curves downward into the lungs, and they impact on or near the **tonsils** and **adenoids,** large collections of immunologically active lymphoid tissue in the back of the pharynx.

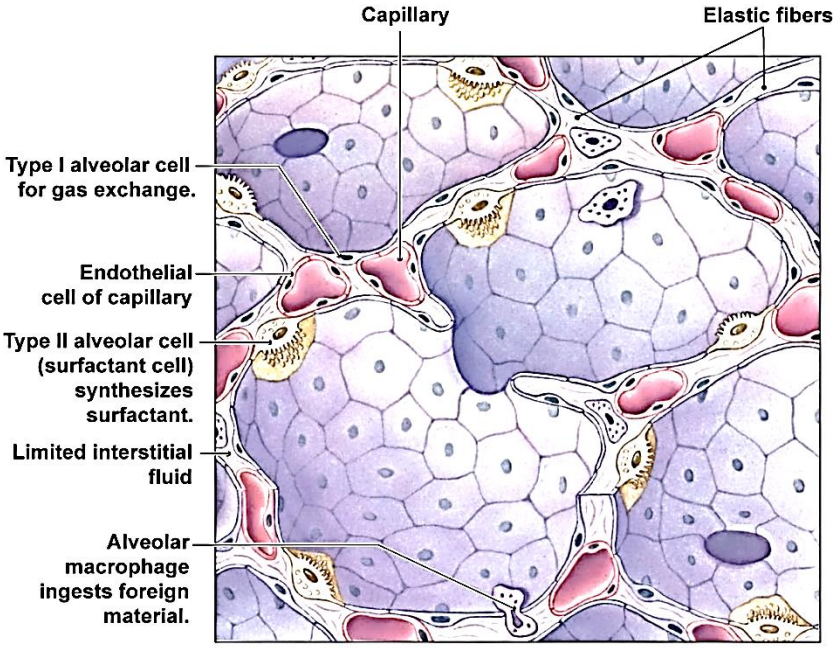
Particles 2–10 μm in diameter generally fall on the walls of the bronchi as the air flow slows in the smaller passages. There they initiate reflex bronchial constriction and coughing.

➢ **Metabolic & Endocrine Functions of the Lungs**

In addition to their functions in gas exchange, the lungs have a number of metabolic functions. They manufacture surfactant for local use. They also contain a fibrinolytic system that lyses clots

in the pulmonary vessels. They release a variety of substances that enter the systemic arterial blood, and they remove other substances from the systemic venous blood that reach them via the pulmonary artery. Prostaglandins are removed from the circulation, but they are also synthesized in the lungs and released into the blood when lung tissue is stretched.

- **Surfactant**:
  - More concentrated in smaller alveoli
  - Mixture containing proteins and phospholipids
  - Newborn respiratory distress syndrome
    - Premature babies
    - Inadequate surfactant concentrations
  - Decreases surface tension
  - Increases alveolar stability
  - Prevents transudation of fluid

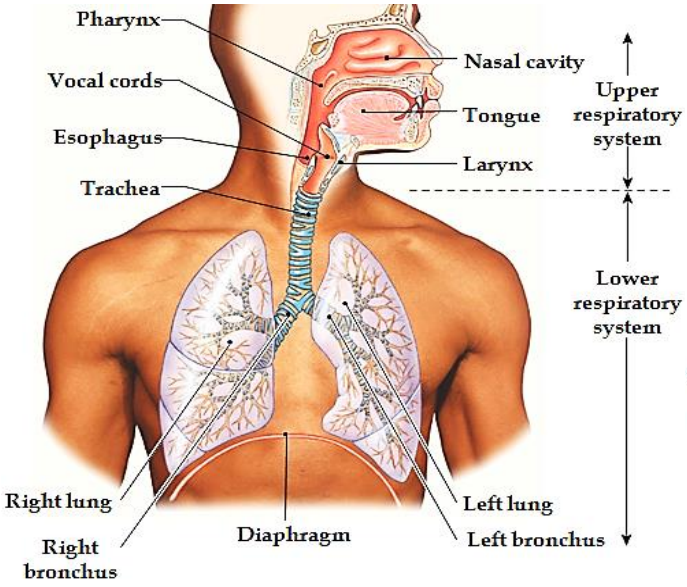


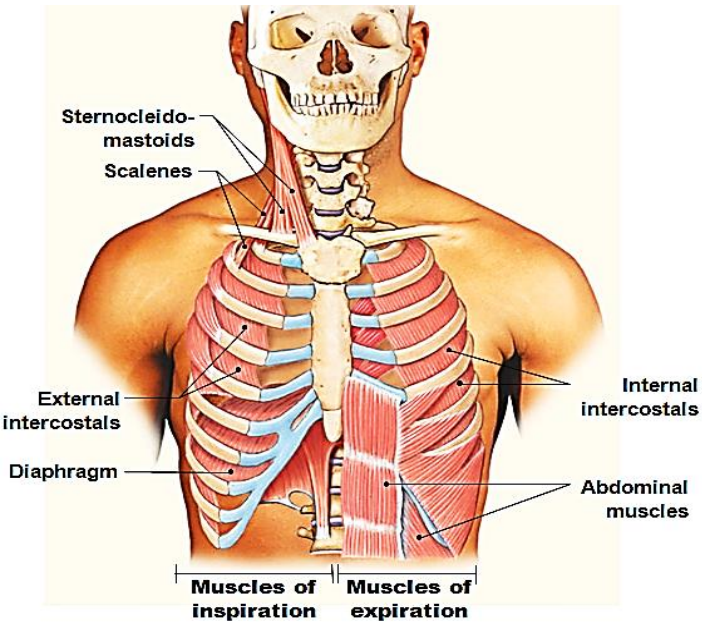
The lungs also activate one hormone; the physiologically inactive decapeptide angiotensin I is converted to, aldosterone-stimulating octapeptide angiotensin II in the pulmonary circulation. The reaction occurs in other tissues as well, but it is particularly prominent in the lungs.
Removal of serotonin and norepinephrine reduces the amounts of these vasoactive substances reaching the systemic circulation.
However, many other vasoactive hormones pass through the lungs without being metabolized. These include epinephrine, dopamine, oxytocin, vasopressin, and angiotensin II.
The respiratory system is the anatomical system of an organism that introduces respiratory gases to the interior and performs gas exchange.
Molecules of O2 and CO2 are passively exchanged, by diffusion, between the gaseous external environment and the blood. This exchange process occurs in the alveolar region of the lungs

Respiratory System Components

- Conducting system
- Alveoli
- Bones and muscle of thorax

Muscles Used for Ventilation

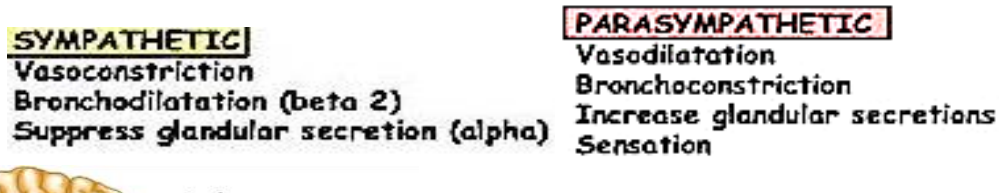


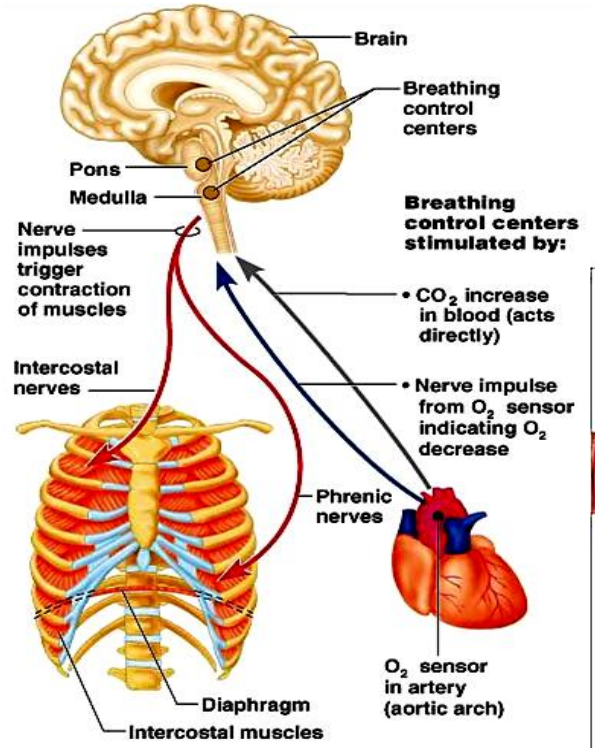


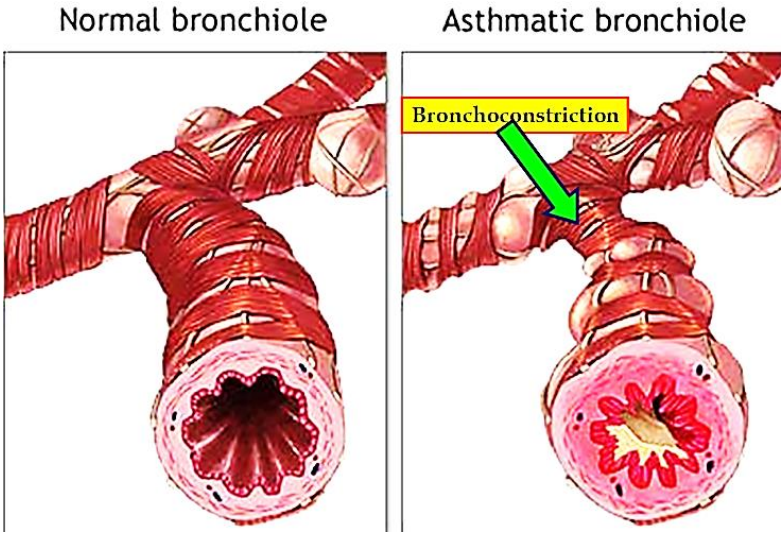
1- Air conducting part: Where no gas exchange occurs, it consists of nasal cavity (each lined with a ciliated, glandular (mucous) epithelium), pharynx (common for both food and air), larynx (provided with vocal cords), trachea (provided with supporting cartilaginous rings), main bronchi (each bronchus passing to the corresponding lung) and their branches till the terminal bronchioles (each repeatedly branching, until it ends into a huge number of tiny saccular " alveoli").

The muscular walls of bronchioles receive a double nerve supply:

- from the " vagus nerve " , which bring about their contraction ,
- from the " sympathetic " , which causes their dilatation .







The contraction of bronchiolar muscles leads to reduction of the bronchiolar lumen (accordingly, resistance to the flow of air into or out of the alveoli). Therefore, an intense spasm of these of these muscles caused the distressing condition known as "asthma".

Respiratory part: A pair of lungs, where gas exchange occurs, it consists of respiratory bronchioles, alveolar ducts and alveoli.

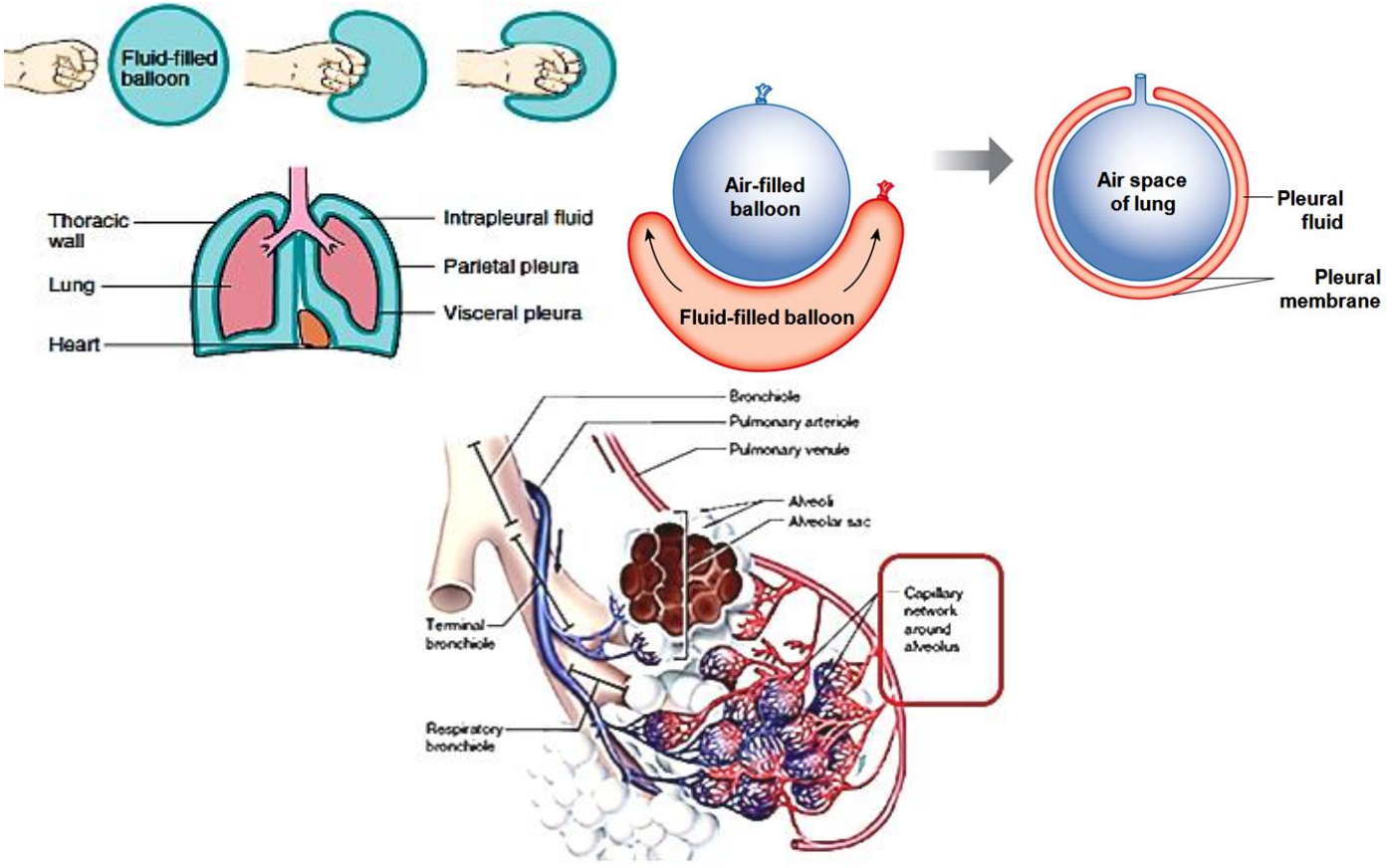
In health, both pleurae are everywhere in contact with one another, with only so little fluid in between as will ensure the minimum of friction and the easy gliding of lung in its expansion and retraction.

On entering a lung lobule, the walls of the bronchiolar tree become thinner and thinner, until at length they are lined by a single layer of squamous epithelium.

Outside the alveoli, a network of similarly thin – walled capillaries is spread out so densely. Thus between the air in the lungs and the blood in the capillaries nothing intervenes but a thin partition, which is only two squamous cells in thickness.

In this way, the exposure of blood to air is the most complete. Moreover, between the adjacent alveoli, there exist only a single layer of capillaries (both sides of which are thus exposed simultaneously to the alveolar air).

Relationship of lungs, pleura, and thoracic wall, shown as analogous to pushing a fist into a fluid-filled balloon. Note that there is no communication between the right and left intrapleural fluids. For purposes of illustration in this figure, the volume of intrapleural fluid is greatly exaggerated. It normally consists of an extremely thin layer of fluid between the pleura membrane lining the inner surface of the thoracic wall (the parietal pleura) and that lining the outer surface of the lungs (the visceral pleura).

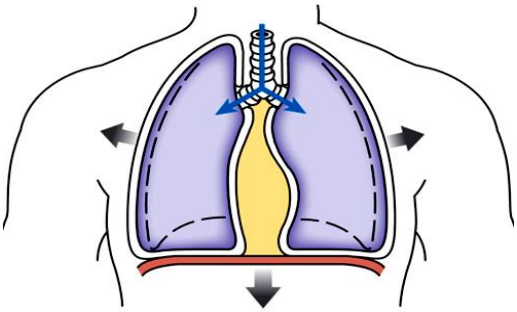


I – Physical Aspect of Respiration

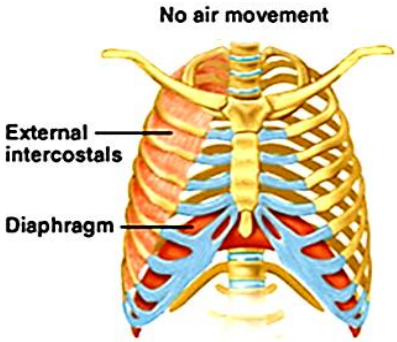
At rest, the inspiratory and expiratory movements alternate rhythmically at a rate of 15-20 / min in healthy adults.

A–During inspiration, all diameters of the thorax are increased:

1. Increase in the vertical diameter is brought about by contraction of the diaphragm (the fibers of which initially arch upwards and inwards). By such contraction, the diaphragm descends (becomes less arched). The diaphragm is the most important muscle in respiration.



Diaphragm contracts, thoracic volume increases.



Relaxed state

2. The thorax is also enlarged in its other diameters by contraction of the "external intercostals muscles", which extend from the lower border of one rib (where the origin of muscle is), to the upper border of the next rib below (muscle insertion).

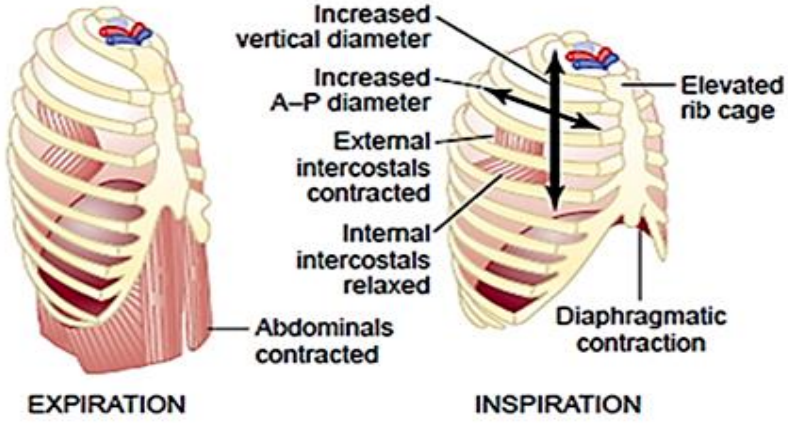
Since the ribs initially slope anteriorly down words, therefore on contraction of the external intercostals, the anterior end of the ribs (of the middle ones in particular) is drawn up words to a front position .The sternum is thus thrust forwards, causing an increase in the anteroposterior diameter of thorax. Meanwhile, the lower ribs are rotated a little outwards, causing an increase in the transverse diameter of chest.

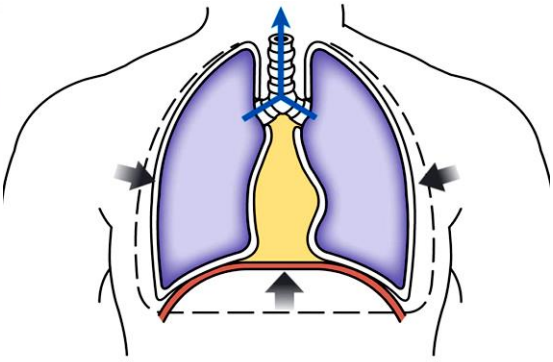
B- Expiration:

(1) **The diaphragm relaxes**, ascending upwards in thus helping to expel the contained air by pressing on the lungs from below.

(2) **The external intercostals also relax**, thus bringing about lowering of the ribs, which results in pressing the lungs from in front backwards, and from outside inwards. Moreover, the thoracic walls return to their initial position.

Due to their own weight. and due to the pull exerted upon them from inside by natural recoil of the elastic wall of lung. In forcible expiration (e.g. in coughing): The factors just mentioned are supplemented the space of the abdominal muscles, by which the spaces of the abdomen is reduced. The contained viscera are thus pushed upwards, and the (relaxed) diaphragm is forced more upwards.

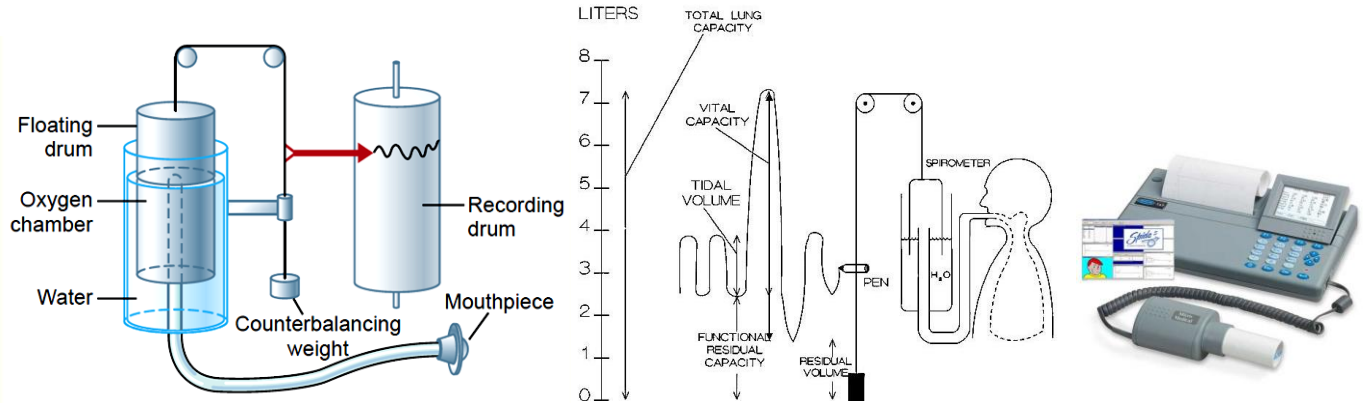




Diaphragm relaxes, thoracic volume decreases.

- **Lung volumes and capacities**

Most of the lung volumes and capacities are measured by the Spirometer which records the changes which occur in the lungs during respiration.



Lung volumes:

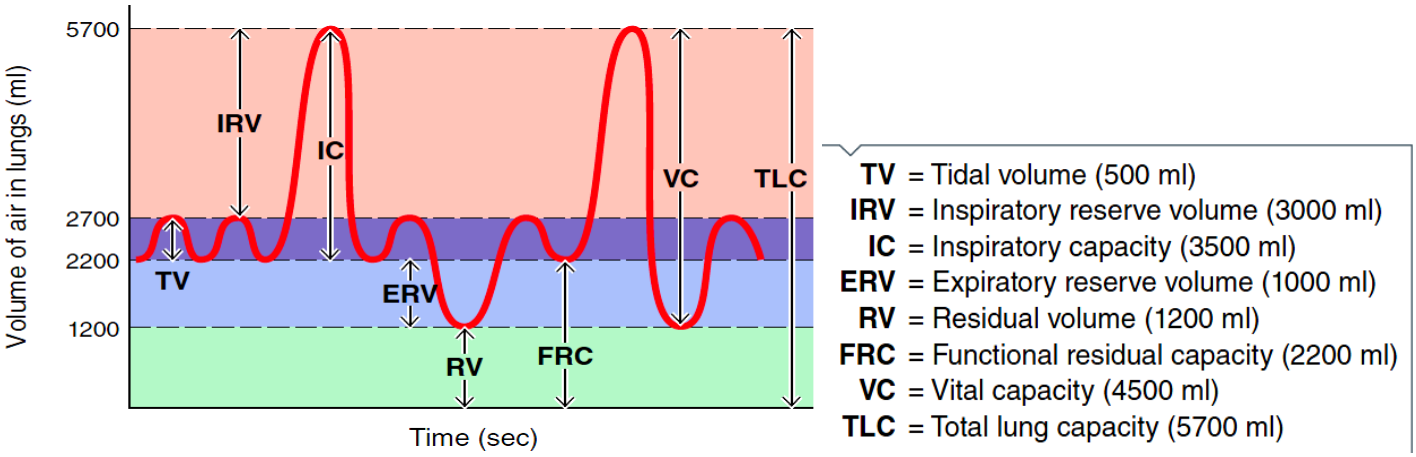
1) **Tidal volume:(T.V)** 500c.c. It is the volume of air inspired or expired during rest.

2) **Inspiratory Reserve volume: (I.R.V)** 3000cc. It is the maximum volume of air which can be inspired after normal inspiration.

3) **Expiratory reserve volume: (E.R.V)** 1000c.c. It is the maximum volume of air which can be expired after normal expiration.

N.B. The above 3 volumes can be measured by the Spirometer.

4) **Residual volume:** (R.V) 1200 c.c. It is the volume of air remaining in the lungs after maximal expiration. It can't be expelled to atmosphere except after opening of the chest wall and squeezing the lungs.



Lung capacities: (It equals 2 or more of lung volumes).

1. Inspiratory capacity: 3500 c.c. It is the maximum volume of air which can be inspired after normal expiration. It equals T.V (500) + I.R.V (3000) = 3500 c.c.

2. Functional Residual capacity: 2200 c.c. It is the volume of air remaining in the lung after normal expiration.

It equals E.R.V (1000) + R.V (1200) = 2200 c.c.

3. Total lung capacity: (T.L.C) 5700 c.c. It is the volume of air contained in the lungs after deepest inspiration. It equals all lung volumes = 5700 c.c.

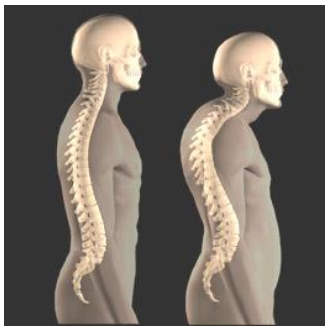
4. Vital capacity: (V.C) 4500 c.c. It is the volume of air given out by maximal expiration after maximal inspiration.

It equals I.R.V (3000) + T.V (500) + E.R.V (1000) = 4500 c.c.

- **Factors affecting vital capacity:**

1. **Condition of the respiratory muscles:** In athletes respiratory muscles are strong; so, V.C ↑ . In poliomyelitis respiratory muscles are paralyzed so, V.C↓ .

2. **Condition of thoracic cage:** In deformities of thoracic cage e.g. Kyphosis, Lordosis and Sclerosis, V.C↓ .



3. **Elasticity of the lungs:** In emphysema and pulmonary fibrosis, elasticity of the lungs↓. So, V.C↓.
4. **Resistance to air flow in Respiratory passage:** In Bronchial asthma, Resistance ↑. So, V.C ↓.
5. **Volume of blood in pulmonary circulation:** In left side heart failure and in mitral stenosis the volume of blood in pulmonary circulation ↑ which take the place of air. So, V.C↓.
6. **Posture:** In the recumbent position V.C is less than in the standing or sitting positions due to ↑ blood volume in the pulmonary circulation. due to ↑R.V because blood takes place of air. Viscera press in the diaphragm and limits its descent.
7. **Intra-abdominal pressure:** In pregnancy and ascites it increases and limits the descend of diaphragm. So, V.C↓.

❖ **Ventilation of Alveoli**

- **Inspired air is divided into 2 parts:**

a) Part of it remains in the air conducting part where no gas exchange takes place. This part is called the Dead space, it is about 150c.c.
b) The other part is present in the respiratory part where gas exchange takes place. It is called Alveolar air, it is about 350c.c.

Expired air is a mixture of Alveolar air and Dead space air.

Difference between Alveolar air and Expired air:

♣ **Alveolar air:**

It is the air present in the respiratory part. It undergoes gas exchange with the venous blood. So, it is relatively poor in O_2 14% and rich in CO_2 6%.

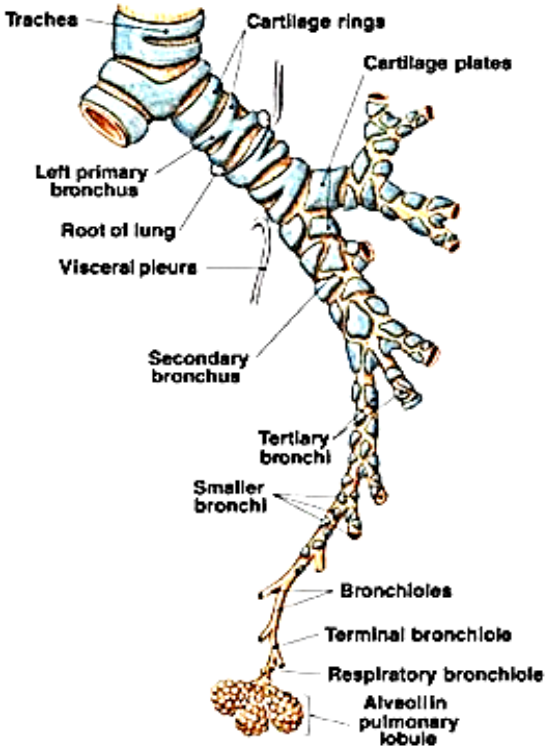
♣ **Expired air:**

It is a mixture of Alveolar air and air present in the Dead space, which remain Atmospheric (i.e. rich in O_2 20% and poor in CO_2 0.1%). So, Expired air contains more O_2 16% and less CO_2 4% compared with Alveolar air.

Types of Dead space (D.S)

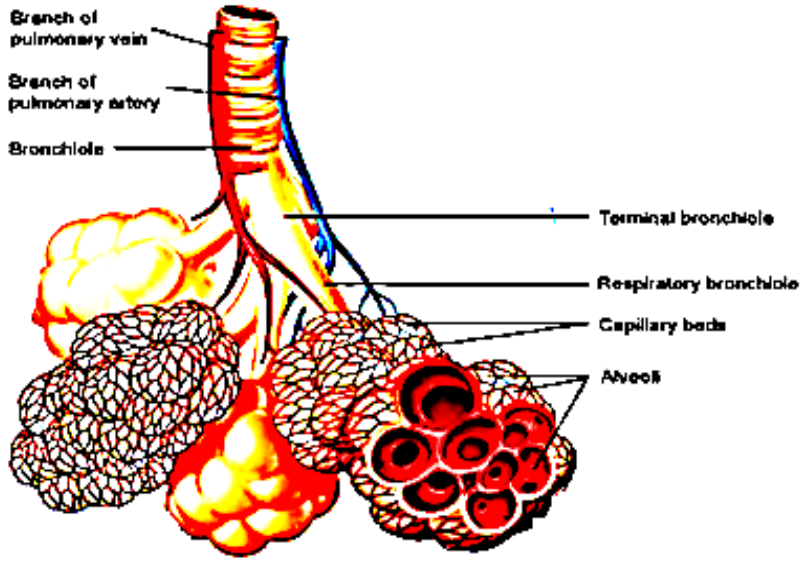
1. Anatomical Dead space:

It is the volume of air present in the air conducting part down to terminal bronchiole. It is about 150c.c.



2. Alveolar Dead space:

It includes the alveoli which are ventilated but do not receive blood supply. So, there is also no gas exchange.



Physiological Dead space:

It equals Anatomical D.S + Alveolar D.S. Normally Anatomical D.S = Physiological D.S. i.e. no Alveolar D.S because all alveoli receive blood supply.

Functions of the Dead space:

1- It adjusts the temperature of the inspired air before it reaches the alveoli.

2- It moistens the inspired air before it reaches the alveoli.

3- It filters the inspired air from particulate matters. Large particles are caught in the hairs of the nose or stick to its mucous covering. Smaller particles which pass the nose stick to the mucous covering of the trachea or bronchi.

4- It is provided with many mechanisms to remove the particulate matters that stick to the wall e.g.:

a) **Movement of the cilia** of the epithelial lining the D.S.
b) **Sneezing reflex:** It results from irritation of the nasal mucosa. It consists of deep inspiration followed by strong and rapid expiration to remove the irritant. (Glottis is continuously open).
c) **Cough Reflex:** It results from irritation of the larynx or trachea or bronchi. It consists of moderate inspiration followed by strong Exp. with glottis at first closed then opened to remove the irritant.

II – Chemical Aspect of Respiration

Transport of the respiratory gases:

❖ **Concerning oxygen:**

➔ **O_2 carriage by Blood; O_2 present in blood in 2 forms:**

I- In physical solution:

It is dissolved in the plasma. It is very little in amount but it is very important because:

- It determines the O_2 tension of blood (*Tension means pressure of gas in aliquid*).
- Its tension determines the direction and rate of diffusion of O_2. Its amount in the arterial blood is 0.3 cc / 100 cc blood, in the venous blood it is 0.13 cc /100 cc blood. So, it is not enough for tissue oxygenation.

II- In chemical combination:

It is loosely combined with hemoglobin (HB). Each 100 cc blood contains 15 gm HB. Each gm can carry 1.33 O_2. So, 100 cc blood can carry 15 × 1.33 = 20 cc O_2 (when fully saturated).

➢ **O_2 content:**

It is the volume of O_2 present in chemical combination with HB of 100 cc blood. It varies according to O_2 tension, and HB content. In arterial blood 19 c.c/100 c.c, and in venous blood 14 c.c/100 c.c blood.

➢ **O_2 capacity:**

It is the volume of O_2 present in chemical combination with HB of 100 cc of blood when HB is fully saturated with O_2. It equal 20 cc/100 cc blood. Blood inside the body is never becomes fully saturated because:

1. O_2 tension inside the lung alveoli is not enough to saturate blood.
2. Not all venous blood go to lungs undergo gas exchange, little amount of blood supplies the bronchi and returns venous.
3. In left atrium and ventricle some of the coronary venous blood is mixed with the oxygenated blood coming from lungs.

It is measured outside the body by exposing blood to high O_2 pressure e.g. atmospheric air.

► **Oxygen Transport**

Oxygen Delivery to the Tissues

The O_2 delivery system in the body consists of the lungs and the cardiovascular system. O_2 delivery to a particular tissue depends on the amount of O_2 entering the lungs, the adequacy of pulmonary gas exchange, the blood flow to the tissue, and the capacity of the blood to carry O_2. The blood flow depends on the degree of constriction of the vascular bed in the tissue and the cardiac output. The amount of O_2 in the blood is determined by the amount of dissolved O_2, the amount of hemoglobin in the blood, and the affinity of the hemoglobin for O_2.

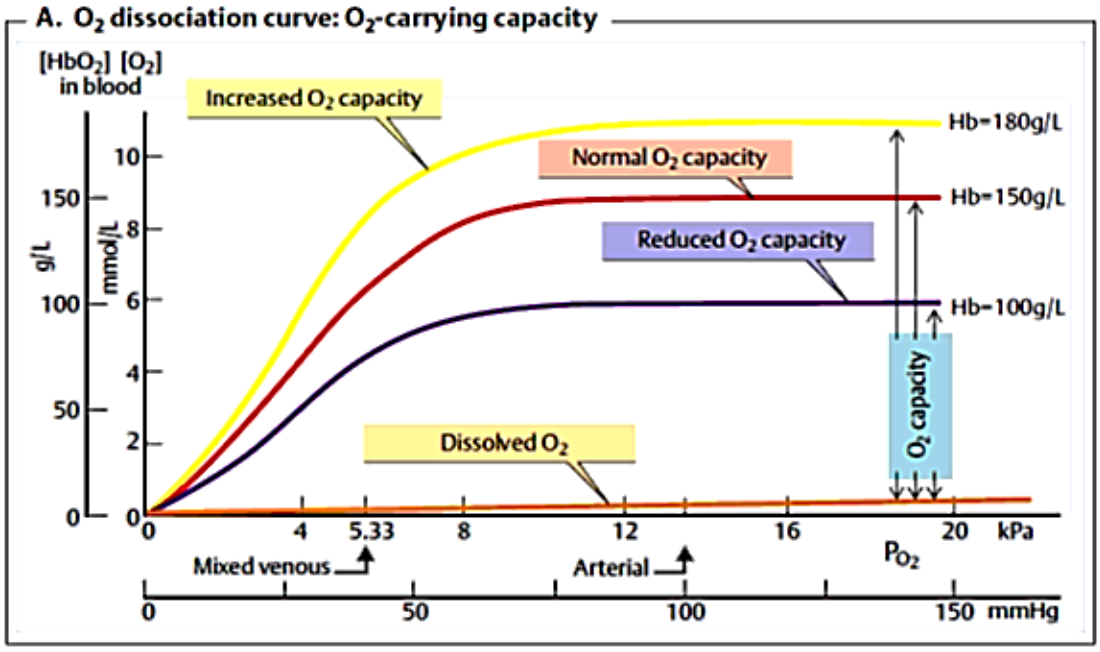
In arterial blood $= \frac{19}{20} \times 100 = 95\%$

In venous blood $= \frac{14}{20} \times 100 = 70\%$

► **O_2 dissociation curve:**

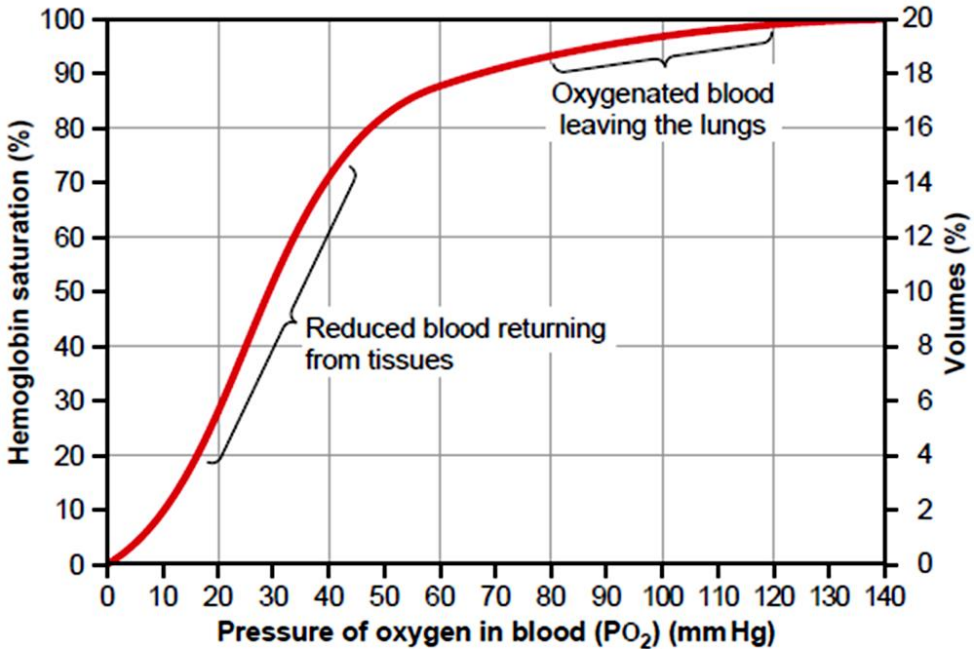
A.O2 dissociation curve of blood HB:

It represents the relation between O_2 tension and O_2 % saturation. O_2 capacity is determined by exposing a sample of blood to a high O_2 pressure e.g. atmospheric air. O_2 content varies according to O_2 tension. It is determined by exposing samples of blood to various O_2 tensions from 0 to 100 mmHg. O_2% saturation of each sample is determined, and O_2 dissociation curve is drown between O_2 tension and O_2% saturation



The oxygen-hemoglobin dissociation curve, which demonstrates a progressive increase in the percentage of hemoglobin, bound with oxygen as blood Po_2 increases, which is called the per cent saturation of hemoglobin. Because the blood leaving the lungs and entering the systemic arteries usually has a Po_2 of about 95 mmHg, one can see from the dissociation curve that the usual oxygen saturation of systemic arterial blood averages 97 %.

Conversely, in normal venous blood returning from the peripheral tissues, the Po_2 is about 40 mmHg, and the saturation of hemoglobin averages 75 %.



Physiological significance of O_2 Dissociation Curve:

1) **Arterial point:**

- **At sea level:** O_2 tension in the alveoli and arterial blood is about 100 mm Hg. HB is highly saturated 95%.
- **At high altitude:** O_2 tension in the alveoli and arterial blood decreases to 80 mm Hg. HB is still highly saturated 93%, this enables the persons living at high altitude to get enough O_2. So the arterial point moves in the horizontal part with little ↓ in O_2 % saturation.

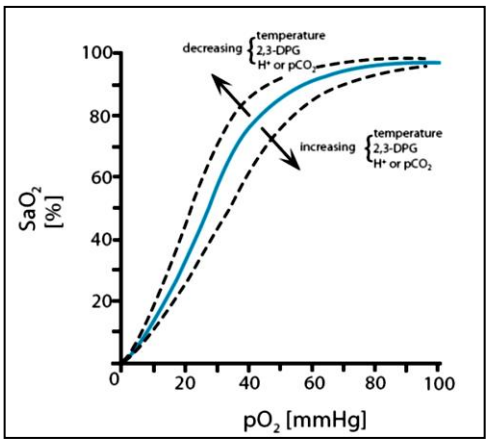
2) **Venous point:**

- **At rest:** O_2 tension in the tissues and venous blood is 40 mm Hg. HB is 70% saturated. So, the tissues take 25% of arterial O_2 which is enough at rest.
- **During muscular exercise:** O_2 tension in the tissues and venous blood decreases to 20 mm Hg. HB becomes 20% saturated. So, the tissues take 75% of arterial O_2 which enough to supply the active muscles. So, venous point moves on the vertical part with marked ↓ in O_2 % saturation.

Factors affecting O_2 dissociation curve:

I. Factors shifting the curve to right:

1) ↑ Co_2 tension (as in muscular exercise).
2) ↑$[H^+]$ (as in diabetic acidosis).
3) ↑Temperature of blood. (as in fevers).
4) ↑HB concentration (as in Polycythemia).
5) ↑2,3 Di-phosphoglycerate (2,3 DPG), helps release of O_2 from HG to the tissues. 2,3 DPG results from anaerobic glycolysis which occurs inside the RBC. ↑ 2,3 DPG occurs in muscular exercise and hypoxia.



- **Significance of shift to right:**

Shift to right means that the affinity of HB towards O_2 is decreased i.e. helps unloading from O_2. So, HB gives its O_2 more easily to the tissues. During muscular exercise there is ↑ CO_2 , ↑ H^+ , ↑ temperature , ↑ HB and ↑ 2,3 DPG. So the curve is shifted to right to provide the active muscle with enough O_2.

II. Factors shifting the curve to left:

1) ↓ CO_2 tension (as during sleep).
2) ↓ $[H^+]$ (as in high altitude).
3) ↓ temp of blood (as during sleep).
4) ↓ HB concentration (as in anemia).
5) ↓ 2,3 DPG.

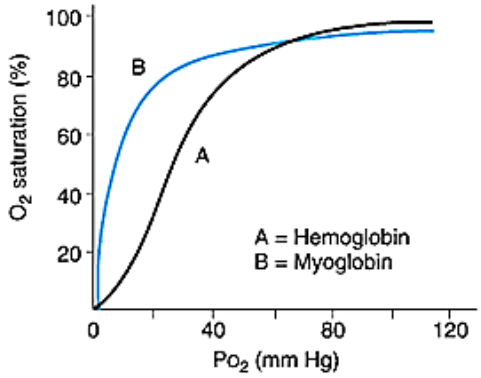
- **Significance of shift to left:**

Shift to left means that the affinity of HB towards O_2 is increased i.e. helps loading with O_2. So, HB gives its O_2 with difficulty to the tissues. During rest or sleep there is ↓ CO_2, ↓ H^+, ↓ temperature & ↓ HB. So, the curve is shifted to left i.e. gives little O_2 to the tissues.

➢ **Myoglobin**

Myoglobin is present in the muscle. It contains one Fe atom which can combine with one O_2 molecule. O_2 D. curve of the Myoglobin is rectangular hyper polar i.e. it remains horizontal till very low O_2 tension then suddenly descends vertically.

It does not give its O_2 except at very low O_2 tension e.g. during severe muscular exercise. After exercise Myoglobin takes O_2 rapidly and becomes saturated again. So, Myoglobin acts as O_2 store to be used only during severe muscular exercise.



Carbon Dioxide Transport As regards CO2:

(a) When CO_2 is taken up by the blood of tissues it diffuses from the plasma into the corpuscles, where:

(i) a large part of it (60-65 %) is hydrated – through the catalytic action of a specific enzyme known as " Carbonic anhydrase " into carbonic acid , which gives $HCO3^-$ ions. The majority of these ions exchange their position in the RBCs with the Cl^- ions of plasma, resulting in: $NaHCO_3$ in the venous plasma and KCl inside the RBCs (Chloride shift).

Mechanism of Gas Exchange

(A) Gas Exchange at the lungs:

1- In physical solution:

It occurs by diffusion across the pulmonary membrane which consists of alveolar membrane, capillary membrane and interstitial fluid in between. The rate of diffusion depends on the following:

i. **Pressure gradient on both sides of the membrane:** O_2 diffuses from the alveoli (O_2 T. 100 mmHg) to the venous blood (O_2 T. 40 mmHg), CO_2 diffuses from the venous blood (CO_2 T. 46 mmHg) to the alveoli (CO_2 T. 40 mmHg).

ii. **Molecular weight of the gas:** The smaller the molecular weight the more the diffusion. CO_2 has a larger molecular weight than O_2.

iii. **Solubility of the gas in the membrane:** The respiratory gases are soluble in the membrane but CO_2 is more soluble than O_2. This makes CO_2 more diffusable than O_2 in spite of its less pressure gradient, and its larger molecular weight.

iv. **Surface area of the respiratory membrane:** In the lungs it is large 40 m^2, this helps in rapid diffusion of gases.

v. **Thickness of the respiratory membrane:** It is formed of alveolar epithelium, capillary endothelium, and interstitial space in between. The membrane is very thin, so it helps easy diffusion of gases.

❖ **During muscular exercise the diffusion of gases increases due to:**

i. ↑ **Pressure gradient of gases:** because O_2 T ↑ in the alveoli due to hyperventilation, and ↓in the venous blood due to↑ O_2 consumption. CO_2 ↑ in the venous blood due to ↑ metabolism, and ↓ in the alveoli due to hyperventilation.

ii. ↑ **Surface area of pulmonary membrane:** due to dilatation of the pulmonary capillaries and more expansion of the lung alveoli.

iii. ↓ **Thickness of the pulmonary membrane:** due to dilatation of the pulmonary capillaries and expansion of the lung alveoli.

2- In chemical combination:

O_2 diffuses from the lung to the R.B.Cs and combine with HB to form HBO_2. HBO_2 is a strong acid , so it combines with K^+ to form $K.HBO_2$. HCO_3^- combines with H^+ to form H_2CO_3. The later dissociates to CO_2& H_2O. CO_2 produced pass from R.B.Cs to the alveolar air. This process is repeated until CO_2& O_2 in the alveolar air reach equilibrium with the arterial blood.

(B) Gas Exchange at the tissues:

(i) In physical solution: It occurs by diffusion through capillary cellular membrane. The rate of diffusion depends upon the same factors as in the lungs.

(ii) In chemical combination: It occur in opposite direction to what occurs in the lungs.

Nervous Aspect of respiration

Respiration is essentially on involuntary out (automatic) act. It cannot be suspended for > 35 seconds or so.

Nervous control of respiration:

The automatic character of respiration is due to rhythmical discharge of impulses, from a pair of groups of neurons situated in the floor of 4th ventricle (in medulla oblongata of the brain), known as respiratory centers (R.C).

R.C is composed of the following centers:

1- Inspiratory center (I.C) &

2- Expiratory center (E.C):

They lie in the medulla oblongata, I.C is ventral while E.C is dorsal to it. They show reciprocal innervation i.e. when one is stimulated the other is inhibited. Stimulation of I.C & E.C together will produce inspiration because I.C is more powerful than E.C. I.C is continuously active unless it is inhibited by the vagal fibers from the lungs and Pneumotaxic center in the pons.

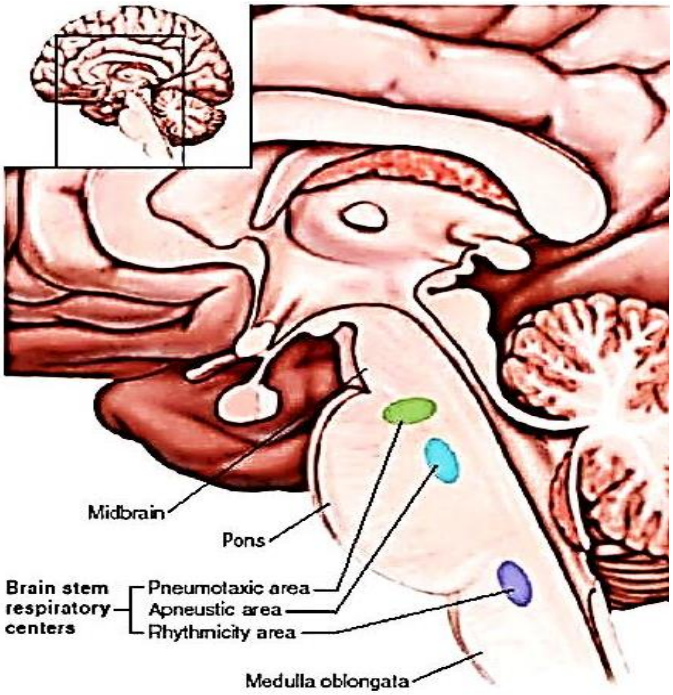
3- Apneustic center: (Ap.C) It is present in the lower part of the pons. It is the pace maker of respiration. It stimulates I.C and inhibits E.C. It is inherently active unless inhibited by the vagal fibers from the lungs and Pneumotaxic center in the pons.

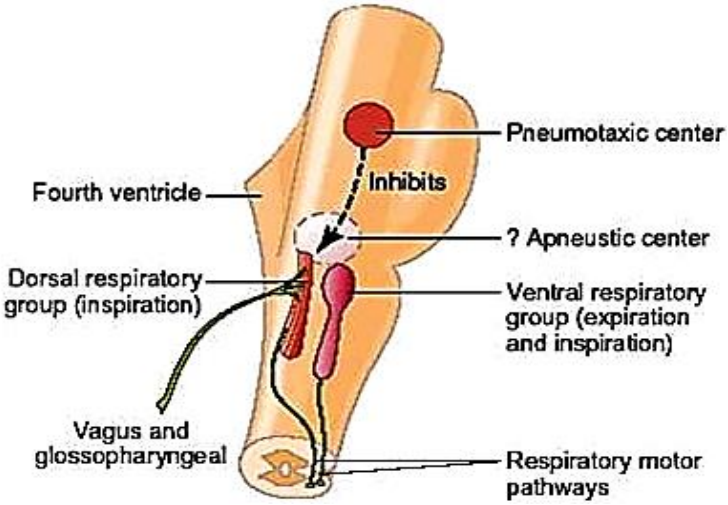
4- Pneumotaxic center: (Pn.C) It is present in the upper part of the pons. It is stimulated from I.C &Ap.C. When Pn.C is stimulated, it inhibits I.C & Ap.C and stimulates E.C. It is stimulated in only in deep inspiration and after cutting of both pulmonary vagi.

5- Gasping center: It is present in the lower part of the medulla. It has no function during normal respiration because it is continuously inhibited from the Ap.C. It only acts before death after failure of Ap.C. It produces gasping respiration in the form of deep spasmodic inspirations which is repeated for few times then death occurs.

The mechanism of (R.C) is summarized as follows:

a) The Inspiratory (In) center discharges impulses spontaneously and continuously to the inspiratory muscles. But, this activity is periodic or "rhythmic" (not continuous with no interruption), as inhibitory impulses reach the inspiratory center, coming from the distended lungs (due to inspiration) and from the Pneumotaxic center.
b) The inhibitory discharges from distended lungs are created when inspiration is at its height and the bronchiolar muscles are much stretched, because this stretching affects "stretch receptors" in the lungs, whereby inhibitory impulses ascend along sensory (afferent) vagal fibers, reaching the inspiratory center, to inhibit it.





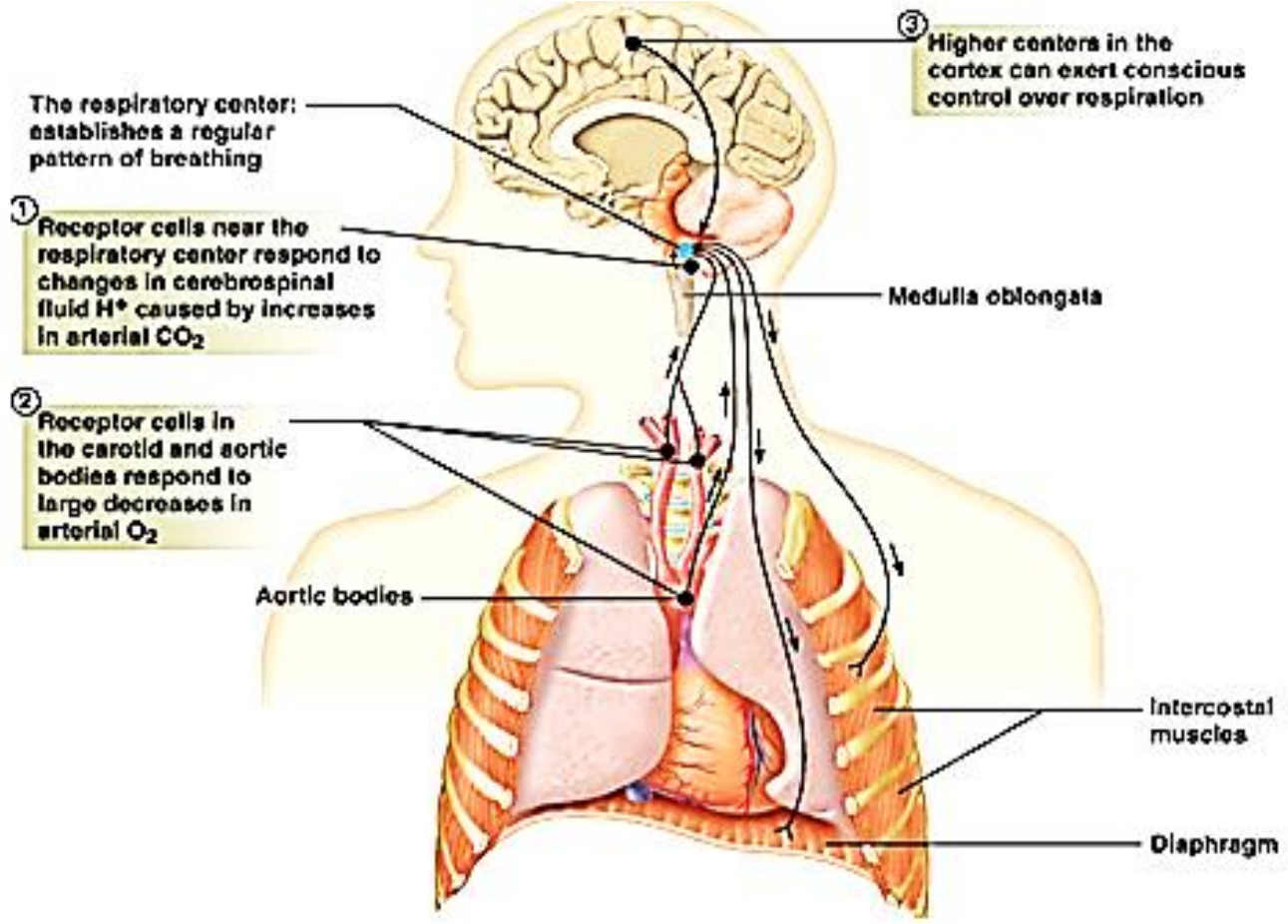
Regulation of respiration

(A) Nervous Regulation:

The (R.C) should be supplied constantly with information's from the lungs and other parts, so that it may react to alter the character of respiration as needed , Such information's concern:

(i) The state of distention of lungs.

(ii) The gaseous content of great arteries (specially CO_2 content) and others. This constitutes the sensory side of nervous control of respiration. On the motor side, reflex impulses are went down (in response to the information's) along motor (efferent) fibers of the nerves supplying the respirator any muscles, so as to modify the action of these muscles.



(B) Chemical Regulation:

Respiration is chemically controlled by the influence of changes in the content of gases on the (R.C).

(1)Changes in CO_2 tension in the alveolar air and arterial blood :

CO_2 tension in these sites is the natural stimulus for normal respiration, as the (R.C.) is very sensitive to it. While the decrease in CO_2 – tension (termed "hypocapnia", as occurs in over breathing voluntarily) inhibits respiration the increase in tension ("hypercapnia" occurring after muscular exercise) bring about increased respiration (called hyperpnoea). The increased CO_2-tension has a direct action on the neurons of (R.C.) through their arterial blood supply, and has also a reflex action on (R.C.) through "chemo – receptors" found in the carotid and aortic bodies.

(2)Change in O_2 – tension:

O_2 – lack (hypoxemia) stimulates breathing, e.g. in high altitude, after performing voluntary hyperventilation, and in the diseases the (R.C.) becomes insensitive to CO_2.

Coughing & Sneezing

Coughing begins with a deep inspiration followed by forced expiration against a closed glottis. This increases the intrapleural pressure to 100 mm Hg or more. The glottis is then suddenly opened, producing an explosive outflow of air at velocities up to 965 km (600 miles) per hour. Sneezing is a similar expiratory effort with a continuously open glottis. These reflexes help expel irritants and keep airways clear.

Respiratory Components of Visceral Reflexes

Hiccup is a spasmodic contraction of the diaphragm and other inspiratory muscles that produces an inspiration during which the glottis suddenly closes. The glottic closure is responsible for the characteristic sensation and sound. Hiccups occur in the fetus in utero as well as throughout extrauterine life. Their function is unknown. Most attacks of hiccups are usually of short duration, and they often respond to breath-holding or other measures that increase arterial PCO_2. Intractable hiccups, which can be debilitating, sometimes respond to dopamine antagonists and perhaps to some centrally acting analgesic compounds.

Yawning is a peculiar "infectious" respiratory act whose physiologic basis and significance are uncertain. Like hiccupping, it occurs in utero, and it occurs in fish and tortoises as well as mammals. The view that it is needed to increase O_2 intake has been discredited. Underventilated alveoli have a tendency to collapse, and it has been suggested that the deep inspiration and stretching open them and prevent the development of atelectasis. However, in actual experiments, no atelectasis-preventing effect of yawning could be demonstrated. Yawning increases venous return to the heart, which may benefit the circulation. It has been suggested that yawning is a nonverbal signal used for communication between monkeys in a group, and one could argue that on a different level, the same thing is true in humans.

Artificial Respiration

In acute asphyxia due to drowning, CO or other forms of gas poisoning, electrocution, anesthetic accidents, and other similar causes, artificial respiration after breathing has ceased may be lifesaving. It should always be attempted, because respiration stops before the heart stops. There are numerous methods of emergency artificial respiration, but the method presently recommended to produce adequate ventilation in all cases is mouth-to-mouth breathing. The advantages of mouth-to-mouth resuscitation lie not only in its simplicity but also in the fact that it works by expanding the lungs.

Mouth-to-Mouth Breathing :In this form of resuscitation, the operator first places the victim in the supine position and opens the airway by placing a hand under the neck and lifting, while keeping pressure with the other hand on the victim's forehead. This extends the neck and lifts the tongue away from the back of the throat. The victim's mouth is covered by the operator's mouth while the fingers of the hand already on the forehead occlude the nostrils . About 12 times a minute, the operator blows into the victim's mouth a volume about twice the tidal volume, then permits the elastic recoil of the victim's lungs to produce passive expiration.

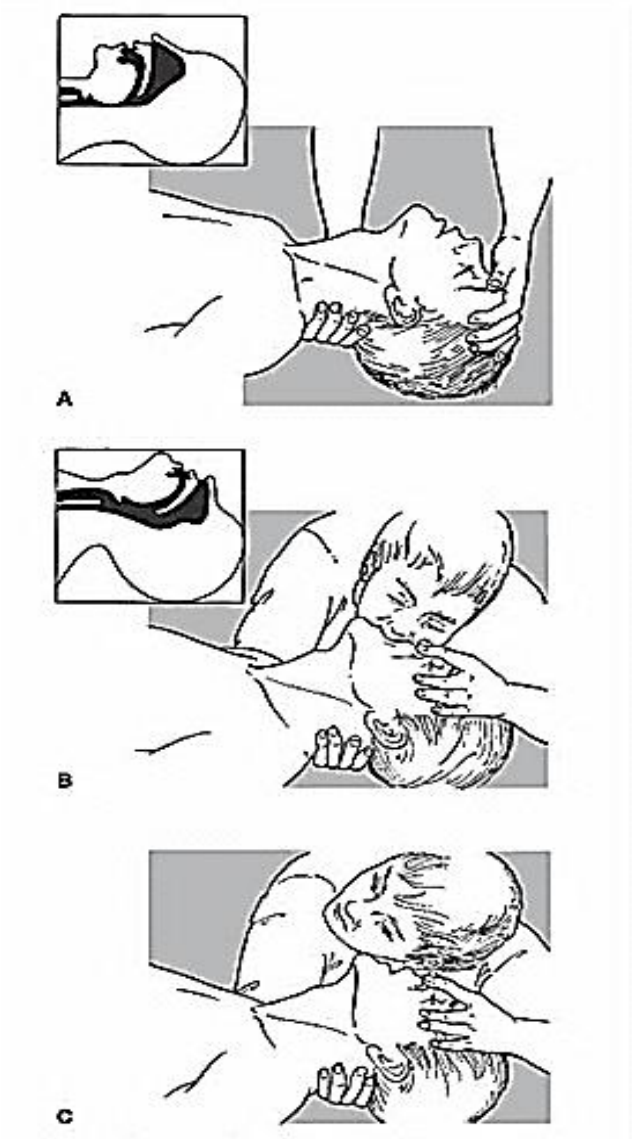
The victim's neck is kept extended. Any gas blown into the stomach can be expelled by applying upward pressure on the abdomen from time to time. In apneic individuals in whom no heartbeat is detectable, mouth-to-mouth breathing should be alternated with cardiac massage.

Proper performance of mouth-to-mouth resuscitation

A: Open airway *by positioning neck anteriorly in extension. Inserts show airway obstructed when the neck is in resting flexed position and open when neck is extended.*

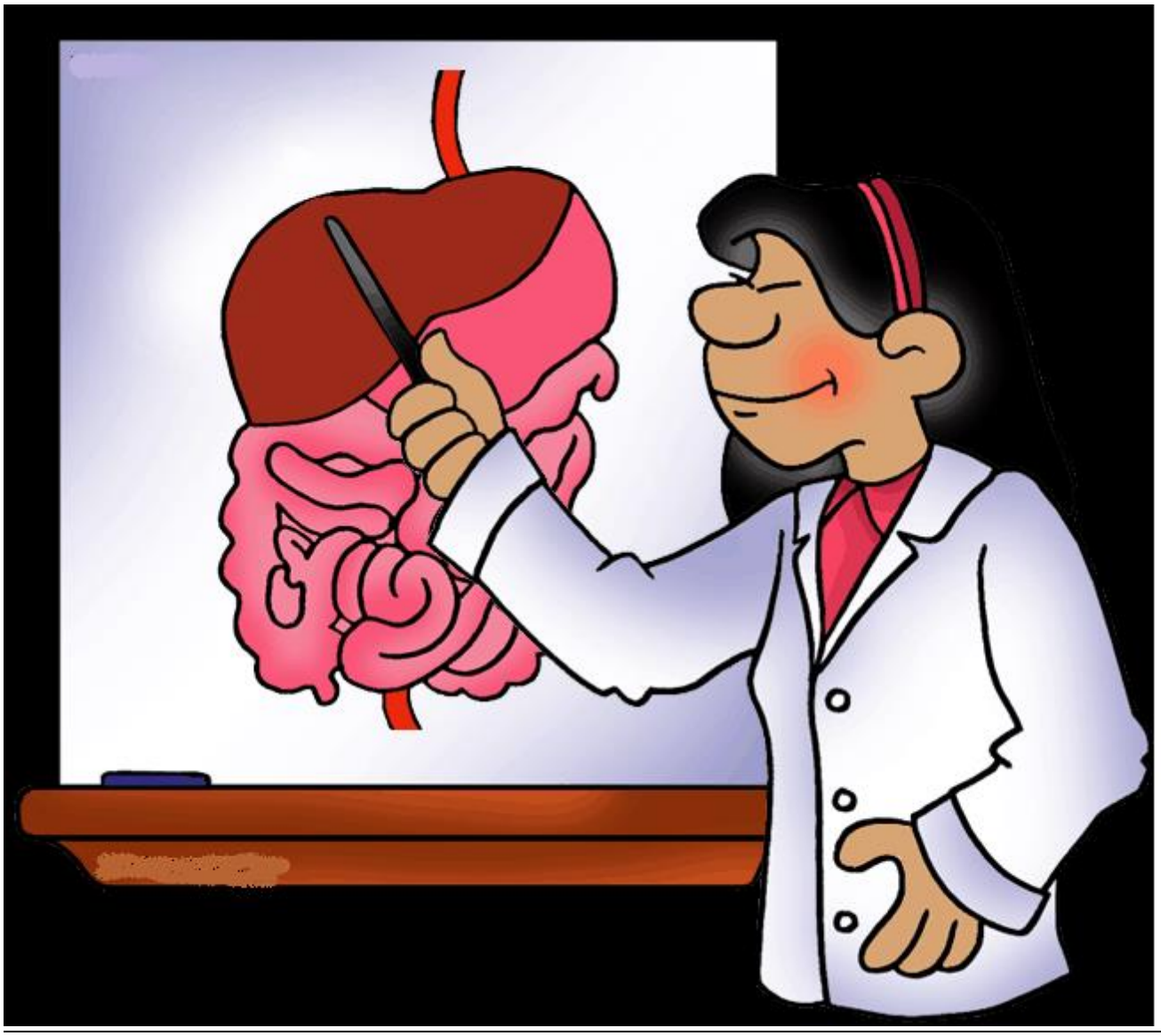
B: Close victim's nose with fingers*, seal mouth around victim's mouth, and deliver breath by vigorous expiration.*

C: Allow victim to exhale passively *by unsealing mouth and nose. Rescuer should listen and feel for expiratory air flow. Repeat B and C 12 times per minute.*



CHAPTER ELEVEN

DIGESTIVE SYSTEM



CHAPTER ELEVEN DIGESTIVE SYSTEM

Structure and Function of Organs of Digestion and Accessory Organs

The digestive system may be divided into two groups of organs:

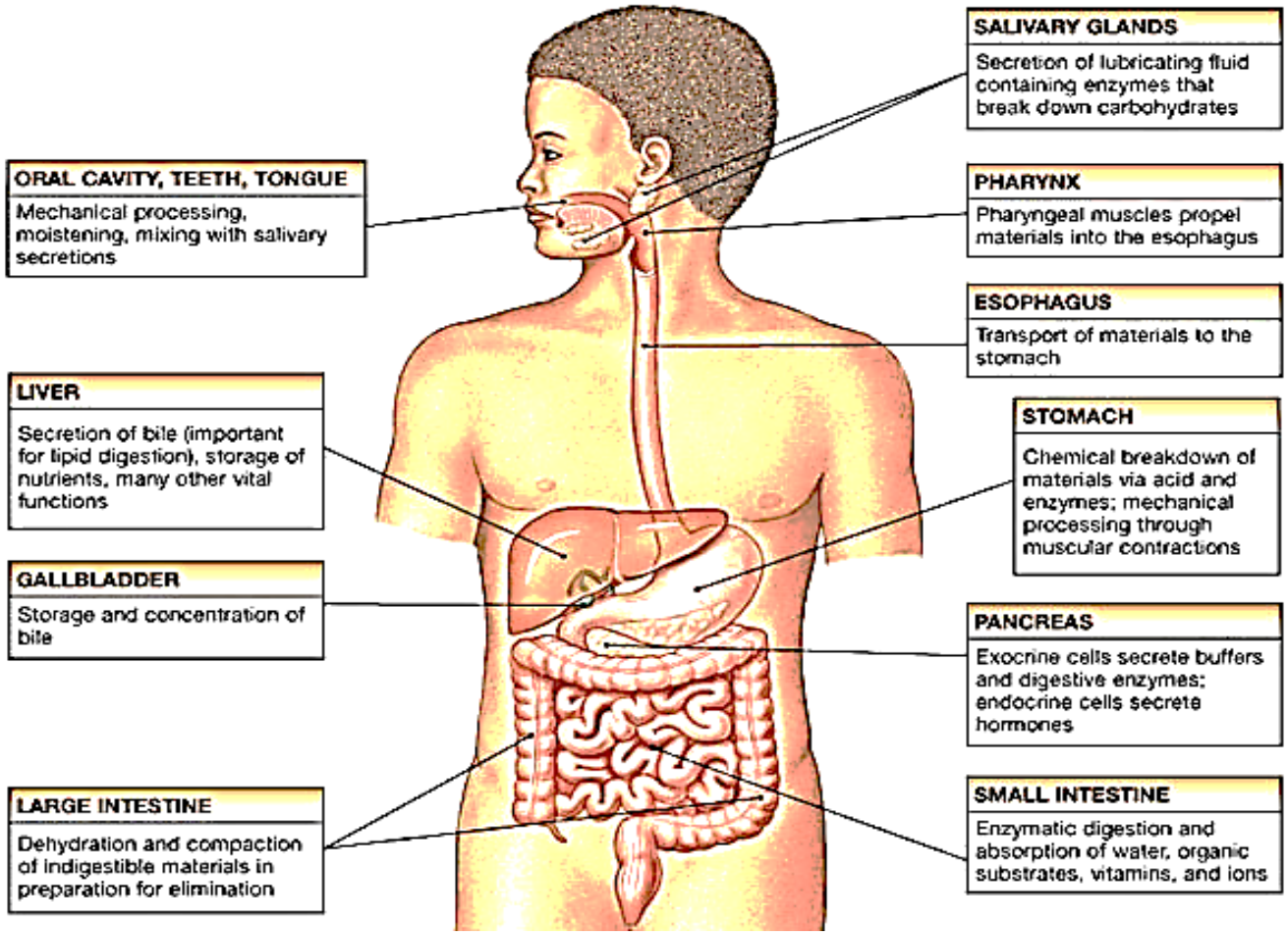
1. The digestive tract, a continuous passageway beginning at the mouth, where food is taken in, and terminating at the anus, where the solid waste products of digestion are expelled from the body

2. The accessory organ, which are necessary for the digestive process but are not a direct part of the digestive tract. They release substances into the digestive tract through ducts.

The Digestive Tract

The digestive tract, is a muscular tube extending through the body. It is composed of several parts: the mouth, pharynx, esophagus, stomach, small intestine, and large intestine.

The digestive tract is sometimes called the alimentary tract, derived from a Latin word that means "food".It is more commonly referred to as the gastrointestinal (Gl) tract because of the major importance of the stomach and intestine in the process of digestion.



The Walls of the Digestive Tract

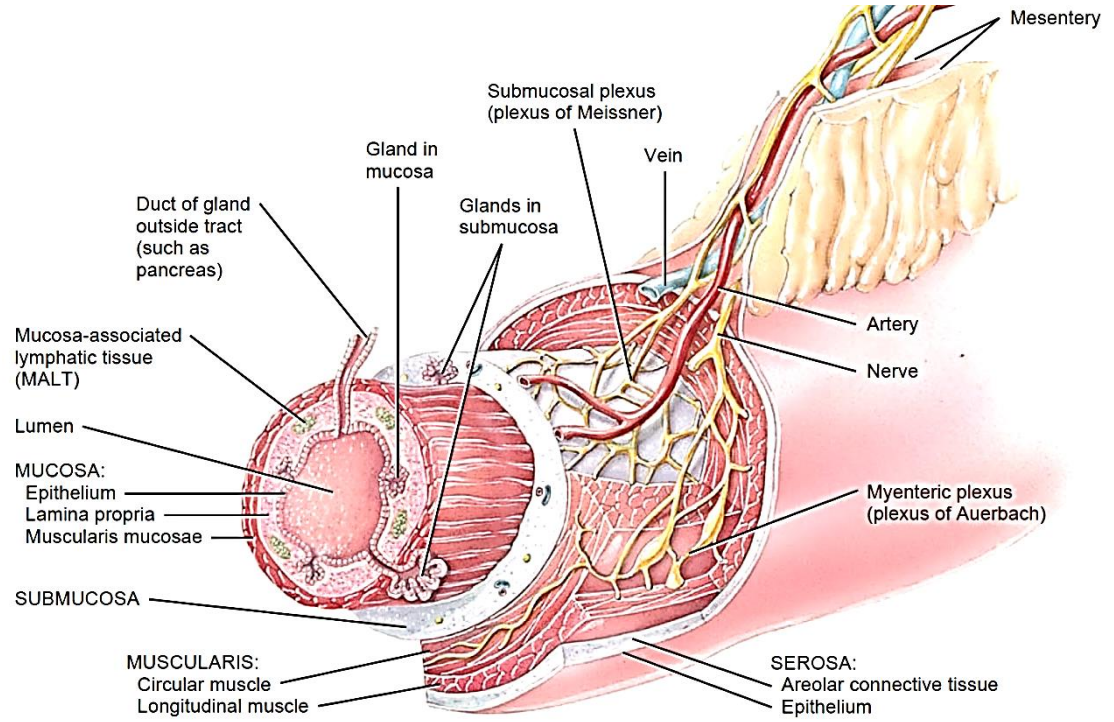
Although specified for specific tasks in different organs, the wall of the digestive tract, from the esophagus to the anus, is similar in structure throughout. Follow the diagram of the small intestine The layers of this wall from the inner most to the outermost surface.

First is the mucous membrane, so called because its epithelial layer contains many mucus-secreting cells. The type of epithelium is simple columnar.

The layer of connective tissue beneath this, the submucosa, contains blood vessels and some of the nerves that help regulate digestive activity.

Next are two layers of smooth muscle. The inner layer has circular fibers, and the outer layer has longitudinal fibers. The alternate contractions of these muscles create the wavelike movement that propels food through the digestive tract and mixes it with digestive juices. This movement is called

peristalsis. The outermost layer of the wall consists of fibrous connective tissue. Most of the abdominal organs have an additional layer of serous membrane that is part of the peritoneum.

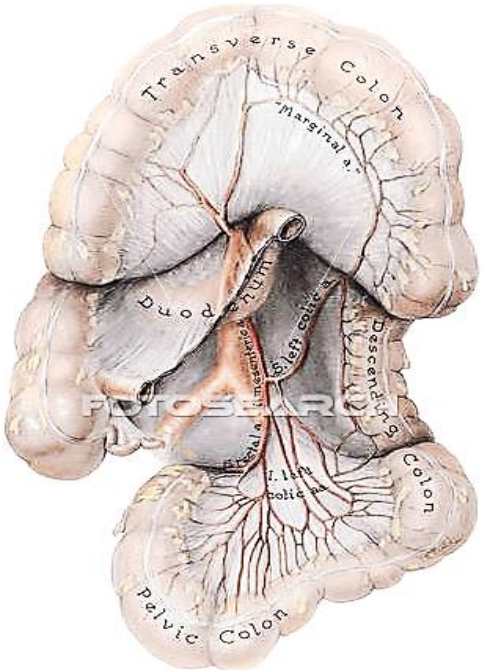


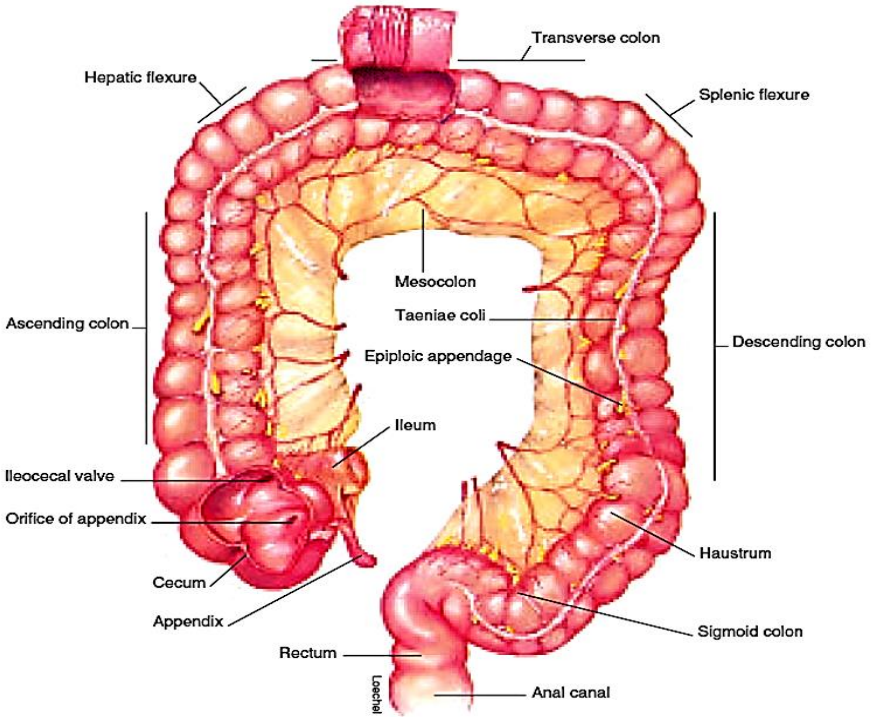
The Peritoneum

The abdominal cavity is lined with a thin, shiny serous membrane that also covers most of the abdominal organs. The portion of this membrane that lines the abdomen is called the **parietal peritoneum;** that covering the organ is called the **visceral peritoneum**.

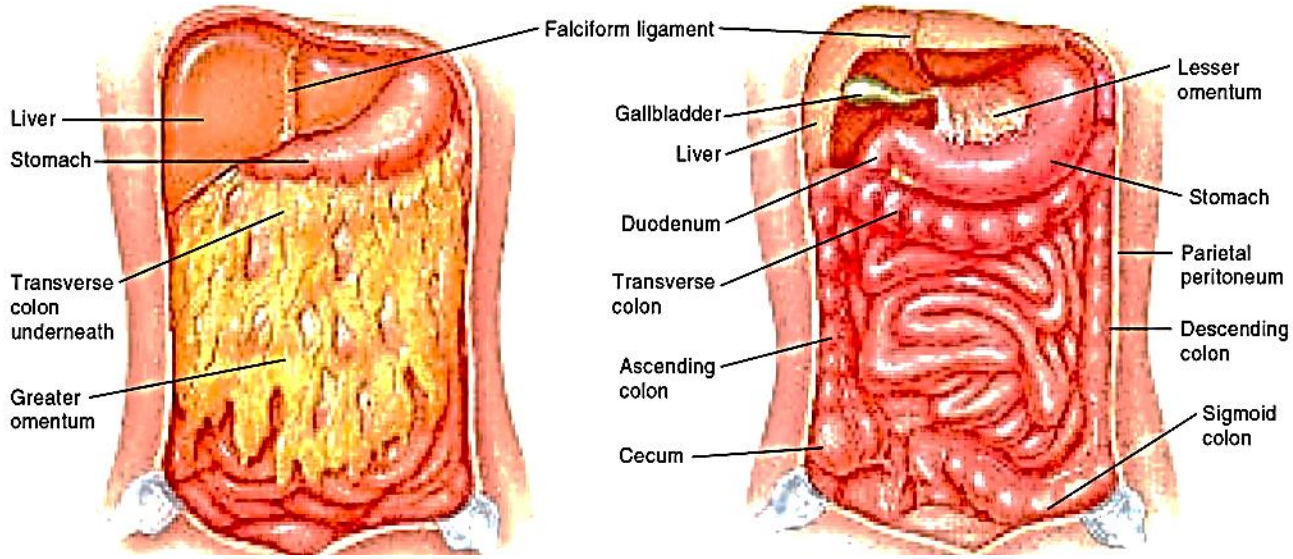
In addition to these single layered portions of the peritoneum there are a number of double-layered structures that carry blood vessels, lymph vessels, and nerves, and sometimes act as ligaments supporting the organs.

The mesentery is a double-layered portion of the peritoneum shaped somewhat like a fan. The handle portion is attached to the back wan, and the expanded long edge is attached to the small intestine. Between the two layers of membrane that of the mesentery are the blood vessels, lymphatic vessels, and nerves that supply the intestine.



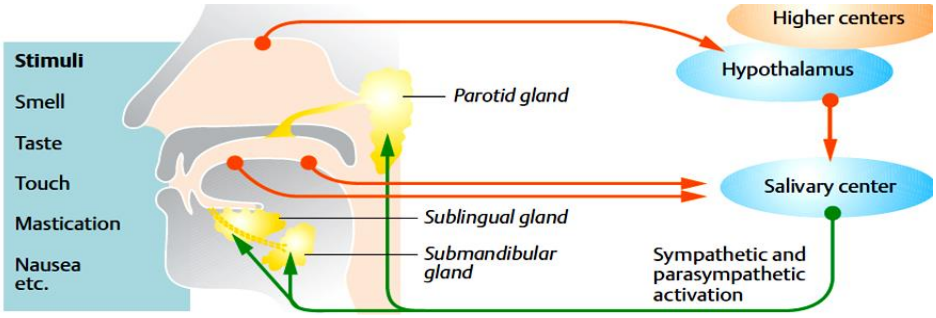


The section of the peritoneum that extends from the colon to the back wall is the mesocolon.
A large double layer of the peritoneum containing much fat hangs like an apron over the front of the intestine. **This greater omentum** extends from the lower border of the stomach into the pelvic part of the abdomen and then loops back up to the transverse colon. There is also a smaller membrane, called **the lesser omentum** that extends between the stomach and the liver.



Gastrointestinal Physiology

The alimentary tract provides the body with a continual supply of water, electrolytes, and nutrients.
To achieve this requires :
(1) Movement of food through the alimentary tract;
(2) Secretion of digestive juices and digestion of the food;
(3) Absorption of water, various electrolytes, and digestive products;
(4) Circulation of blood through the gastrointestinal organs to carry away the absorbed substances;
(5) Control of all these functions by local, nervous, and hormonal systems.
The first step in the digestive system can actually begin before the food is even in your mouth. When you smell or see something that you just have to eat, you start to salivate in anticipation of eating, thus beginning the digestive process.
Food is the body's source of fuel.



❖ **During digestion two main processes occur at the same time.**

•**Mechanical digestion:** larger pieces of food get broken down into smaller pieces while being prepared for chemical digestion.

Mechanical digestion starts in the mouth and continues into the stomach.

•**Chemical digestion:** several different enzymes break down macromolecules into smaller molecules that can be more efficiently absorbed.

Chemical digestion starts with saliva and continues into the intestines.

Ingestion of Food

The amount of food that a person ingests is determined principally by intrinsic desire for food called hunger.

The type of food that a person preferentially seeks is determined by appetite.

These mechanisms in themselves are extremely important automatic regulatory systems for maintaining an adequate nutritional supply for the body.

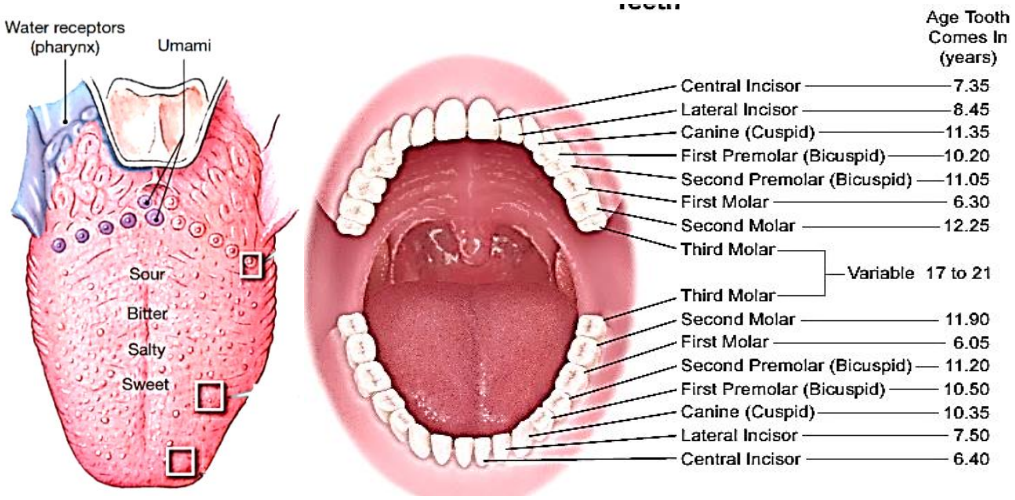
The Mouth

The mouth, also called the ***oral cavity***, is where a substance begins its travels through the digestive tract .

The mouth has three digestive functions:

1. To receive food, a process called ingestion
2. To prepare food for digestion
3. To begin the digestion of starch.

In to this space projects a muscular organ, the tongue, which is used for chewing and swallowing, and is one of the principal organs of speech. The tongue has on its surface a number of special organs, called taste buds, by means of which taste sensations (bitter, sweet, sour, or salty) can be differentiated.



The oral cavity also contains the teeth. A child between 2 and6 years of age has 20 teeth; an adult with a complete set of teeth has 32. Among these, the cutting teeth, or incisors, occupy the front part of the oral cavity, whereas the lager grinding teeth, the molars, are in the back.

Deciduous, or Baby Teeth

The first eight deciduous teeth to appear through the gums are the incisors. Later the canines (eye teeth) and molar appear. Usually, the 20 baby teeth have all appeared by the time a child has reached the age of 2 or 21/2 years. During the first 2 years the permanent teeth develop within the jaw bones from buds that are present at birth. The first permanent tooth to appear is the important 6-year molar. This permanent tooth comes in before the baby incisors are lost. Because decay and

infection of adjacent deciduous molars may spread to and involve new, permanent teeth, deciduous teeth need proper care.

Permanent Teeth

As a child grows, the jawbones grow, making space for additional teeth. After the 6-year molars have appeared, the baby incisors loosen and are replaced by permanent incisors.

Next, the baby canines are replaced by permanent canines, and finally, the baby molars are replaced by the bicuspids (premolars) of the permanent teeth.

Now the larger jawbones are ready for the appearance of the12-year, or second, permanent molar teeth. During or after the late teens, the third molars, or so-called wisdom teeth, may appear. In some cases the jaw is not large enough for these teeth or there are other abnormalities, so that the third molars may have to be removed.

► **Mastication (Chewing)** :The teeth are admirably designed for chewing, the anterior teeth (incisors) providing a strong cutting action and the posterior teeth (molars), a grinding action.
Chewing is important for digestion of all foods, but especially important for most fruits and raw vegetables because these have indigestible cellulose membranes around their nutrient portions that must be broken before the food can be digested.

Also, chewing aids the digestion of food for still another simple reason:

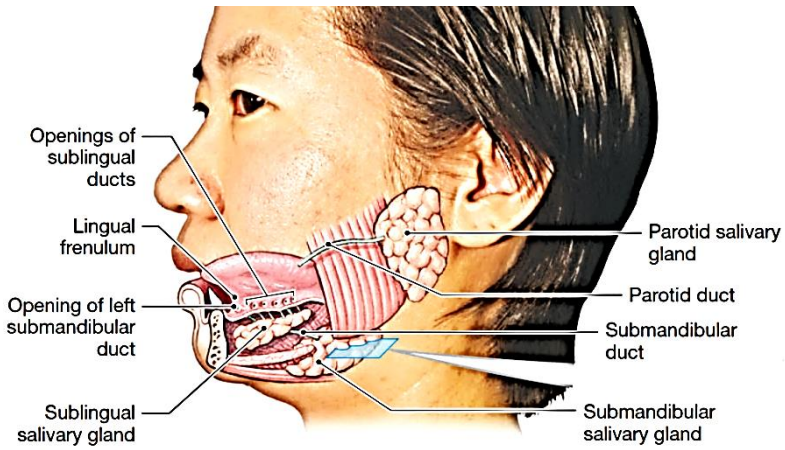
- Digestive enzymes act only on the surfaces of food particles; therefore, the rate of digestion is absolutely dependent on the total surface area exposed to the digestive secretions.
- In addition, grinding the food to a very fine particulate consistency prevents excoriation of the gastrointestinal tract and increases the ease with which food is emptied from the stomach into the small intestine, then into all succeeding segments of the gut.

The Salivary Glands

While food in the mouth, it is mixed with ***saliva***, one purpose of which is to moisten the food and facilitate the processes of chewing, or mastication, and swallowing, or deglutition.

Saliva also helps keep the teeth and mouth clean and reduce bacterial growth.

This watery mixture contains mucus and an enzyme called salivary amylase, which begins the digestive process by converting starch to sugar. It is manufactured mainly by three pairs of glands that function as accessory organs:



1. The parotid glands, the largest of the group, are located below and in front of the ear.

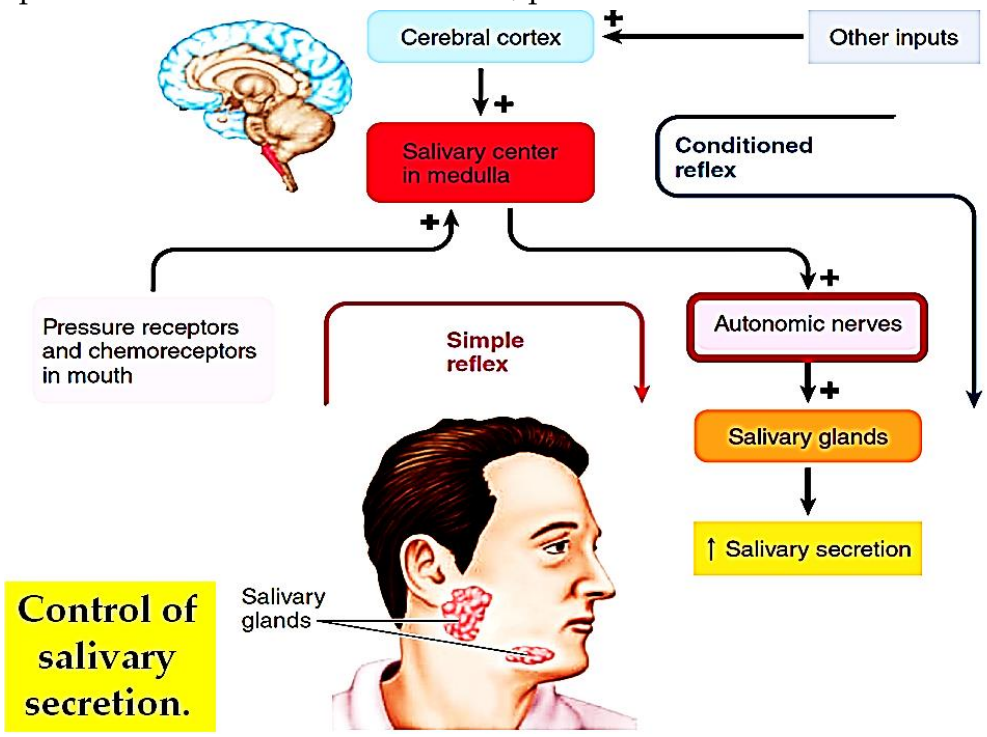
2. The submandibular, or sub maxillary, glands are located near the body of the lower jaw

3. The sublingual glands are under the tongue.

All these glands empty by means of ducts into the oral cavity.

Saliva is a colorless turbid and viscid slightly alkaline fluid (pH is 6.7 with a range of 5.6 to 7.6). It consists mainly of water and contains about 0.5 % solid material formed of organic and inorganic substances.

- **The organic** substances (60%) include :
  mucin, albumin, urea and salivary amylase (ptyalin) , maltase and Lysozyme.
- **The inorganic** substances are chiefly :
  chlorides, phosphates and bicarbonates of sodium, potassium and calcium.



The secretion of saliva is mainly controlled by the autonomic nervous system.
This secretion is affected by psychic stimuli brought about by sight, smell, mastication of food or even by thinking of food.
The daily flow of saliva in case of a normal person is about 1 – 1.5 L.

❖ **Functions of the saliva**:

1- Saliva moistens and lubricates the food to be easily swallowed.
2- It has some cleaning action on the mouth and teeth.
3- It dissolves solid food stuffs due to its high water content and thus subserves the sense of taste.
4- Certain substances, such as mercury, lead and iodine salts, are excreted from the body through the saliva.
5- It plays a role in regulating the water balance of the body.
6- Bicarbonates, phosphates and proteins in the saliva have an important buffering action, protecting the mouth against acids and alkalis.
7- It contains the enzyme Lysozyme which has an antibacterial effect thus destroying some microorganisms entering the mouth.

8- It is also contains the enzyme amylase (ptyalin) which digests carbohydrates composed of alpha – glucose.
One can demonstrate the digestive action of ptyalin in the mouth by chewing a piece of bread for several minutes. After this time, the bread tastes sweet because of the maltose that has been liberated from the starches of the bread.

Digestion by the salivary amylase starts in the buccal cavity, but food doesn't remain in the mouth long enough for ptyalin to complete the breakdown of starches into maltose the action of ptyalin can continue for as long as half an hour after the food has entered the stomach, that is until the contents of the fundus are mixed with the stomach secretions.

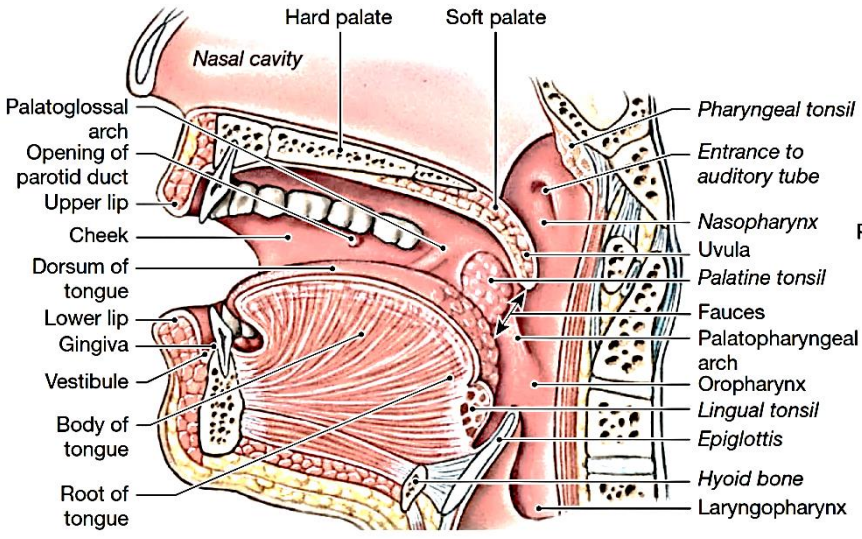
The activity of the salivary amylase is blocked by the acid of the gastric secretions,

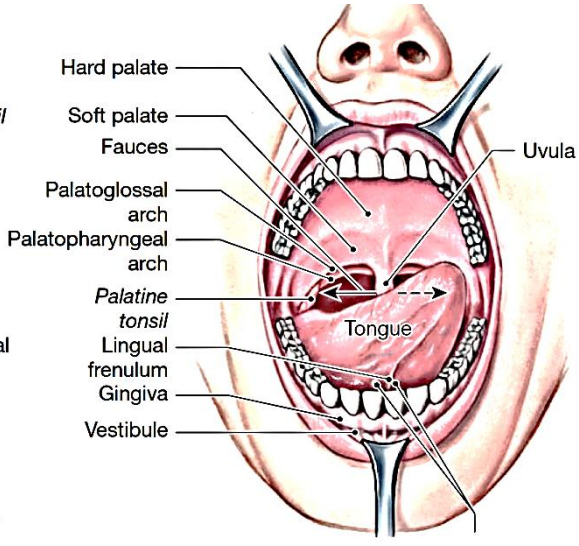
Nevertheless, on the average, before the food becomes completely mixed with the gastric secretions, as much as 30 to 40% of the starches will have been hydrolyzed mainly to maltose.

The Pharynx and Esophagus

The pharynx is commonly referred to as the throat. The oral part of the pharynx is visible when you look into an open mouth and depress the tongue. The palatine tonsils may be seen at either side. The pharynx also extends upward to the nasal cavity and downward to the level of the larynx.

The soft palate is tissue that forms the back of the roof of the oral cavity. From it hangs a soft, fleshy, V -shaped mass called the uvula.





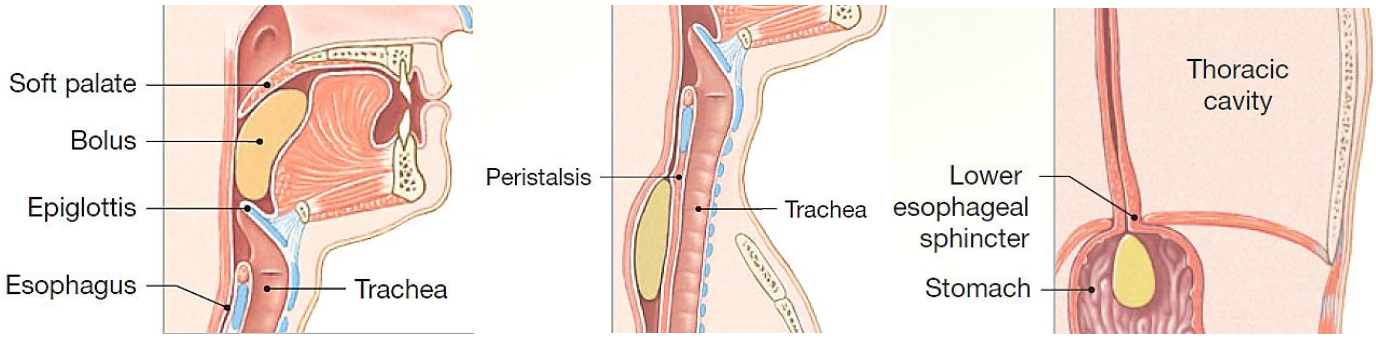
► **Swallowing (Deglutition):** Swallowing is a complicated mechanism, principally because the pharynx subserves respiration as well as swallowing.
The pharynx is converted for only a few seconds at a time into a tract for propulsion of food.

It is especially important that respiration not be compromised because of swallowing.

In general, swallowing can be divided into:
(1) A voluntary stage, which initiates the swallowing process;
(2) A pharyngeal stage, which is involuntary and constitutes passage of food through the pharynx into the esophagus;

(3) An esophageal stage, another involuntary phase that transports food from the pharynx to the stomach.
The upper third of the esophageal wall consists of striated muscle, the rest contains smooth muscle. During the process of swallowing, or deglutition, the tongue pushes a bolus of food into the throat

In swallowing, a small portion of chewed food mixed with saliva, called a bolus, is pushed by the tongue into the pharynx, swallowing occurs rapidly by an involuntary reflex action. At the same time, the soft palate and uvula are raised to prevent food and liquid from entering the nasal cavity, and the tongue is raised to seal the back of the oral cavity. The entrance of the trachea is guarded during swallowing by a leaf-shaped cartilage, the epiglottis, which covers the opening of the larynx. The swallowed food is then moved by peristalsis into esophagus, a muscular tube about 25cm (10 inches) longthat carries food into the stomach.



Before joining the stomach, the esophagus must pass through the diaphragm. It passes through a space in the diaphragm called the esophageal hiatus. If there is a weakness in the diaphragm at this point, a portion of the stomach or other abdominal organ may protrude through the space, a condition called hiatal hernia.

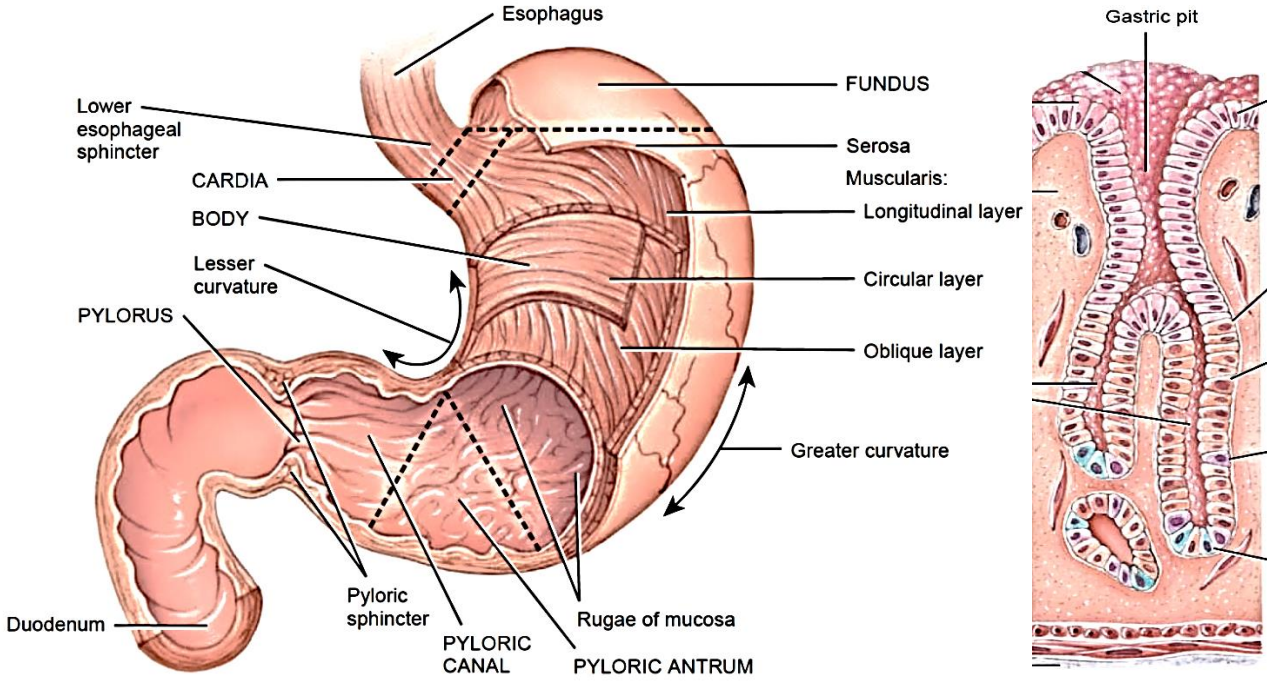
The Stomach

The stomach is an expanded J-shaped organ in the upper left region of the abdominal cavity.

In addition to the two muscle layers already described, it has a third, inner oblique (angled) layer that aids in grinding food and mixing it with digestive juices. The left-facing arch of the stomach is the greater curvature, whereas the right surface forms the lesser curvature. Each end of the stomach is guarded by a muscular ring, or sphincter, that permits the passage of substances in only one direction. Between the esophagus and the stomach is the lower esophageal sphincter (LES). This valve has also been called cardiac sphincter because it separates the esophagus from the region of the stomach that is close to the heart.

We are sometimes aware of the existence of this sphincter; sometimes it does not relax as it should, produce a feeling of being unable to swallow past that point. Between the distal, or far, end of the stomach and the small intestine is the pyloric sphincter. The region of the stomach leading into this sphincter, the pylorus, is important in regulating how rapidly food moves into the small intestine.

When the stomach is empty, the lining forms many folds called rugae. These folds disappear as the stomach expands.



Special cells in the lining of the stomach secrete substances that mix together to form gastric juice, the two main components of which are:

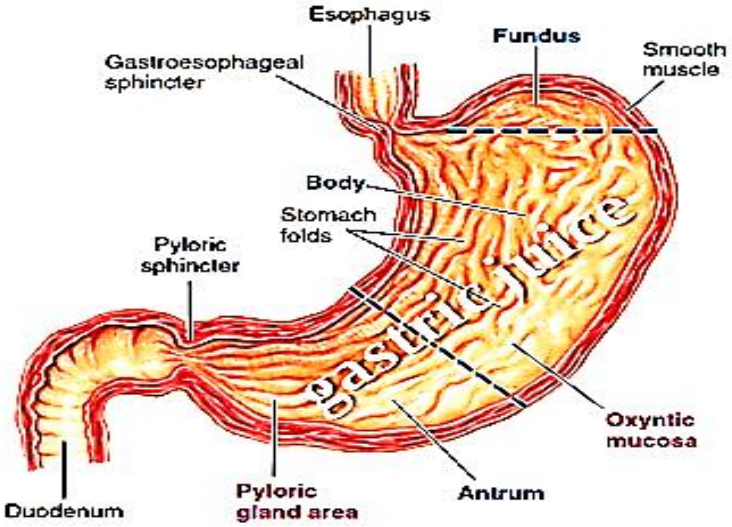
1. Hydrochloric acid (HCL), a strong acid that softens the connective tissue in meat and destroys foreign organisms

2. Pepsin, a protein-digesting enzyme. This enzyme is produced in an inactive form and is activated only when food enters the stomach and HCL is produced. The semi-liquid mixture of gastric juice and food that leaves the stomach to enter the small intestine is **called chyme.**

Secretions from the Oxyntic (Gastric) Glands: It is composed of types of cells:

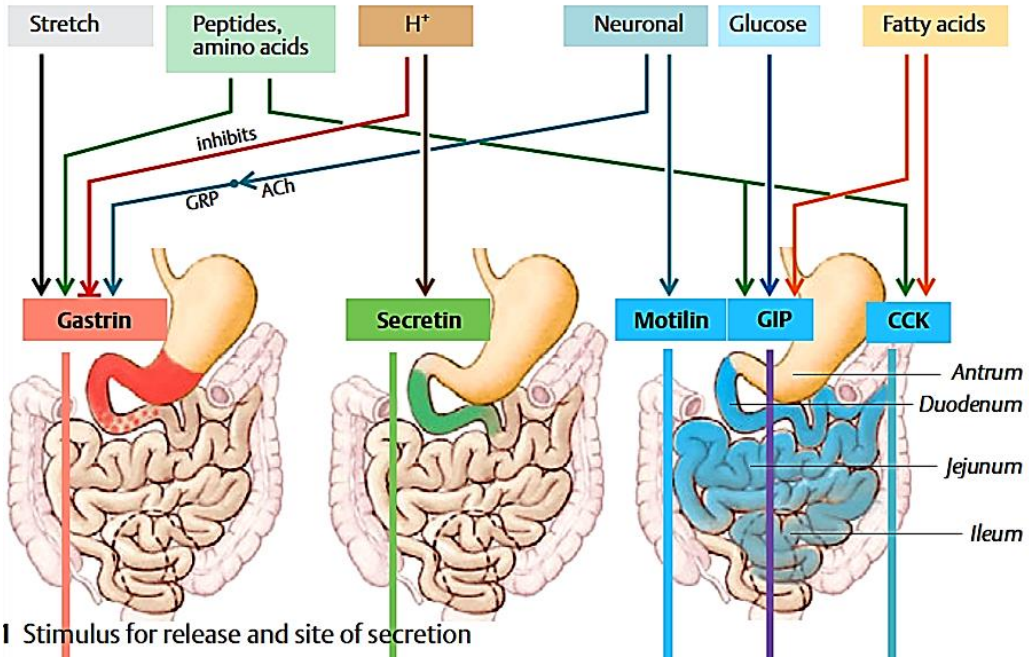
1. **Mucous neck cells**, which secrete mainly mucus;
2. **Peptic (or chief) cells**, which secrete large quantities of pepsinogen; and
3. **Parietal (or oxyntic) cells**, which secrete hydrochloric acid and intrinsic factor.
4. **Argentaffin cells**: secrete stomach gastrin

Gastric juice is the whole secretion of gastric glands. When food reaches the stomach, the gastric juice is secreted in response of nervous as well as hormonal stimulation.



The nervous stimulation is initiated by the taste, smell or sight of food, or even by thinking of food. Upon contact of food with the pyloric mucosa , the mucosa releases a hormone called gastrin which is passed into the blood stream , circulates with the blood and when it returns to the wall of the stomach it stimulates the gastric cells to produce their secretions.

Gastrin has a variety of actions, but its principal physiologic actions are stimulation of gastric acid and pepsin secretion. Gastrin stimulates insulin secretion

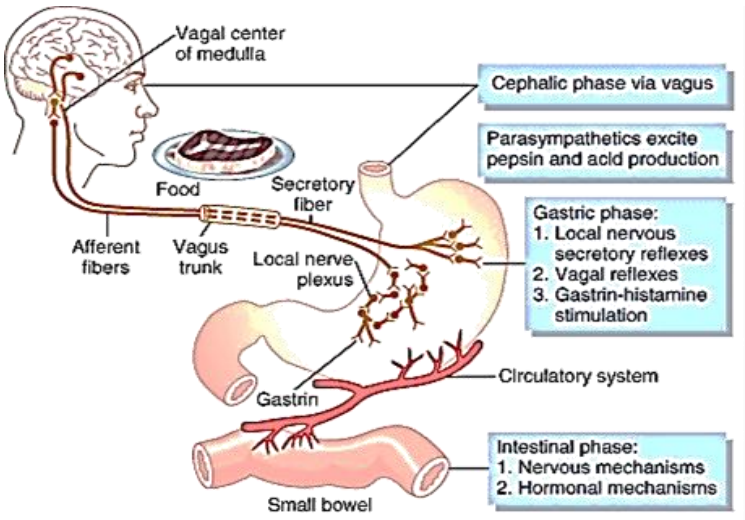


Phases of Gastric Secretion

Gastric secretion is said to occur in three phases; a *cephalic phase*, a *gastric phase*, and an *intestinal phase*.

1- Cephalic Phase:

The cephalic phase of gastric secretion occurs even before food enters the stomach, especially while it is being eaten. It results from the sight, smell, thought, or taste of food, and the greater the appetite, the more intense is the stimulation. This phase of secretion normally accounts for about 20 % of the gastric secretion associated with eating a meal.



2- Gastric Phase : Once food enters the stomach, it excites

(1) long vagovagal reflexes from the stomach to the brain and back to the stomach,
(2) local enteric reflexes, and
(3) the gastrin mechanism

All of which in turn cause secretion of gastric juice during several hours while food remains in the stomach. The gastric phase of secretion accounts for about 70% of the total gastric secretion associated with eating a meal and therefore accounts for most of the total daily gastric secretion of about 1500 milliliters.

3- Intestinal Phase: The presence of food in the upper portion of the small intestine, particularly in the duodenum, will continue to cause stomach secretion of small amounts of gastric juice, probably partly because of small amounts of gastrin released by the duodenal mucosa.

Inhibition of Gastric Secretion by Other Post-Stomach Intestinal Factors

Although intestinal chyme slightly stimulates gastric secretion during the early intestinal phase of stomach secretion, it paradoxically inhibits gastric secretion at other times. This inhibition results from at least two influences:

1- **The presence of food in the small intestine** initiates a reverse enterogastric reflex, transmitted through the mesenteric nervous system as well as through extrinsic sympathetic and vagus nerves, that inhibits stomach secretion. This reflex can be initiated by distending the small bowel, by the presence of acid in the upper intestine, by the presence of protein breakdown products, or by irritation of the mucosa.

2- **The presence of acid, fat, protein breakdown products**, hyperosmotic or hypo-osmotic fluids, or any irritating factor in the upper small intestine causes release of several intestinal hormones. One of these is secretin, which is especially important for control of pancreatic secretion. However, secretin opposes stomach secretion. Three other hormones-gastric inhibitory peptide, vasoactive intestinal polypeptide, and somatostatin-also have slight to moderate effects in inhibiting gastric secretion. The functional purpose of inhibitory gastric secretion by intestinal factors is presumably to slow passage of chyme from the stomach when the small intestine is already filled or already overactive. In fact, the enterogastric inhibitory reflexes plus inhibitory hormones usually also reduce stomach motility at the same time that they reduce gastric secretion.

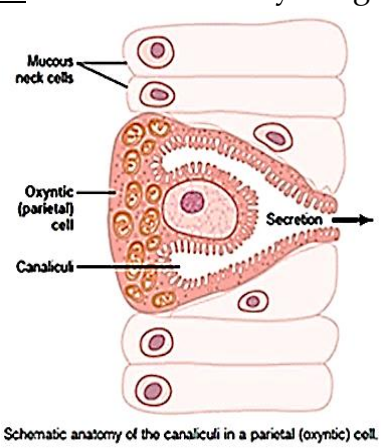
- **Gastric Juice:**

The gastric secretion is a watery solution containing HCL, pepsin and lipase; young mammals have rennin instead of pepsin.

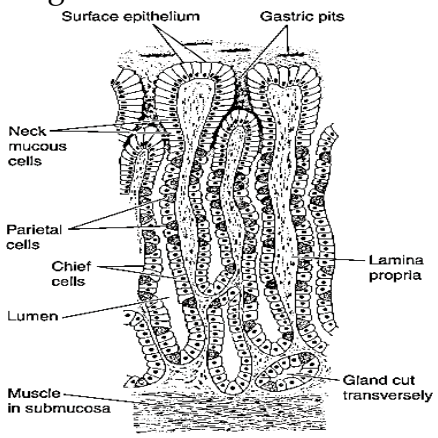
I. Hydrochloric acid: HCL serves the following functions:

1. It kills the majority of the bacteria taken in with the food, thus preventing putrefaction in the stomach.
2. It converts the inactive pepsinogen into active pepsin and provides a favorable medium for it action.
3. It causes partial hydrolysis of some sugars.
4. It changes protein into acid metaproteins, a preparatory step necessary for protein digestion.
5. It regulates the evacuation of the stomach by stimulating the pyloric sphincter to relax.
6. When HCL leaves the stomach to the duodenum, it stimulates the mucosa of the latter release the hormone secretin.

Hydrochloric Acid SecretionHCL is secreted by the gastric gland cells of the mucosa of the stomach in the following manner:

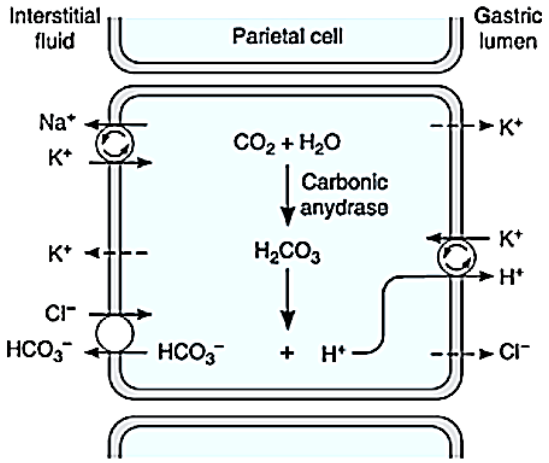


Schematic anatomy of the canaliculi in a parietal (oxyntic) cell.



The parietal cell secretion is probably an isotonic solution of HCl that contains 150 meq of Cl^- and 150 meq of H^+ per liter. Yet the pH of the cytoplasm of the parietal cells, like that of other cells, is 7.0–7.2, and the comparable concentrations per liter of plasma are about 100 meq of Cl^- and 0.00004 meq of H^+.

It is H^+–K^+ ATPase in the apical membrane of the parietal cells that pumps H^+ against a concentration gradient of this magnitude (Figure). The parietal cells are polarized, with an apical membrane facing the lumen of the gastric glands and a basolateral membrane in contact with the interstitial fluid. Canaliculi extend from the apical surface into the cell. At rest, the cells contain abundant tubulovesicular structures with H^+–K^+ ATPase molecules in their walls. When the parietal cells are stimulated, the tubulovesicular structures move to the apical membrane and fuse with it, thus inserting many more H^+–K^+ ATPase molecules into the membrane. These ATPase molecules are now exposed to the K^+ in the ECF, and H^+–K^+ exchange begins.



HCl secretion by parietal cells in the stomach. Active transport by ATPase is indicated by arrows in circles. H^+ is secreted into the gastric lumen in exchange for K^+ by H^+–K^+ ATPase. HC_3^- is exchanged for Cl^- in the interstitial fluid by an antiport, and Na^+–K^+ ATPase keeps intracellular Na^+ low

Cl^- is also extruded down its electrochemical gradient through channels that are activated by cAMP. The H^+ that is extruded comes from H_2CO_3, and H_2CO_3 in turn is formed by the hydration of CO_2 (Figure). This latter reaction is catalyzed by carbonic anhydrase, and the parietal cells are particularly rich in this enzyme. The HCO_3^- formed by dissociation of H_2CO_3 is extruded by an antiport in the basolateral membrane of the parietal cells that exchanges HCO_3^- for another anion, and since Cl^- is the most abundant anion in interstitial fluid, the exchange is mainly for Cl^-. When gastric acid secretion is elevated after a meal, sufficient H^+ may be secreted to raise the pH of systemic blood and make the urine alkaline **(postprandial alkalinetide).**

Secretion of Intrinsic Factor:

The substance intrinsic factor, essential for absorption of vitamin B12 in the ileum, is secreted by the parietal cells along with the secretion of hydrochloric acid.

When the acid-producing parietal cells of the stomach are destroyed, which frequently occurs in chronic gastritis, the person develops not only achlorhydria (lack of stomach acid secretion) but often also pernicious anemia because of failure of maturation of the red blood cells in the absence of vitamin B12 stimulation of the bone marrow.

Pepsin

This enzyme is first secreted in an inactive form called pepsinogen. This enzyme is a proteolytic enzyme which works in an acid medium. Its optimum pH is 1-2. It accelerates the breakdown of proteins into free amino acids. But, during the short period of gastric digestion in the stomach, it usually breaks down proteins only partially into polypeptides (proteoses and peptones).

pepsinogen $\xrightarrow{HCl}$ **pepsin**

proteins $\xrightarrow{pepsin}$ **proteoses& peptones (polypeptides)**

Rennin

This is a proteolytic enzyme which is unique to mammals. It is secreted in an inactive form known as prorennin which is converted by HCl into the active rennin.

Rennin changes the protein caseinogen of milk into insoluble calcium caseinate, in presence of calcium ion. Its optimum pH is 5.6; thus it acts only in the stomach of young mammals because the medium is far more acidic in older mammals (pH 1-2).

Gastric lipase: this is a weak lipolytic enzyme of optimum pH of 5-6 & like rennin; it acts in the stomach of young's. It hydrolyses emulsified fasts into glycerol and fatty acids.

During gastric digestion, the pyloric sphincter muscle remains closed, but as the food becomes progressively more acidic, the sphincter relaxes from time to time to allow some of the semi – fluid contents of the stomach (chyme) to pass into the duodenum.

- **Factors affecting Stomach Emptying:**
  - **1. Innervation of stomach :**sympathetic :relax the wall of stomach and contract pyloric sphincter and delay emptying of stomach
  - **2. Distention of intestine**: inhibits gastric motility by increasing sympathetic and decreasing vagal.
  - **3. Degree of filling the stomach:**-empty stomach has least motility and emptying power-slight distention stimulate gastric movements-severe distention inhibit gastric movements
  - **4.Consistency of the food-**fluids leave in few minute-carbohydrate is more rapidly leave than proteins because they are easily liquefied
  - **5. Nature of meal**-fat, hypertonic solutions of glucose when come in contact with doudenal mucosa secrete Enterogastron - if Hcl or fat are placed in duodenum, it will initiate reflex inhibition of gastric motility ,by vagus (Enterogastric Reflex)
  - **6. Emotional conditions :**in nervousness :gastric motility increase-fear, pain, and exercise inhibit motility. at not easily liquefied so, stay longer in stomach

The Small Intestine

- **Small intestine**

The small intestine is the longest (>6 m). It is divided into three segments:

• Duodenum: first 20 cm• Jejunum: next 2.5 m• Ileum: remaining 3.5 m

Most digestion and absorption take place in the small intestine: • Plicae circulares

• Villi

Function of the small intestine

- ☺ Forward propulsion of contents(chym into colon)
- ☺ Major site of digestion (upper half)
- ☺ 90% absorption (lower half)
- ☺ Protection against infection
- ☺ Hormone secretion
- ☺ Intestinal juice secretion

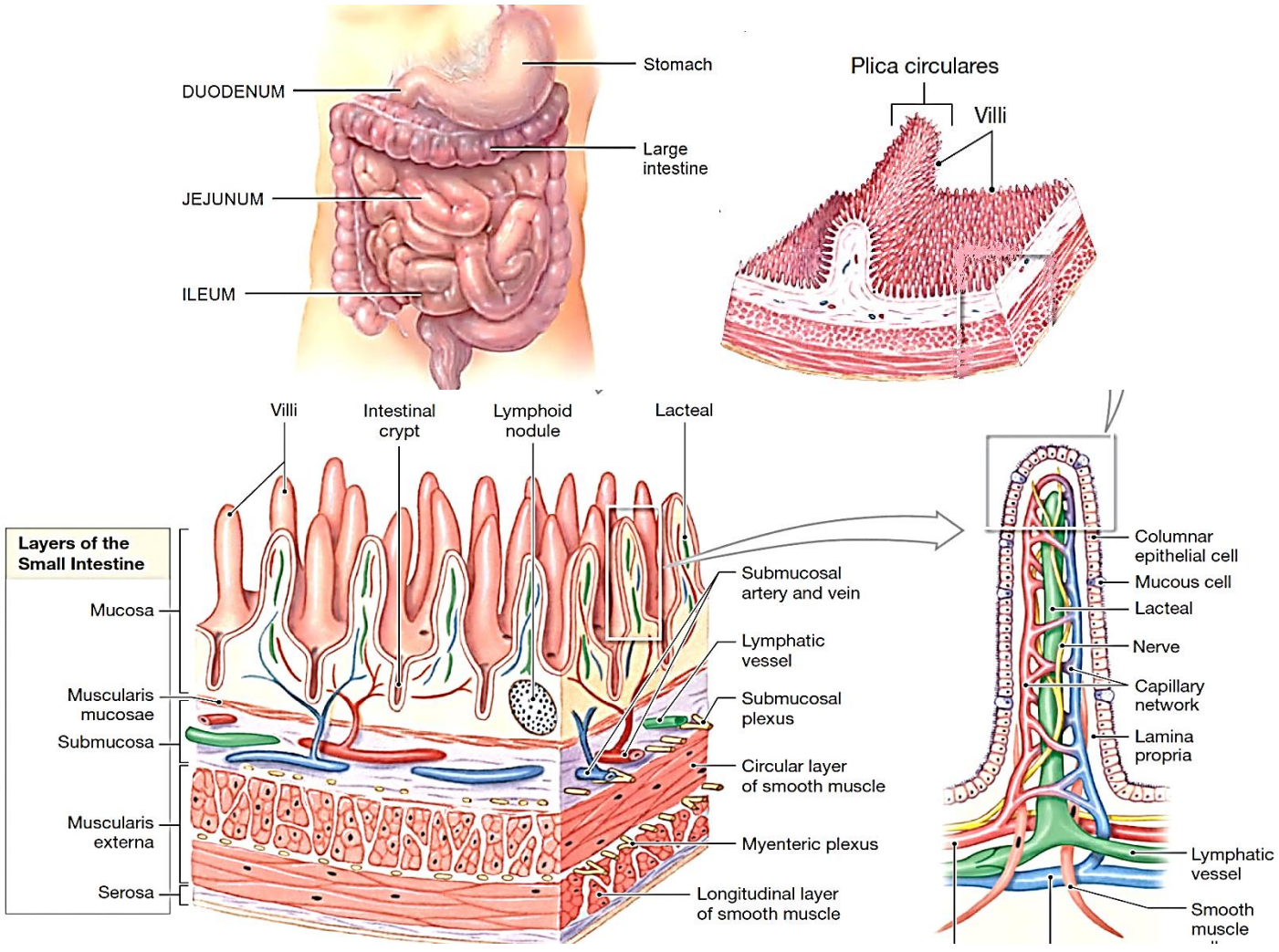
The small intestine is the longest part of the digestive tract. It is known as the small intestine because, although it is longer than the large intestine, it is smaller in diameter, with an average width of about 2.5 cm (1 inch). When relaxed to its full length, the small intestine is about 6 m (20 feet) long. The first 25 cm (10 inches) or so of the small intestine make up the duodenum. Beyond the duodenum are two more divisions:

the jejunum, which forms the next two fifths of the small

intestine, and the **ileum**, which constitutes the remaining portion.

The wall of the duodenum contains glands that secrete large amounts or mucus to protect the small intestine from the strongly acid chyme entering from the stomach. Cells of the small intestine also secrete enzymes that digest proteins and carbohydrates. In addition, digestive juices from the liver and pancreas enter the small intestine through a small opening in the duodenum. Most of the digestive process takes place in the small intestine under the effects of these juices.

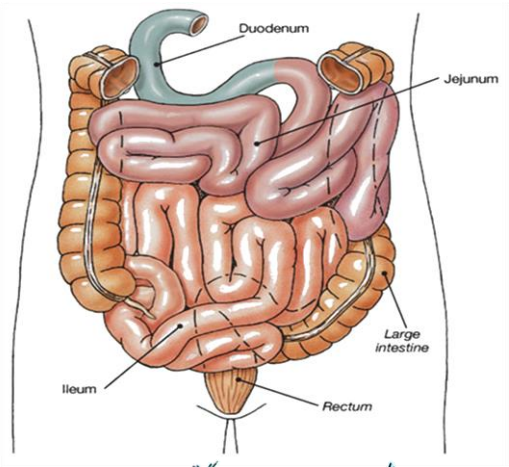
Most absorption of digested food also occurs through the walls of the small intestine. To increase the surface area of the organ for this purpose, the mucosa is formed into millions of tiny, finger-like projections, called villi ,Which give the inner surface a velvety appearance.



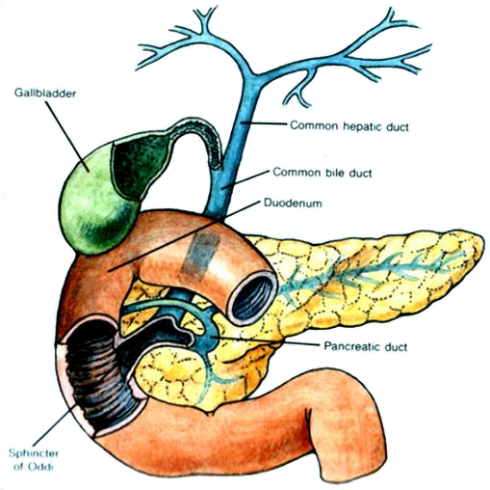
In addition, each epithelial cell has small projecting folds of the cell membrane known as microvilli. These create are markable increase in the total surface area available in the small intestine for the absorption of nutrients.

INTESTINAL DIGESTION

The anterior part of the small intestine is called duodenum which is truly an important organ, because so much digestion occurs in this relatively small region. The duodenal secretions are also involved in the elimination of waste products, particularly when the kidneys are impaired.



As the chyme enters the duodenum in small amount by means of pyloric sphincter, it is quickly mixed with the flow of pancreatic, bile and intestinal juices because the pancreatic and bile ducts are only a few inches away from the pylorus. These juices are alkaline and neutralize the acidity of chyme; thus, they make the intestine alkaline.



The combined effects of pancreatic and bile juices cause the chief chemical changes of the intestinal digestion. Intestinal digestion is also dependent upon the intestinal juice secreted by several types of glands which are located abundantly in the mucus lining of the intestine.

Mechanism of intestinal secretion

1.Local axon reflex: stimulated by chemicals (acid chyme, pancreatic juice) and mechanical (pressure of food)

2.Hormonal mechanism :intestinal hormones increase when Hcl comes in contact with duodenum.

3.Extrinsic nerves: parasympathetic (vagal) stimulation

Propulsive Movements

Peristalsis in the Small Intestine Chyme is propelled through the small intestine by peristaltic waves. These can occur in any part of the small intestine, and they move toward the anus at a velocity of0.5 to 2.0 cm/sec, faster in the proximal intestine and slower in the terminal intestine.

Control of Peristalsis by Nervous and Hormonal :

Signals Peristaltic activity of the small intestine is greatly increased after a meal.

This is caused partly by the beginning entry of chyme into the duodenum causing stretch of the duodenal wall. Also, peristaltic activity is increased by the so-called **gastroenteric reflex** that is

initiated by distention of the stomach and conducted principally through the mesenteric plexus from the stomach down along the wall of the small intestine.

As the chyme enters the intestines from the stomach and elicits peristalsis, this immediately spreads the chyme along the intestine; and this process intensifies as additional chyme enters the duodenum.

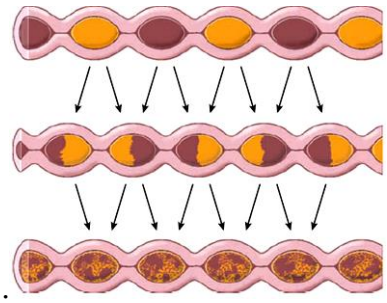
On reaching the ileocecal valve, the chyme is sometimes blocked for several hours until the person eats another meal; at that time, a gastroileal reflex intensifies peristalsis in the ileum and forces the remaining chyme through the ileocecal valve into the cecum of the large intestine.

Propulsive Effect of the Segmentation Movements

The segmentation movements, although lasting for only a few seconds at a time, often also travel 1 centimeter or so in the anal direction and during that time help propel the food down the intestine.

Segmentation consists of oscillating, ring like contractions of the circular smooth muscle along the small intestine's length; between the contracted segments are relaxed areas containing a small bolus of chyme .

The contractile rings occur every few centimeters, dividing the small intestine into segments like a chain of sausages

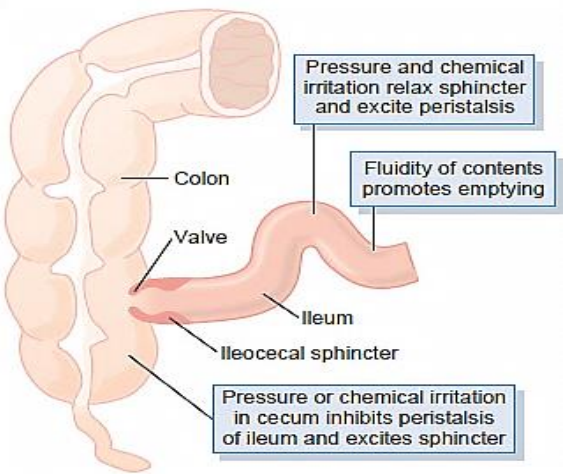


Rather, after a brief period, the contracted segments relax, and ring like contractions appear in the previously relaxed areas.

The new contraction forces the chyme in a previously relaxed segment to move in both directions into the now relaxed adjacent segments.

Function of the Ileocecal Valve A principal function of the ileocecal valve is to prevent backflow of fecal contents from the colon into the small intestine.

The ileocecal valve itself protrudes into the lumen of the cecum and therefore is forcefully closed when excess pressure builds up in the cecum and tries to push cecal contents backward against the valve lips. The valve usually can resist reverse pressure of at least 50 to 60 centimeters of water.



Absorption in the Small Intestine90% occur in the small intestine through villi by diffusion, facilitated diffusion, osmosis , and active transport

Trypsin acting on native proteins causes them to pass through several intermediate stages such as globulins, alkaline metaproteins, peptones, polypeptides and finally converting them into mono-amino acids (leucine, aspartic acid) and di-amino acids (as lysine and arginine, etc.)

INTESTINAL JUICE This is a watery and slightly alkaline fluid. It has a pH about 7.6 and contains some 1.6 % of solid matters out of which about 1% is inorganic and the rest are organic.

It contains the following enzymes:

a) Disaccharidases:

These hydrolyse disaccharides into monosaccharides, thus ending the process of digestion of carbohydrates. They include:

(i) **Maltase**, which split maltose into 2 molecules of glucose.

maltose $\xrightarrow{\text{maltase}}$ glucose + glucose.

(ii) **Sucrase**, which acts on sucrose

sucrose $\xrightarrow{\text{sucrase}}$ glucose + fructose

(iii) **Lactase**, which acts on lactose

Lactose $\xrightarrow{\text{lactase}}$ glucose + galactose.

b) Erepsin:

This is a complex of proteolytic enzymes including Carboxypeptidase, aminopeptidase and dipeptidase, which hydrolyse polypeptides and dipeptides of different structures into free amino acids, thus bringing digestion of proteins begun by pepsin and trypsin to an end. (i.e. free aminoacids).

c) Intestinal lipase: This is an esterase which hydrolyses fats into glycerol and fatty acids.

d) Nuclease and Nucleotidase:

These enzymes act on nucleic acid components of the nucleoproteins, finally splitting them into corresponding purine or pyrimidine nucleotides.

There are other enzymes such as esterase, phosphatase, arginase, catalase, proteinase and tyrosinase which act mostly on the broken down food stuffs during the passage through the epithelial cells of the small intestine on their way to the blood vessels.

GASTROINTESTINAL HORMONES

1- **Gastrin:** has a variety of actions, but its principal physiologic actions are stimulation of gastric acid and pepsin secretion. Gastrin stimulates insulin secretion**.**

2- **Enterogastrone:**a putative hormone that inhibits gastric acid secretion and motility.

3- **Cholecystokinin- Pancreozymin CCK:** it was formerly thought that a hormone called cholecystokinin produced contraction of the gallbladder whereas a separate hormone called pancreozymin increased the secretion of pancreatic juice rich in enzymes.

It is now clear that a single hormone secreted by cells in the mucosa of the upper small intestine has both activities, and the hormone has therefore been named cholecystokinin-pancreozymin.

It is also called CCK-PZ or, most commonly, CCK causing contraction of the gallbladder and secretion of a pancreatic juice rich in enzymes.

Gastrin and CCK stimulate glucagon secretion, and since the secretion of both gastrointestinal hormones is increased by a protein meal, either or both may be the "gut factor" that stimulates glucagon secretion. The secretion of CCK is increased by contact of the intestinal mucosa with the products of digestion.

4- **Secretin** increases the secretion of bicarbonate by the duct cells of the pancreas and biliary tract. It thus causes the secretion of a watery, alkaline pancreatic juice. Its action on pancreatic duct cells is mediated via cAMP. It also augments the action of CCK in producing pancreatic secretion of digestive enzymes.

It decreases gastric acid secretion and may cause contraction of the pyloric sphincter.

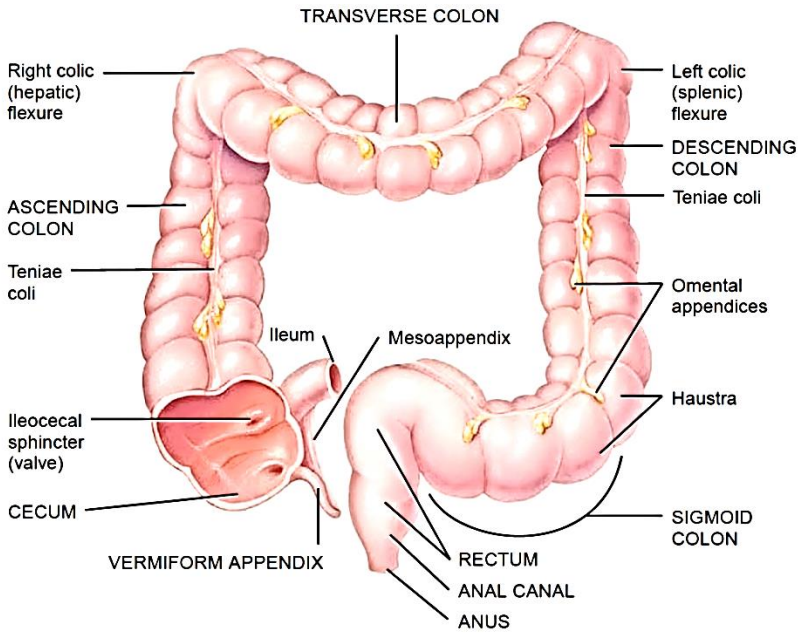
The Large Intestine

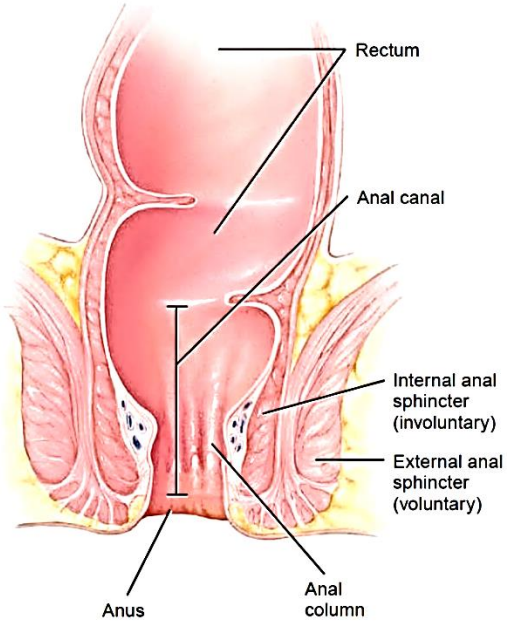
The storage and elimination of undigested waste and the reabsorption of water are the functions of the large intestine.

The large intestine is about 6.5 cm (2.5 inches) in diameter and about 1.5 m (5 feet) long. The outer longitudinal muscle fibers form three separate bands on the surface. These bands draw up the wall of the organ to give it its distinctive puckered appearance.

The large intestine begins in the lower right region of the abdomen. The first part is a small pouch called the cecum.

Between the ileum of the small intestine and the cecum is a sphincter, the ileocecal valve that prevents food from traveling backward into the small intestine. Attached to the cecum is a small, blind tube containing lymphoid tissue; it is called the vermiform appendix (vermiform means "wormlike").





Inflammation of this tissue as a result of infection or obstruction is appendicitis.

The second portion, the **ascending colon,** extends upward along the right side of the abdomen toward the liver. The large intestine then bends across the abdomen, forming the **transverse colon.** At this point it bends sharply and extends downward on the left side of the abdomen into the pelvis ,forming the **descending colon**. The lower part of the colon bends posteriorly in an S shape and continues downward as the sigmoid colon.

The sigmoid colon empties into the rectum, which serves as a temporary storage area for indigestible or unabsorbable food residue.

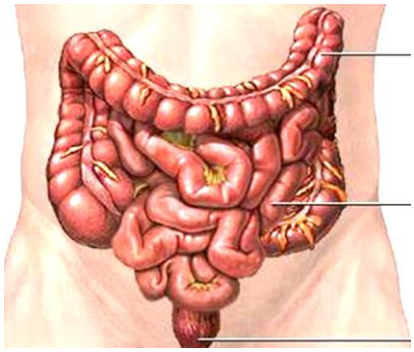
Enlargement of the veins in this area constitutes hemorrhoids'. A narrow portion of the distal large intestine is called the anal canal. This leads to the outside of the body through an opening called the anus.

Functionally divided into:

1. Proximal colon :from cecum to middle of transverse colon ,for absorption

2.Distal colon :from middle of transverse colon to the junction of pelvic colon with rectum for storage

3. Rectum and anal canal ,for defecation



About 1500 milliliters of chyme normally pass through the ileocecal valve into the large intestine each day.

Most of the absorption in the large intestine occurs in the proximal one half of the colon, giving this portion the name absorbing colon, whereas the distal colon functions principally for feces storage until a propitious time for feces excretion and is therefore called the storage colon.

The large intestine can absorb a maximum of 5 to 8 liters of fluid and electrolytes each day

The mucosa of the large intestine secretes bicarbonate ions while it simultaneously absorbs an equal number of chloride ions in an exchange transport process.

Absorption of sodium and chloride ions creates an osmotic gradient across the large intestinal mucosa, which in turn causes absorption of water.

Bacterial Action in the Colon.

Numerous bacteria, especially colon bacilli, are present even normally in the absorbing colon.

They are capable of digesting small amounts of cellulose, in this way providing a few calories of extra nutrition for the body.

Bacterial infections often cause the crypts in the terminal ileum and in the large intestine to secrete 10 or more liters of fluid each day, leading to severe and sometimes lethal diarrhea.

Other substances formed as a result of bacterial activity are vitamin K, vitamin B12, thiamine, riboflavin, and various gases that contribute to flatus in the colon, especially carbon dioxide, hydrogen gas, and methane.

The bacteria-formed vitamin K is especially important because the amount of this vitamin in the daily ingested foods is normally insufficient to maintain adequate blood coagulation.

Composition of the Feces.

- Three-fourths water and
- one-fourth solid matter that itself is composed of about
- 30 per cent dead bacteria,
- 10 to 20 per cent fat,
- 10 to 20 per cent inorganic matter,
- 2 to 3 per cent protein, and
- 30 per cent undigested roughage from the food and dried
- constituents of digestive juices, such as bile pigment and sloughed epithelial cells.

The brown color of feces is caused by stercobilin and urobilin, derivatives of bilirubin.

The odor is caused principally by products of bacterial action; these products vary from one person to another, depending on each person's colonic bacterial flora and on the type of food eaten.

The actual odoriferous products include indole, skatole, mercaptans, and hydrogen sulfide.

The process of digestion does not take place in the large intestine as the wall of this region does not have any digestive gland. Large intestine simply secretes mucus from the goblet cells in the epithelium which serves as lubricant for easy passage of the undigested food.

At the end of the journey of food along the alimentary canal , feces which consists of a semisolid mass containing unmacerrated plant tissue , pieces of cellulose and other undigested food remains together with small amounts of water , is egested or passed out of the body at intervals through the anal opening .

Large quantities of mucus, but no enzymes, are secreted by the large intestine. At intervals, usually after meals, the involuntary muscles within the walls of the large intestine propel solid waste material, called feces or stool, toward the rectum. This material is then eliminated from the body by both voluntary an involuntary muscle actions, a process called defecation.

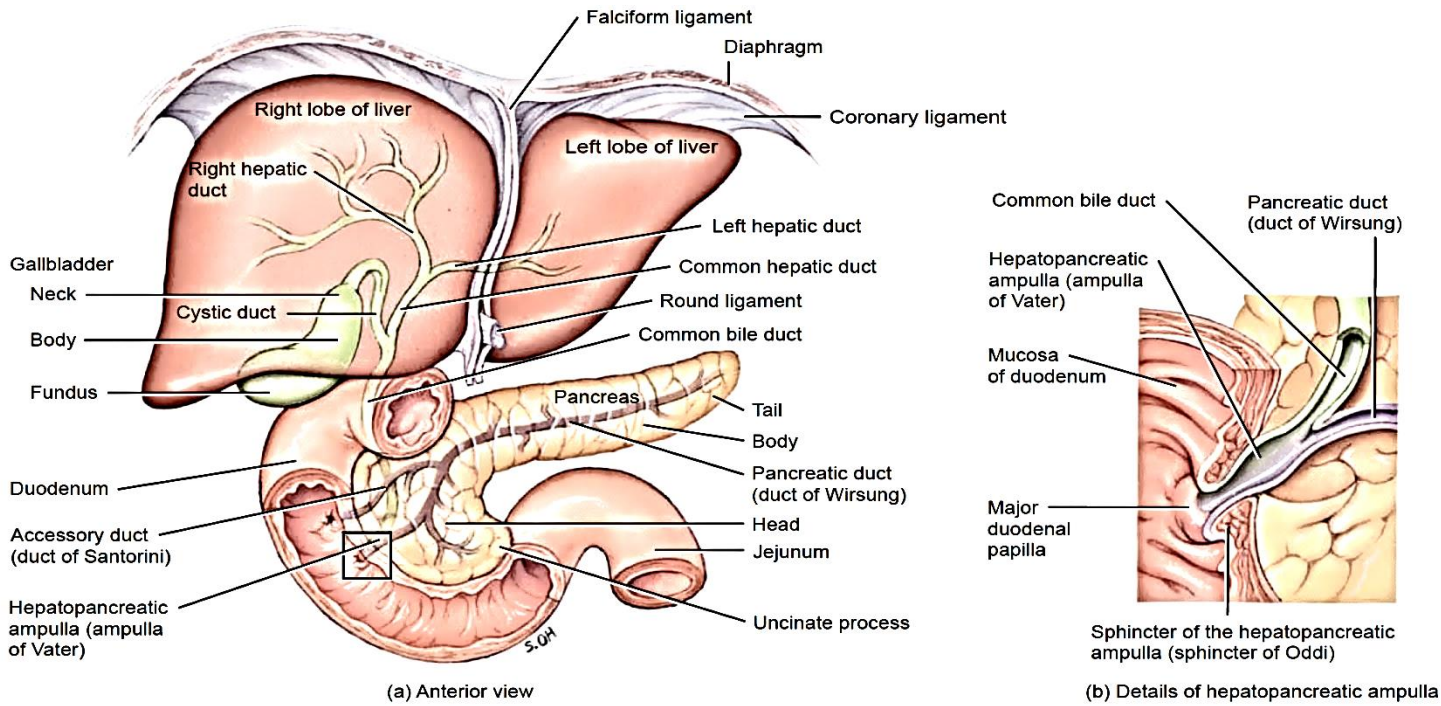
The Accessory Structures

The Liver

The liver, is the largest glandular organ of the body .

It is locate din the upper right portion of the abdominal cavity under the dome of the diaphragm. The lower edge of a normal-sized liver is level with the lower margin of the ribs. It has a large right lobe and a smaller left lobe; the right lobe includes two inferior smaller lobes.

The liver is supplied with blood through two vessels: the portal vein and the hepatic artery.



(a) Anterior view

(b) Details of hepatopancreatic ampulla

The hepatic artery carries oxygenated blood, whereas the portal system of veins carries blood that is rich in the end products of digestion.

This most remarkable organ has so many functions that only some of its major activities **can list here:**

1. The storage of glucose (simple sugar) in the form of glycogen, in animal starch. When the blood sugar level falls below normal, liver cells convert glycogen to glucose and release it into the bloodstream; this serves to restore the normal concentration of blood sugar.

2. The formation of blood plasma proteins, such as albumin, globulins, and clotting factors

3. **The synthesis of urea,** a waste product of protein metabolism. Urea is released into the blood and transported to the kidneys for elimination.

4. The modification of fats, so cells all over the body can use them more efficiently

5. The manufacture of bile

6. The destruction of old red blood cells. The pigment released from these cells in both the liver and the spleen is eliminated in the bile. This pigment (bilirubin) gives the stool its characteristic dark color.

7. The detoxification (removal of the poisonous properties)of harmful substances such as alcohol and certain drugs

8. The storage of some vitamins and iron

The main digestive function of the liver is the production of bile. The salts contained in bile act like a detergent to emulsify fat, that is, to break up fat into small droplets that can be acted on more effectively by digestive enzymes. Bile also aids in the absorption of fat from the small intestine.

Bile leaves the lobes of the liver by two ducts that merge to foln1 the common hepatic duct.

After collecting bile from the gallbladder, this duct, now called common bile duct, delivers bile into the duodenum.

With respect to the excretory function of liver, bile – pigments represent the excretory part of bile.

Human bile is a yellowish – orange fluid which owes its color to two different bile pigments: "bilirubin" a red pigment, and "biliverdin" which is green.

Bilirubin is the substance that gives bile its color.

It is formed during the breakdown of senescent red blood cells.

In the process of degradation, the heme portion of the hemoglobin molecule is oxidized to form biliverdin, which is then converted to free bilirubin .

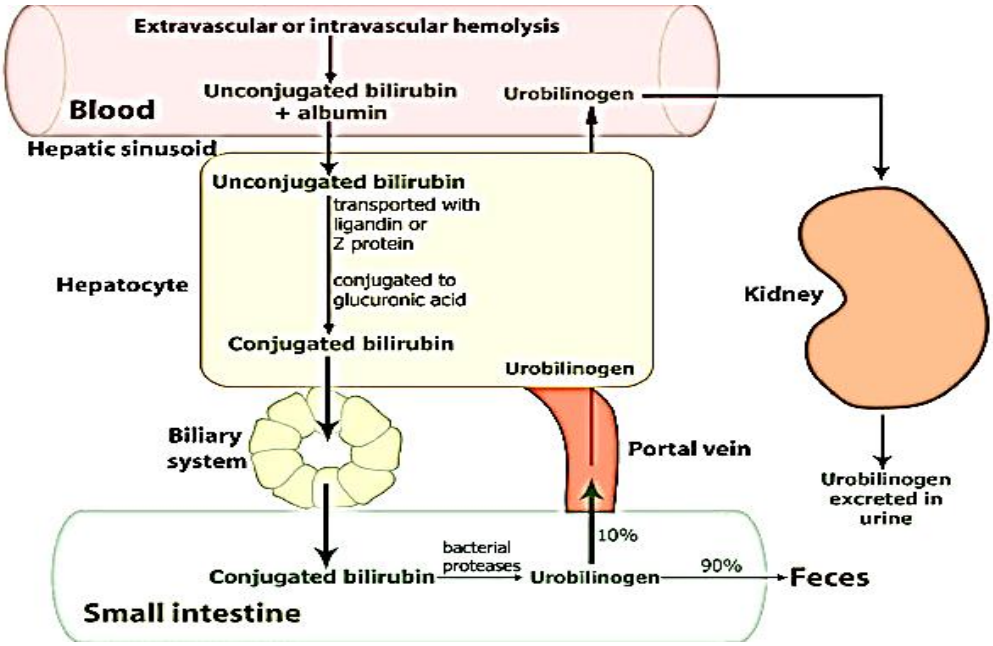
Free bilirubin, which is insoluble in plasma, is transported in the blood attached to plasma albumin.

Even when it is bound to albumin, this bilirubin is still called free bilirubin.

As it passes through the liver, free bilirubin is released from the albumin carrier molecule and moved into the hepatocytes.

Inside the hepatocytes, free bilirubin is converted to conjugated bilirubin, making it soluble in bile.

Conjugated bilirubin is secreted as a constituent of bile, and in this form it passes through the bile ducts into the small intestine.

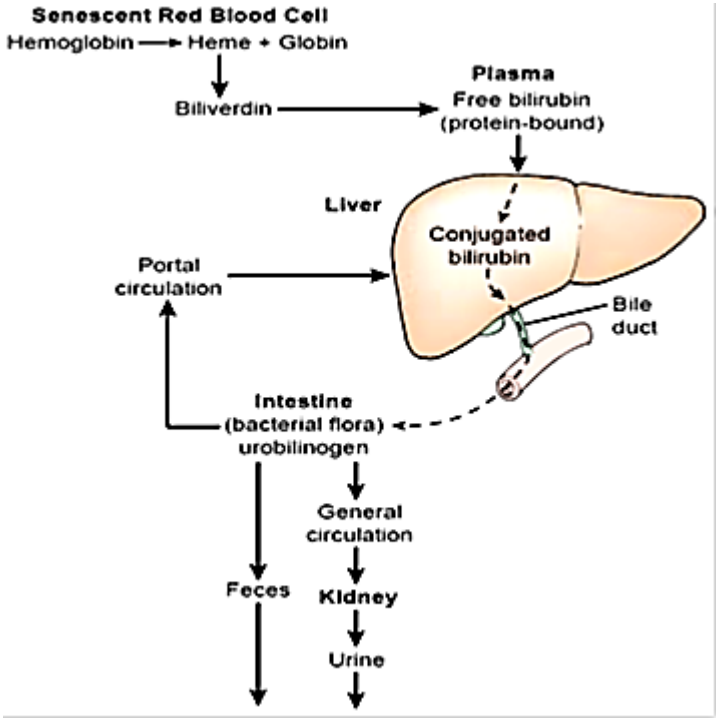


In the intestine, approximately one half of the bilirubin is converted into a highly soluble substance called urobilinogen by the intestinal flora.

Urobilinogen is either absorbed into the portal circulation or excreted in the feces.

Most of the urobilinogen that is absorbed is returned to the liver to be re-excreted into the bile.

A small amount of urobilinogen, approximately 5%, is absorbed into the general circulation and then excreted by the kidneys.



Usually, only a small amount of bilirubin (0.1 to 1.2 mg/dL) is found in the blood.

Laboratory measurements of bilirubin usually measure the free and the conjugated bilirubin as well as the total bilirubin.

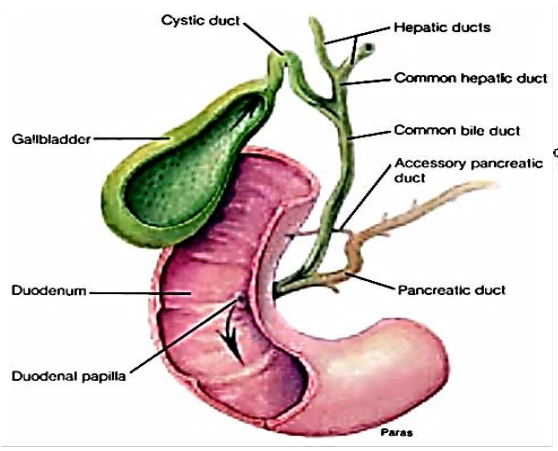
These are reported as the direct (conjugated) bilirubin and the indirect (unconjugated) bilirubin.

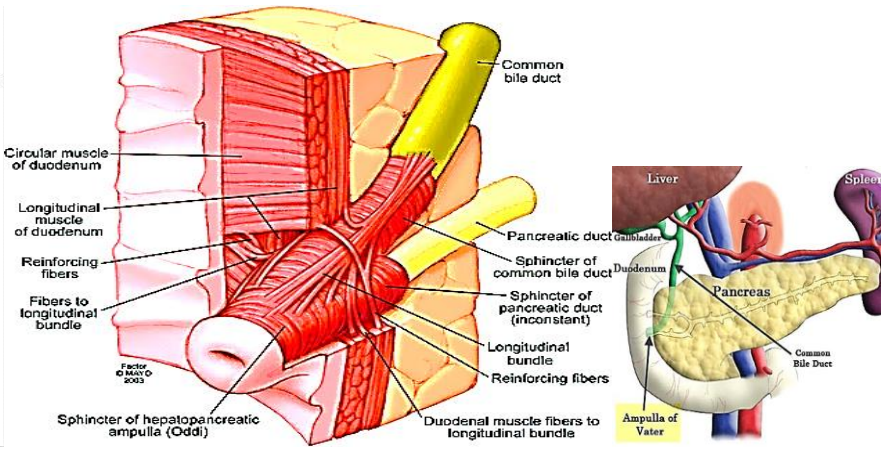
In man, the former pigment (bilirubin) is the chief one, while the latter is only the oxidative product of former.

These two pigments are formed as a result of the breakdown of the red pigment, hemoglobin, of the worn – out red blood corpuscles.

The Gallbladder

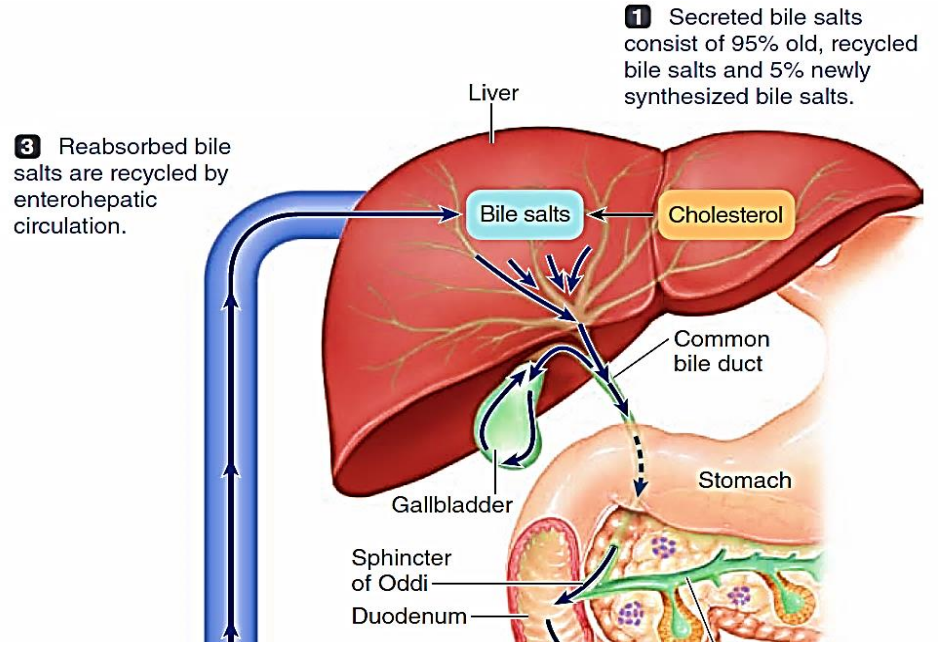
The gallbladder is a muscular sac on the inferior surface of the liver that serves as a storage pouch for bile. Although the liver may manufacture bile continuously, the body is likely to need it only a few times a day. Consequently, bile from the liver flows into the hepatic ducts and then up through the cystic duct connected with the gallbladder. When chyme enters the duodenum, the gallbladder contracts, squeezing bile through the cystic duct and into the common bile duct leading to the duodenum.





Composition of bile:

The bile, collected directly from the liver contains over 97.5% of water but the bile collected from the gall bladder as little as 92% of water In this way the bladder bile is less watery in comparison to liver bile .



Bile whether, hepatic or gall bladder is always isosmotic with respect of blood plasma. The chief solid constituents of the bile are proteins, inorganic salts, bile – pigments, bile salts, cholesterol,

lecithin and other lipid constituents. Bile also contains a little amount of bile acids including glycholic acid and taurocholic acid.

The composition of human bile, as secreted by the liver and as obtained from the gall bladder, is shown in the following table:

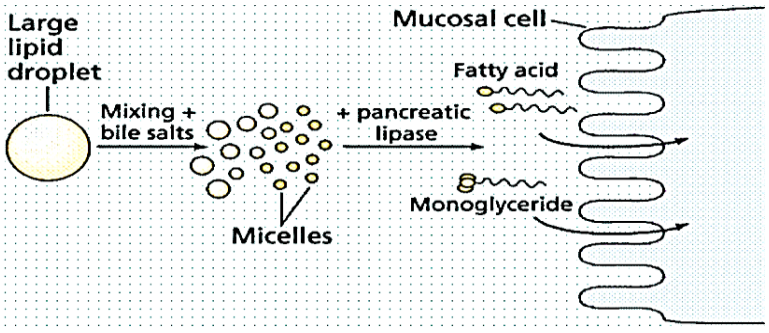
Role of bile:

In the formation of bile, the liver has both secretory and excretory activities. The secretory function is represented by the role of bile – salts in the digestion and absorption of fats.

The most important constituents of bile are 3- kinds of substances "bile salts ", "bile pigments ", & "cholesterol".

Bile salts, are manufactured by the hepatic cells, and are the sodium and potassium salts of glycocholic and taurocholic acids.

1) On reaching the intestine, they break the large globules of fats into very small droplets, a process called "emulsification". This process greatly increases the surface area of the fat, so that it will be subjected more completely to the enzymatic action of lipases.
2) Bile salts were also found to activate the pancreatic lipase (steapsin), the action of the enzyme is at least three fold in the presence of bile. After use in the digestive process, about 90% of the bile salts are re- absorbed by the portal circulation and returned to the liver to be used again.
3) Moreover , bile salts play an important role in fat – absorption , since they combine with insoluble fatty acids produced by digestion of fats , forming water – soluble compounds which can readily diffuse into the intestinal cells. In the intestinal cells, the compounds break down liberating again the fatty acids and bile salts.



The bile salts have two important actions in the intestinal tract

First, they have a detergent action on the fat particles in the food.

This decreases the surface tension of the particles and allows agitation in the intestinal tract to break the fat globules into minute sizes.

This is called the emulsifying or detergent function of bile salts.

► **Function, bile salts help in the absorption of:**

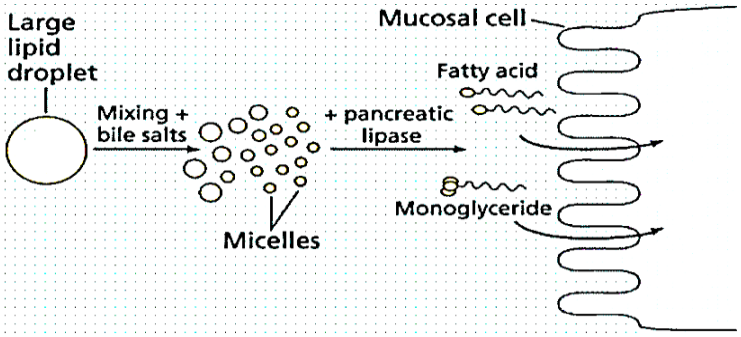
(1) fatty acids,

(2) monoglycerides,

(3) cholesterol, and

(4) other lipids from the intestinal tract.

They do this by forming very small physical complexes with these lipids; the complexes are called micelles, and they are semi-soluble in the chyme because of the electrical charges of the bile salts.



The Pancreas

The pancreas is a long gland that extends from the duodenum to the spleen. The pancreas produces enzymes that digest fats, proteins, carbohydrates, and nucleic acids. The protein digesting enzymes are produced in inactive forms, which must be converted to active forms in the small intestine by other enzymes. The pancreas also produces large amounts of alkaline fluid, which neutralizes the chyme in the small intestine, thus protecting the lining of the digestive tract.

These juices collect in a main duct that joins the common bile duct or empties into the duodenum near the common bile duct. Most persons also have an additional smaller duct that opens into the duodenum.

The pancreas also functions as an endocrine gland, producing the hormones insulin and glucagon that regulate sugar metabolism. These secretions of the islets cells are released directly into the blood.

PANCREATIC JUICE

Pancreatic juice is probably one of the most potent digestive juices available in the body. In .man some 500- 800 ml of it are secreted daily

The pancreatic juice is composed of some 98% of water and 2% solid substances. The solid substances include some potentially powerful enzymes which act on proteins, fats and carbohydrates. It also has high proportion of sodium bicarbonate (NaHCO3) and di-sodium hydrogen phosphate (Na2HPO4) these basic salts make the juice quite alkaline, within the 7.5 – 8 pH range Activated pancreatic juice contains trypsin, chymotrypsin, carboxypeptidase, pancreopeptidase E or elastase, amylopsin (amylase) and steapsin (pancreatic lipase). Other enzymes such as sucrase, lactase and maltase, may be present in ..trace quantities

Nature and actions of pancreatic enzymes

1. Trypsin :

This enzyme is secreted as the inactive proenzymetrypsinogen. This is rapidly activated into trypsin by the enzyme enteropeptidase (formally called enterokinase) secreted into the intestinal lumen by the duodenal mucosa. This type of activation is called autocatalytic .

Trypsin is the most effective proteolytic enzyme which acts upon all proteins that are attached by pepsin. It acts more rapidly and completely if the proteins have been partly hydrolyzed by pepsin.

Trypsin acting on native proteins causes them to pass through several intermediate stages such as globulins, alkaline metaproteins, peptones, polypeptides and finally converting them into mono-amino acids (leucine, aspartic acid) and di-amino acids (as lysine and arginine, etc)

2. Chemotrypsin :

It is another proteolytic enzyme secreted as inactive proenzyme called chemotrypsinogn. It is activated by trypsin to form the active enzyme chemotrypsin.

$$\text{chemotrypsinogen} \xrightarrow{\text{trypsin}} \text{chemotrypsin .}$$

It acts in very much the same way as trypsin but in addition it possesses the power of clotting milk. It completes the protein digestion at an optimum pH of 7-8.

3. Carboxypeptidase:

This enzyme acts on the polypeptides resulting from gastric and pancreatic digestions. It's splitting action occurring at the peptide linkage adjacent to the free terminal carboxylic group (COOH) of the polypeptide, converting it into dipeptides.

4, Pancreatic amylase "amylopsin":

It is the most important starch splitting enzyme which converts starch, dextrin and glycogen into maltose.

It requires the presence of chloride and phosphate as coenzyme and acts best at pH 7.

5,Pancreatic lipase or steapsin

This is the most important fat spitting enzyme of pancreatic juice so that, an extremely small amount of it is needed to bring about normal fat digestion, whereas if it is lacking almost all the ingested fat is eliminated undigested in the feces. It acts on the fats that have been emulsified, converting them into glycerol and fatty acids.

Absorption :

After food is digested, it is absorbed; that is, it moves through the mucous membrane lining of the small intestine into the blood and lymph. In other words, food absorption is the process by which molecules of amino acids, glucose, fatty acids, and glycerol goes from the inside of the intestines into the circulating fluids of the body. Absorption of foods is just as essential as digestion of foods. The reason is fairly obvious.

As long as food stays in the intestines, it cannot nourish the millions of cells that compose all other parts of the body. Their lives depend on the absorption of digested food and its transportation to them by the circulating blood.

Structural adaptations of the digestive tube, including folds in the lining mucosa, villi, and micro villi, increase the absorptive surface and the efficiency and speed of transfer of materials from the intestinal lumen to body fluids. Many salts such ass odium are actively transported through the

intestinal mucosa. Water follows by osmosis. Other nutrients are also actively transported into the blood of capillaries in the intestinal villi.

Fats enter the lymphatic vessels or lacteals found in intestinal villi.

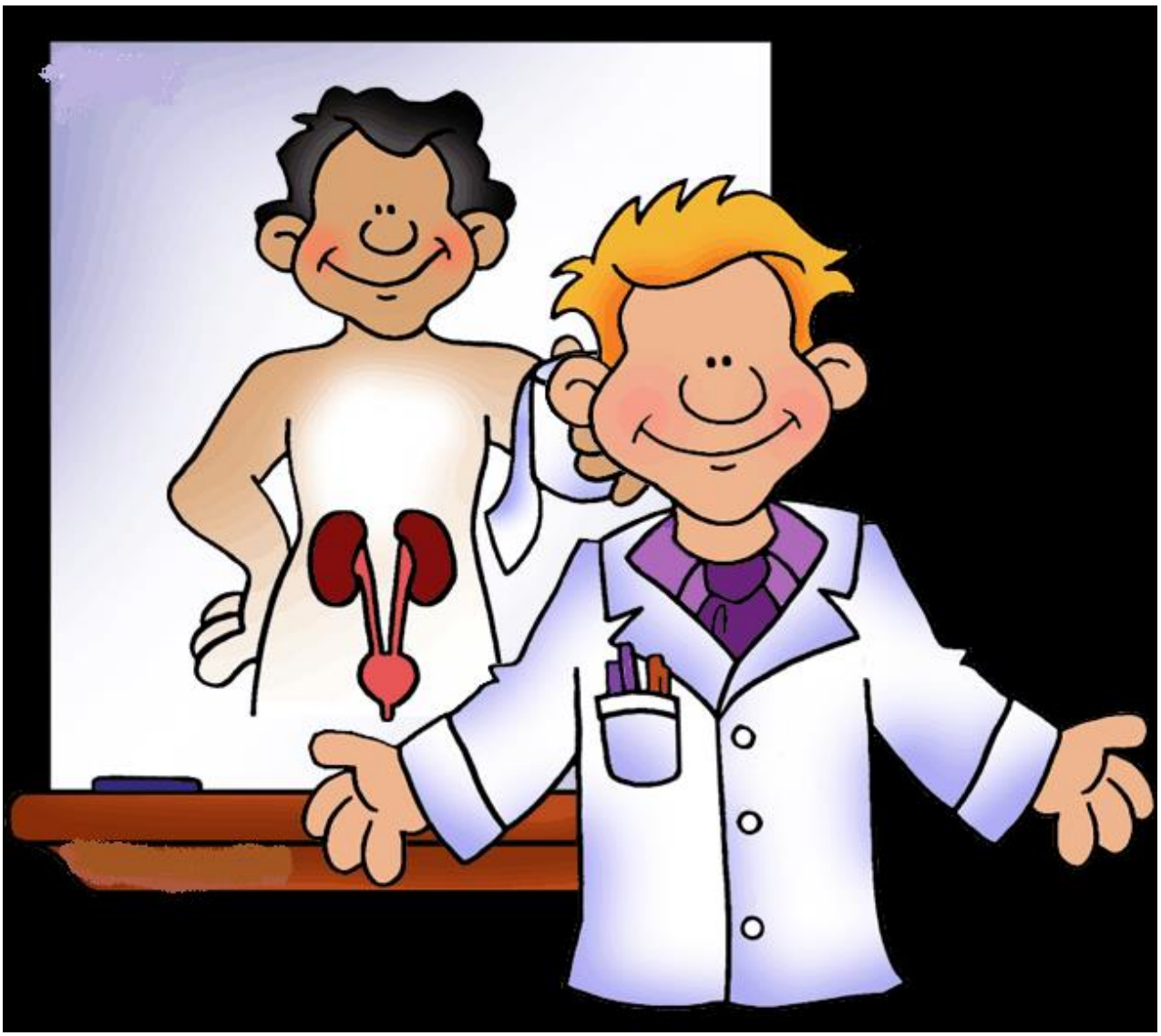
Digestive juices and enzymes	**Substance Digested (or hydrolysed)**	**Resulting Products***
Saliva Amylase	Starch (Polysaccharide)	Maltose (disaccharide)
Gastric Juice Protease (Pepsin) plus hydrochloric acid	Proteins	Partially digested proteins
Pancreatic Juice Protease (trypsin) and Lipase Amylase	Proteins (intact of partially digested) Fats emulsified by bile Starch	Peptides **Fatty acids, amino acids** and **glycerol** Maltose
Intestinal Juice Peptidases Sucrase Lactase Maltase	Peptides Sucrose (cane sugar) Lactase (Milk sugar) Maltase (malt sugar)	**Amino acids** Glucose and fructose (simple sugars) **Glucose** and **galactose** (Simple sugars **Glucose**

*Substances underlined are end products of digestion (that is, completely digested foods ready for absorption)

CHAPTER TWELVE

THE

URINARY SYSTEM



CHAPTER TWELVETHE URINARY SYSTEM

The urinary system is also called the excretory system of the body because one of its functions is to remove waste products from the blood and eliminate them from the body.

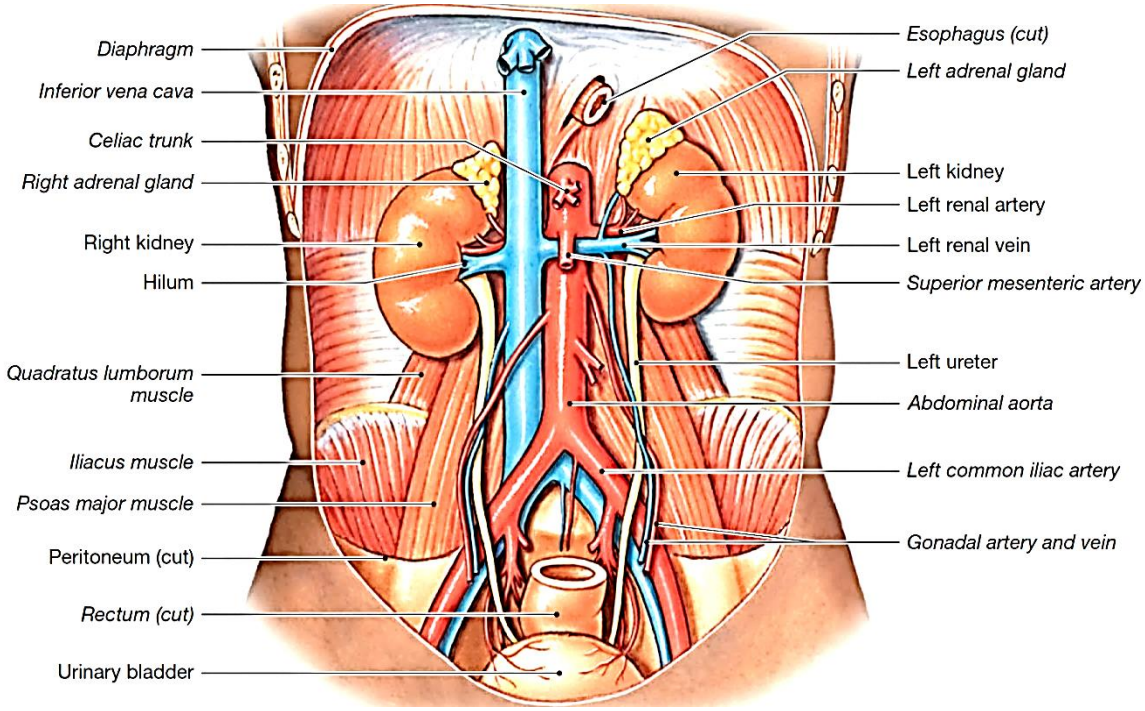
The urinary system consists of:

- Two kidneys: this organ extracts wastes from the blood ,balance body fluids and form urine.

- Two ureters: this tube conducts urine from the kidneys to the urinary bladder.

- The urinary bladder: this reservoir receives and stores the urine brought to it by the two ureters.

- The urethra: this tube conducts urine from the bladder to the outside of the body for elimination.



Major functions of the urinary system:

1. Excretion of wastes

2. Hormonal production (rennin-angiotensin , calcitriol and erythropoietin)

3. Acid base balancing

4. Regulate ion concentrations in the blood

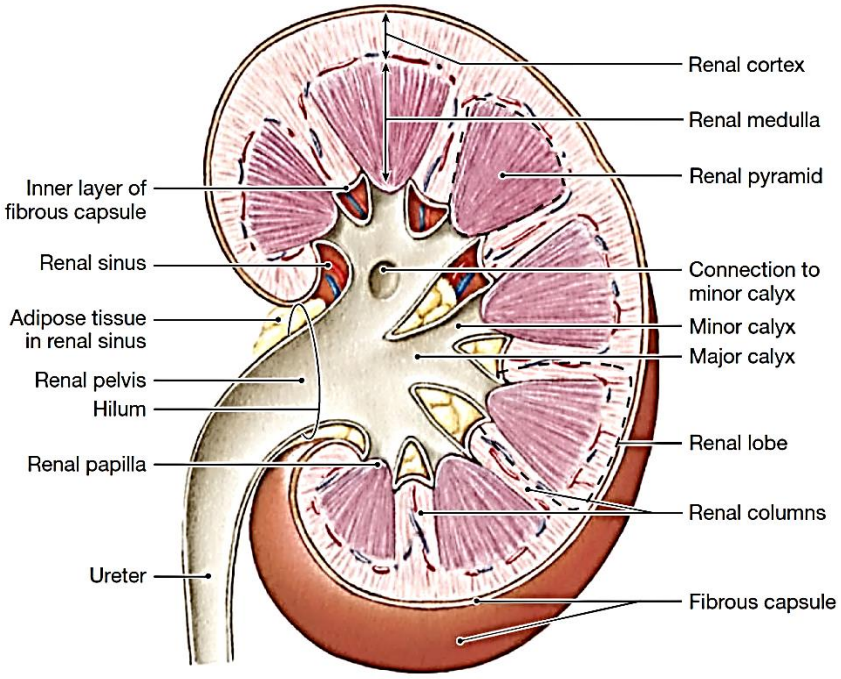
The kidney

External Anatomy of the kidney

A pair of reddish brown, bean shaped organ located in the posterior wall of the abdominal region, one in each side of the vertebral column. They usually span between T12 to L3. They are protected at least partially by the last pair of ribs and capped by the adrenal gland. The bean shape of the kidney is medially concave and laterally convex. On the medial concave border is **the hilus** (small indented area) where blood vessels, nerves & ureters enter and leave the kidney.

Covering and supporting each kidney are three layers of tissue:

- Renal capsule – innermost, tough, fibrous layer
- Adipose capsule – the middle layer composed of fat, giving the kidney protective cushion.
- Renal fascia – is outer sub-serous membrane, connective tissue layer.



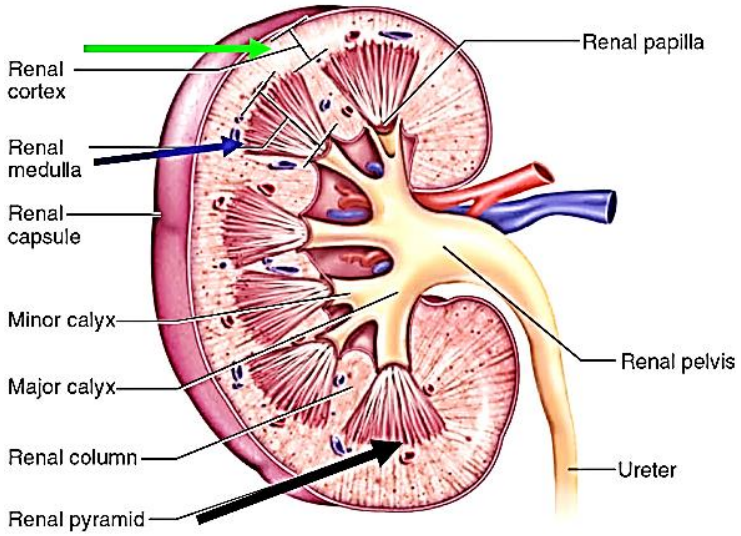
Internal Anatomy of the kidney

A sagittal section of the kidney reveals three distinct regions called:

- **Pelvis,**
- **Medulla and**
- **Cortex (from inside out).**

The renal pelvis is the large collecting space within the kidney formed from the expanded upper portion of the ureters.

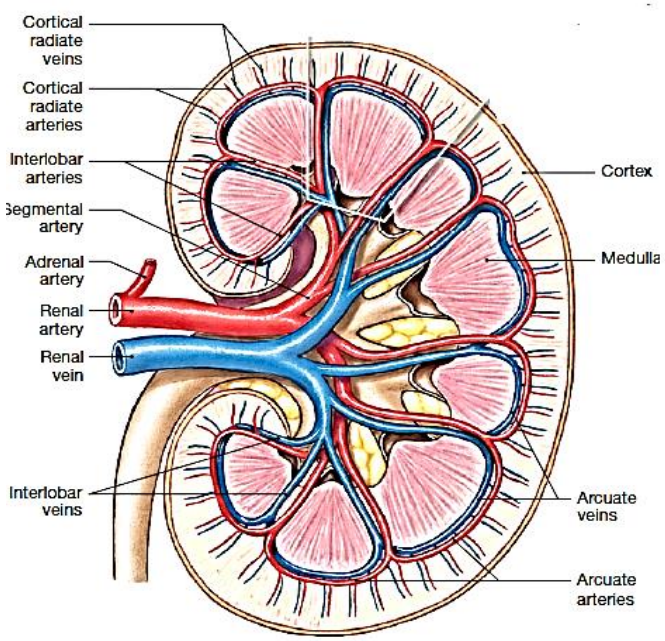
The pelvis branch to two cavities, these are 2-3 major calyces and8 to 18 minor calyces.

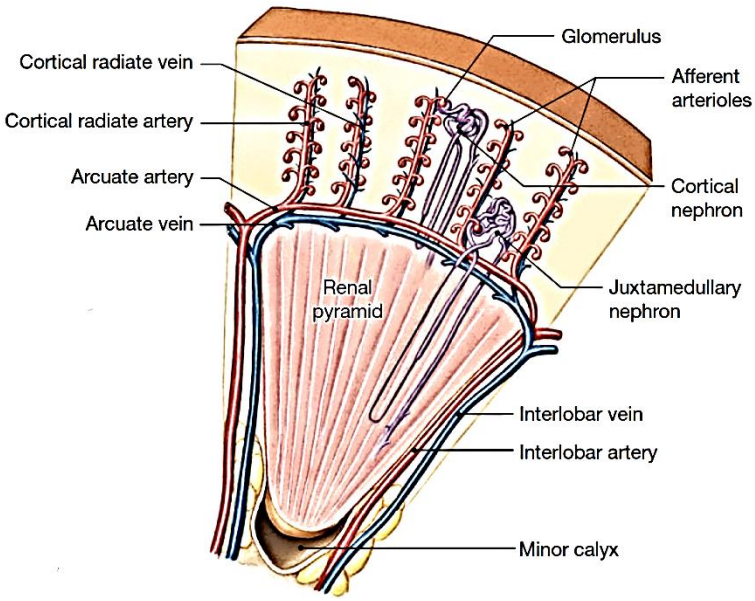


The Renal medulla is the middle portion of the kidney. It consists of 8 to 18 renal pyramids, which are longitudinally striped, one cone shaped area.

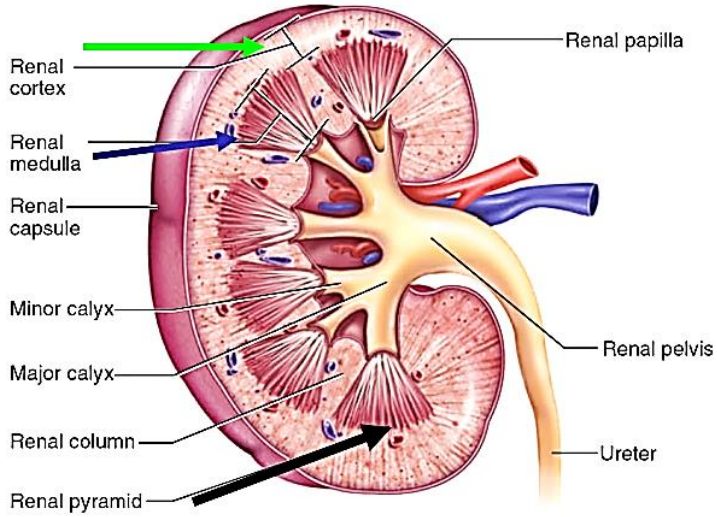
The base of each pyramid is adjacent to the outer cortex. The apex of each renal pyramid ends in papilla, which opens to a minor calyx.

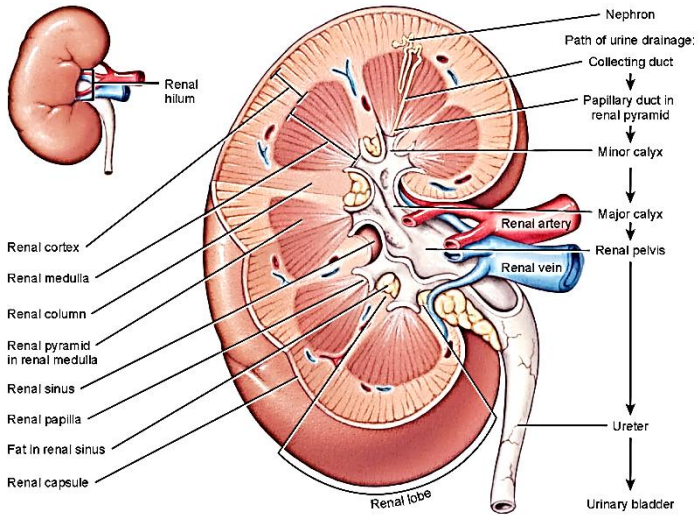
Pyramids contain tubules and collecting ducts of the nephron. Tubules involved in transportation and re-absorption of filtered materials.





The renal cortex is the outermost portion of the kidney. It is divided in to two region the outer cortical and the inner juxtamedullary region. The cortical tissue that penetrates between pyramids forms Renal Columns. The renal columns composed of mainly collecting tubules.





Blood circulation:

Blood is supplied to the kidneys by renal artery and drainage is by renal vein.

Nerve supply

By renal plexus of autonomic nervous system.

The nephron

The basic functional unit of the kidney is the nephron. Each nephron is an independent urine-forming unit.

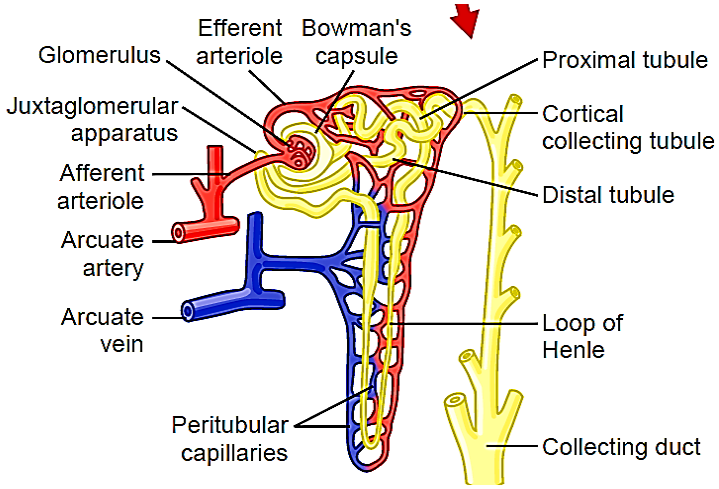
Each kidney contains approximately one million nephrons.

The kidney cannot regenerate new nephrons.

Therefore, with renal injury, disease, or normal aging, there is a gradual decrease in nephron number.

After age 40, the number of functioning nephrons usually decreases about 10 per cent every 10 years

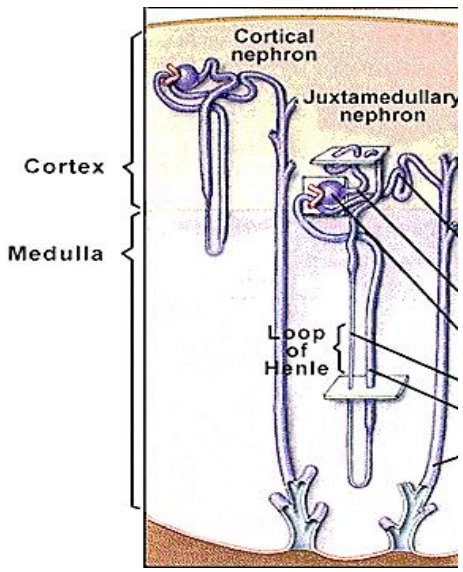
Each nephron contains a tuft of glomerular capillaries called the glomerulus, through which large amounts of fluid are filtered from the blood, and a long tubule in which the filtered fluid is converted into urine on its way to the pelvis of the kidney



► **Two Types of Nephrons**

1.Cortical nephrons (85%) shorter, mostly in cortex of kidney, produce "standard" urine

2.Juxtamedullary nephrons (15%), "juxta-next-to" the medulla - responsive to ADH, can concentrate urine



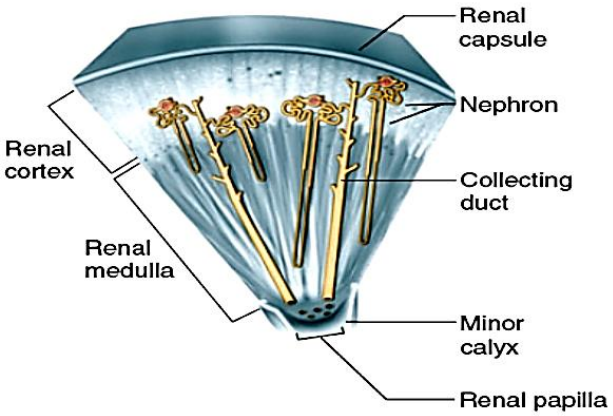
THE URETERS

Are a tubes about 26 cm long.

Extend from the kidneys to the posterior surface of the bladder.

Ureter enlarges as it enters kidney, becoming renal pelvis, which expands into several branches called calyces.

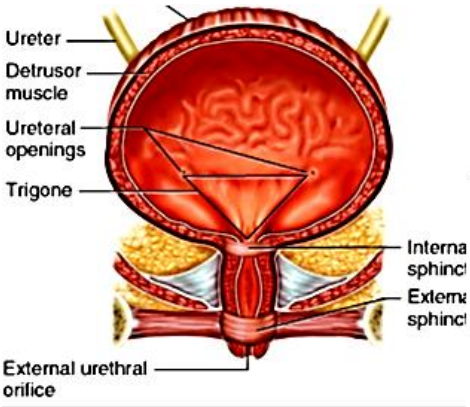
Each calyx contains a renal papilla.

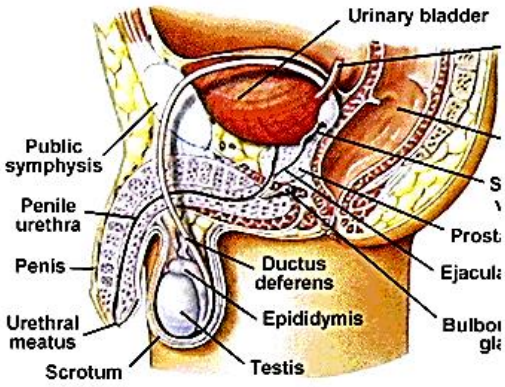


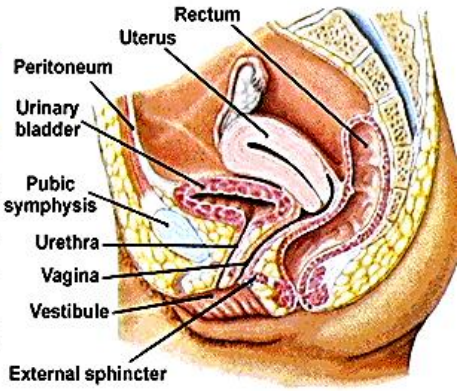
THE URINARY BLADDER

Is a collapsible bag situated in the pelvic cavity at the back of the symphysis pubis and in the front of the rectum in male ,but in the female is separated from the rectum by the uterus and vagina .

It has three openings (two ureters and one urethra) a ranged in the form of triangle, called the **TRIGONE** of the bladder with the apex below at the urethral opening .



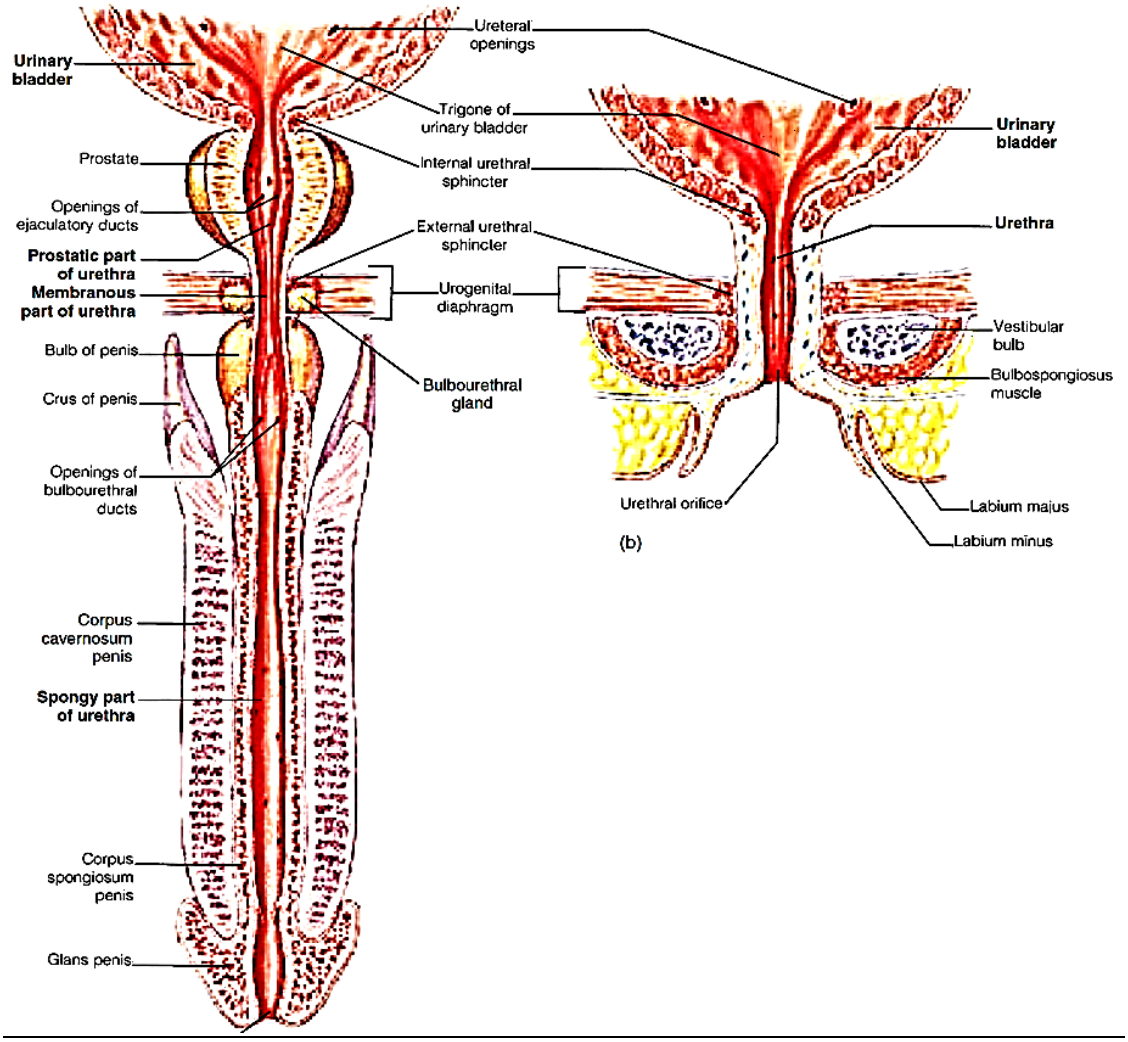




THE URETHRA

Is a canal conveying the urine from the bladder to the exterior (about 16-20cm in male and it is part of the reproductive functions and about 4 cm in female**) have two sphincters :**

- **internal, involuntary**
- **external, voluntary sphincter**



It has 3 parts :

1.The pelvic or prostatic portion :the first part from the bladder side.

2.The perineal or membranous portion

3.The penile or spongy portion have two sphincters :-

- ♣ internal,involuntary
- ♣ external,voluntary sphincter

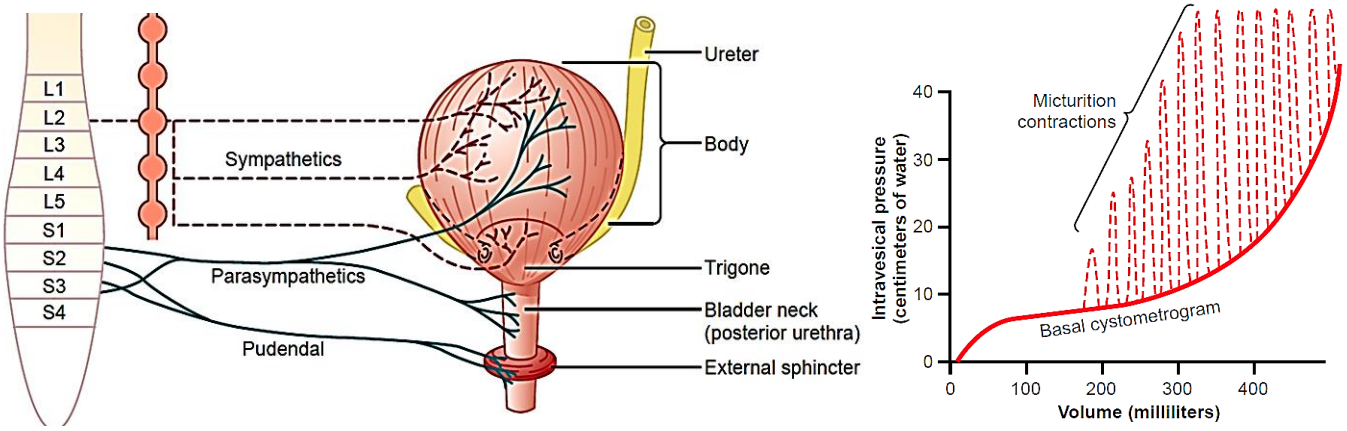
Physiologic Anatomy of the Bladder

The urinary bladder, is a smooth muscle chamber composed of two main parts: The body, and the neck.

The smooth muscle of the bladder is called the detrusor muscle.

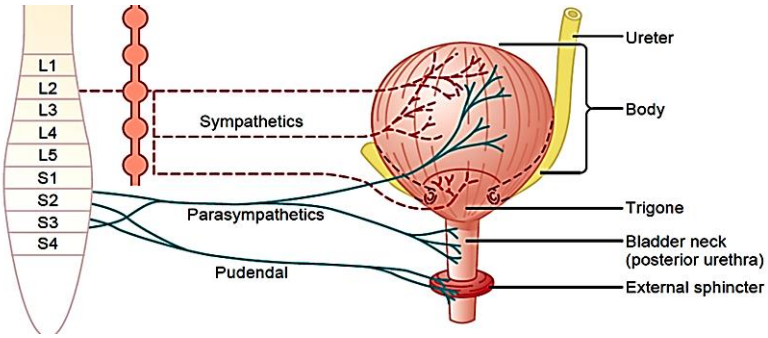
Its muscle fibers extend in all directions and, when contracted, can increase the pressure in the bladder to 40 to 60 mm Hg.

Thus, contraction of the detrusor muscle is a major step in emptying the bladder.



Smooth muscle cells of the detrusor muscle fuse with one another so that low-resistance electrical pathways exist from one muscle cell to the other.

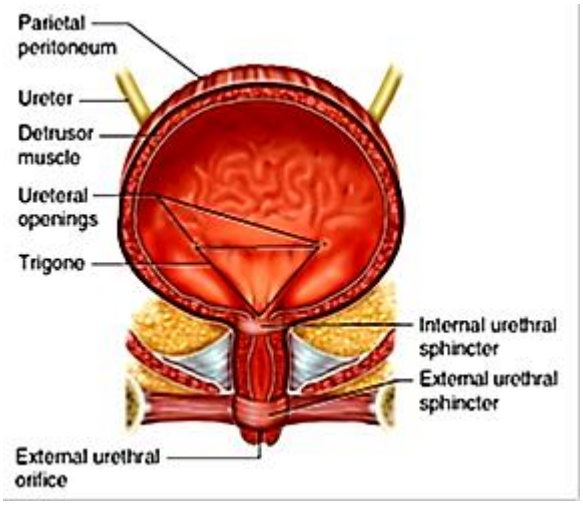
Therefore, an action potential can spread throughout the detrusor muscle, from one muscle cell to the next, to cause contraction of the entire bladder at once.



The bladder neck (posterior urethra) is 2 to 3 centimeters long, and its wall is composed of detrusor muscle interlaced with a large amount of elastic tissue.

The muscle in this area is called the internal sphincter.

Its natural tone normally keeps the bladder neck and posterior urethra empty of urine and, therefore, prevents emptying of the bladder until the pressure in the main part of the bladder rises above a critical threshold.



Beyond the posterior urethra, the urethra passes through the urogenital diaphragm, which contains a layer of muscle called the external sphincter of the bladder.

This muscle is a voluntary skeletal muscle, in contrast to the muscle of the bladder body and bladder neck, which is entirely smooth muscle.

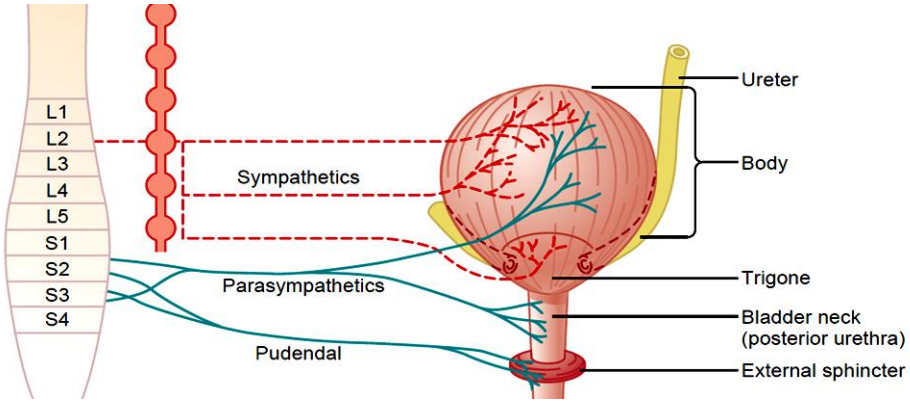
The external sphincter muscle is under voluntary control of the nervous system and can be used to consciously prevent urination even when involuntary controls are attempting to empty the bladder.

Innervation of the Bladder

Sympathetic: relax the wall of the urinary bladder ,but constrict internal urethral sphincter(prevent micturation)

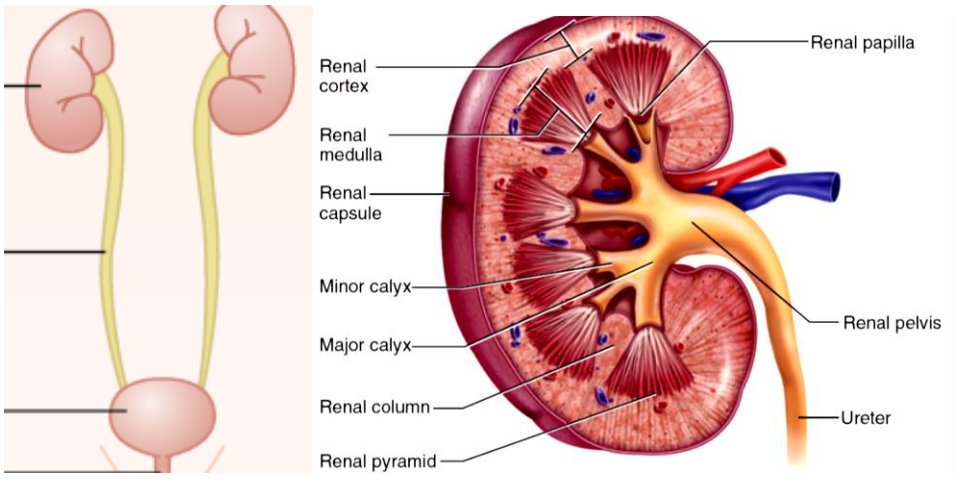
ParaSympathetic: contract wall of the urinary bladder ,but relax urethral sphincter(micturation)

Somatic (pudendal nerve):control the voluntary skeletal muscles of external sphincter



Transport of Urine from the Kidney Through the Ureters and into the Bladder

Urine that is expelled from the bladder has essentially the same composition as fluid flowing out of the collecting ducts; there are no significant changes in the composition of urine as it flows through the renal calyces and ureters to the bladder.



Urine flowing from the collecting ducts into the renal calyces stretches the calyces and increases their inherent pacemaker activity, which in turn initiates peristaltic contractions that spread to the renal pelvis and then downward along the length of the ureter,

The walls of the ureters contain smooth muscle and are innervated by both:

Sympathetic and parasympathetic nerves

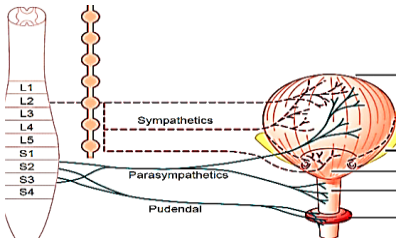
As with other visceral smooth muscle, peristaltic contractions in the ureter are enhanced by parasympathetic stimulation and inhibited by sympathetic stimulation.

Micturation: Process by which urinary bladder empties when it becomes filed . **2 main steps:**

1- Bladder fill progressively until the wall tension rises above threshold .
2- Micturation reflex, sensory stretch receptors in the bladder wall.

They are the result of a stretch reflex initiated by sensory stretch receptors in the bladder wall, especially by the receptors in the posterior urethra when this area begins to fill with urine at the higher bladder pressures.

Sensory signals from the bladder stretch receptors are conducted to the sacral segments of the cord through the pelvic nerves and then reflexively back again to the bladder. Through the parasympathetic nerve fibers by way of these same nerves.



Bladder partially filled lead to contraction lead to relax spontaneously if the bladder does not emptied lead to pressure falls to baseline which lead to bladder continues to fill often and powerful contractions.

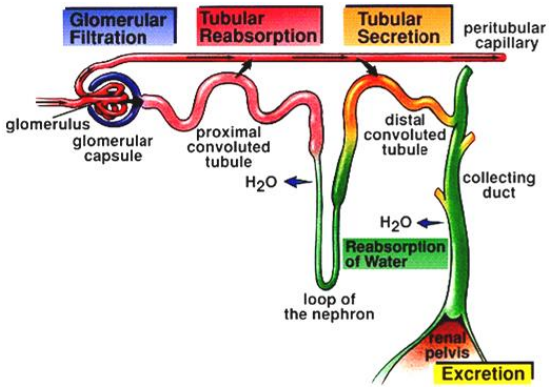
Urine Formation Results from Glomerular Filtration Tubular Reabsorption, and Tubular Secretion

Urinary excretion rate = Filtration rate - Reabsorption rate + Secretion rate:

Urine formation begins when a large amount of fluid that is virtually free of protein is filtered from the glomerular capillaries into Bowman's capsule

Most substances in the plasma, except for proteins, are freely filtered, so that their concentration in the glomerular filtrate in Bowman's capsule is almost the same as in the plasma.

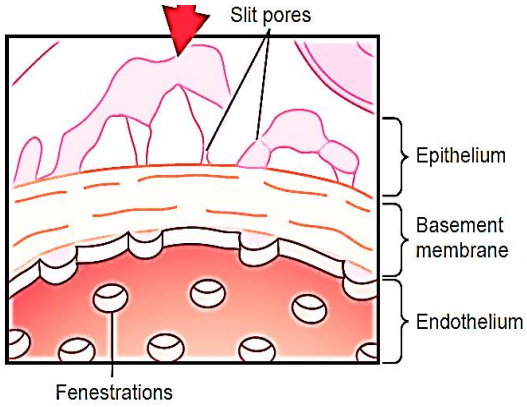
As filtered fluid leaves Bowman's capsule and passes through the tubules, it is modified by reabsorption of water and specific solutes back into the blood or by secretion of other substances from the peritubular capillaries into the tubules.



Glomerular Capillary Membrane

The glomerular capillary membrane is similar to that of other capillaries, except that it has **three major layers:**

1.the endothelium of the capillary, fenestrate ,negatively charged
2.a basement membrane, and
3.a layer of epithelial cells (podocytes) surrounding the outer surface of the capillary basement membrane.



Surrounding the endothelium is the basement membrane, which consists of a meshwork of collagen and proteoglycan fibrillate that have large spaces through which large amounts of water and small solutes can filter.

The basement membrane effectively prevents filtration of plasma proteins, in part because of strong negative electrical charges associated with the proteoglycans.

The epithelial cells: These cells are not continuous but have long foot like processes (podocytes) that encircle the outer surface of the capillaries.

The epithelial cells, which also have negative charges, provide additional restriction to filtration of plasma proteins.

Thus, all layers of the glomerular capillary wall provide a barrier to filtration of Plasma proteins.

Normally, about 20% of the plasma that enters the glomerulus is filtered at the net filtration pressure of10 mm Hg, producing collectively through all glomeruli 180 liters of glomerular filtrate each day for an average glomerular filtration rate (GFR) of 125 ml/min in males (160 liters of filtrate per day for an average GFR of 115 ml/min in females).

Physiologic Control of Glomerular Filtration and Renal Blood Flow

The determinants of GFR that are most variable and subject to physiologic control include

- the glomerular hydrostatic pressure and
- the glomerular capillary colloid osmotic pressure.

These variables, in turn, are influenced by

- the sympathetic nervous system,
- hormones and autacoids (vasoactive substances that are released in the kidneys and act locally), and

- other feedback controls that are intrinsic to the kidneys.

Strong activation of the renal sympathetic nerves can constrict the renal arterioles and decrease renal blood flow and GFR.

Moderate or mild sympathetic stimulation has little influence on renal blood flow and GFR.

For example, reflex activation of the sympathetic nervous system resulting from moderate decreases in pressure at the carotid sinus baroreceptors or cardiopulmonary receptors has little influence on renal blood flow or GFR.

The renal sympathetic nerves seem to be most important in reducing GFR during severe, acute

disturbances lasting for a few minutes to a few hours, such as those elicited by the defense reaction, brain ischemia, or severe hemorrhage.

In the healthy resting person, sympathetic tone appears to have little influence on renal blood flow.

Hormonal and Autacoid Control of Renal Circulation

Autacoids =vasoactive substances released by the kidney and act locally.

Norepinephrine, Epinephrine, and Endothelin

Constrict Renal Blood Vessels and Decrease GFR.

Hormones that constrict afferent and efferent arterioles, causing reductions in GFR and renal blood flow, include norepinephrine and epinephrine released from the adrenal medulla.

1.Thus, norepinephrine and epinephrine have little influence on renal hemodynamics except under

extreme conditions, such as severe hemorrhage.

2. Endothelin, is a peptide that can be released by damaged vascular endothelial cells of the kidneys as well as by other tissues.

Minimizing blood loss

May contribute to renal vasoconstriction and decreased GFR

- **Angiotensin II Constricts Efferent Arterioles.**

A powerful renal vasoconstrictor, angiotensin II, can be considered a circulating hormone as well as a locally

produced autacoid because it is formed in the kidneys as well as in the systemic circulation.

Because angiotensin II preferentially constricts efferent arterioles, increased angiotensin II levels raise glomerular hydrostatic pressure while reducing renal blood flow.

Increases tubular reabsorption of sodium and water, which helps restore blood volume and blood pressure.

- Endothelial-Derived Nitric Oxide Decreases Renal Vascular Resistance and Increases GFR.
- Prostaglandins and Bradykinin Tend to Increase GFR.
- Hormones and autacoids that cause vasodilation and increased renal blood flow and GFR

Hormones and Autacoids That Influence Glomerular Filtration Rate (GFR)

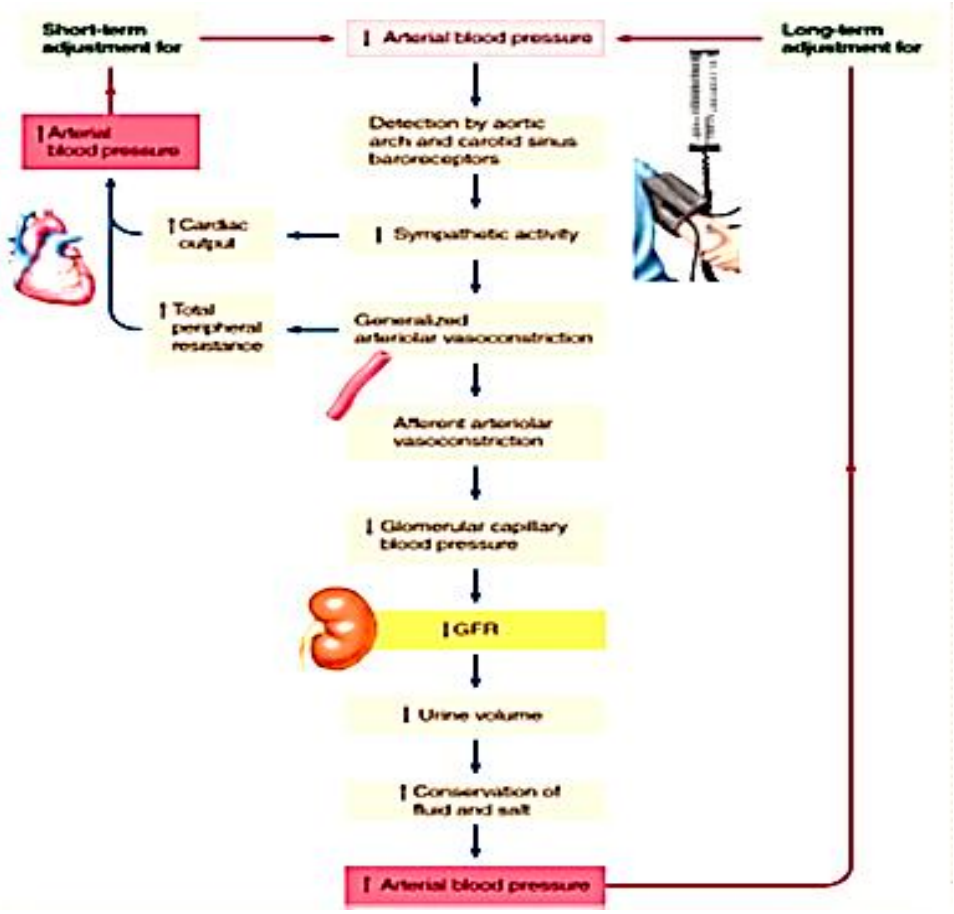
Hormone or Autacoid	Effect on GFR
Norepinephrine	↓
Epinephrine	↓
Endothelin	↓
Angiotensin II	⟷ (prevents ↓)
Endothelial-derived nitric oxide	↑
Prostaglandins	↑

- **<u>Autoregulation of GFR and Renal Blood Flow</u>**

Feedback mechanisms intrinsic to the kidneys normally keep the renal blood flow and GFR relatively constant, despite marked changes in arterial blood pressure.

GFR and renal blood flow is referred to as autoregulation

- **Role of Tubuloglomerular Feedback in Autoregulation of GFR** This feedback helps
  - ensure a relatively constant delivery of sodium chloride to the distal tubule and
  - helps prevent spurious fluctuations in renal excretion that would otherwise occur.

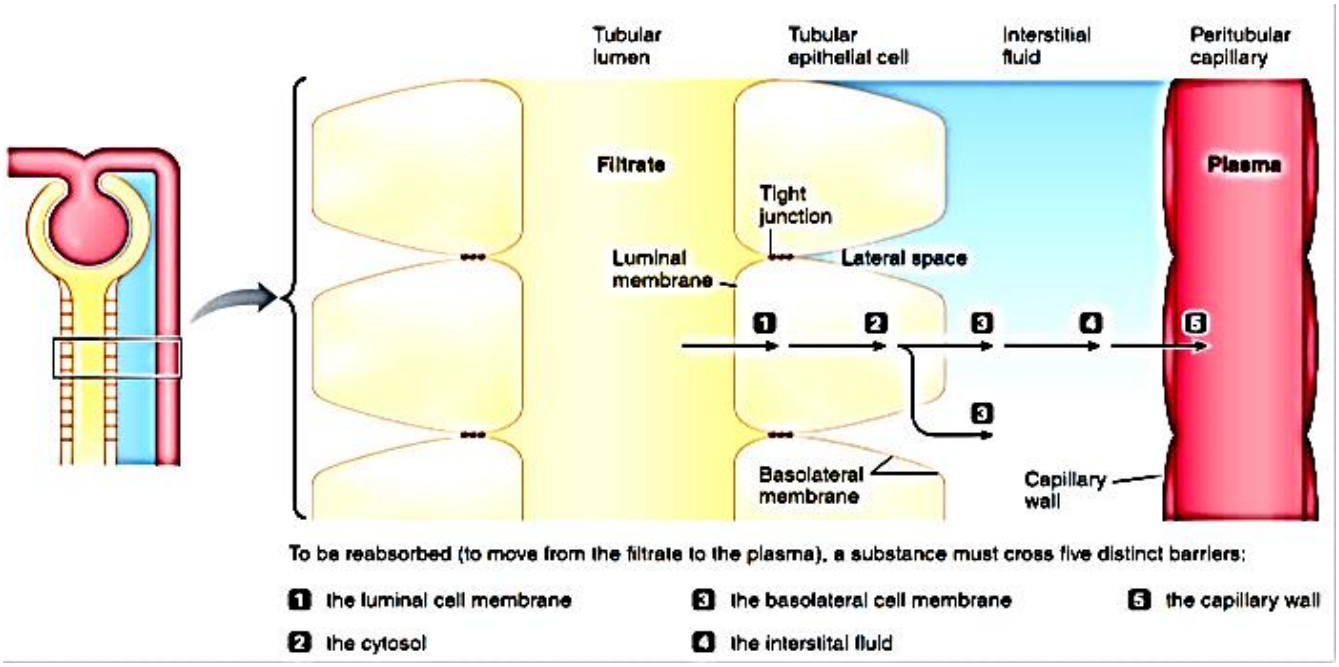


Reabsorption and Secretion by the Renal Tubules

- Tubular Reabsorption Is highly Selective
- Reabsorption and urinary excretion are variable, depending on the needs of the body.

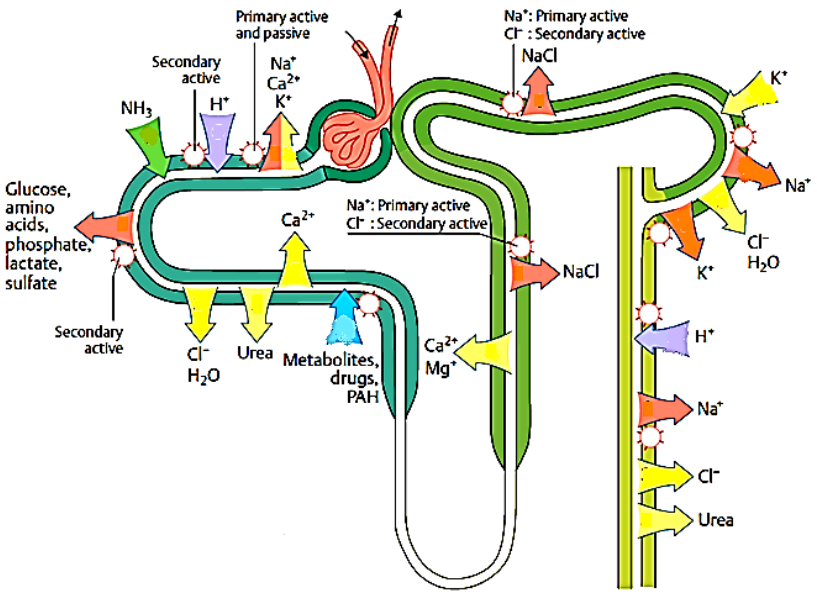
❖ **Ways of reabsorption**: For a substance to be reabsorbed, it must first be transported

- (1) across the tubular epithelial membranes into the renal interstitial fluid and then
  - (2) through the peritubular capillary membrane back into the blood



For instance, water and solutes can be transported either through the cell membranes themselves (transcellular route) or through the junctional spaces between the cells (paracellular route)

Then, after absorption across the tubular epithelial cells into the interstitial fluid, water and solutes are transported the rest of the way through the peritubular capillary walls into the blood by ultrafiltration (bulk flow) that is mediated by hydrostatic and colloid osmotic forces.

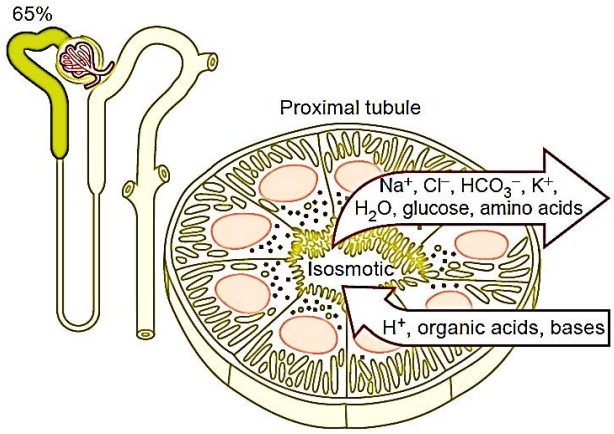


Proximal Tubular Reabsorption

Normally, about 65 per cent of the filtered load of sodium and water and a slightly lower percentage of filtered chloride are reabsorbed by the proximal tubule

The high capacity for active and passive reabsorption of the proximal tubule results from its special cellular characteristics.

Highly metabolic and have large numbers of mitochondria to support potent active transport processes.



- **Causes of high reabsorption rate in proximal tubule**: Have an extensive brush border on the luminal (apical) side of the membrane provide an extensive membrane surface area

  Brush border is also loaded with protein carrier molecules

  In the first half of the proximal tubule, sodium is reabsorbed by co-transport along with glucose, amino acids, and other solutes.

  In the second half of the proximal tubule, little glucose and amino acids remain to be reabsorbed.

  Instead, sodium is now reabsorbed mainly with chloride ions through the intercellular junctions .

  Tubular Processing of the Glomerular Filtrate

  Reabsorption and Secretion by the Renal Tubules

  Urinary excretion = Glomerular filtration

  Tubular reabsorption + Tubular secretion

  Tubular Reabsorption Is Selective and Quantitatively Large

  Glomerular filtration, which is relatively nonselective (that is, essentially all solutes in the plasma are filtered except the plasma proteins or substances bound to them),

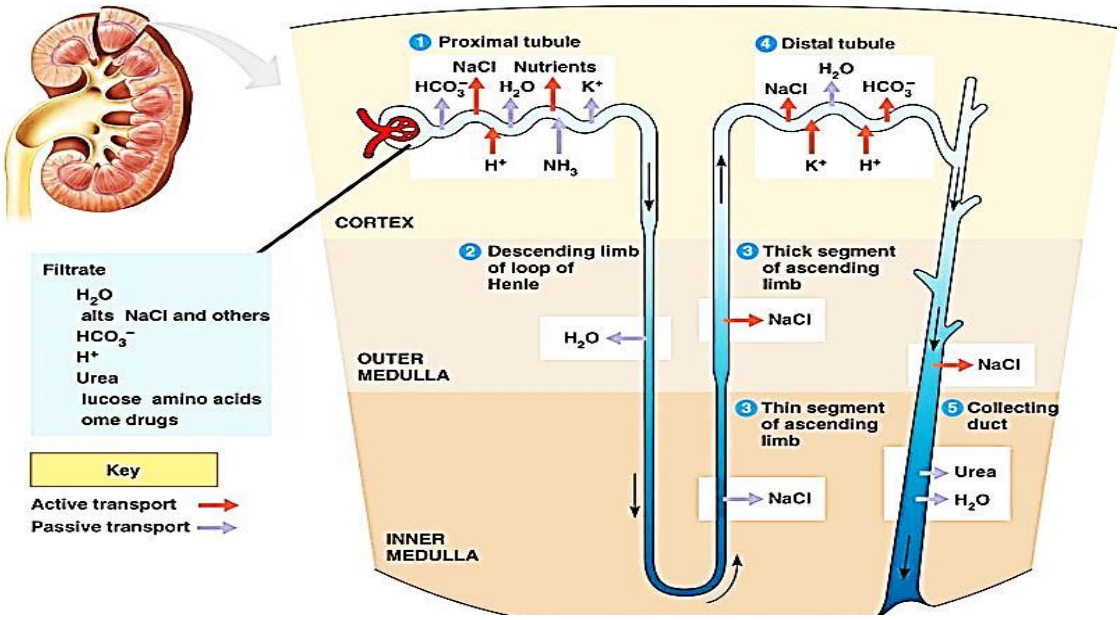
  Tubular reabsorption is highly selective.

  Some substances, such as glucose and amino acids, are almost completely reabsorbed

Solute and Water Transport in the Loop of Henle

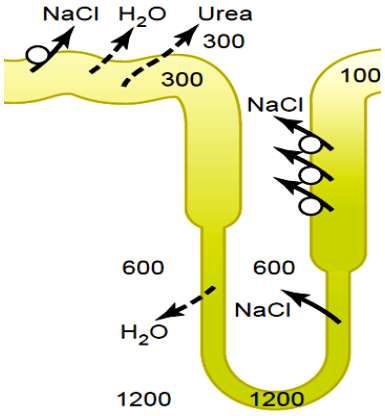
The loop of Henle consists of three functionally distinct segments:

- the thin descending segment,
- the thin ascending segment,
- the thick ascending segment.



The descending part of the thin segment is highly permeable to water and moderately permeable to most solutes, including urea and sodium.

The function of this nephron segment is mainly to allow simple diffusion of substances through its walls.

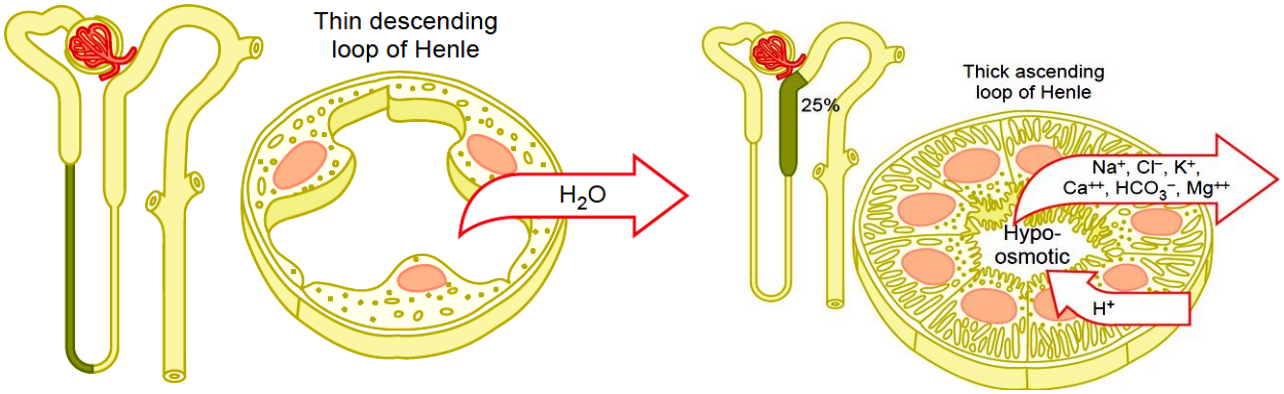


The ascending limb, including both the thin and the thick portions, is virtually impermeable to water, a characteristic that is important for concentrating the urine.

The thick segment of the loop of Henle is an important component of solute reabsorption in the thick ascending limb is the sodium-potassium ATPase pump

The thick ascending limb also has a sodium hydrogen counter-transport mechanism

There is also significant paracellular reabsorption of cations, such as Mg++, Ca++, Na+, and K+, in the thick ascending limb

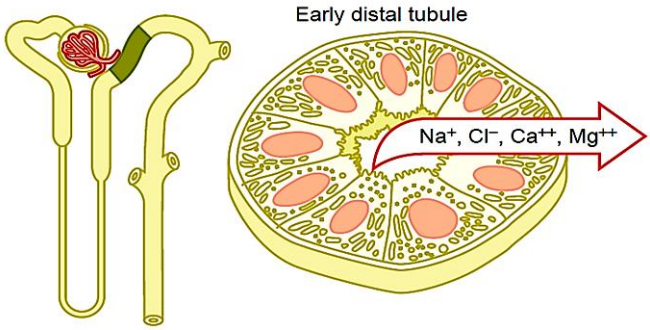


Distal Tubule

Early distal tubule impermeable to water and urea.

For this reason, it is referred to as the diluting segment.

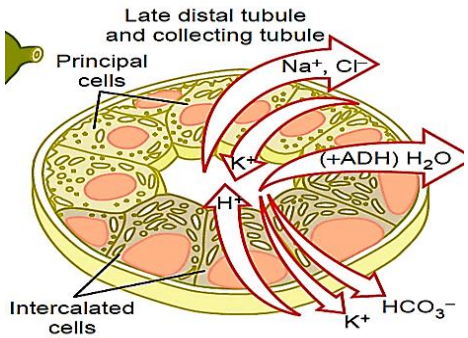
The sodium-chloride co-transporter moves sodium chloride from the tubular lumen into the cell, and the sodium potassium ATPase pump transports sodium out of the cell across the basolateral membrane.



The thiazide diuretics, which are widely used to treat disorders such as hypertension and heart failure, inhibit the sodium chloride co- transporter.

Late Distal Tubule and Cortical Collecting Tubule

Anatomically, they are composed of two distinct cell types : the principal cells and the intercalated cells



- **The functional characteristics of the late distal tubule and cortical collecting tubule can be as follows:**

- The principal cell :Reabsorb Na, secrete K

- The intercalated cells the permeability to water is controlled by the concentration of ADH, which is also called vasopressin
- The intercalated cells play a key role in acid-base regulation of the body fluids ,because it secretes H against a large concentration gradient .
- Impermeable to urea,
- HATPase, H secreted Hco3 reabsorbed

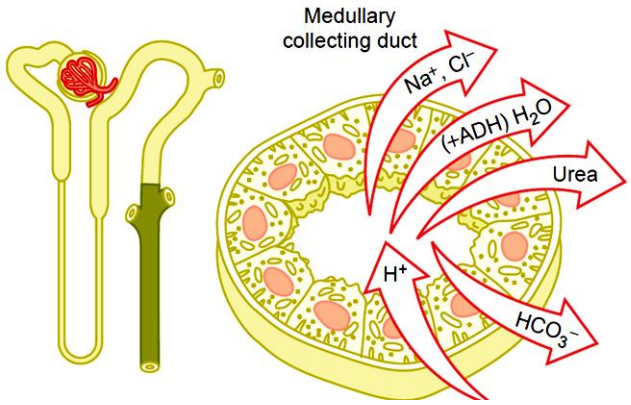
Medullary Collecting Duct

1.The permeability to water is controlled by the level of ADH.

2. Permeable to urea.

3. Capable of secreting hydrogen ions against a large concentration gradient,.

Thus, the medullary collecting duct also plays a key role in regulating acid-base balance.



Regulation of Tubular Reabsorption

- Glomerulotubular Balance

Intrinsic ability of the tubules to increase their reabsorption rate in response to increased tubular load

The importance of glomerulotubular balance is that it helps to prevent overloading of the distal tubular segments when GFR increases.

- **Peritubular Capillary and Renal Interstitial Fluid Physical Forces**

The normal rate of peritubular capillary reabsorption is about 124 ml/min.

Net reabsorption = 10mmHg

Pressure Natriutic and Diuresis: small increases in arterial pressure often cause marked increases in urinary excretion of sodium and water, phenomena that are referred to as pressure natriuresis and pressure diuresis.

Hormonal Control of Tubular Reabsorption:

Aldosterone, secreted by the zone glomerulosa cells of the adrenal cortex, is an important regulator of sodium reabsorption and potassium secretion by the renal tubules.

The primary site of aldosterone action is on the principal cells of the cortical collecting tubule.

Angiotensin II : Most powerful sodium-retaining hormone.

Blood pressure and/or low extracellular fluid volume, such as during hemorrhage or loss of salt and water from the body fluids.

The increased formation of angiotensin II helps to return blood pressure and extracellular volume toward normal by increasing sodium and water reabsorption from the renal tubules

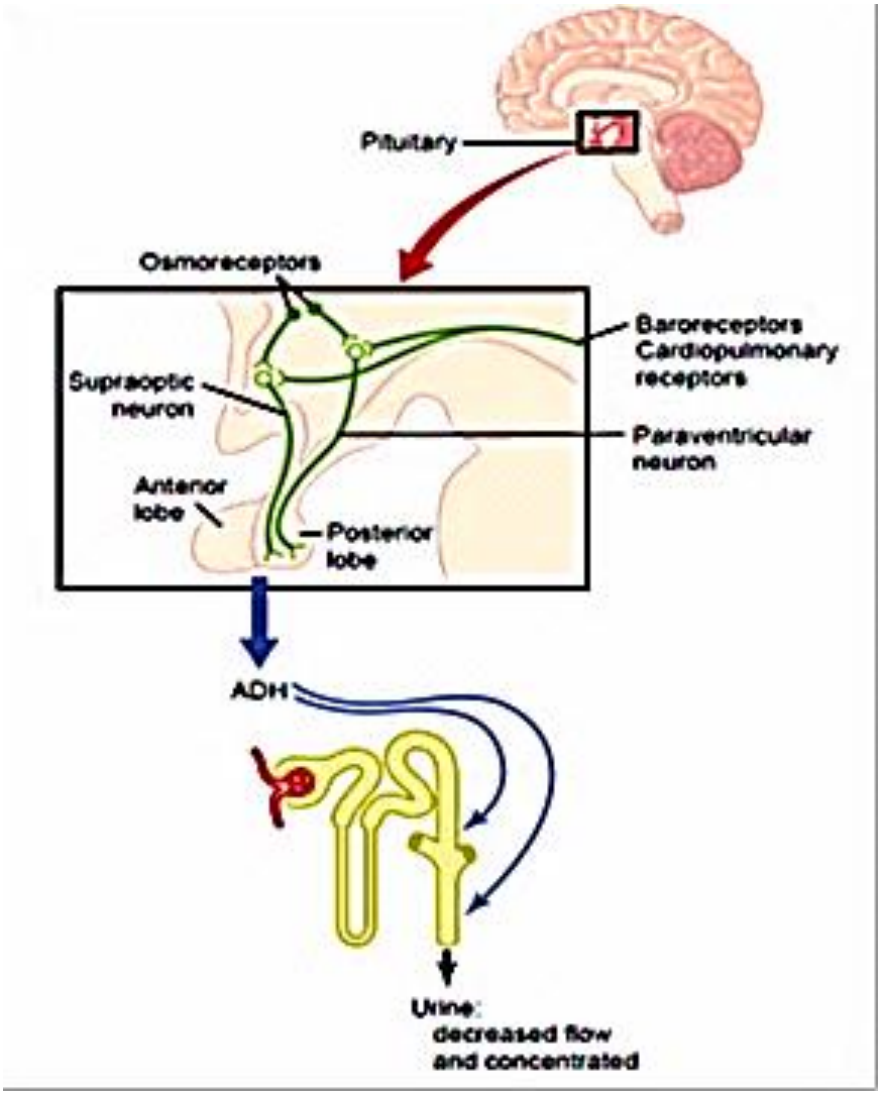
Main effects of :

I. Angiotensin II

1. Stimulates aldosterone secretion, which in turn increases sodium reabsorption.
2. Constricts the efferent arterioles
3. Directly stimulates sodium reabsorption

II . ADH

1. The most important renal action of ADH is to increase the water permeability of the distal tubule, collecting tubule, and collecting duct epithelia.
2. Secreted by posterior pituitary gland

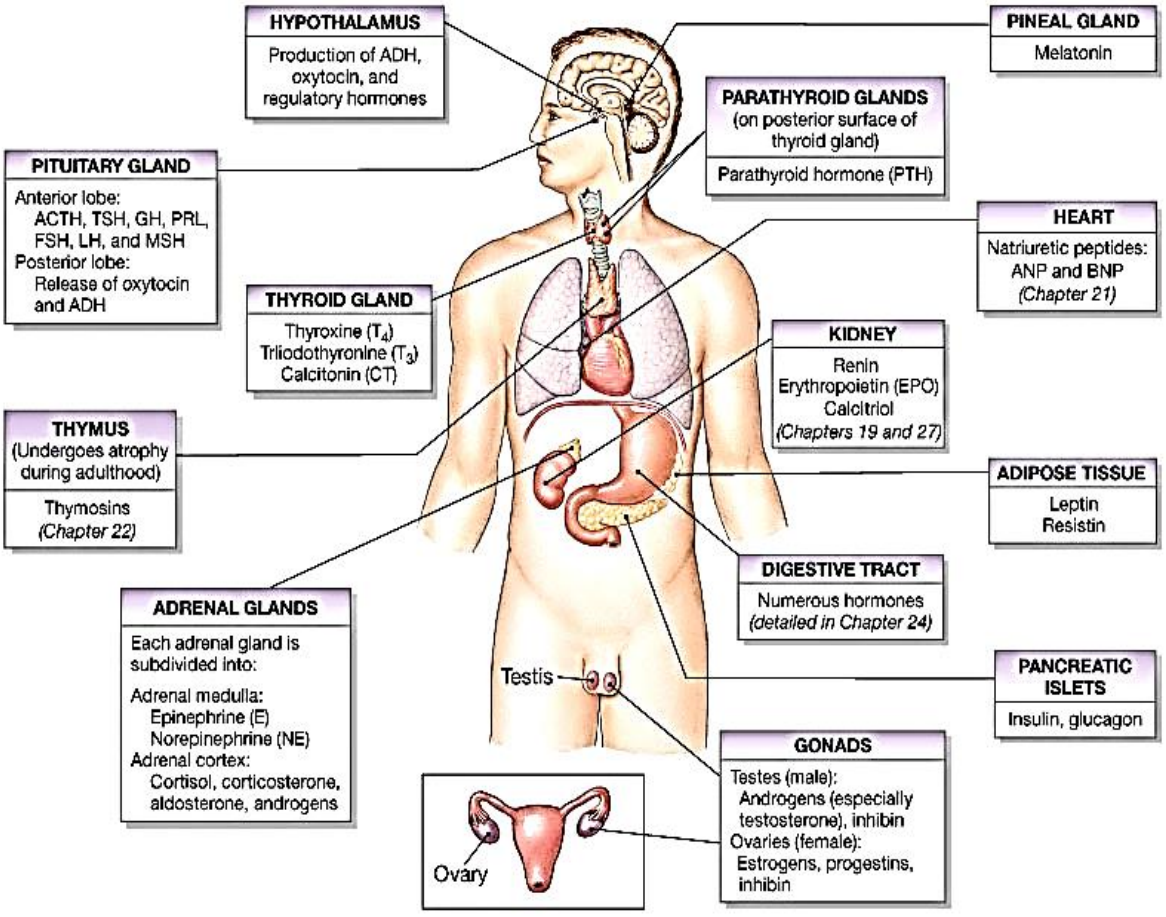


Structure	Major functions
Glomerulus	Vascular (capillary) component of renal corpuscle. Filters (by hydrostatic presure) water, dissolved substances (minus most plasma proteins, blood cells) from blood plasma.
Glomerular capsule	Initial tubular component of nephoron. Transports glomerular filtrate to proximal tubule.
Proximal convoluted tubule	Reabsorbs (by active transport) Na^+, K^+, Ca^{2+}, amino acids, uric acid, ascorbic acid, ketone bodies, glucose. Reabsorbs (as a result of active H^+ secretion) HPO_4^{2-}, Cl^-, SO_4^{2-}, HCO_3^-. Reabsorbs (by osmosis) water. Reabsorbs (by diffusion) urea. Actively secretes substances such as penicillin, histamine, organic acids, organic bases.
Descending loop of the nephron	Reabsorbs (by active transport) Na^+. Reabsorbs (by electrochemical gradient) Cl^-, SO_4^{2-}. Reabsorbs (by osmosis) water. Reabsorbs (by diffusion) urea.
Ascending loop of the nephron	Reabsorbs (by active transport) Na^+, Cl^-. Reabsorbs (by electrochemical gradient) HCO_3^-.
Distal Convoluted tubule	Reabsobs (by active transport) Na+. Reabsorbs (by electrochemical gradient) PO_4^{3-}, Cl^-, SO_4^{2-}, HCO_3^-. Reabsorbs (by osmosis) water. Reabsorbs (by diffusion) urea. Actively secretes H^+.
Collecting duct	Reabsorbs (by active transport) Na^+. Reabsorbs (by osmosis, under control of ADH) water. Actively secrets H^+ and K^+. Actively reabsorbs K^+.

CHAPTER THIRTEEN

THE ENDOCRINE SYSTEM

THIRTEEN THE ENDOCRINE SYSTEM



The endocrine system is a control system of ductless glands that secrete chemical messengers called hormones that circulate within the body via the bloodstream to affect distant cells within specific organs.

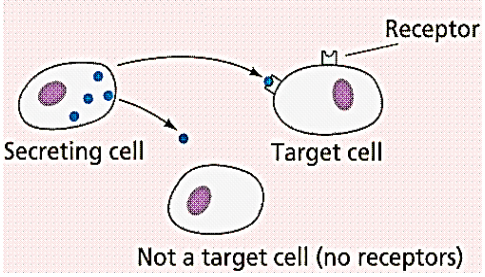
Endocrine glands secrete their products immediately into the blood or interstitial fluid, without storage of the chemical.

Hormones

The chemical messengers released by the glands of the endocrine system are called hormones. Hormones are released directly into the bloodstream and carried to the tissues they affect.

These tissues may be far from where the hormone is produced.

The blood reaches all cells of the body, but only certain cells respond to specific hormones; these responding cells are unique in that they have receptors to which the hormones attach. Only cells that have receptors for a given hormone will respond to that hormone; these cells make up what is called the target tissue.



Chemistry : Definition of hormone:

Hormone is a chemical substance secreted in a minute amounts from a group of cells known as endocrine gland, and carried by the blood from its site of secretion to its site action.

The multiple hormone systems play a key role in regulating almost all body functions, including :

- Metabolism,
- Growth and development,
- Water and electrolyte balance,
- Reproduction, and
- Behavior.

For instance,

- Without growth hormone, a person would be a dwarf.
- Without thyroxine and triiodothyronine from the thyroid gland, almost all the chemical reactions of the body would become sluggish, and the person would become sluggish as well.
- Without insulin from the pancreas, the body's cells could use little of the food carbohydrates for energy.
- Without the sex hormones, sexual development and sexual functions would be absent.

Chemical Structure and Synthesis of Hormones

- **There are three general classes of hormones:**
  a. **Proteins and polypeptides,** including hormones secreted by the anterior and posterior pituitary gland, the pancreas (insulin and glucagon), the parathyroid gland (parathyroid hormone), and many others
  b. **Steroids** secreted by the adrenal cortex (cortisol and aldosterone), the ovaries (estrogen and progesterone), the testes (testosterone), and the placenta (estrogen and progesterone).
  c. **Derivatives of the amino acid tyrosine,** secreted by the thyroid (thyroxine and triiodothyronine) and the adrenal medullae (epinephrine and norepinephrine).

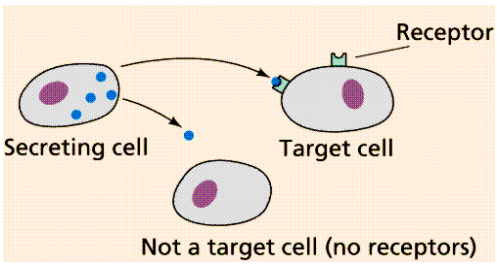
Hormone Receptors and Their Activation

❖ The first step of a hormone's action is to bind to specific receptors at the target cell.

❖ Cells that lack receptors for the hormones do not respond.

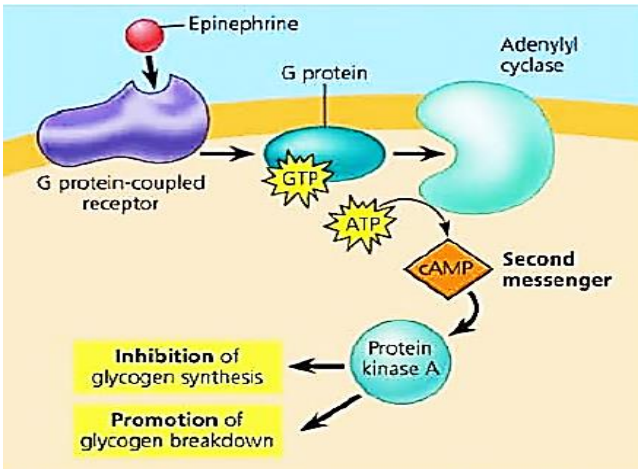
Mechanisms of Action of Hormones

1. Most hormones circulate in blood, coming into contact with essentially all cells.
2. However, a given hormone usually affects only a limited number of cells, which are called target cells. A target cell responds to a hormone because it bears receptors for the hormone.



Receptors for some hormones are located on the target cell membrane, whereas other hormone receptors are located in the cytoplasm or the nucleus.

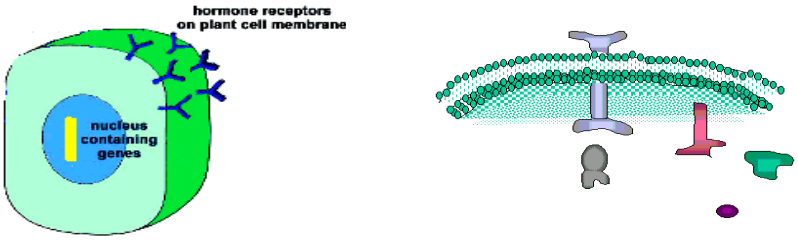
When the hormone combines with its receptor, this usually initiates a cascade of reactions in the cell, with each stage becoming more powerfully activated so that even small concentrations of the hormone can have a large effect.



- Hormonal receptors are large proteins, and each cell that is to be stimulated usually has some 2000 to 100,000 receptors.

Also, each receptor is usually highly specific for a single hormone; this determines the type of hormone that will act on a particular tissue.

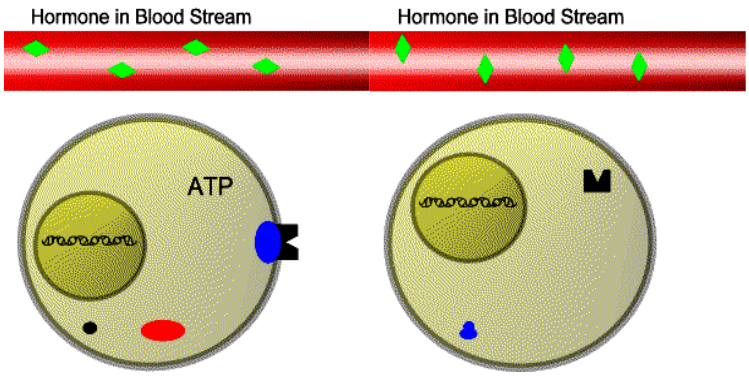
The target tissues that are affected by a hormone are those that contain its specific receptors.



The locations for the different types of hormone receptors are generally the following:

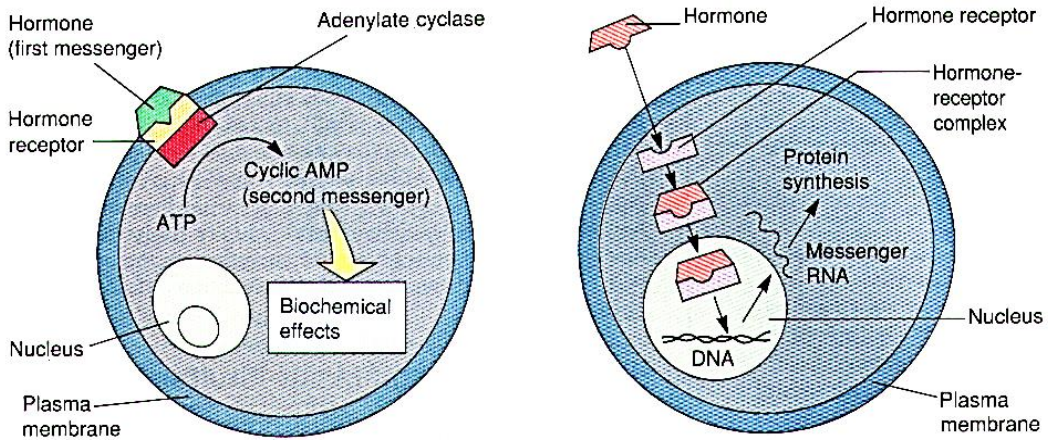
1. **<u>In or on the surface of the cell membrane:</u>**

The membrane receptors are specific mostly for the protein, peptide, and catecholamine hormones.



<u>2. In the cell cytoplasm:</u> The primary receptors for the different steroid hormones are found mainly in the cytoplasm.

3. In the cell nucleus: The receptors for the thyroid hormones are found in the nucleus and are believed to be located in direct association with one or more of the chromosomes.



Hormone produces its action by one of the following mechanisms:

1. **It increases permeability of cell membrane** towards a certain substance e.g.:
   - Insulin increase permeability of the tissue cells to glucose.
   - ADH increase permeability of the distal tubules and collecting ducts to H2O.
   - Growth hormone increase transport of A.A into cells.

2. It stimulates the genes of the cells which increase protein synthesis: This leads to increase synthesis of enzymes which in turn increase metabolic reactions inside the cell.

3. It affects the intra cellular hormone mediators:

Some hormones stimulate the formation of cyclic AMP which stimulates most of the intra cellular enzymes, which in turn increase the rate of metabolic reactions.

Types of hormones:

A) General hormones: They act on the whole body e.g. thyroxin, glucocorticoids etc.

B) Local hormones: They act on the same site of their secretion e.g.

Gastro-intestinal hormones which are secreted from the G.I.T and act on the GIT. e.g. gastrin, secretin etc.

Serotonin: is formed in the GIT and in brain tissues. It causes local V.C of the injured vessels,

increase motility of GIT.

It acts as a chemical transmitter in CNS and plays a role in producing sleep.

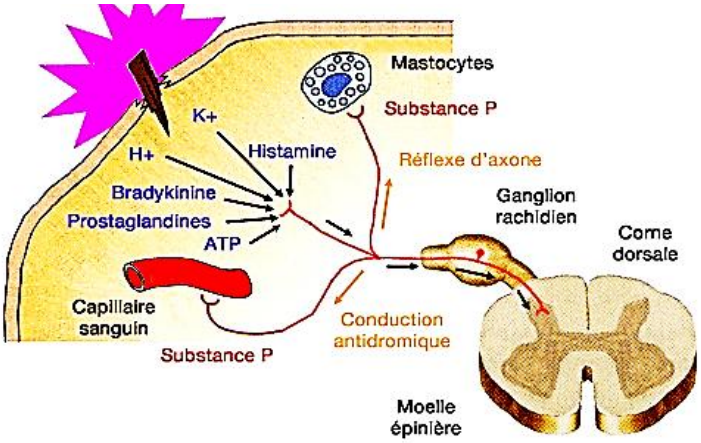
C) Hormones essential for life:

Essential for life means that in absence of these hormones death will occur e.g.:

Insulin: In absence of insulin body can't utilize glucose but instead utilize fat →increase formation of ketone bodies □→ □ Diabetic acidosis □ □→ Diabetic coma □→ □ death.

Parathormone: In absence of parathormone serum Ca++ level decrease → tetany which affect respiratory muscles → Asphyxia → Death.

Aldosterone: In absence of aldosterone there is marked Na+ and H2O loss from the body and marked K+ retention. This leads to acidosis, dehydration and muscle weakness → Death.

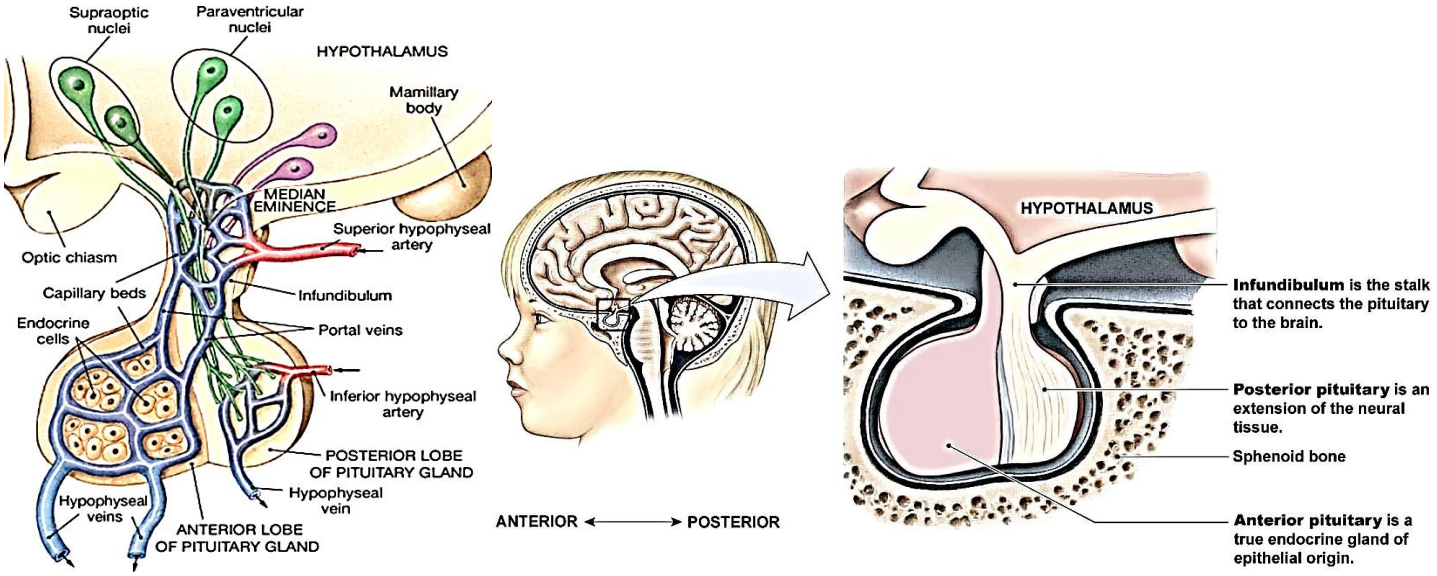


The Pituitary Gland

The pituitary gland or hypophysis, formerly called the "Master gland", secretes several polypeptide hormones that directly or indirectly regulate a wide variety of metabolic and physiologic processes essential to normal growth and development as well as to the maintenance of homeostasis. Many of the hormones secreted by the pituitary gland are critical to the activity of target glands, including the thyroid, adrenal and gonads.

Anatomy

The pituitary gland is located at the base of the brain, resting with in the sella turcica of the sphenoid bone. The pituitary gland maintains elaborate neural and vascular connections with the hypothalamus of the brain, which plays a central role in the integration of neuroendocrine activity .



The pituitary gland has two major divisions:

The anterior lobe (adenohypophysis) and the posterior lobe (neurohypophysis).

Adenohypophysis

The adenohypophysis is served by an elaborate vascular system, including the hypothalamohypophyseal portal system, which transports hypothalamic regulating hormones (hypophyseotropic hormones) to the glandular cells of the adenohypophysis.

The classification of cells in the adenohypophysis is based on specific immunohistochemical techniques. **Accordingly, there are at least five recognized cell types.**

1. ***Somatotrophs***, which secrete growth hormone (hGH) or somatotropin.

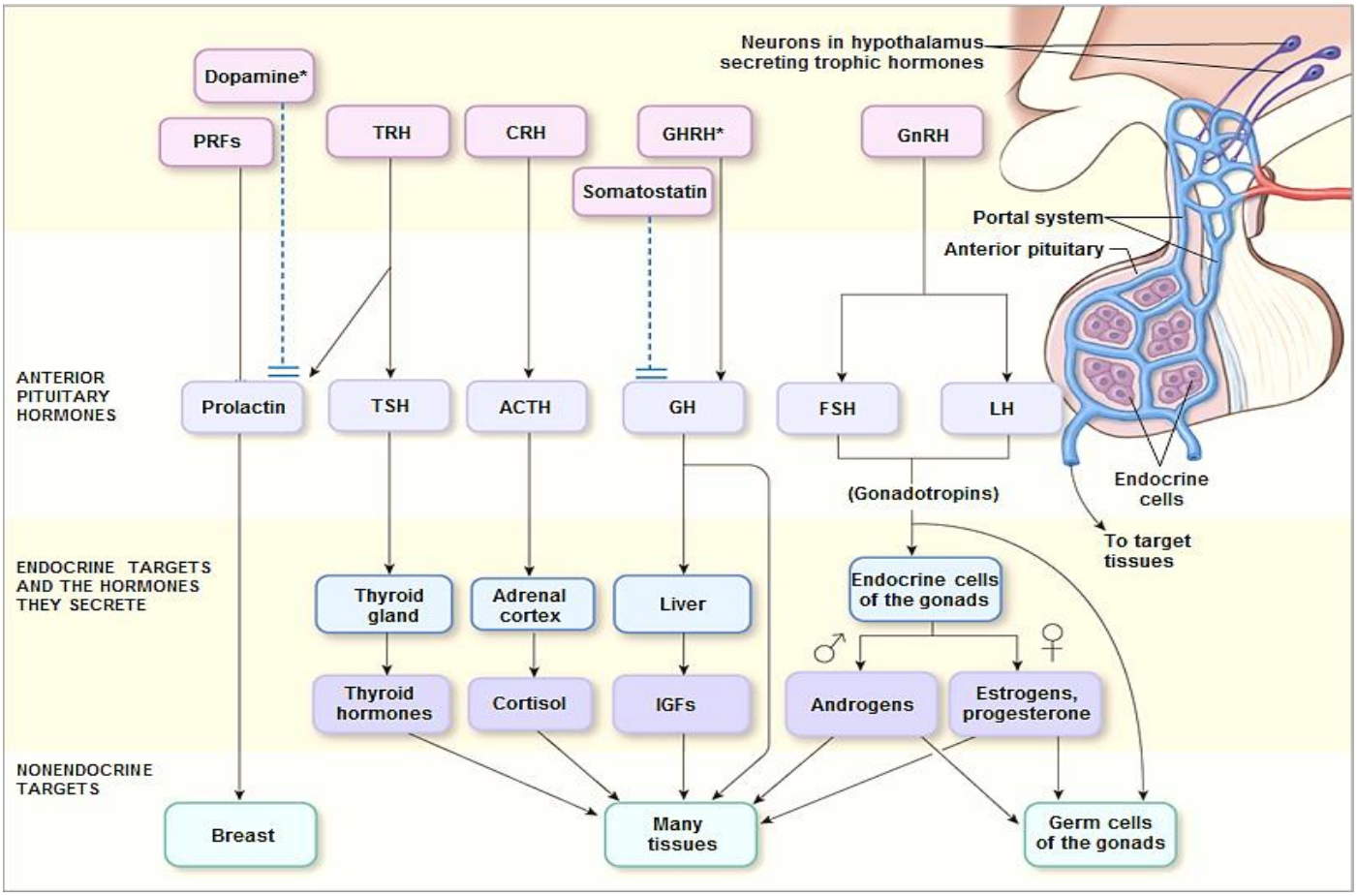
2. ***Lactotrophs***, which secret prolactin (PRL).

3. ***Corticotrophs***, which produce corticotropin (ACTH) and beta-lipotropin (beta-LPH)

4. By splitting a large peptide pro-hormone, proopiomelanocortin(POMC).

5. ***Thyrotrophs***, which secrete thyrotropin (TSH).

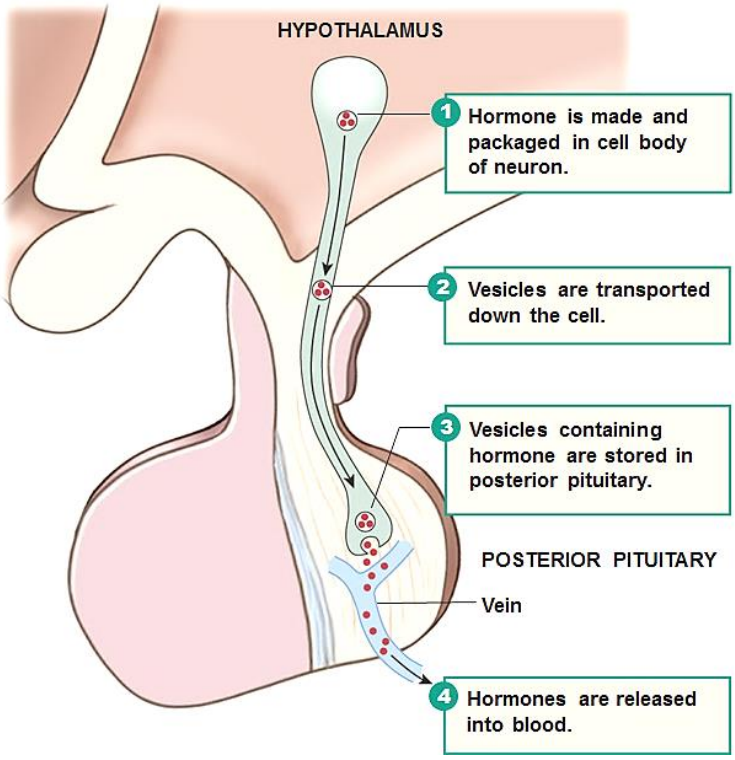
6. ***Gonadotrophs***, which produce follicle stimulating hormone (FSH) and luteinizing hormone (LH)



Neurohypophysis

The neurohypophysis, which is connected directly to the hypothalamus by the infundibular (Pituitary) stalk, is rich in nerve fibers of hypothalamic origin (the hypotahlamohypophyseal tract).

Neurosecretory cells in the supraoptic and paraventricular nuclei of the hypothalamus produce two hormones: **antidiuretic hormone (ADH or Vasopressin) and oxytocin.**



These hormones are then transported along the axons of the hypothalamo hypophyseal tract to the posterior lobe of the pituitary gland for storage and ultimate release under hypothalamic control.

Hormones of the Adenohypophysis

The secretion of hormones by the adenohypophysis is controlled by hypothalamic regulatory (hypophyseotropic) hormones that are transported to the pituitary gland by the hypothalamohypophyseal portal system illustrated in . There are six recognized hypophyseotropic hormones secreted from the median eminence of the hypothalamus:

- ♣ Growth hormone-releasing hormone (GHRH; somatocrinin)
- ♣ Growth hormone-inhibiting hormone (GHIH; somatostatin)
- ♣ Corticotropin-releasing homone (CRH)
- ♣ Thyrotropin-releasing hormone (TRH)
- ♣ Gonadotropin-releasing hormone (GnRH)
- ♣ Prolactin-inhibiting hormone (PIH)

It is probable that a prolactin-releasing hormone (PRH) also exists; however, the exact substance has yet to be clearly identified. Some hypophyseotropic hormones influence the secretion of more than one adenohypophyseal hormone. Gn- RH stimulates secretion of FSH and LH. TRH stimulates the secretion of TSH and prolactin. Somatostatin inhibits the secretion of growth hormone and TSH.

1. Growth Hormone (GH); Somatotropin (STH)

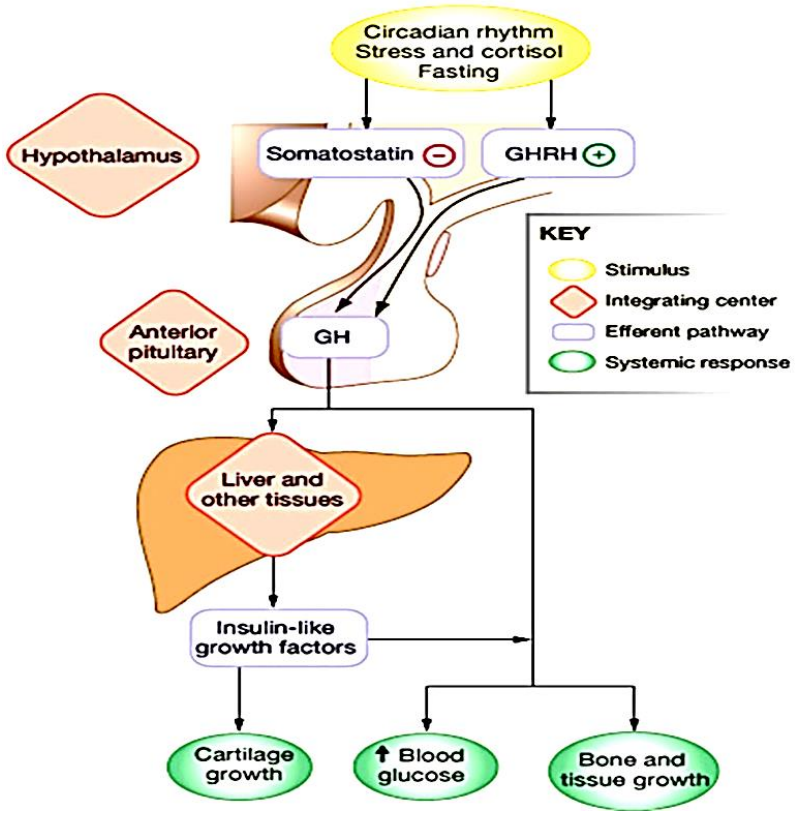
Human growth hormone (hGH) is a peptide hormone composed of 191 aminoacids, and is secreted by the somatotrophs of the adenohypophysis primarily under hypothamic control. The wide variety of factors that may affect GH secretion is summarized below.

Control of Secretion : Factors Promoting GH Secretion

- GHRH (Somatocrinin)
- Hypoglycemia and fasting
- Elevated plasma levels of amino acids (e.g., arginine)
- Stress (physical or psychological)
- Exercise
- Deep sleep
- Levodopa
- Glucagon

Factors Inhibiting GH Secretion

- GHIH (Somatostatin)
- Hyperglycemia
- Elevated plasma levels of free fatty acids
- Rapid eye movement sleep (REM sleep)
- Cortisol
- Alpha-adrenergic blocking agents
- GH (negative feedback mechanism): GH secretion in response to hypoglycemia, fasting, and exercise appears to be reduced by obesity.



Actions : **Effects on Growth:**

Growth is a complex phenomenon influenced by genetic, nutritional, and hormonal factors.

In addition to growth hormone, the thyroid hormones, insulin, androgens, and estrogens play important roles in normal human growth and development at various times of the life cycle.

GH accelerates overall body growth by increasing the mass of both skeletal and soft body tissues through **hyperplasia** (increased cell number) and **hypertrophy** (increased cell size).

The effects of GH are particularly evident in skeletal tissues where chondrogenesis (cartilage formation) and osteogenesis (bone formation) are enhanced, leading to an increase in linear growth and stature before epiphyseal closure and increased bone thickness following closure of the epiphyses.

Growth hormone stimulates certain tissues, notably the liver, to produce somatomedins or insulin like growth factors (IGF-I and IGF-II). These low-molecular-weight peptides mediate the growth-promoting effects of GH, including the stimulation of collagen synthesis and chondrogenesis.

Metabolic Effects

Protein metabolism: GH increases protein synthesis and nitrogen retention by enhancing the incorporation of amino acids into protein. The protein anabolic action results from 1) accelerated entry of amino acids into cells and 2) increased ribonucleic acid (RNA) synthesis. In muscle and liver, the protein anabolic effects are attributed directly to GH.

However, in cartilage, bone, and other body tissues, the protein anabolic and growth-promoting actions are mediated by insulinlike growth factors (somatomedins).

Lipid metabolism: GH stimulates the mobilization and utilization of fats by promoting lipolysis in adipose tissue, thus enabling the body to use stored fats as an energy source.

The elevation of plasma levels of free fatty acids resulting from the hydrolysis of triglycerides (stored neutral fats) is potentially ketogenic.

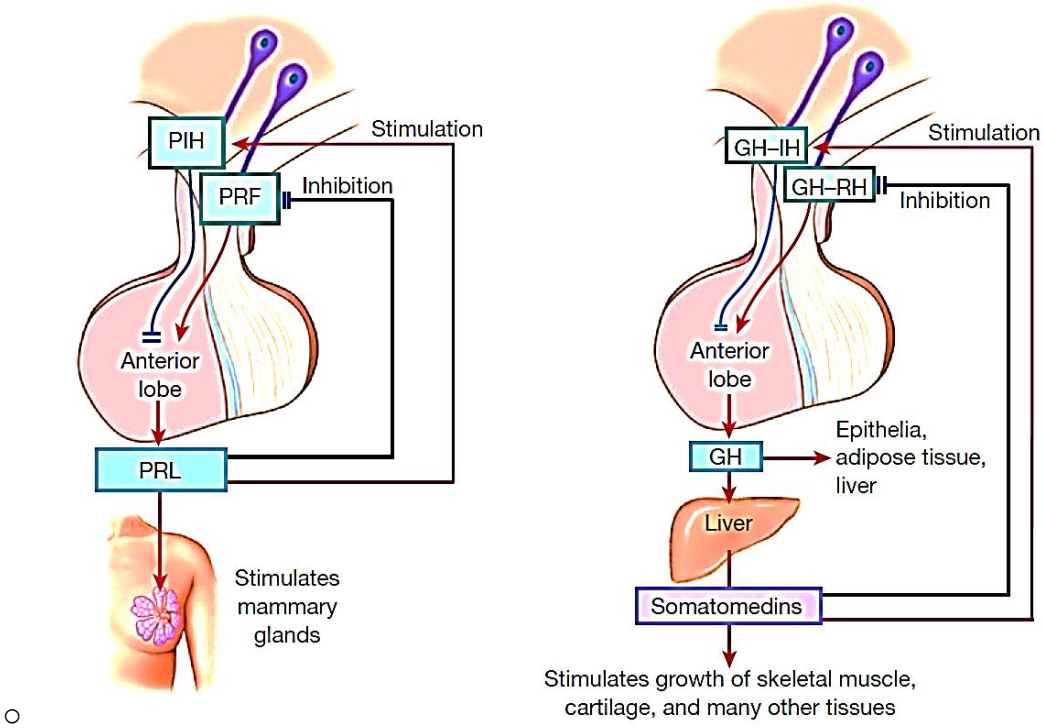
Carbohydrate metabolism: GH elevates blood glucose levels by increasing the hepatic output of glucose (gluconeogenesis) and impairing glucose transport into muscle and adipose tissue ("anti-insulin" action). Excessive secretion of GH may precipitate or increase the severity of clinical diabetes mellitus ("diabetogenic" effect).

2. Prolactin (PRL) : Control of Secretion

Prolactin (PRL) is a peptide hormone composed of 199 amino acids. Its secretion by the lactotrophs of the adenohypoiphysis is tonically suppressed by dopamine (also known as PIH) of hypothalamic origin. Dopamine antagonists (eg, antipsychotic drugs) promote PRL secretion by blocking dopamine receptors, whereas dopamine agonists (eg, bromocriptine) inhibit PRL secretion by activating dopamine receptors.

Secretion of PRL increases during :

- Pregnancy,
- Peaking near the time of parturition.
- Sucking and tactile stimulation of the nipple increase PRL secretion.
- Prolactin facilitates the secretion of dopamine in the hypothalamus, thereby regulating its own secretion by a negative feedback mechanism.



Actions

Prolactin initiates and maintains milk secretion from breasts primed for lactation by other hormones such as estrogens, progesterone, and insulin. It also appears to inhibit the effects of the gonadotropins and may prevent ovulation in lactating women.

Excessive production of PRL (hyperprolactinemia), which may accompany some pituitary tumors, may cause an ovulation and amenorrhea in women, and may lead to impotence and infertility in men

3. Follicle-Stimulating Hormone (FSH) : Control of Secretion

Follicle-stimulating hormone (FSH) is a glycoprotein gonadotropic hormone whose secretion is stimulated by hypothalamic GnRH. Inhibit, a polypeptide produced by testicular sertoli cells in the male and follicular granulosa cells in the female, acts directly on the adenohypophysis to inhibit FSH secretion.

Actions

Follicle-stimulating hormone directly stimulates the sertoli cells in testicular seminiferous tubles, there by promoting spermatogenesis in the male. In the female,

FSH stimulates follicular growth and development within the ovaries. The actions of FSH are mediated by cyclic AMP.

4. Luteinizing Hormone (LH; Interstitial Cell Stimulating Hormone; ICSH)

Control of Secretion : Like FSH, LH is a glycoprotein hormone whose secretion is stimulated by GnRH. Testosterone inhibits LH secretion through a direct action on the adenohypophysis, as well as indirectly by inhibiting hypothalamic GnRH production.

The effects of female hormones on LH secretion are more complex. Constant, moderate levels of estrogen (without progesterone) have a negative feedback effect on LH, whereas high estrogen

levels exert a positive feedback that leads to a surge in LH production. High levels of progesterone and estrogen (luteal phase of the ovulatory cycle) inhibit LH secretion.

Actions

In the male, this hormone stimulates testosterone production by testicular interstitial cells (of Leydig); hence the alternate name, interstitial cell stimulating hormone (ICSH).

In the female, LH promotes maturation of ovarian follicles and sustains their secretion of estrogens. LH is also responsible for ovulation and the formation of the corpus luteum.

The actions of LH are mediated by cyclic AMP.

5. Thyroid-Stimulating Hormone (Thyrotropin; TSH)

Control of Secretion

Thyrotropin-releasing hormone (TRH) and cold (especially in infants) promote secretion of TSH by the thyrotrophs of the adenohypophysis. Elevated plasma levels of free thyroid hormones (T3 and T4) inhibit thyrotropin secretion. Stress also inhibits TSH secretion.

Actions

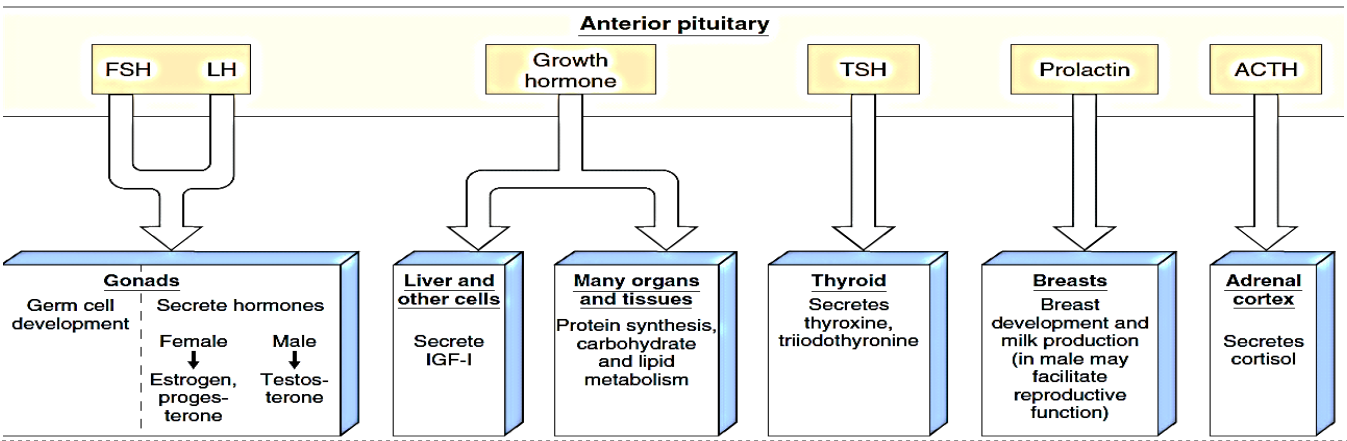
Thyroid-stimulating hormone maintains the structural integrity of the thyroid gland and promotes the synthesis and release of thyroid hormones thyroxine (T4) and triiodothyronine (T3).

The actions of TSH on the thyroid gland are mediated by cyclic AMP, and they are detailed in the section on the thyroid gland.

6. Adrenocorticotropic Hormone (Corticotropin; ACTH) : Control of Secretion

Adrenocorticotropic hormone (ACTH) is secreted in irregular bursts that follow a diurnal circadian rhythm, with peak production in the early morning. ACTH secretion is regulated by the hypophyseotropic hormone corticotropin-releasing hormone (CRH) the increased production of ACTH in response to many stressors appears to be mediated through the hypothalamus and CRH. Elevated plasma free glucocorticoid (cortisol) levels inhibit both CRH and ACTH secretion.

Action Adrenocorticoptropic hormone exerts its tropic effects on the adrenal glands, promoting structural integrity and steroidogenesis in the adrenal cortex. The stimulation of corticosteroid production (Steroidogenesis) in response to ACTH is mediated by the second messenger, cyclic AMP.

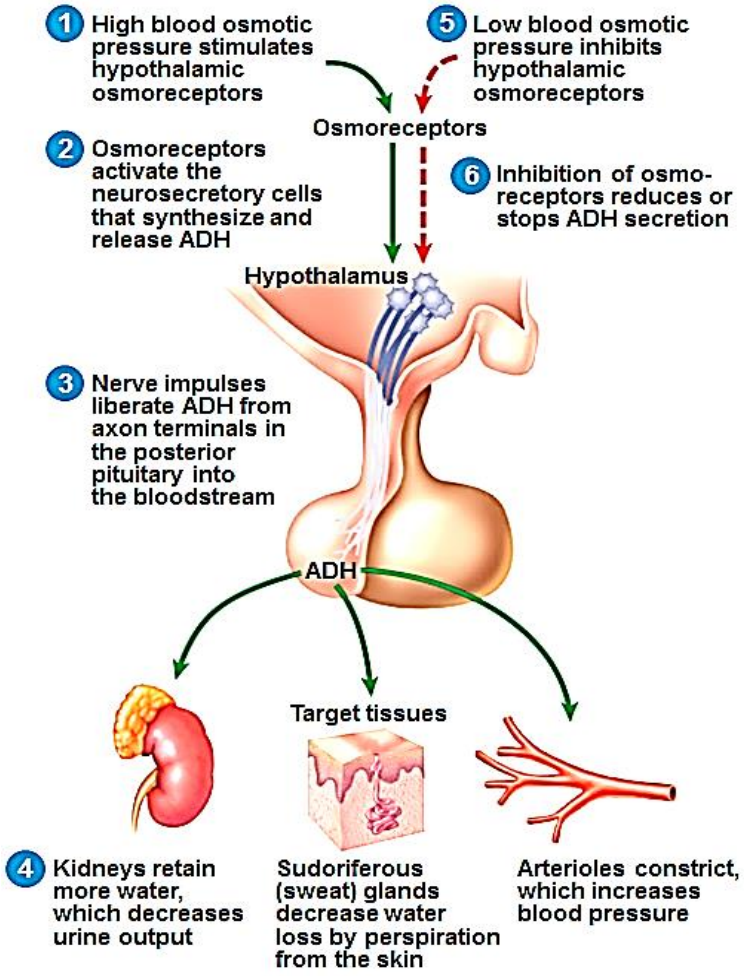


Hormones of the Neurohypophysis

1. Antidiuretic Hormone (ADH; Vasopressin) : Control of Secretion

Antidiuretic hormone (ADH) is a polypeptide hormone of hypothalamic origin that is stored in and released from the neurohypophysis in response to a variety of stimuli. Included among these are increased plasma osmolality, reduced extracellular fluid (ECF) volume, pain, emotional stress.

Osmoreceptors found in the anterior hypothalamus monitor changes in plasma osmolality, whereas ECF volume changes are detected by volume ("Stretch") receptors located in the wall the left atrium. The osmoreceptors and volume receptors work in concert to exert precise control over ADH secretion, thus forming a delicate homeostatic feedback mechanism for the regulation of ECF volume and concentration.



Actions

The principal physiologic role of ADH is to regulate extracellular fluid volume and osmolality by controlling the final volume and concentration of urine. ADH increases the permeability of the distal nephron (late distal convoluted tubules and collecting ducts) to water. The enhanced reabsorption of water from the renal tubules results in the production a concentrated urine that is reduced in volume.

ADH produce a pressor (hypertensive) effect that results from a direct constrictor action of the hormone on vascular smooth muscle. The early observations that posterior pitutary extracts produce a marked elevation of arterial blood pressure led to the initial naming of this hormone as vasopressin.

2. Oxytocin : Control of Secretion and Actions

The two major physiologic actions of oxytocin are exerted on the female breast and uterus. Oxytocin binds to a G-protein coupled receptor that ultimately brings about elevated intracellular calcium levels.

Galactokinetic Action (Milk Ejection Reflex). The ejection of milk from a primed, lactating mammary gland follows a neuroendocrine reflex in which oxytocin serves as the efferent limb.

The reflex is normally initiated by sucking, which stimulates cutaneous receptors in the areola of the breast.

Afferent nerve impulses travel to the supraoptic and paraventricular nuclei of the hypothalamus to effect the release of oxytocin from the neurohypophysis. Oxytocin is carried by the blood to the mammary gland, where it causes contraction of myoepithelial cells surrounding the alveoli and lactiferous ducts to bring about the ejection of milk (milk letdown).

In lactating women, tactile stimulation of the breast areola, emotional stimuli, and genital stimulation may also lead to oxytocin release and activate the ejection of milk.

Oxytocic Action. Oxytocin acts directly on uterine smooth muscle to elicit strong, rhythmic contractions of the myometrium. Uterine sensitivity to oxytocin varies with its physiologic state and with hormonal balance. The gravid (Pregnant) uterus is highly sensitive to oxytocin, particularly in the late stages of gestation.

Uterine sensitivity to oxytocin is greatly enhanced by estrogen and inhibited by progesterone. Oxytocin release appears to follow a neuroendocrine reflex initiated by genital stimulation. It has been suggested that oxytocin may facilitate sperm transport through the female genital tract.

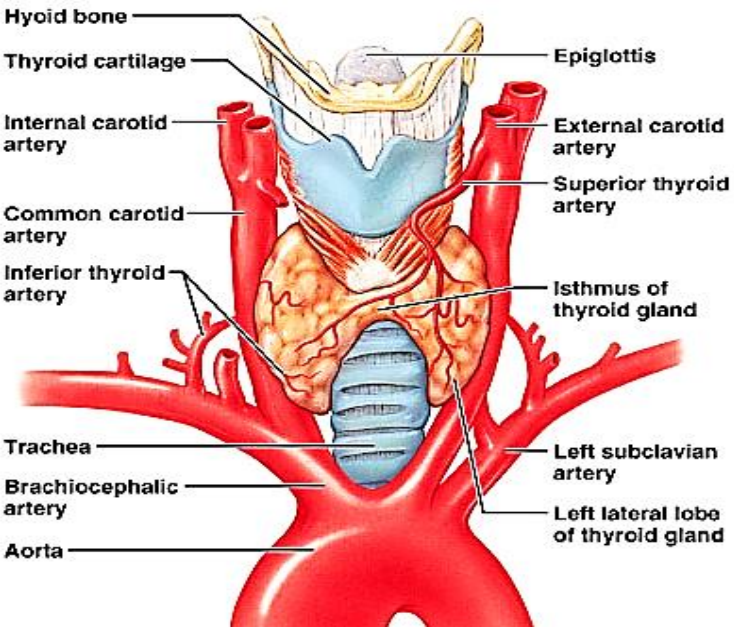
The Thyroid Gland

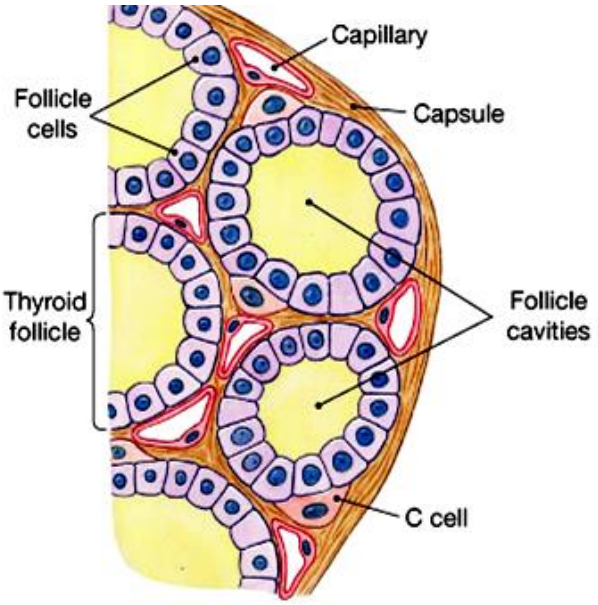
The hormones of the thyroid gland exert a wide spectrum of metabolic and physiologic actions that affect virtually every tissue in the body.

Anatomy

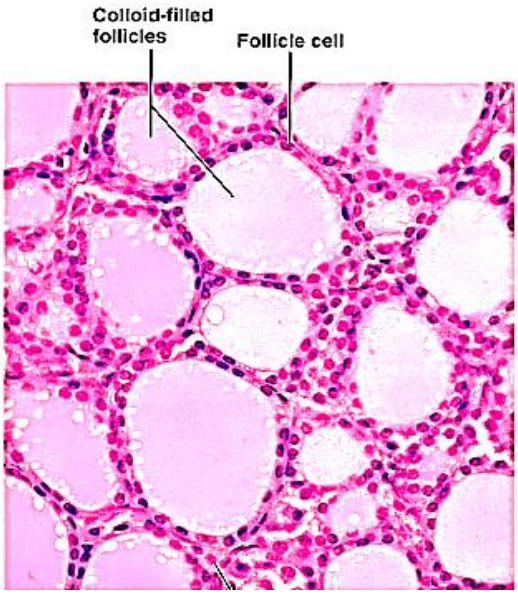
The thyroid gland is a bilobed organ overlying the trachea anteriorly. It is formed of 2 lobes connected by an isthmus. Each lobe pyramidal in shape with an apex, a base & 3 surfaces (medial, lateral & posterior). Each lobe is about 5 x 3 x 2 cm (length x width x thickness).

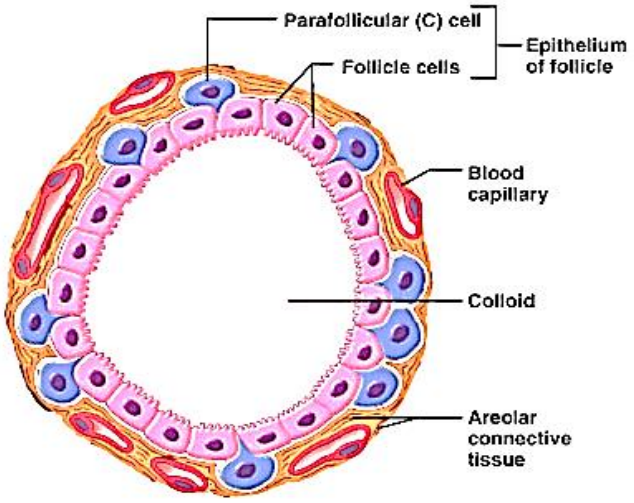
The thyroid gland is composed of numerous closely packed spheres or follicles. Each follicle consists of a simple cuboidal epithelium (follicular cells) enclosing a lumen or cavity containing a viscous hyaline substance termed colloid.





The chief constituent of the colloid is the iodinated glycoprotein thyroglobulin. Interspersed among the follicles are small clusters of parafollicular (C) cells, which secrete calcitonin, a hormone affecting calcium metabolism.





Thyroid Hormones

The follicular cells of the thyroid gland secrete two hormones, thyroxine (3, 3', 5, 5'-tetraiodothyronine or T4) and 3, 3', 5- triiodothyronine (T3). The plasma levels of these hormones are regulated by the hypothalamopituitary axis as outlined in Figure 8-2. Intrinsic (intrathyroidal) mechanisms, as well as bioavailability of iodine, influence thyroid hormone production.

Biosynthesis of Thyroid Hormones

1. Iodide uptake: Ingested iodine is readily absorbed from the GI tract in the reduced iodide state. Iodide ions are actively transported from the blood into the thyroid follicles by an energy-requiring "trapping" mechanism often called the iodide pump.

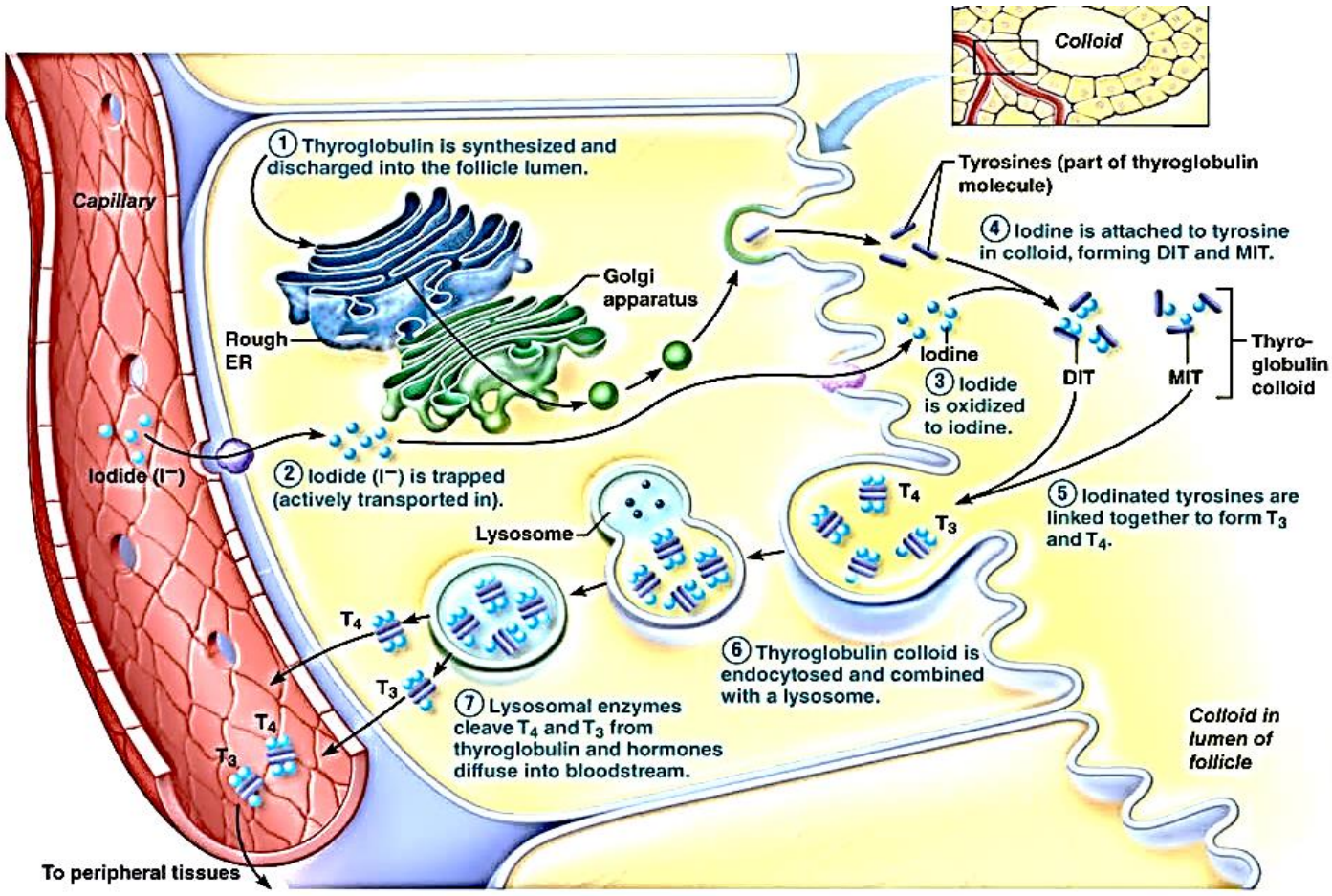
The normal thyroid: serum ratio of iodide is 25:1. The uptake of iodide is enhanced by TSH and may be blocked by anions such as perchlorate and thiocyanate.

2. Oxidation to iodine: On entering the colloid, iodide is rapidly oxidized to iodine in the presence of peroxidase enzymes. Thiouracil appears to inhibit peroxidase activity.

3. Iodination of tyrosine: Free molecular iodine spontaneously combines with tyrosine residues on the thyroglobulin (TGB) to form 3-monoiodotyrosine (MIT) and3, 5-diiodotyrosine (DIT).

4. Coupling reaction: Two iodinated tyrosines combine to form either T3 or T4. The coupling occurs within the thyroglobulin molecule, and the reaction appears to be promoted by TSH.

5. Storage and release of thyroid hormones: T3 and T4 remain stored within the colloid bound to thyroglobulin until a stimulus for secretion arrives.



On stimulation by TSH, portions of the TGB (colloid) are engulfed by microvilli that extend from the apical surface of the follicular cells.

Droplets of the engulfed colloid fuse with lysosomes, and proteolytic enzymes release T3 and T4 from the TGB. The lipophilic hormones (T3 and T4) readily diffuse to nearby capillaries and enter the bloodstream.

Thyroid-stimulating hormone, acting through cyclic AMP, increases the production of thyroid hormones by promoting virtually every step in the biosynthetic process, including the synthesis of TGB and the eventual release of T3 and T4 from storage.

Transport

Circulating thyroid hormones bind specifically with thyroxine binding globulin and thyroxine-binding prealbumin, and nonspecifically with serum albumin. The extent of plasma protein binding can be measured as protein-bound iodine (PBI).

Only the small fraction of circulating thyroid hormones that is in the free (unbound) form is physiologically active and inhibitory to TSH secretion.

Fate

Much of the iodine released during biodegradation is recycled and reused for synthesis of new hormones. The remainder is excreted in the urine.

Metabolism occurs chiefly in the liver, and excretion is mainly through the kidneys.

The conjugated hormones are excreted through the bile and eliminated in the stool.

Actions

Normal function of thyroid hormones (T_3 & T_4):

- **I .ON METABOLISM:**

General metabolism: T3 and T4 stimulate metabolism of all tissue cells which is manifested by

- ↑ O2 consumption,
- ↑ CO2 production,
- ↑ heat production and ↑ Basal Metabolic Rate (BMR).

Carbohydrate metabolism:

- ↑ blood glucose level due to stimulation of glycogenolysis, gluconeogenesis (i.e. formation of glucose from non carbohydrate sources),
- ↑absorption of glucose from the intestine.
- T3 &T4 also ↑ glucose uptake by tissue cells due to ↑ metabolism.

Fat metabolism:

- ↑ Fat mobilization and ↑ F.A oxidation due to ↑ metabolism. ↑Fat utilization, □↓ cholesterol, ↓ phospholipids & triglycerides.

Protein metabolism:

Normally T3 and T4 stimulate protein anabolism through stimulation of growth hormone. Excess T3 and T4 stimulate protein catabolism through stimulation of glucocorticoids.

Vitamin metabolism:

T3 and T4 ↑ □absorption of carotenes and □ ↑ their conversion to vitamin A.

Water and mineral metabolism:

T3 and T4 ↑ " excretion of water, Ca++ and phosphates.

- **II- ON GROWTH AND MATURATION:**

T3 and T4 are essential for growth and maturation of tissues.

This is because T3 and T4 stimulate secretion and action of growth hormone.

- **III- ON NERVOUS SYSTEM:**

T3 and T4 are essential for normal development and normal functions of the nervous system.

- **IV- On Body system:**

On Cardiovascular System (CVS): "
↑HR due to ↑ □ metabolism, □ ↑ COP due to □ ↑HR. Atrial blood pressure (ABP): □ ↑ systolic BP and □ ↓diastolic BP (□ ↑ pulse pressure).

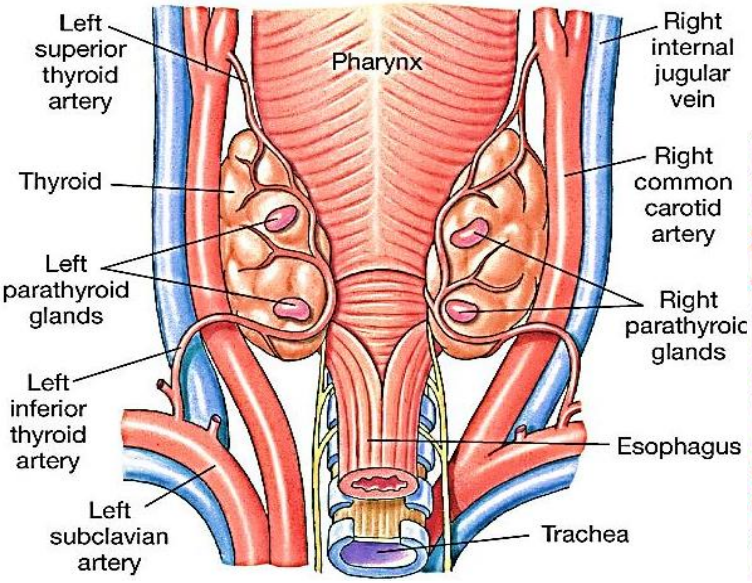
- **V- On respiratory system:** Hyperventilation (" ↑ rate and depth of respiration) due to " ↑ metabolism.
- **VI - On GIT: "** ↑ appetite, " ↑ motility, " ↑ secretion of digestive juice and " ↑ absorption.
- **VII - On sexual functions:**

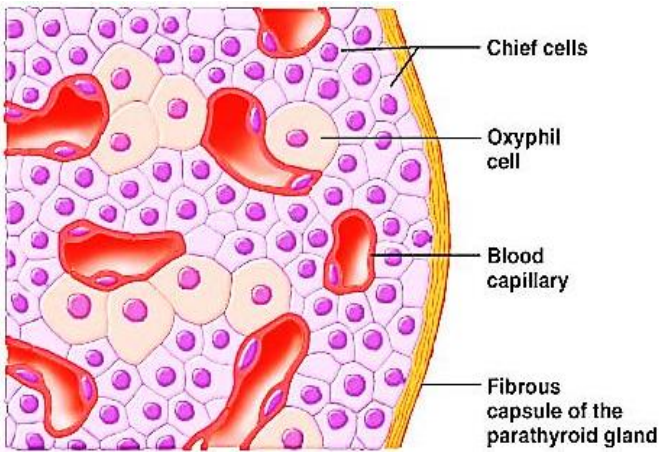
T3 and T4 are essential for normal development and normal functions of the sex organs.
They are essential for normal menstrual cycle, spermatogenesis and fertility.
They are also needed for milk secretion.

- **VIII - On blood:** T3 and T4 stimulate erythropoiesis due to stimulation of bone marrow metabolism.
- **IX - On skin:** T3 and T4 are essential for normal structure and normal functions of the skin.

The Parathyroid Glands

- Most people have four On posterior surface of thyroid gland. In response to hypocalcemia (low plasma calcium), the chief cells of the parathyroid glands secrete a single 84- amino-acid polypeptide hormone known as parathyroid hormone (PTH). **Parathyroids (two types of cells): Rare chief cells :** Abundant oxyphil cells **Chief cells produce PTH :**





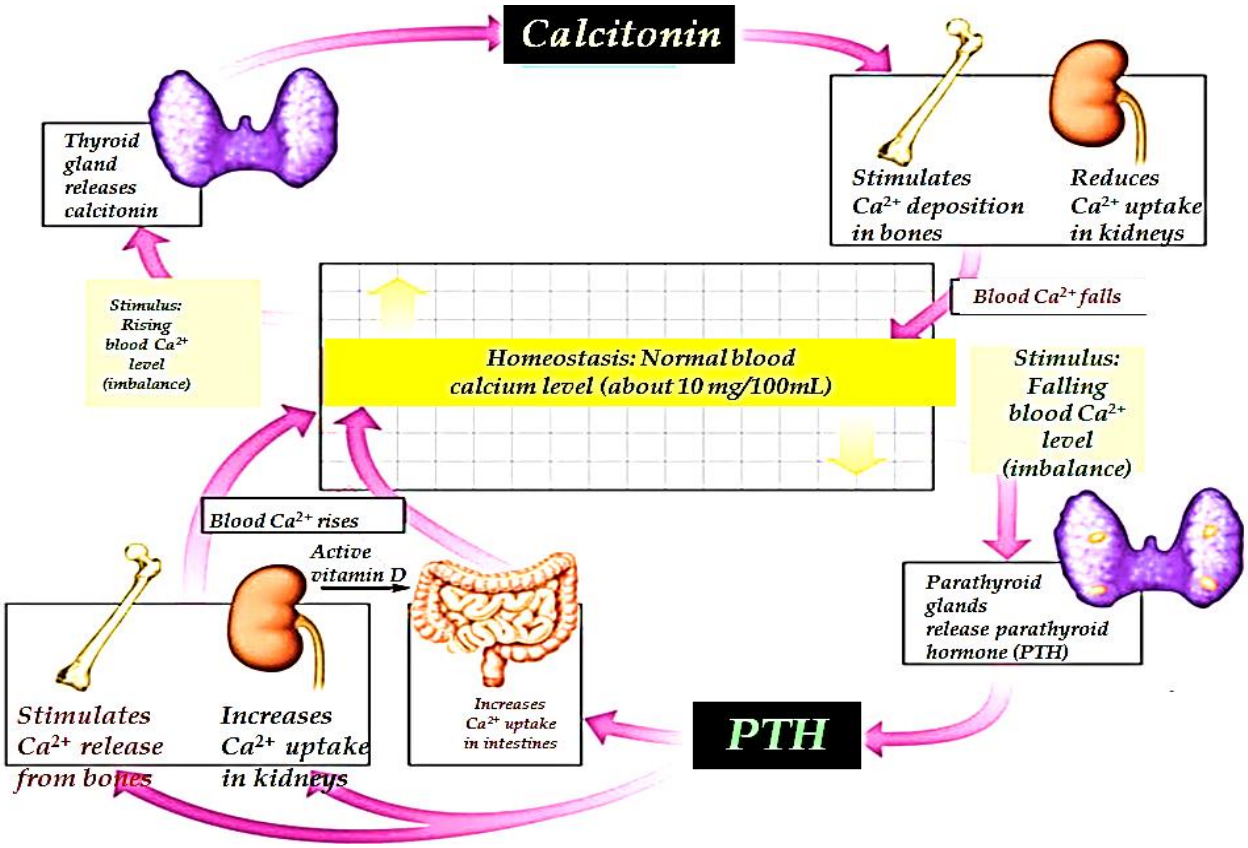
Parathyroid hormone regulates serum calcium levels by exerting its effects on the following target tissues:

(1) On the bone:

Parathormone helps mobilization of calcium and phosphates from the bone to the blood because it stimulates osteoclastic activity.

(2) On the kidneys:

Parathormone inhibits reabsorption of phosphates from the renal tubules which leads to increase its excretion in urine. So, phosphate ions decrease in blood.



This leads to increase Ca++ in blood to maintain solubility product constant.

It is known that the concentration of Ca++ is inversely proportional to the concentration of PO4-- ions.

Parathormone also increases reabsorption of Ca++ from the renal tubules leading to increase Ca++ in blood.

(3) On the intestine:

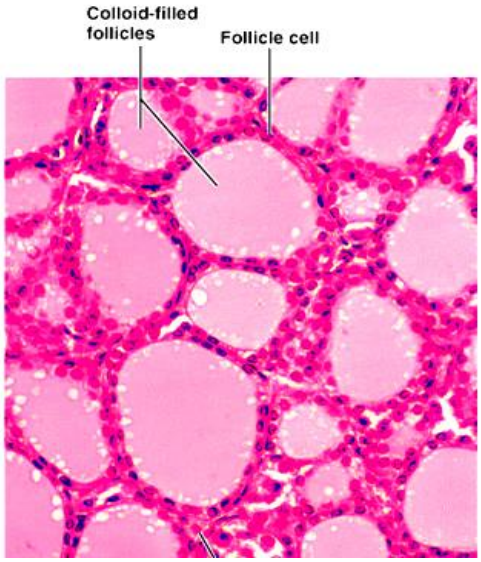
Parathormone helps absorption of Ca++ and PO4-- ions from the intestine.

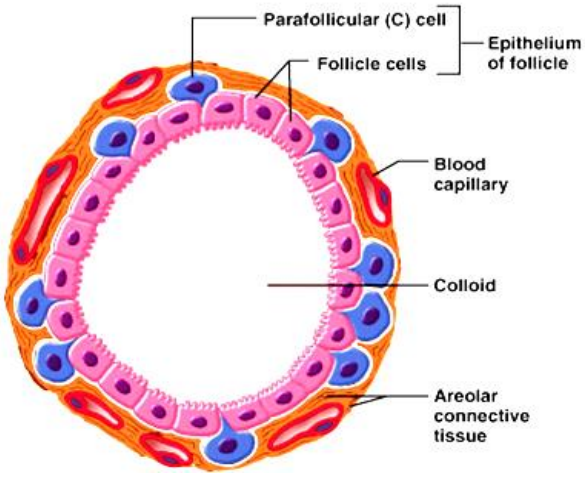
Thyrocalcitonine

It is secreted from the parafolicular cells of the thyroid gland.

Its main action is to lower Ca++ ions when these ions increase above normal.

Thyrocalcitonine produces this effect by its action on the following sites:





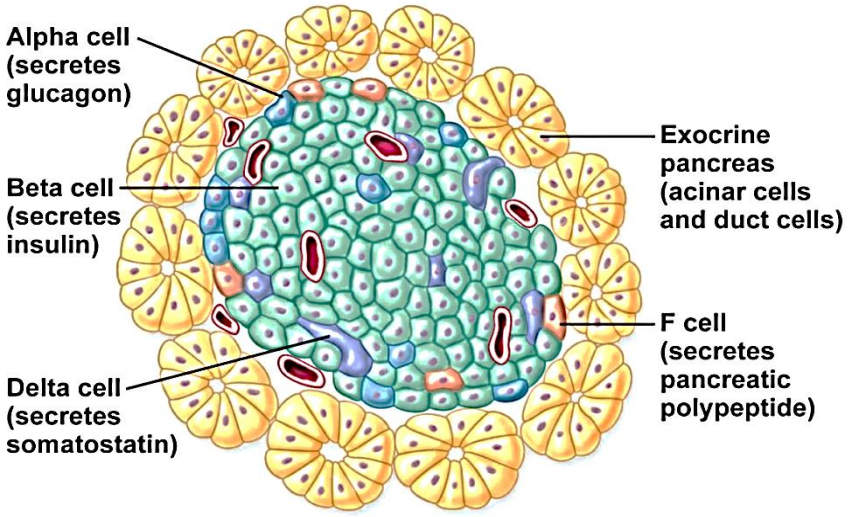
Thyrocalcitonine decreases absorption of Ca++ and PO4-- ions from the intestine.

Calcitriol elevates plasma calcium and phosphate levels primarily by promoting the intestinal absorption of both ions, but also by increasing renal tubular reabsorption of calcium and phosphate.

The major actions of PTH are mediated by cyclic AMP.

The Pancreas

The endocrine functions of the pancreas are performed by the islets of langerhans (also called pancreatic islets) –small, highly vascularized masses of cells scattered throughout the pancreas and representing only 1% to 3% of the entire organ.



The Islets of Langerhans contain four types of secretary cells, as follows:

- Alpha (A) cells, which secrete glucagons
- Beta (B) cells, which secrete insulin
- Delta (D) cells, which secrete somatostatin
- PP (F) cells, which secrete pancreatic polypeptide

Insulin-secreting beta cells are the most numerous, making up to 75% of the islet cell population. The A cells containing glucagons comprise approximately 20% of islet cell mass, whereas the somatostatin–containing D cells account for 3% to 5% of pancreatic islet cells.

The F cells make up less than2% of islet cells and secrete a polypeptide that slows food absorption .

The paracrine relationship exists within the pancreatic islets, with one hormone affecting the secretion of other pancreatic hormones.

Somatostatin inhibits the secretion of insulin, glucagons, and pancreatic polypeptide. Insulin inhibits the secretion of glucagons, whereas glucagon stimulates the secretion of insulin and somatostatin.

1. Glucagon

Glucagon is a 29-amino-acid polypeptide hormone secreted by the alpha cells of the pancreatic islets primarily in response to hypoglycemia (low blood sugar). Glucagon is essentially a catabolic hormone that decreases carbohydrate and lipid energy stores and increases the amount of glucose and fatty acids available for oxidation.

Control of Secretion

The plasma glucose concentration is the major physiologic regulator of glucagons secretion. In addition to hypoglycemia and fasting, the following factors promote glucagons secretion: amino acids, exercise, stress, gastrin, cortisol, CCK, acetylcholine, and beta-adrenergic stimulation.

The rate of glucagons secretion is inhibited by elevated blood levels of glucose and free fatty acids, and by somatostatin, insulin, secretin, phenytoin, and alpha-adrenergic stimulation.

Major Actions

- Carbohydrate metabolism: Glucagon stimulates hepatic glycogenolysis, thereby promoting the release of glucose from liver glycogen stores. The action is mediated by cyclic AMP, which stimulates protein kinase activity, leading to the activation of phosphorylase, the glycogenolytic enzyme. Glucagon also interacts with hormone receptors coupled to the activation of phospholipase C, which eventually leads to calcium influx and stimulation of glycogenolysis. In addition to stimulating hepatic glycogenolysis, glucagon inhibits glycogenesis and raises the rate of hepatic gluconeogenesis.

The net effect is an elevation of blood glucose (hyperglycemia).

- Lipid metabolism: Glucagon stimulates lipolysis, thereby increasing the release of free fatty acids and glycerol from adipose tissue. Glucagon also enhances hepatic ketogenesis by facilitating conversion of fatty acids to ketone bodies.

- Protein metabolism: Glucagon exerts a catabolic action on hepatic proteins and inhibits the incorporation of amino acids into hepatic protein.

- Cardiac effects: large amounts of exogenous glucagons produce appositive inotropic effect on the heart by increasing myocardial levels of cyclic AMP.

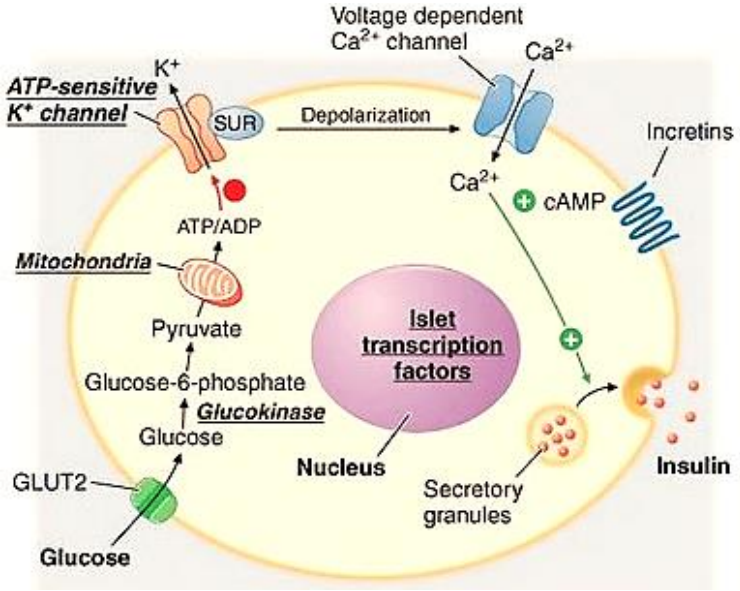
2. Insulin

Structure, Biosynthesis, and Secretion

Insulin is a polypeptide hormone composed of 51 amino acids arranged in two chains (A and B), linked by disulfide bridges.

Insulin is derived from a large polypeptide precursor-proinsulin which is synthesized in the endoplasmic reticulum of beta cells and packaged into membrane-bounded granules within the Golgi complex.

Insulin secretion occurs through exocytosis (emiocytosis), a calcium-dependent process that is enhanced by cyclic AMP and potassium. On entering the circulation, insulin is transported largely in free molecular form, not bound to plasma proteins.



Control of Secretion

The secretion of insulin is regulated primarily by the blood glucose level, with an elevation of blood glucose (hyperglycemia), increasing both production and release of insulin. Ingested glucose effects a far greater secretion of insulin than an equivalent amount of intravenously administered glucose because several gastrointestinal hormones, including gastrin, secretin, CCK, gastric inhibitory

polypeptide (GIP), and glucagons, stimulate insulin secretion.

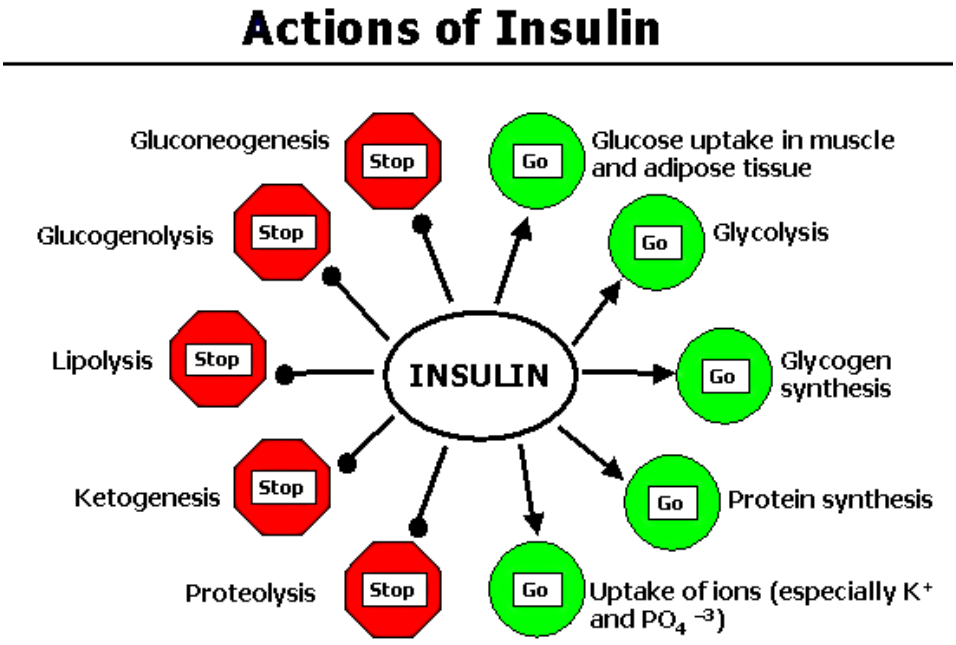
Major Actions

· Cellular membrane permeability:

Insulin facilitates the transport of glucose across selected cell membranes, thereby accelerating the entry of glucose into muscle, adipose tissue, fibroblasts, leukocytes, mammary glands, and the anterior pituitary.

The transport of glucose into the liver, brain, renal tubules, intestinal mucosa, and erythrocytes is independent of insulin.

- ♣ **Carbohydrate metabolism:** Insulin effectively lowers the level of blood glucose by enhancing the transport and peripheral utilization of glucose. Insulin increases muscle and liver glycogen while inhibiting those that produce glycogenolysis. Glycolytic enzymes are also activated by insulin, whereas several enzymes involved in gluconeogenesis are inhibited.
- ♣ **Protein metabolism:** Insulin is strongly anabolic, increasing protein synthesis and inhibiting protein catabolism. Insulin increases the incorporation of amino acids into protein by accelerating the entry of amino acids into the cell and possibly by increasing RNA synthesis.
- ♣ **Lipid metabolism:** Insulin stimulates formation of triglycerides (lipogenesis) and inhibits their breakdown (lipolysis). Insulin accelerates synthesis of fatty acid and glycerol phosphate and enhances cellular permeability to fatty acids, leading to increased deposition of triglycerides in adipose tissue.



3. Pancreatic Somatostatin

Unlike glucagons and insulin, pancreatic somatostatin does not directly regulate intermediary metabolism. Rather, it slows digestion and absorption of nutrients by exerting several inhibitory effects on digestive processes (eg, reduced gastric acid secretion decreased gastric emptying and inhibition of CCK secretion and gallbladder contraction).

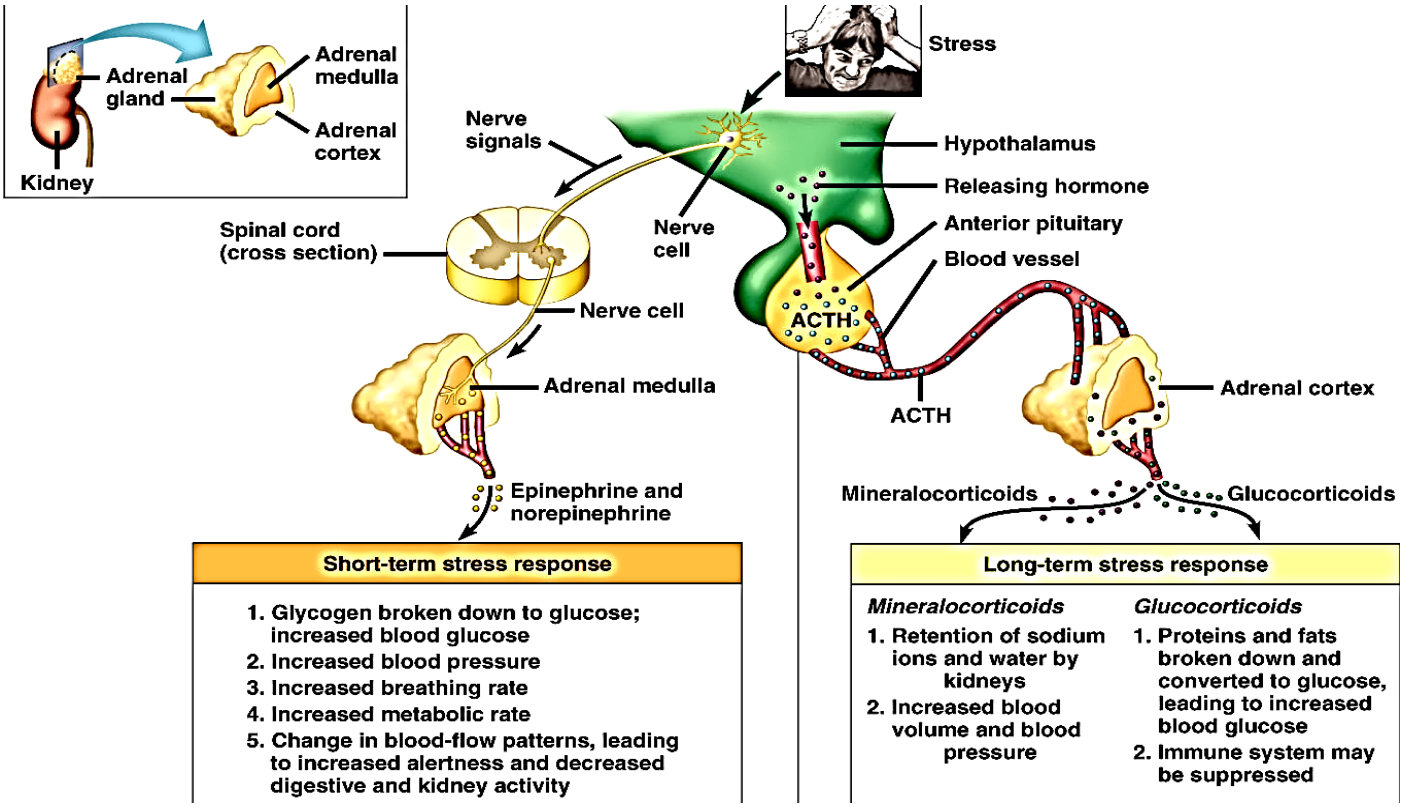
Pancreatic somatostatin secretion is stimulated by elevations and gastrointestinal hormones (eg, CCK).

The Adrenal Glands

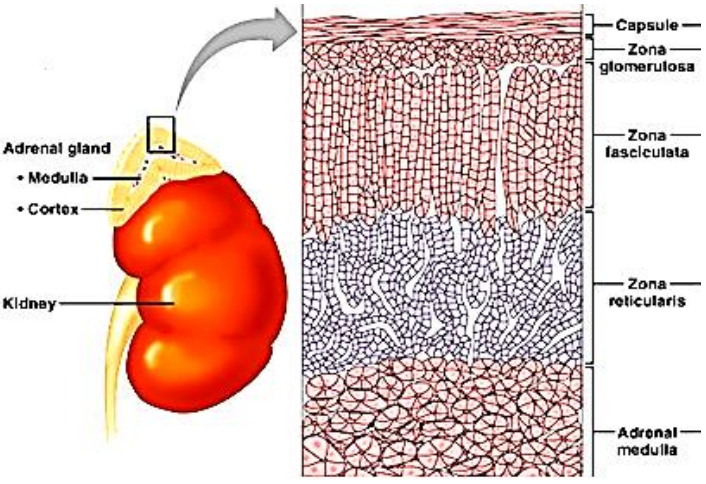
The adrenal (suprarenal) glands are paired yellowish masses of tissue situated at the superior pole of each kidney. Each gland consists of two distinct entities–an outer adrenal cortex and an inner adrenal medulla–that differ in embryologic origin, character, and function.

Adrenal Medulla

The adrenal medulla develops from the embryonic ectoderm. It remains functionally associated with the sympathetic nervous system, being essentially a modified sympathetic ganglion whose postganglionic neurons have lost their axons and become secretory.



Histologically : The medullary cells, often termed chromaffin cells because their granules possess affinity for chromium salts, secrete the catecholamine hormones epinephrine (adrenaline) and norepinephrine (noradrenalin). The principal secretory product is epinephrine, with norepinephrine normally accounting for only 20% of the total secretion.



Adrenal medullary secretion of the catecholamines is physiologically controlled by the posterior hypothalamus. The hormones are stored in cellular granules, and are released in response to the following stimuli: sympathetic nervous system activation, hypoglycemia, pain, hypoxia, hypotension, cold, emotional stress, acetylcholine, histamine, and nicotine.

Actions of Adrenal Modullary Hormones

Epinephrine and norepinephrine mimic the effects of sympathetic nerve discharge, producing the following effects :

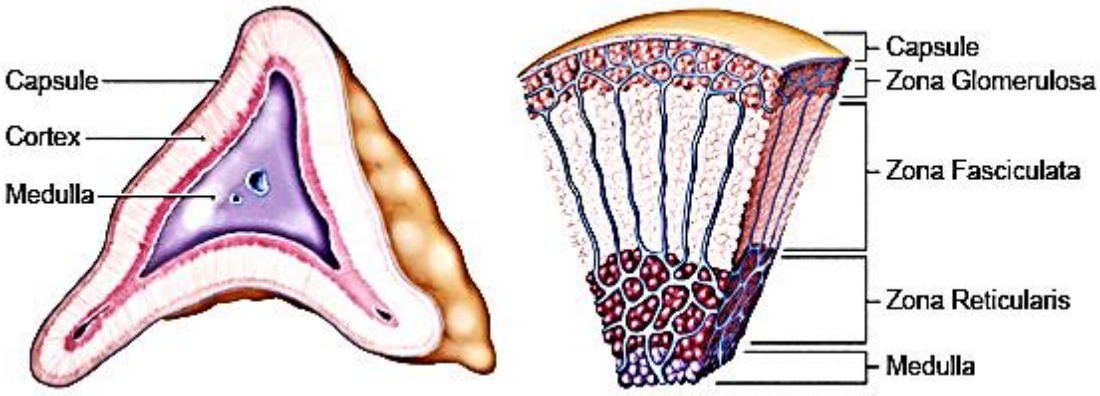
- ☼ Direct increase in cardiac rate and myocardial force of contraction
- ☼ Dilation of coronary and skeletal muscle blood vessels
- ☼ Constriction of the cutaneous and visceral vasculature
- ☼ Relaxation of respiratory smooth muscle
- ☼ Inhibition of GI motility
- ☼ Pupillary dilation (mydriasis)
- ☼ Glycogenolysis in liver and muscle
- ☼ Lipolysis

The cardiac excitatory effects and the metabolic actions of lipolysis and glycogenolysis are mediated by cyclic AMP, the latter involving the activation of phosphorylase enzyme by protein kinase.

The catecholamines also elevate the metabolic rate (calorigenic action), stimulate the central nervous system, increase alertness, and stimulate respiration.

Adrenal Cortex

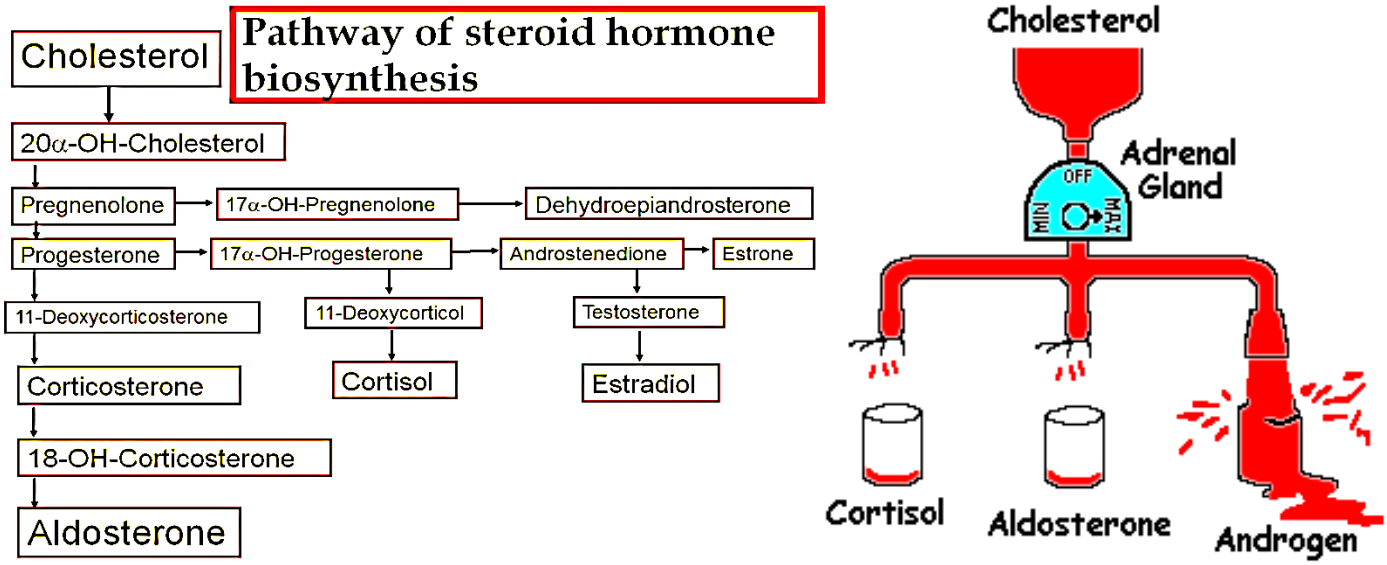
Adrenal cortical tissue is structurally arranged into three concentric regions or zones: a thin outer zona glomerulosa, a thick middle zona fasciculate, and an inner zona reticularis



Chemically, the steroid hormones of the adrenal cortex, the adrenocorticoids, are all derivatives of cholesterol. The adrenocorticoid hormones are usually divided into three functional groups: the mineralocorticoids, such as aldosterone, which regulate electrolyte and water balance; the glucocorticoids, such as cortisol, which affect carbohydrate, protein, and fat metabolism; and the adrenogenital steroids or sex hormones.

The adrenogenital steroids are of three types: androgens(such as dehydroepiandrosterone), estrogens (such as estradiol), and progestins (such as progesterone).

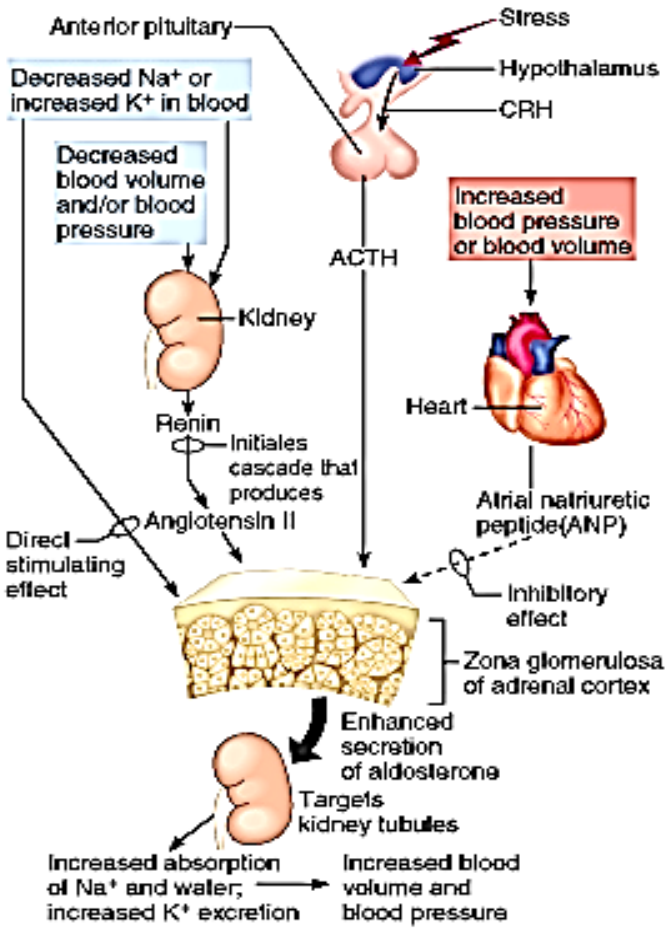
Under normal physiologic conditions the adrenogenital steroids are secreted (under ACTH control) in minute amounts, and therefore they exert minimal effects on reproductive functions. Excessive secretion of adrenal androgens results in precocious pseudopuberty in boys, and causes masculinization of females (adrenogenital syndrome).



1. Mineralocorticoids

Control of Secretion and Actions.

Aldosterone is the principal physiologic mineralocorticoid secreted by the zona glomerulosa. Its secretion is regulated primarily by the reninangiotensin mechanism. The plasma concentrations of sodium and potassium are involved in the control of aldosterone secretion.



Hyperkalemia (elevated plasma potassium) exerts a direct stimulating effect on the zona glomerulosa, whereas hyponatremia (low plasma sodium) activates the rennin-angiotensin mechanism. Atrial natriuretic hormone (ANH) inhibits rennin secretion, and it also reduces the responsiveness of the cells in the zona glomerulosa to angiotensin II. Other factors contributing to the control of aldosterone secretion include blood volume and ACTH, the latter exerting a limited, nonselective stimulatory effect.

Aldosterone plays a major physiologic role in the maintenance of electrolyte and fluid balance by promoting the renal tubular reabsorption of sodium and the secretion of potassium and hydrogen.

2. Gllucocorticoids

Control of Secretion and Actions.

Glucocorticoid secretion, which occurs primarily in the zona fasciculate, is controlled by ACTH.

A variety of stressful stimuli, including anxiety, fear, hypoglycaemia, hypotension, and haemorrhage, increase secretion of adrenocorticotropic hormone-releasing hormone (CRH) from the hypothalamus.

CRH promotes ACTH secretion by the adenohypophysis, and ACTH stimulates adrenal cortical secretory activity, thereby elevating blood levels of cortisol (the principal physiologic glucocorticoid). Elevated blood levels of free cortisol normally exert a negative feedback control over further secretion of CRH and ACTH. Prolonged ACTH secretion results in hypertrophy and hyperplasia of the adrenal cortex and excessive secretion of all adrenocorticoid hormones.

The metabolic and physiologic actions of the glucocorticoids are :

- **Carbohydrate metabolism:** Glucocorticoids stimulate hepatic gluconeogenesis and inhibit peripheral uptake and utilization of glucose by skeletal muscle and adipose tissue, thereby promoting hyperglycemia. Hepatic glycogenesis is also enhanced.
- **Protein metabolism:** Glucocorticoids exert protein catabolic and antianabolic actions, promoting the breakdown of existing proteins while inhibiting the incorporation of amino acids into new proteins.
- **Lipid metabolism:** Glucocorticoids inhibit lipogenesis and favor mobilization of fats from adipose tissues. When present in large amounts, these hormones favor redistribution of adipose stores by promoting loss of fat from the extremities, and accumulation of fat depots in central body regions (e.g., "moon face" and "buffalo hump" formation).
- **Permissive Actions:** Clucocorticoids are essential for normal vascular reactivity to the catecholamines. Other permissive effects include support for the metabolic actions (particularly lipolysis) of the catecholamines and glucagons.
- **Blood and immunologic effects**: Glucocorticoids inhibit the immune response, cause involution of lymphoid tissue, and reduce blood levels of lymphocytes, eosinophils, and basophils. These hormones also stimulate erythropoiesis and elevate circulating levels of platelets and neutrophils.
- **GI tract effects:** Glucocorticoid hormones stimulate gastric acid and pepsin secretion and inhibit the production of protective mucus, thereby favouring development of gastric ulcers.

The Sex Glands

The sex glands, the ovaries of the female and the testes of the male, not only produce the sex cells but are also important endocrine organs. The hormones produced by these organs are needed in the development of the sexual characteristics, which usually appear in the early teens and for the maintenance of the reproductive apparatus once full development has been attained.

The main androgen produce by the male sex glands, testosterone, is responsible for the growth of and functioning of the male reproductive tract.

Those structures directly concerned with reproduction are considered primary sex characteristics. Testosterone is also responsible for male secondary sex characteristics such as a deep voice and facial hair.

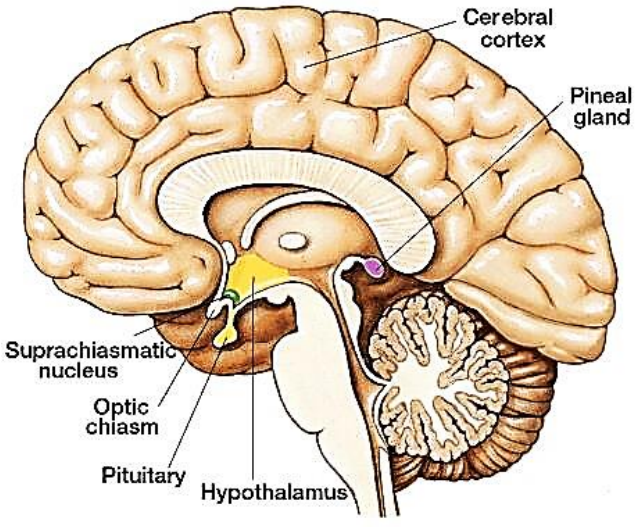
In the female, the hormones that most nearly parallel testosterone in their actions are the estrogens. Estrogens contribute to the development of the female secondary sex characteristics and stimulate the development of the mammary glands, the onset of menstruation, and the development and functioning of the reproductive organs.

The Thymus Gland

The thymus gland lies in the upper part of the chest above the heart. Through the production of a hormone called thymosin, it helps in the development of certain white blood cells, called Tlymphocytes that help protect the body against foreign organisms. Because this gland is most active before birth and early in life, it is large during childhood but gradually shrinks in size as one reach adulthood.

The Pineal Gland

The pineal gland, a small, flattened, cone-shaped structure located between the two parts of the thalamus, produces a hormone called melatonin in a number of animals and also in humans. Melatonin, or some other hormone from the pineal, is thought to regulate the release of certain substances from the hypothalamus, which may in turn regulate the secretion of gonadotropins from the pituitary. There is evidence from animal studies that the production of melatonin is regulated by the amount of light in the environment.



CHAPTER FOURTEEN

THE REPRODUCTIVE SYSTEM

CHAPTER FOURTEEN THE REPRODUCTIVE SYSTEM

Males and females have anatomically distinct reproductive organs that are adapted for producing gametes, facilitating fertilization, and, in females, sustaining the growth of the embryo and fetus.

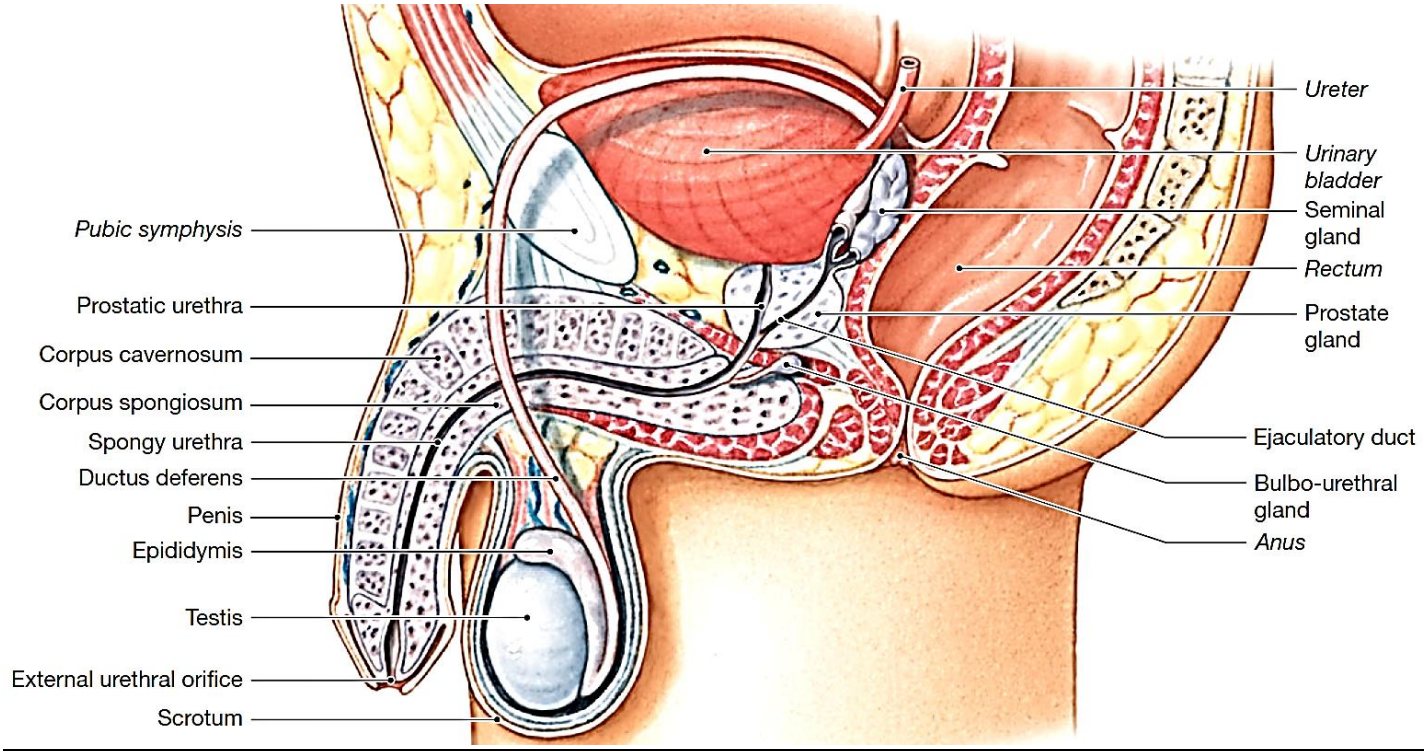
Male reproductive system

The male's reproductive role is to produce sperm and deliver to the vagina.

These functions require four different structures.

These are the:

- Testes,
- Accessory glands,
- Accessory ducts and
- Penis.



Male internal reproductive organs

Male internal reproductive organs include the tastes, series of ducts and accessory glands.

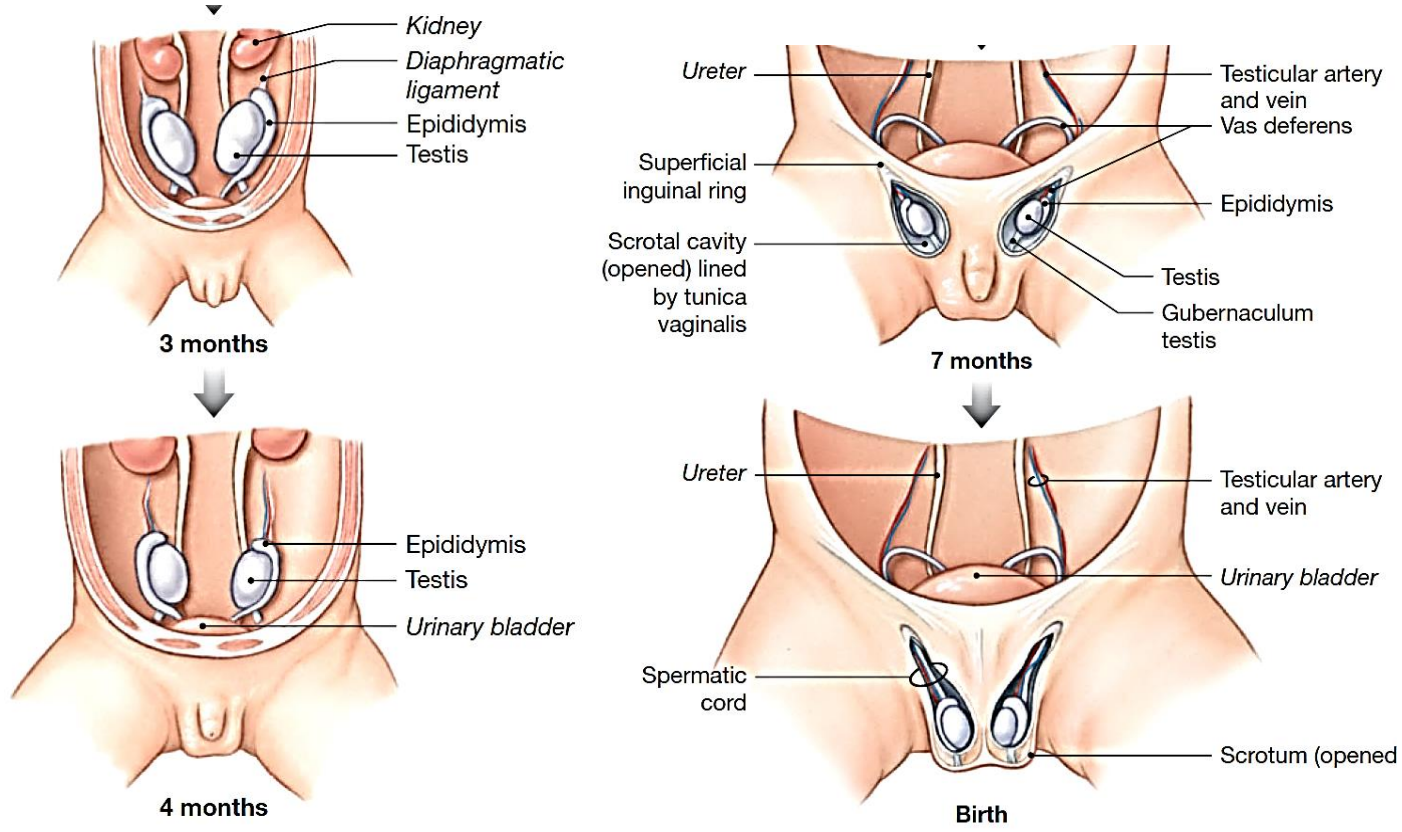
Testes

The testes :are suspended outside the abdomen in the scrotum, which is divide internally into 2 sacs one for each testis.

A testis is un oval shaped The left testis is generally located about 1cm lower than the right.

Testes are paired male reproductive organs (gonads), which produce sperm. During fetal life, tests

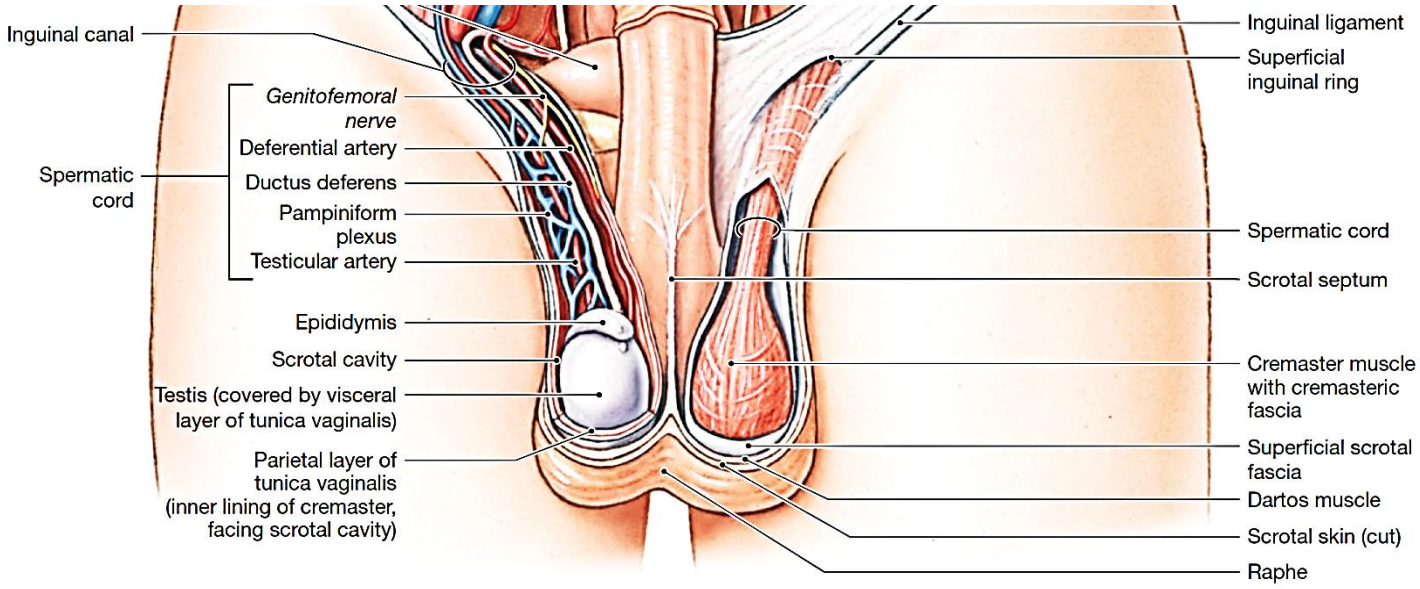
are formed just below the kidneys inside the abdomino-pelvic cavity. By third fetal month it stats is to descend and by the seventh month of fetal life it passes through the inguinal canal.



The inguinal canal is a passageway leading to scrotum.

Scrotum is a paunch of skin that hangs between the thighs.

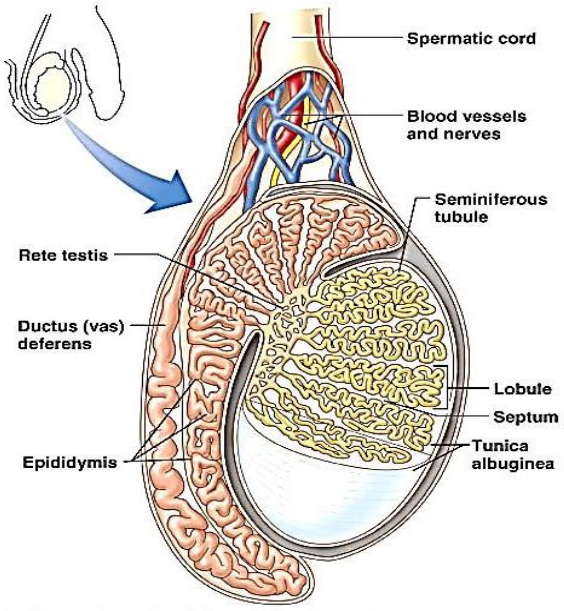
The tests complete their descent in to the scrotum shortly before or after birth. Because the tests hang in scrotum out side the body their temperature is of cooler than the body temperature by 3 Degree Fahrenheit. This lower temperature is needed for sperm production and survival. Under the skin of scrotum, Cremaster muscle elevates tests during cold weather.

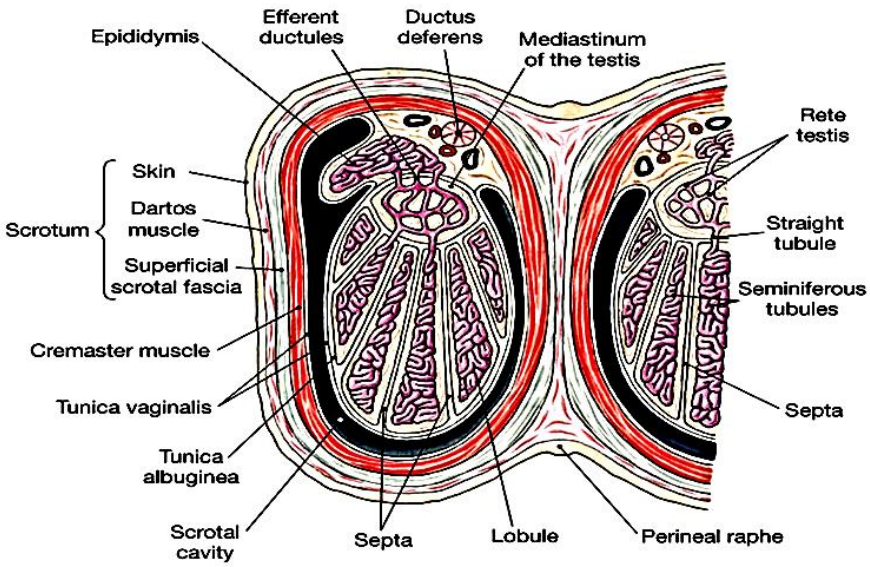


The interior of the scrotum is divided in to two by median septum. One tests in each compartment. Each test is oval-shaped measuring 4.5 c.m. long, 2.5 c.m wide in adult. It is enclosed in fibrous sac **called Tunica Albuginea.**

The sac extendes into a test to divide the test in to lobules. Next to tunica albuginea is Tunica Vaginals, which is a continuation of membrane of abdomino-pelvic cavity.

Each testis contain 800 lightly coiled **Semniferous Tubules which produce thousands of sperm each second.** The walls of the semniferous tubes are lined with *germinal* tissue.

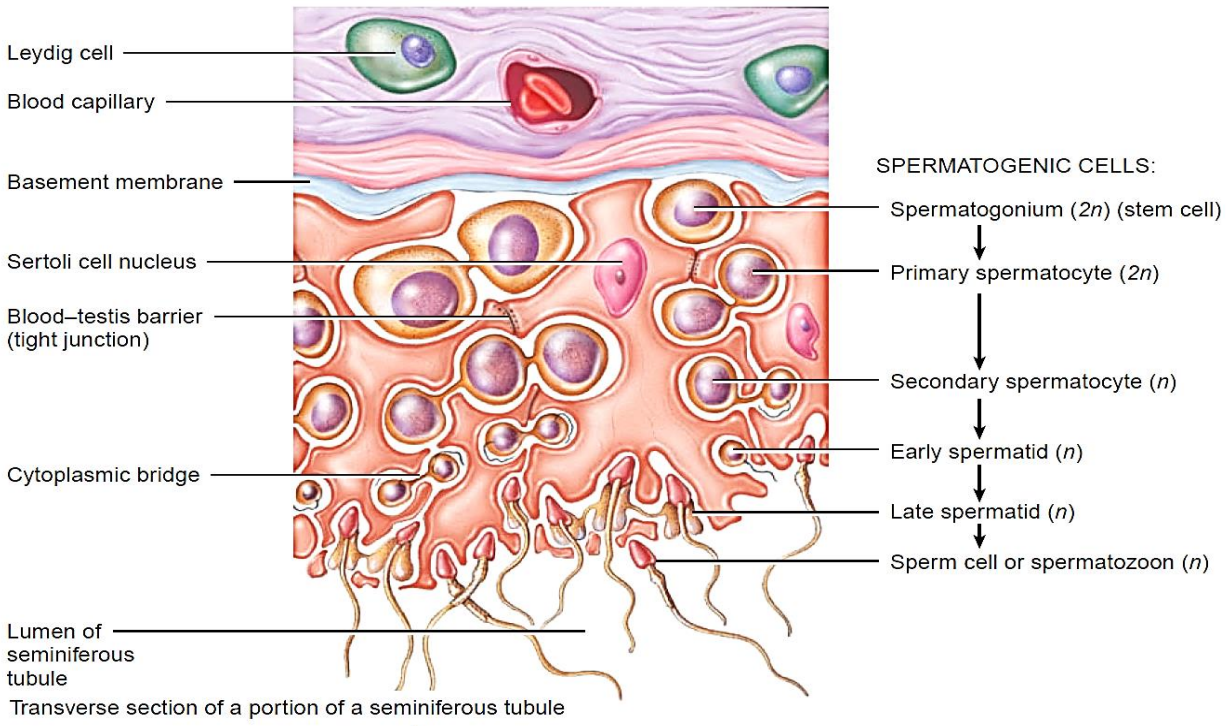




The germinal tissue contains two types of cells:

- ♣ Spermatogenetic cell producing or developing the sperm cell and
- ♣ The sustentacular cell, which provide nourishment for the germinal sperm.

Between the semniferous tubules clusters of endocrine cells called interstitial endocrinocytes (Leydig cell) secret male sex hormone (Androgens) where testosterone is the most important.



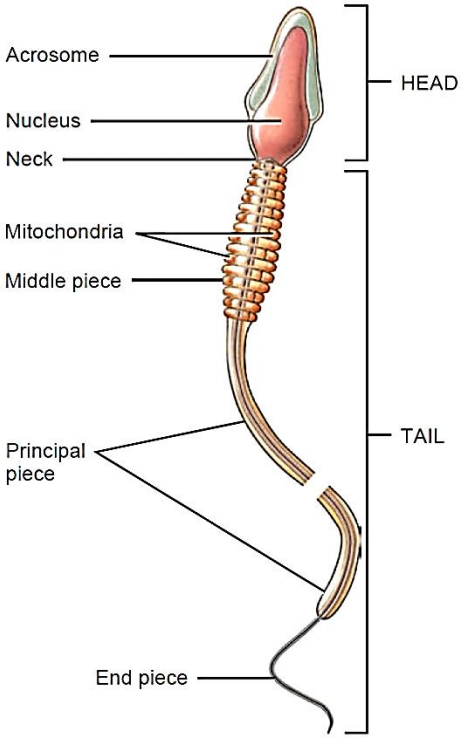
Transverse section of a portion of a seminiferous tubule

Sperm

The male genetic carrying cell is the sperm. Matured spermatozoa have head, middle piece, and tail At the tip of the head is acrosome containing several enzymes that help the sperm to penetrate the egg.

In the center of head is a nucleus-containing chromosome. This is a genetic constitution. The middle piece consist mainly coiled mitochondria for energy production for tail motility.

Sperm is one of the smallest cells in the body. For complete development each sperm need 2 months.



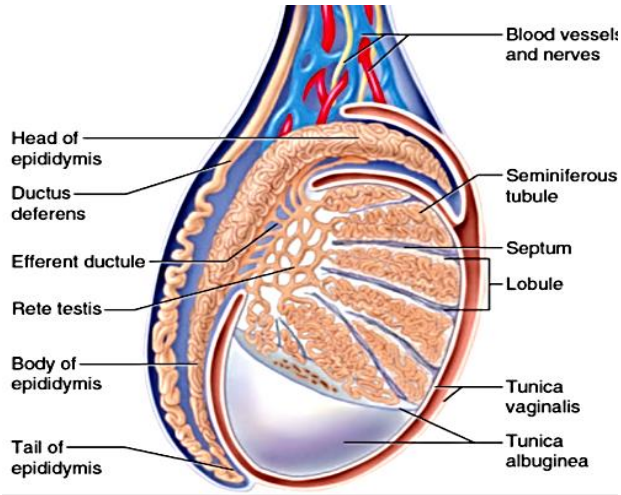
Normally 300 to500 million sperms are released during ejaculation. A male who releases 20 to 30 million normal sperms tends to be infertile.

Accessory Ducts

The sperm produced in the testes are carried to the point of ejaculation from penis by a system of ducts. These are :

- Epididymis,
- Ductus defference,
- Ejaculatory duct
- Urethra.

Epididymis: The semniferous tubules merge in the central posterior portion of the testes as epydidimis . It is a tube about 4 cm long , comma-shaped, tightly coiled tube.



Found on the superior part of the testis and along the posterior lateral side.

The efferent ducts extend up ward in to a convoluted mass of tubes that forms a crescent shape along

its side. This coiled tube is epididymis. **It has three main functions:**

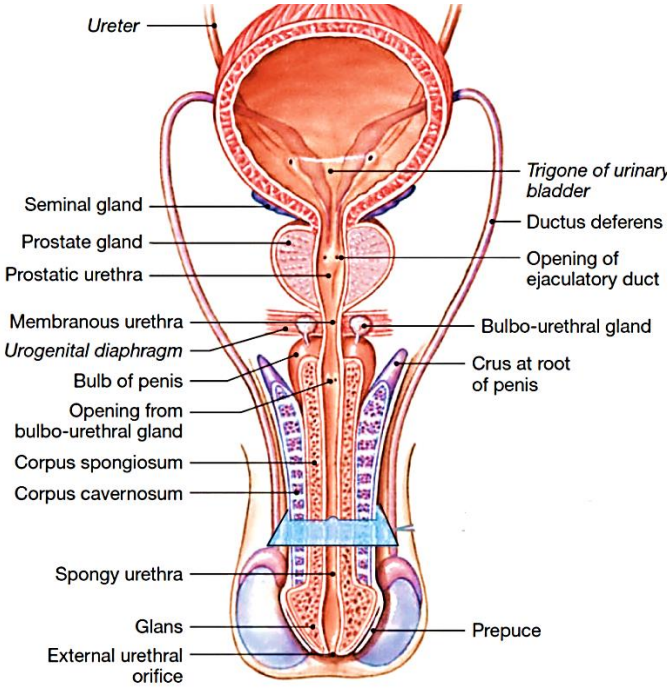
- Store sperm until maturity (at least 20 days) and facilitates maturation.
- Expels sperm with the contraction of muscles in the epididymis walls to the vas deferens
- Passage way of sperm .Recycles damaged spermatozoa

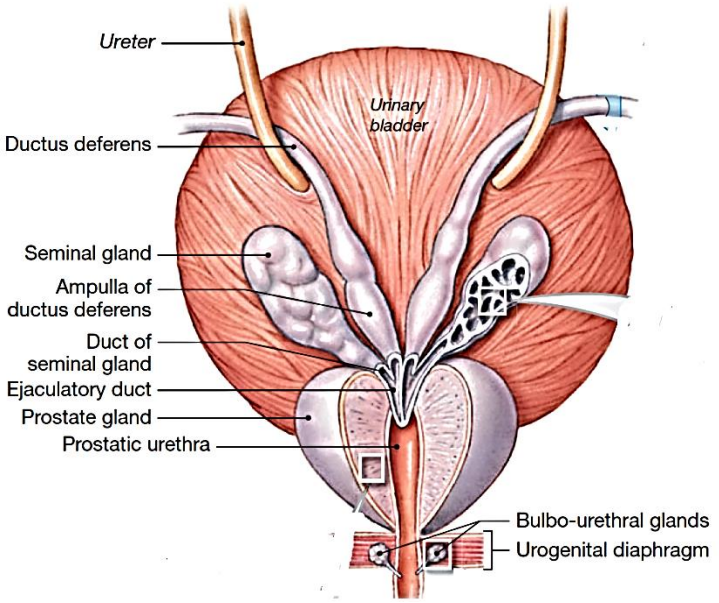
Epididymis divided into 3 parts : head, body & tail.

The head join with efferent ductule and tail continues as vas deference (ductus deference).

Ductus deferens:

It is the dilated continuation of epididymis. It is located easily over the spermatic cord; hence male permanent (surgical) contraceptive method (vasectomy) is usually performed over it. As it passes from tail of epididymis it is covered by spermatic cord containing testicular artery, vein, autonomic nerves, lymphatic and connective tissue.





After the ducts deferens pass through the inguinal canal it free from spermatic cord and pass behind the urinary bladder, where it travels along side an accessory gland, the seminal vesicle and becomes ejaculatory duct. Just before reaching the seminal vesicle, it widens in to an enlarged portion, Ampulla.

Ampulla is sperm storage sites before ejaculation. It transports sperm by peristalsis during peak stimulation.

Ejaculatory duct : The ducts deference joins to the duct of seminal vesicle at ejaculatory duct. It is about 2 c.m. long. They receive secretion from the seminal vesicles and pass through the prostate where they receive additional secretion.

Urethra

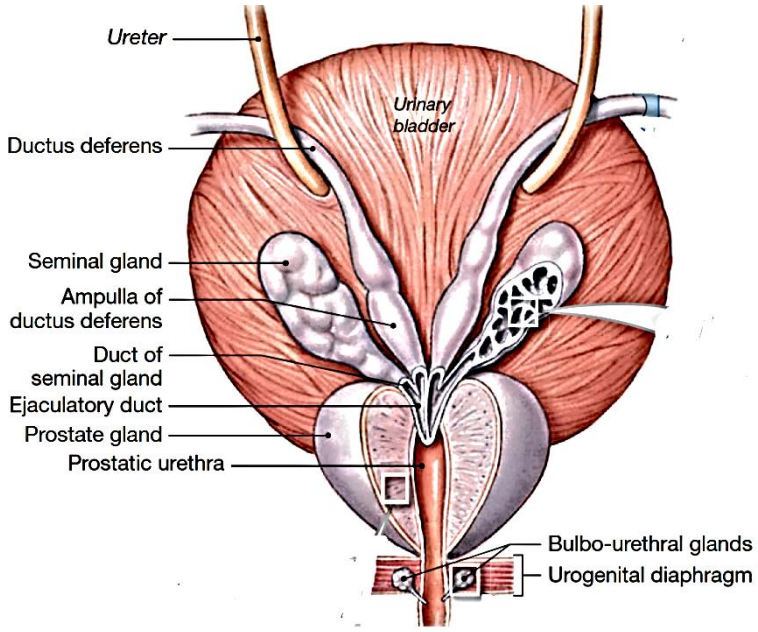
It is the final section of the reproductive system. It leads from the urinary bladder through the prostate gland and to the tip of the penis. Its reproductive function is to transport semen out of penis.

Accessory Glands These are :

- Seminal vesicles,
- Prostate and
- Bulbo-uretral glands.

Seminal vesicles

Seminal vesicles are paired; secretary sacs lie next to the Ampulla of the ducts deference. It produces alkaline secretion. It contains mostly water, fructose, prostaglandin & vit. C.



Prostate gland

Prostate gland lies inferior to the urinary bladder surrounding the first portion of urethra. It is a rounded mass about the size of chestnut. The secretion of prostate makes sperm motile and neutralizes vaginal acidity. A thin but firm capsule of fibrous connective tissue surrounds the prostate. The secretion of prostate mainly contains water, acid phosphate & cholesterol.

Bulbo-uretheral gland

Each gland is about shape & size of pea. They lie directly Bulbo urethral glands are pair; one in each side of the urethra. below prostate. Bulbo urethral glands secrets clearly alkaline fluid to neutralize the acidity of urine during the onset of sexual excitement and it also act as a lubricant.

Male external reproductive organ

- **Penis**

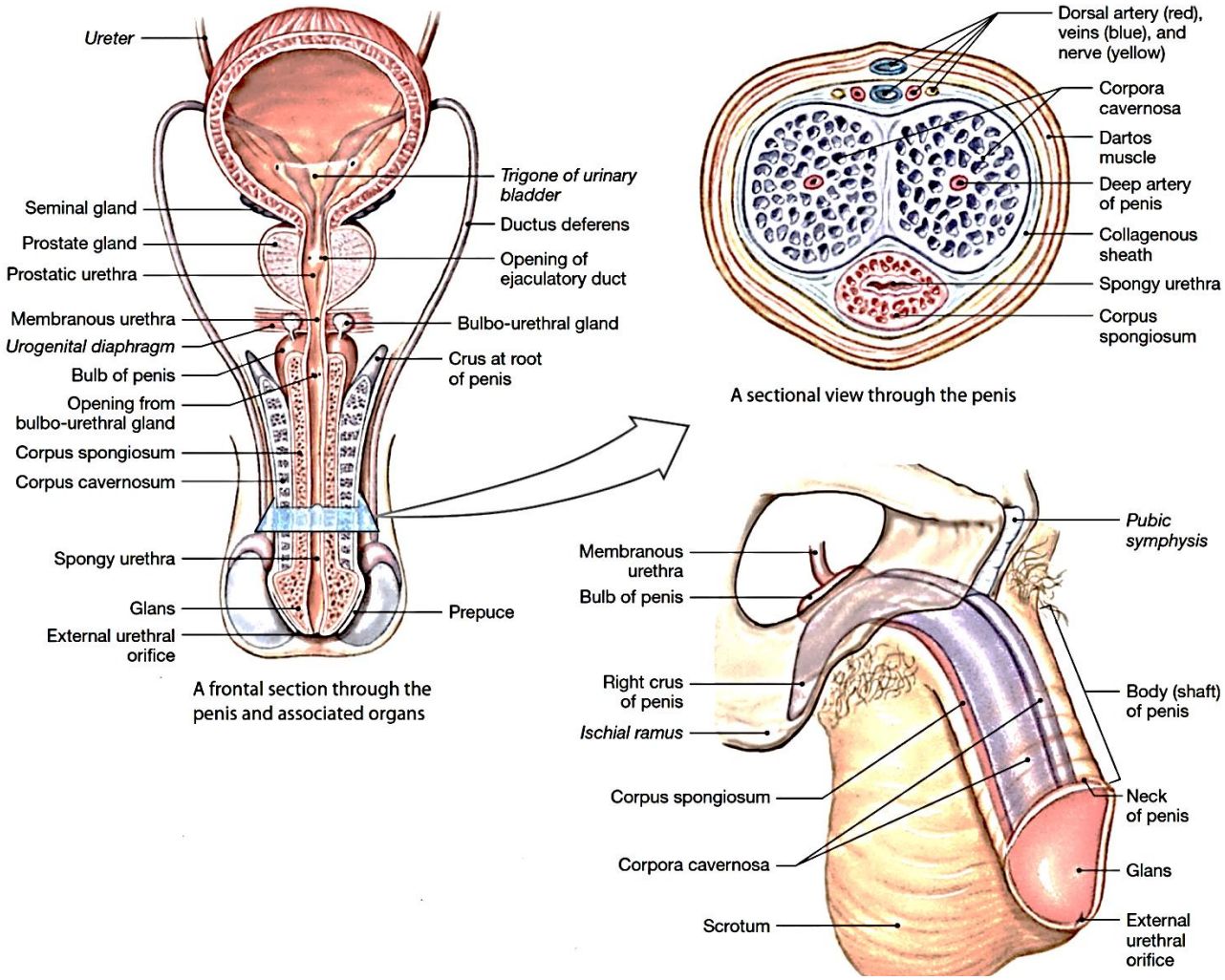
Penis is an extension from the perineal area of male. It is soft and erective tissue. Has two main function,

- It caries urine through urethra to the outside during urination
- Transports semen through the urethra during ejaculation.

In addition to urethra penis contains three cylindrical strands of erectable tissue.

- Two corpora cavernosa, which run parallel to the dorsum of penis.
- Corpus spongiosum which contain urethra

The corpora cavernosa are surrounded by a dense, relatively inelastic connective tissue called tunica albuginea. It also contains numerous vascular cavities called venous sinusoids.



The corpus spongiosum extends distally beyond the corpora cavarnosa & becomes expanded at the tip of the penis, which is called glans penis. The loosely fitting skin of the penis is folded forward

over the glans to form the prepuce or foreskin, which usually excised during circumcision. Corona of the glans is a ridge at behind margin of glans.

- **Semen**

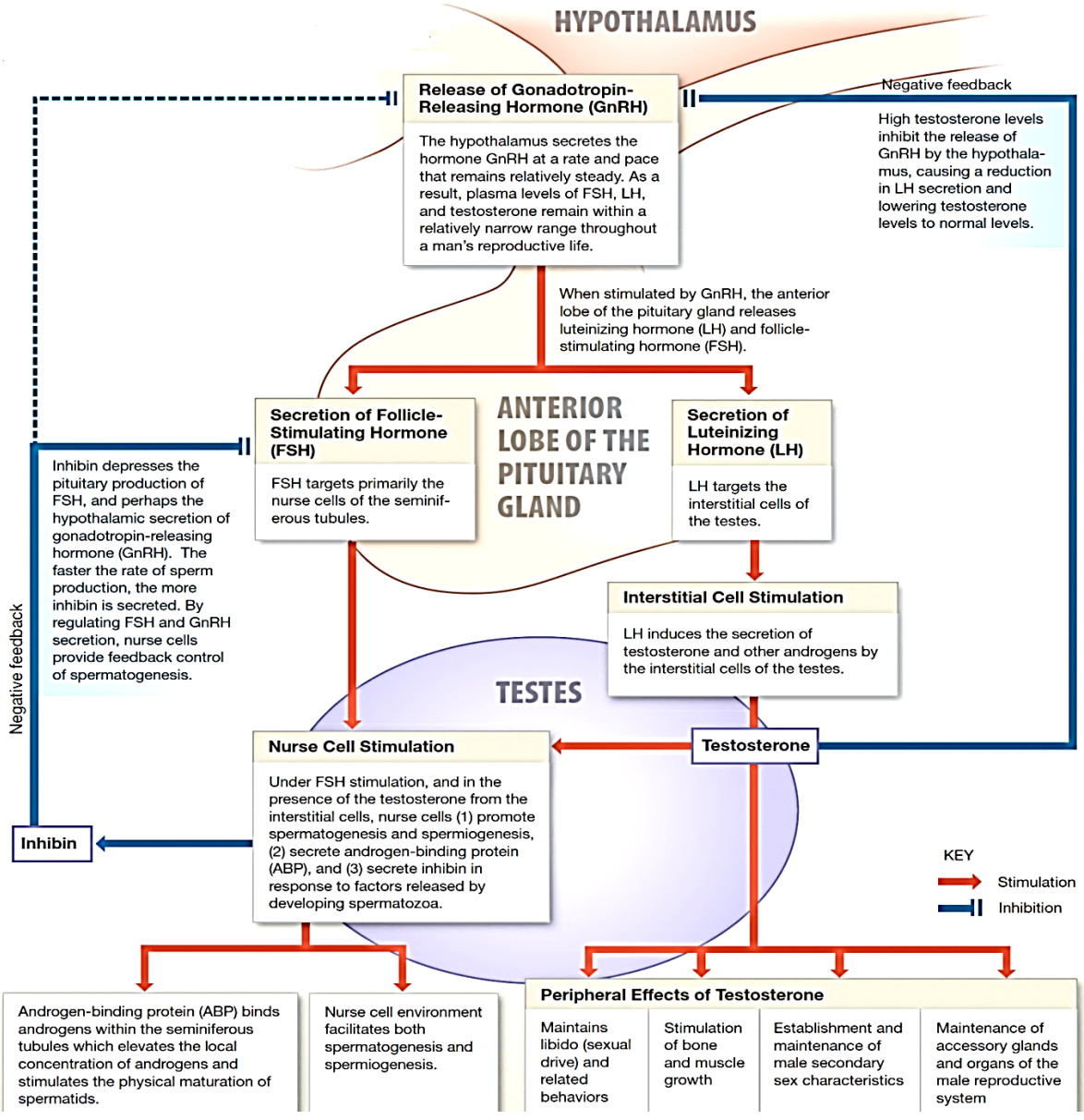
Secretions from epididymis, seminal vesicles, prostate and bulbourethral gland together with sperm make up semen.

Sperm make up only 1% of semen. The rest is a fluid secretion from accessory glands, which provide fructose to nourish sperm and alkaline medium to neutralize urethral (acidity because of urine) & vaginal acidity. It also contains vitamin C, and trace elements like calcium, Zinc, magnesium, Copper and sulfur.

Hormonal regulation in the male

Follicular Stimulating Hormone (FSH) and Leutinizing Hormone (LH) produced by the anterior pituitary gland are chiefly responsible for stimulating spermatogenesis and testosterone secretion. Follicular Stimulating Hormone (FSH) and Leutinizing Hormone (LH) affect only tests whereas;

Testosterone affects not only spermatogenesis but also sex organ development and appearance of secondary male sexual characteristics.



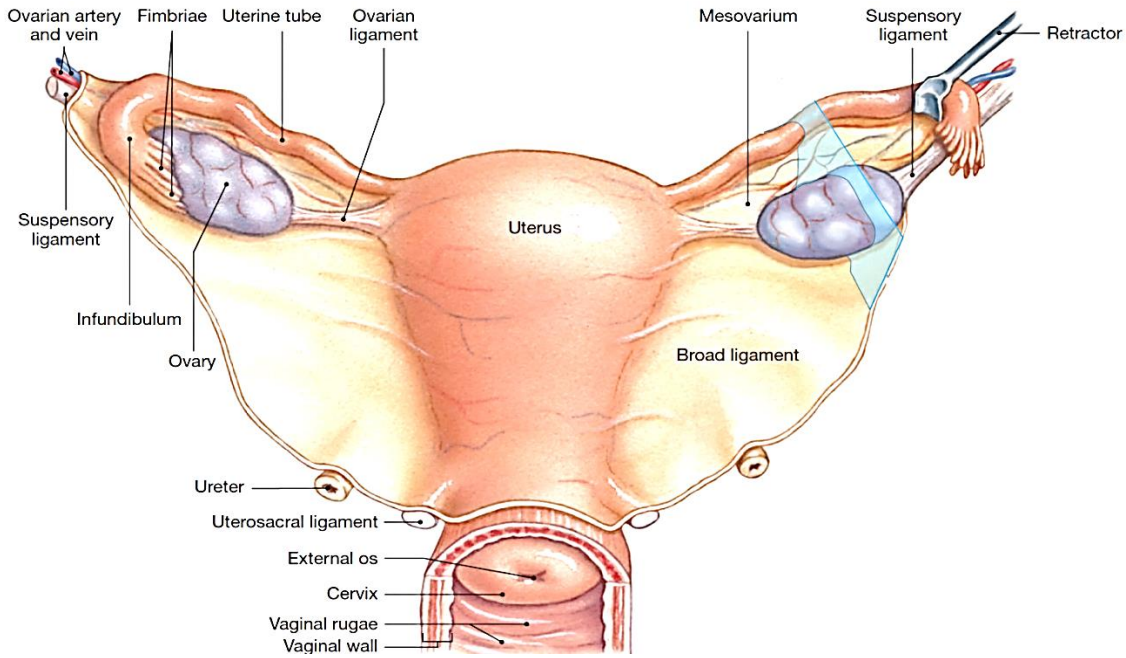
Female Reproductive System

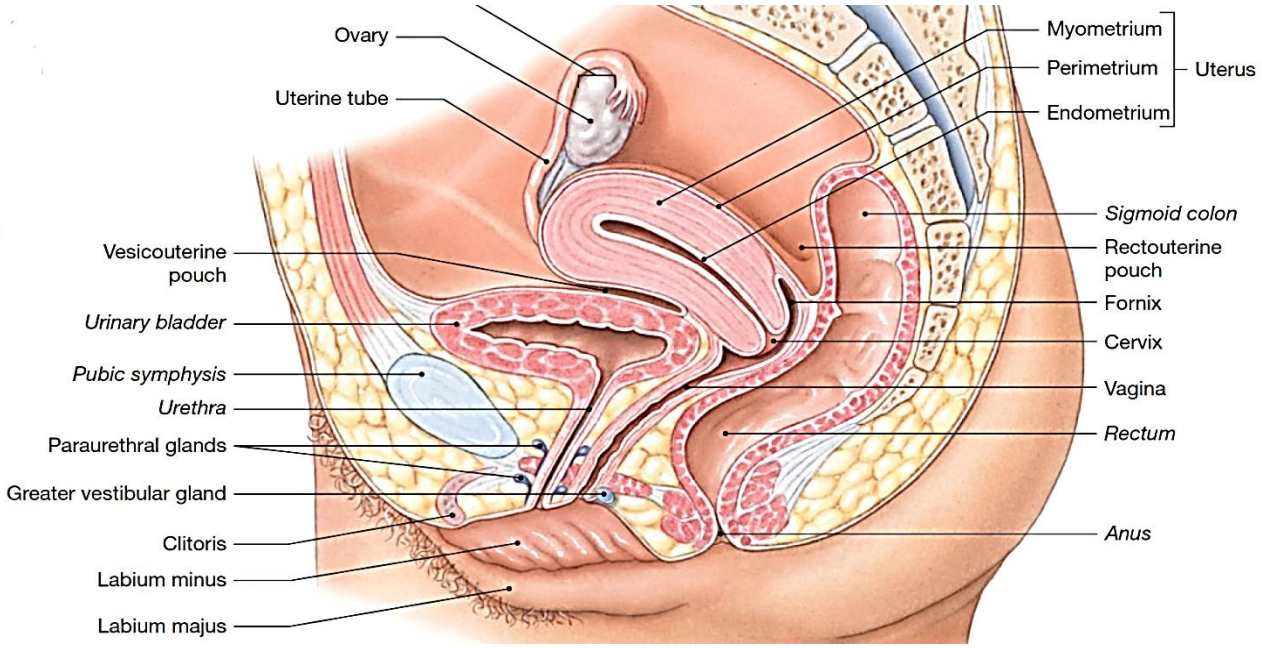
The reproductive role of females is far more complex than that of male. They are :

- Produce egg
- After fertilization they also carry and protect the developing embryo.
- Production of milk by mammary glands

The female reproductive system consists of :

- ♣ Ovaries,
- ♣ Uterine tube,
- ♣ Uterus,
- ♣ Vagina,
- ♣ External genital organ (vulva)
- ♣ Mammary glands.





Females internal reproductive organs

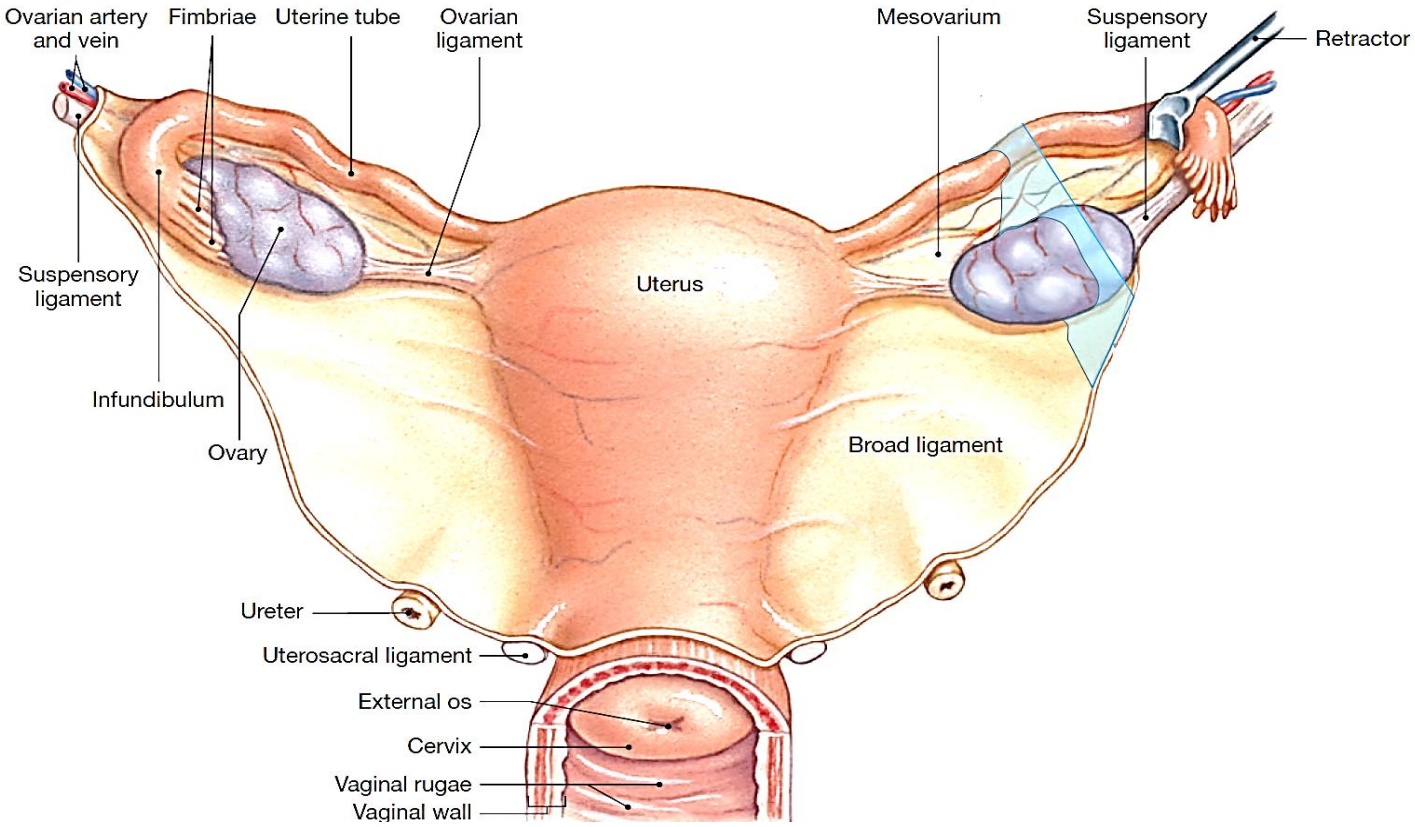
Females' internal reproductive organs include :

- ♣ The ovary
- ♣ The uterine tube,
- ♣ The uterus,
- ♣ The vagina.

Ovaries

Ovaries are females' gonads that produce ova and female hormones. It is about the size of unshelled almond. The ovaries are located in the pelvic cavity one in each side of the uterus. Each ovary is attached by a mesentery called ***mesovarium*** to the backside of each ***broad ligament***.

A thickening border of the mesovarium called ovarian ligament extends from the ovary to the uterus. The ovary is suspended from the pelvic wall by the suspensory ligament.

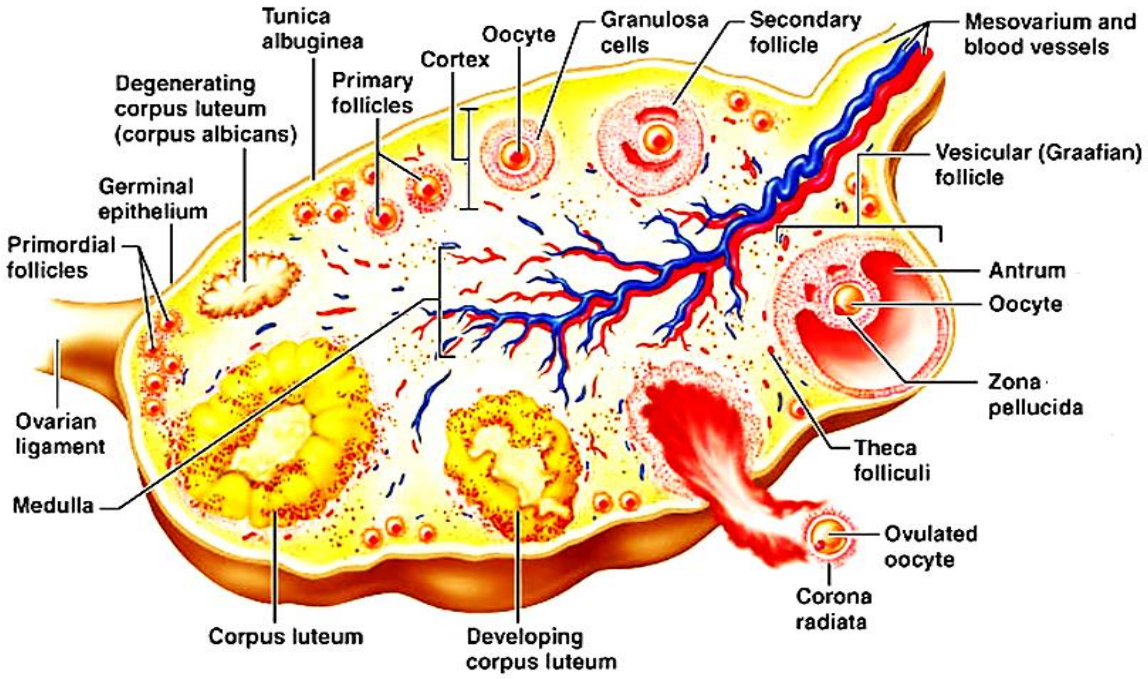


A layer of specialized epithelial cells called germinal layer covers the ovaries. Beneath it, is the storma mass of connective tissue, which contains ova in various stage of maturity.

A cross section of ovary reveals a cortex and a vesicular medulla. The cortex contains round epithelial vesicle or follicles; follicles are actual center of ovum production or oogenesis.

Follicles are usually located directly beneath the cortex of the ovary. They secret estrogen. But once they begin to mature they migrate towards the inner medulla. The medulla consists of layers of soft stromal tissue. The stromal tissue contains rich supply of blood vessels, nerves and lymph useless. After ovulation the lining of the follicles grow in ward, forming corpus luteum (yellow body), which temporary function as endocrine tissue. It secret estrogen & progesterone which

stops additional ovulation, thickening of uterine wall & stops additional mammary glands in anticipating prognoses.



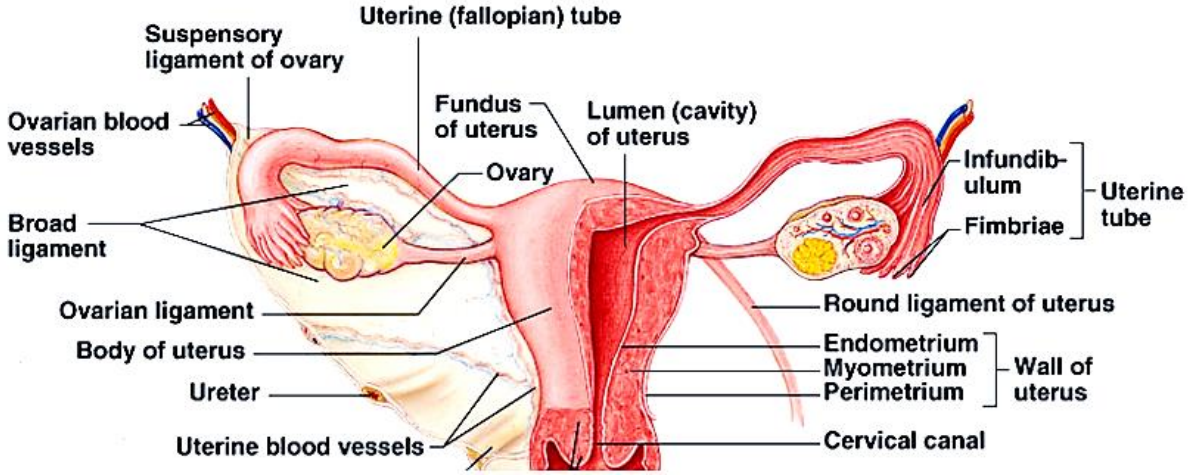
If pregnancy does not occur within 14 days corpus luteum degenerates and menstruation follows. If pregnancy occurs it continue to work for 2 to 3 months & eventually degenerate when placenta takes the duty.

Ovulation

Ovulation is the process by which the mature ova rupture from the follicle and eventuality travel to the fallopian tube. In response to high concentration of FSH & LH from anterior pituitary, the mature follicle ruptures this process. About 20 ovarian follicles mature each month beginning at puberty. However, only one ruptures to release ova (oocyte).

Uterine tubes

Uterine tubes are a pair of tubes one in each side of the uterus. The uterine tube (fallopian tube conveys the oocyte to the uterus. One of the paired tubes receives the oocyte from the ovary to take to the uterus. The tubes are not directly connected to ovary. But it open in abdominal cavity very close to ovary.



It has 3 distinct portions :

- **Infundibulum:** funnel shaped, near the ovary
- **Ampulla:** thin walled middle portion
- **Isthmus:** opens in to the uterus

The tube is made of three layers, the outer serous membrane, the middle muscular (smooth muscle) and the inner mucous membrane.

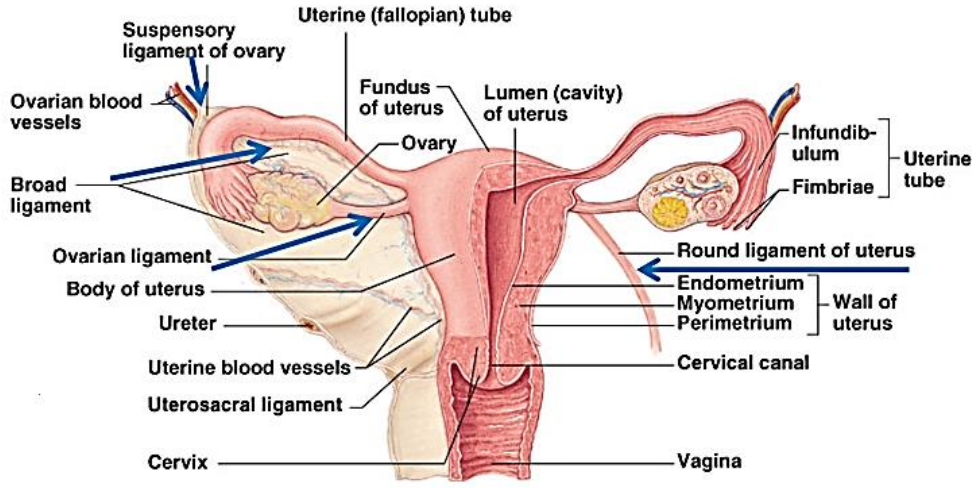
The Infundibulum is fringed with feathery fimbrea, which may over lapse the ovary. Unlike sperm the oocyte is unable to move by its own instead it is carried along the uterine tube towards uterus by peristaltic contraction of the fallopian tube.

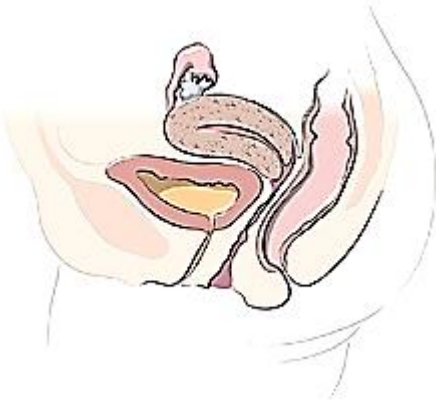
Uterus

The uterine tubes terminate in the uterus. A hallow muscular organ located in front of the rectum and behind the urinary bladder. It is like an inverted pear when viewed anterior and is pear size as well, during pregnancy it increases 3-6 times.

In the adult measures about 8 cm in length ,5 cm width, 3 cm thickness.

Round ligament help the uterus tilted forward over the bladder. It is attached to the lateral wall of the pelvis by two broad ligaments. Two utero-sacral ligaments extend from the upper part of the cervix to the sacrum. The posterior & anterior ligaments attach uterus to rectum and urinary bladder respectively.





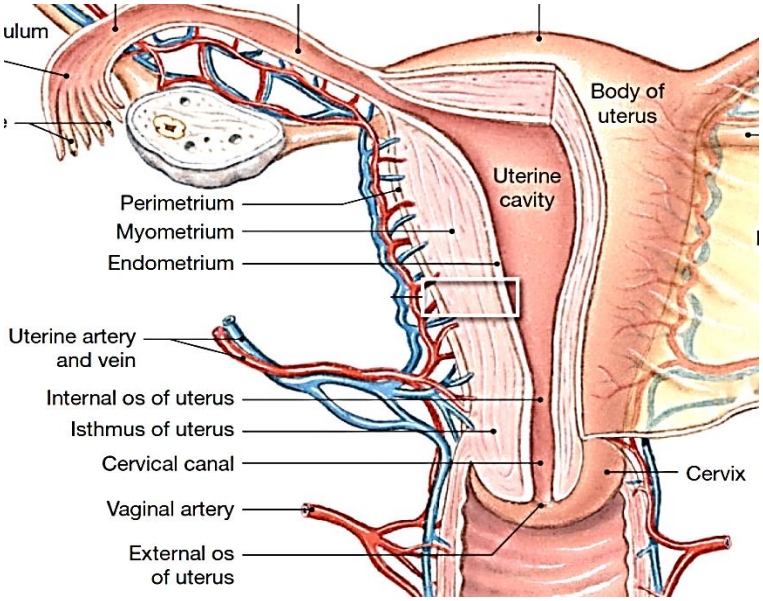
Uterus has three parts:

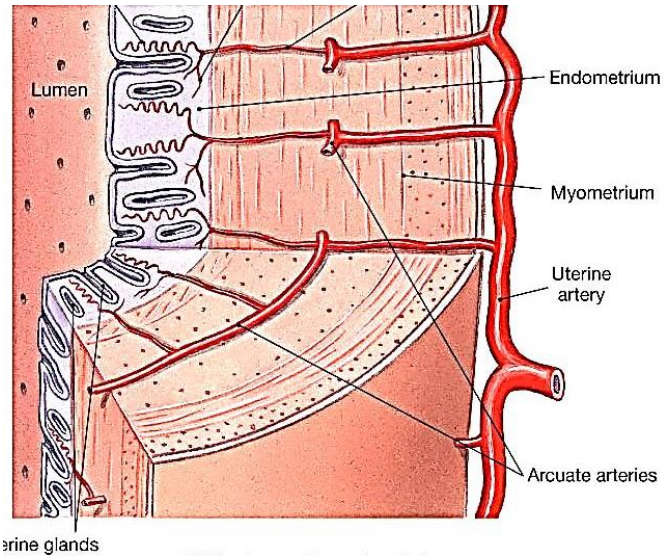
- **Fundus:** the upper, doom shaped part
- **Body:** the tapering middle portion
- **Cervix:** the body terminate to narrow portion

The constricted region between the body and cervix is the isthmus. The interior of the cervix is the cervical canal, which opens in to vagina.

The uterus is made up of three layers of tissue. The outer serosa layer, the middle muscular layer called myometrium which make the bulky uterine wall composed of 3 layers of smooth muscle and the innermost, composed of specialized mucous membrane, endometrium.

The endometrium contains abundant blood supply. It is composed of two layers. These are stratum functionalist that shed during every menstruation and if pregnancy occurs it continues to be site of attachment and nourishment for morula (fertilized zygote) and the second layer of endometrium is stratum basal that attaches to myometrium.

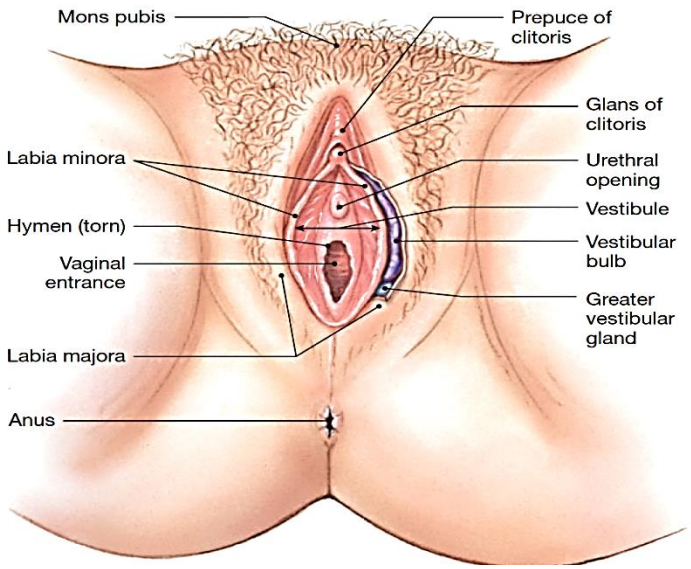


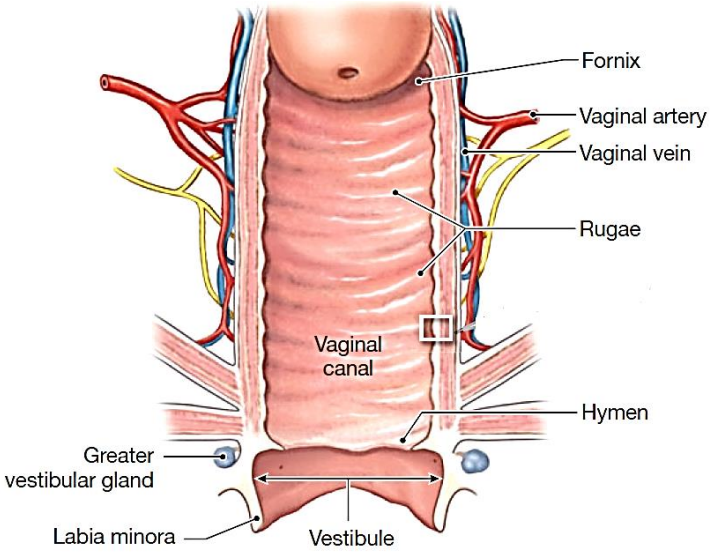


Vagina

A muscle lined tube 8 to 10 c.m long. It is the site where semen deposits during sexual intercourse; it is a canal for menstrual flow and a birth canal during childbirth. It lies behind urinary bladder and rectum & anterior to rectum and anus. Its wall is mainly composed of smooth muscle & fibroelastic connective tissue.

The mucous that lubricates the vagina comes from glands in the cervix and the acidic environment is because of the fermentation action of bacteria. A fold of vaginal mucosa called hymen that partially blocks the vaginal entrance in virgin.



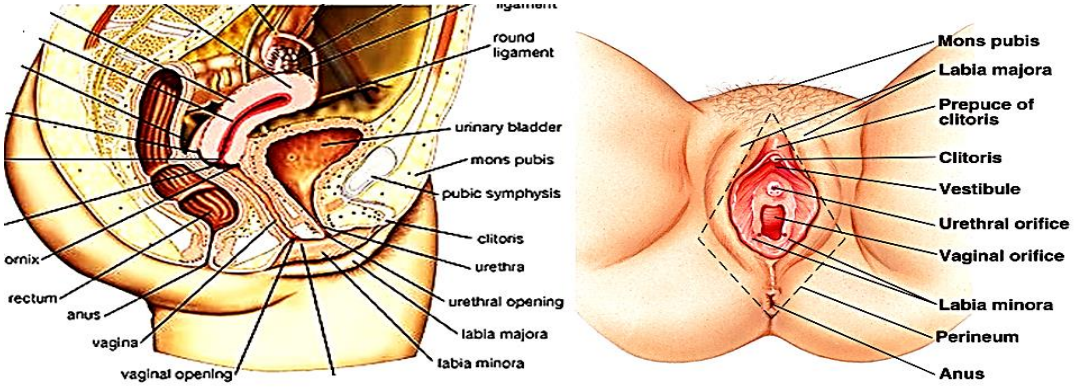


Females External reproductive organs

This includes: Mons pubis, Labia majora, Labia minora, Vestibular glands, Clitoris and Vestibule of vagina, which collectively called vulva.

The mons pubis

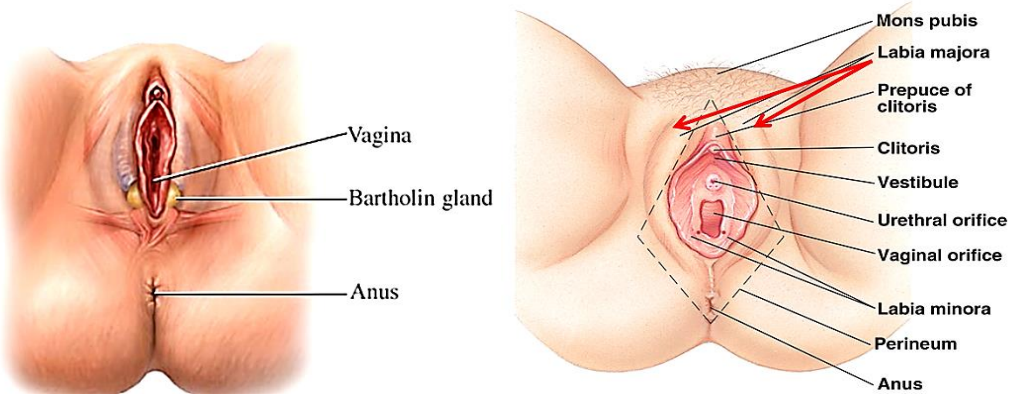
It is a pad of fat in front of the symphysis pubis in the female. At puberty it becomes covered with pubic hair.



Labia majora

They are 2 rounded folds of skin which pass backwards from the mons veneris in acurve and unite at their posterior ends in the perineum,in front of anus.

At the posterior end of each is a small gland which secretes mucus, called BARTHOLIN,S GLAND (Vestibular Glands)



The outer surface of lips is covered by hair, but in the inner is smooth and free from hair.

Labia minora

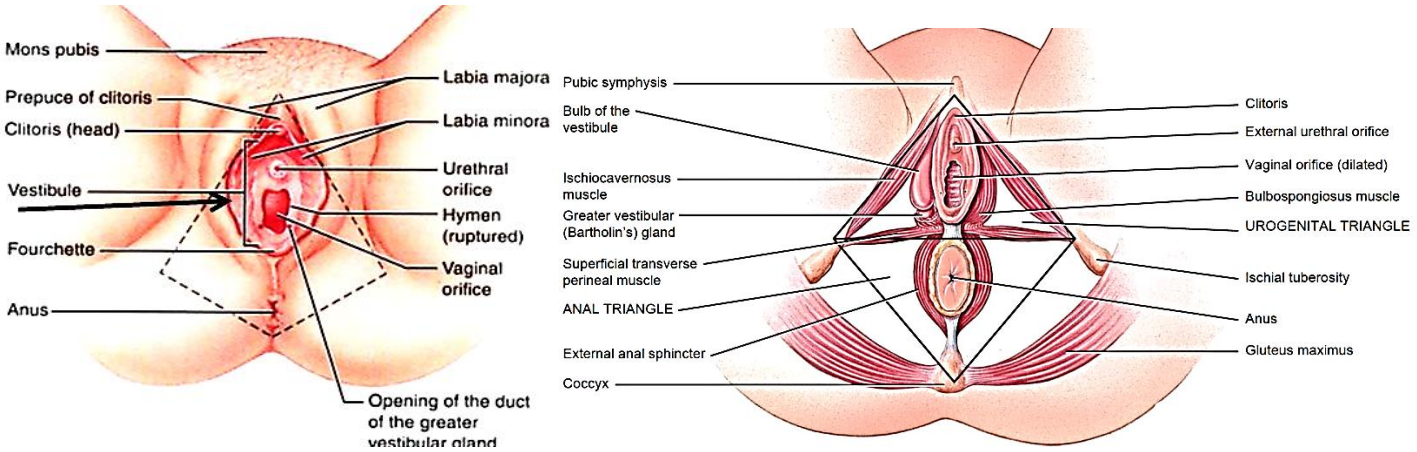
Labia minora is a smaller fold of skin that lies between the labia majora. Labia minora and labia majora surround vaginal& urethral orifices. The labia merge at the top to form the foreskin or prepuce called the clitoris.

The clitoris

Clitoris is a small erectile organ at the upper end of vulva below the mons pubis, where the two folds of labia minora meet. Like the penis it contain many nerve endings, hence it is very sensitive part.

The vestibule

Vestibule is a space between the labia minora. Its floor contains the greater vestibular glands and the opening for the urethra & vagina. During sexual arousal the greater and lesser vestibular gland secrete alkaline mucous for lubrication to aid penetration.



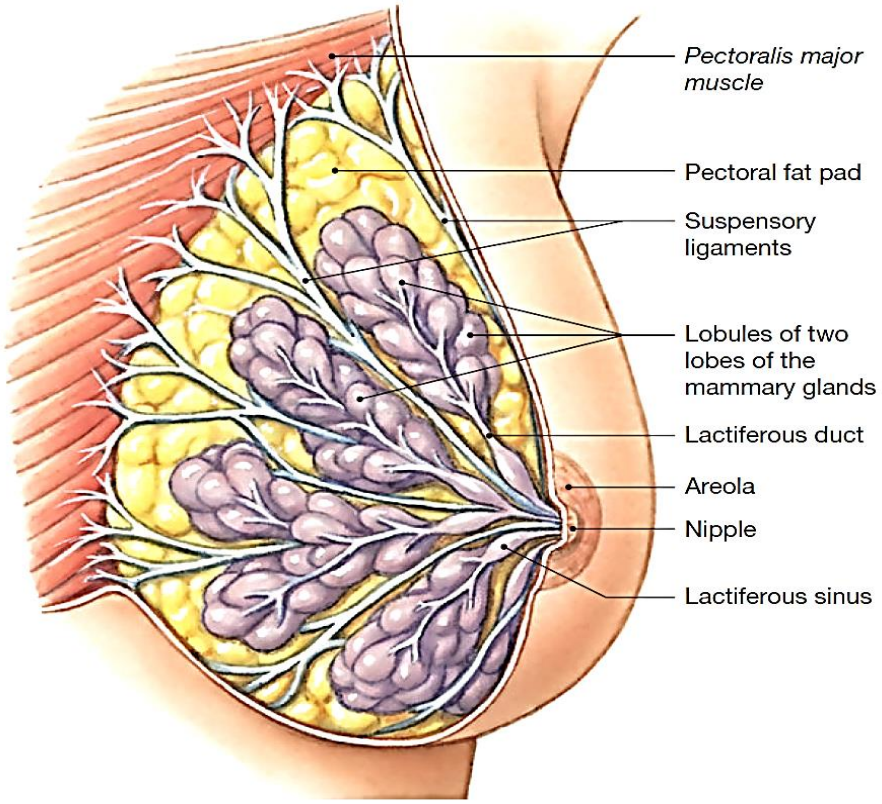
Perineum

Perineum is a diamond shaped region. Is the mass of tissue and skin situated between the external genitals and the anus. It stretches to a remarkable extent during labor, but in spite of this elasticity is frequently torn and requires surgical repair.

Mammary Glands

Within the paired breasts of women are modified sweat glands that produce and secrete milk. Each breast extends from the lateral surface of sternum to middle of axilla. It is held in place by suspensory ligaments of breast. The amount of adipose tissue matters the size of the breast not the mammary cells. Each breast is composed of 15-20 lobes of areolar gland that radiate from the nipple. This cluster of glands with lactiferous ducts that carry milk from the gland look like bunches of grapes. **Lactiferous sinuses** are dilated portion of **lactiferous ducts.**

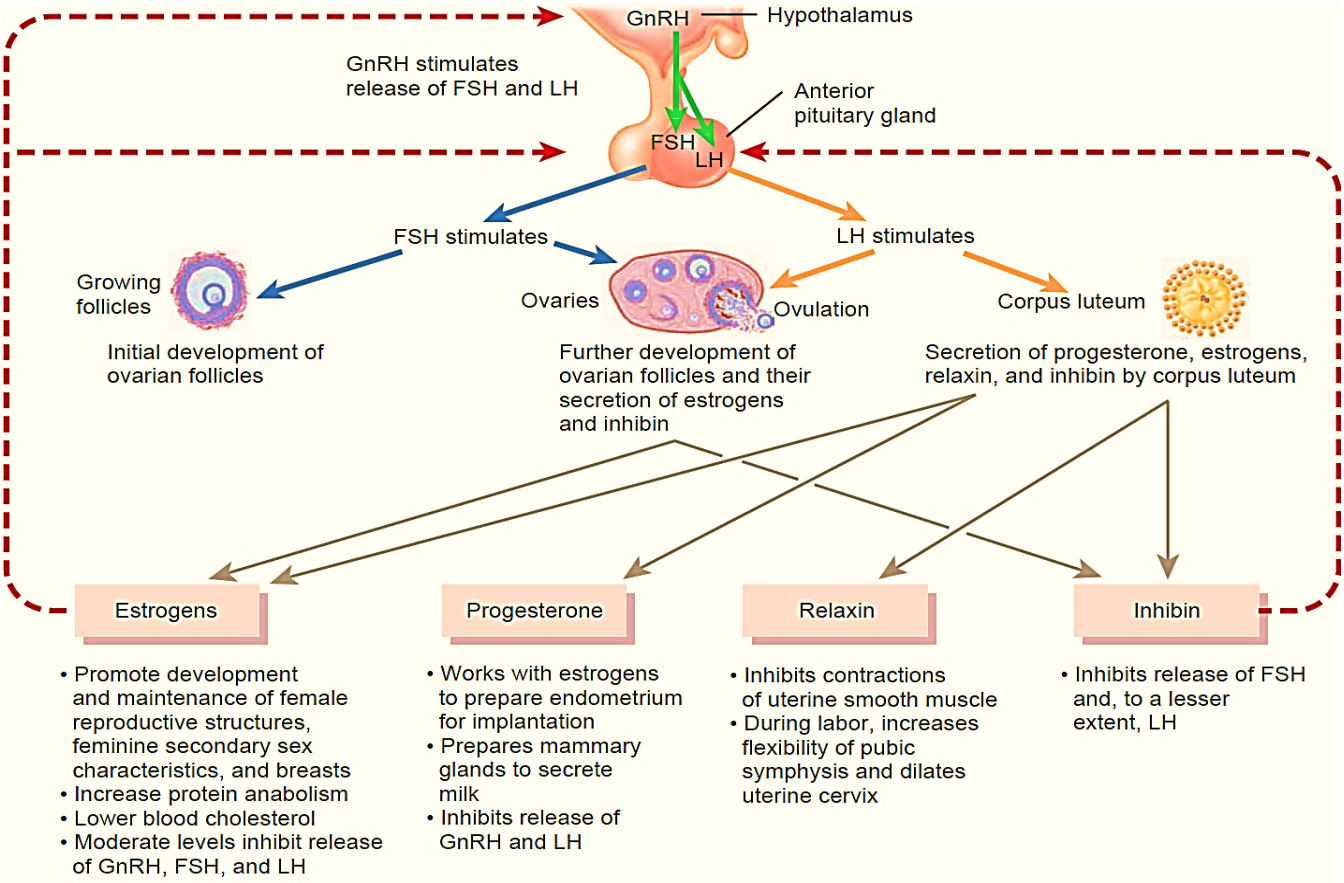
Areola is the pigmented area around the nipple. It enlarges & darkness during pregnancy.



Hormonal regulation

Females are fertile a few days each month by the action of Gonadotrophin Realizing Hormone (GnRH) and FSH & LH.

In pregnant women additional hormone called Human Chorionic Gonadotrophin (HCG), which affects uterine and other body metabolism is released by placenta.



The menstrual cycle: Menstrual Phase

The menstrual phase (MEN-stroo-al), also called menstruation(men_-stroo-A¯ -shun) or menses (MEN-se¯z _ month), lasts for roughly the first 5 days of the cycle. (By convention, the first day of menstruation is day one of a new cycle.)

- **EVENTS IN THE OVARIES:**

Under the influence of FSH, several primordial follicles develop into primary follicles and then into secondary follicles. This developmental process may take several months to occur. Therefore, a follicle that begins to develop at the beginning of a particular menstrual cycle may not reach maturity and ovulate until several menstrual cycles later.

- **EVENTS IN THE UTERUS:**

Menstrual flow from the uterus consists of 50–150 mL of blood, tissue fluid, mucus, and epithelial cells shed from the endometrium. This discharge occurs because the declining levels of progesterone and estrogens stimulate release of prostaglandins that cause the uterine spiral arterioles to constrict. As a result, the cells they supply become oxygen-deprived and start to die. Eventually, the entire stratum functionalis sloughs off. At this time the endometrium is very thin, about 2–5 mm, because

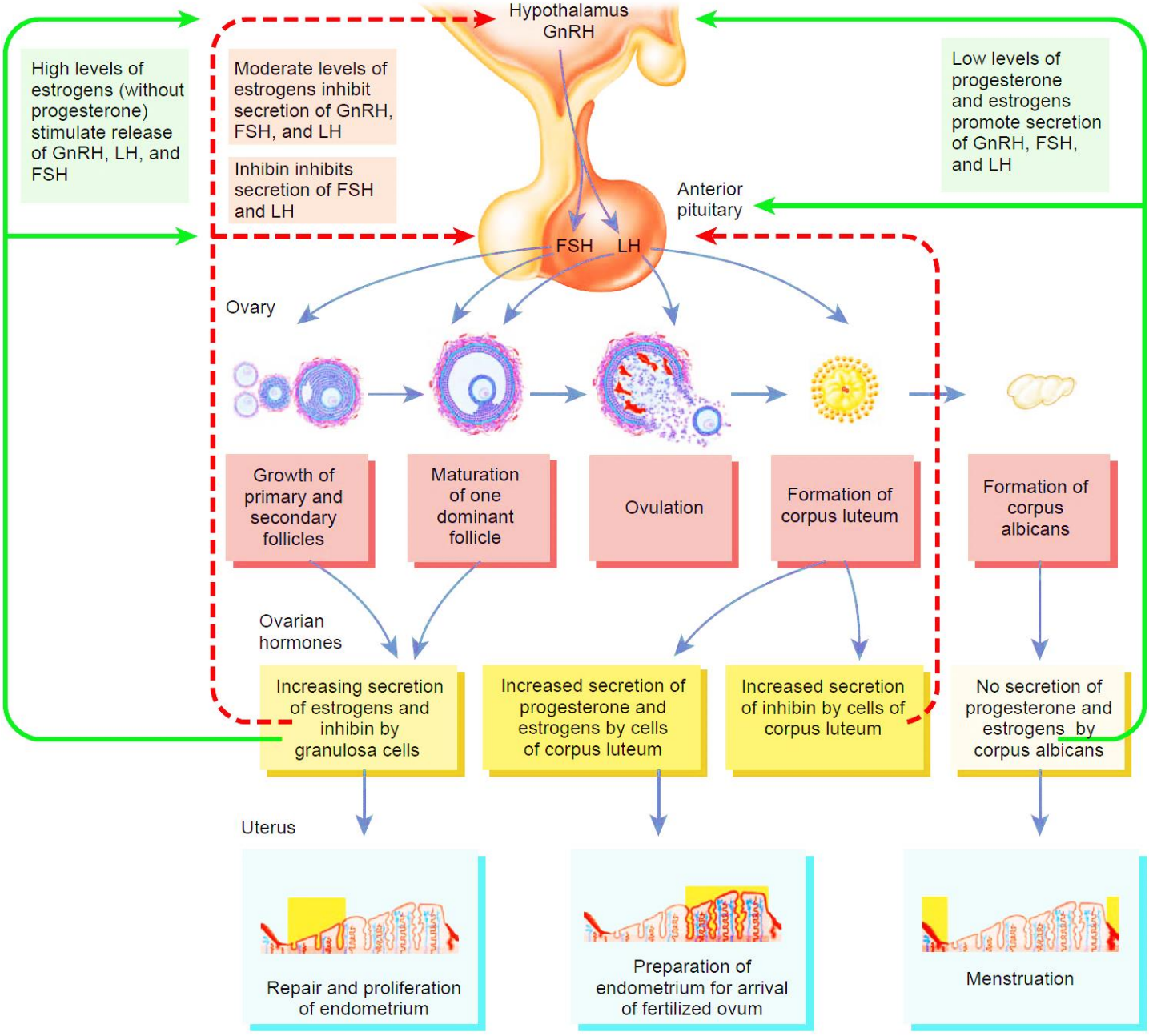
only the stratum basalis remains. The menstrual flow passes from the uterine cavity through the cervix and vagina to the exterior.

Preovulatory Phase: The preovulatory phase is the time between the end of menstruation and ovulation. The preovulatory phase of the cycle is more variable in length than the other phases and accounts for most of the differences in length of the cycle. It lasts from days6 to 13 in a 28-day cycle.

- **EVENTS IN THE OVARIES**

Some of the secondary follicles in the ovaries begin to secrete estrogens and inhibin. By about day 6, a single secondary follicle in one of the two ovaries has outgrown all the others to become the **dominant follicle.**

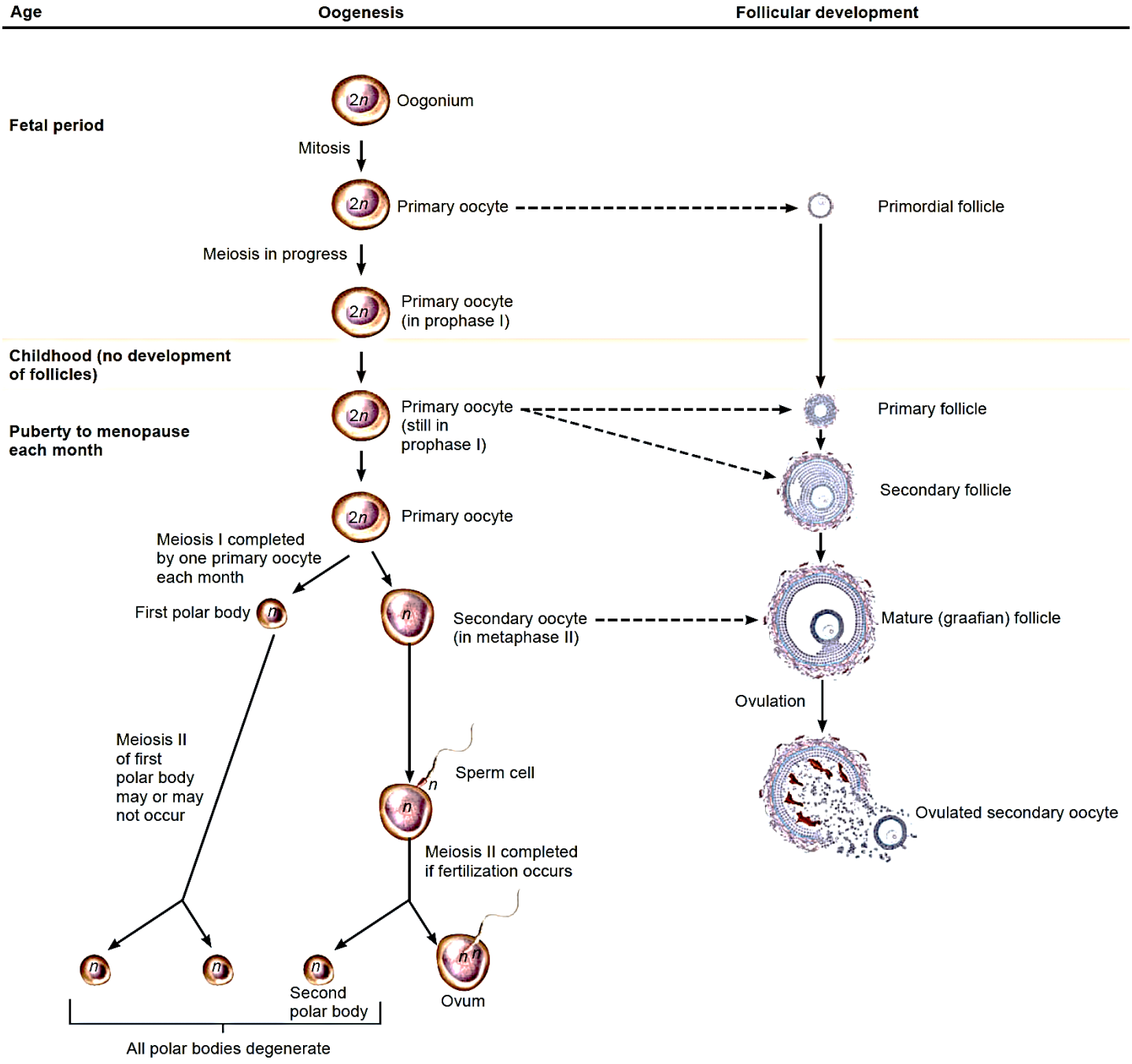
Estrogens and inhibin secreted by the dominant follicle decrease the secretion of FSH, which causes other, less well-developed follicles to stop growing and undergo atresia. Fraternal (non identical) twins or triplets result when two or three secondary follicles become codominant and later are ovulated and fertilized at about the same time.



Gametogenesis

Gametogenesis is the process of formation of sex cells. It is said spermatogenesis in male & oogenesis in females. Both involve meiosis, a form of nuclear division.

Summary of Oogenesis and Follicular Development



Conception

Conception is the process of fertilization and subsequent establishment of pregnancy. Pregnancy begins with fertilization of an ovum and ends with delivery of the fetus after birth. When sperm is deposited in the vagina, the many spermatozoa wriggle about in all direction. Some travel in to the uterus and oviduct to meet the ova. If ova and sperm join this is called fertilization. From the day of fertilization on wards pregnancy starts.

Summary of major reproduction hormones

Hormone	Function	Source
FEMALE Estrogen	Stimulates thickening of uterine wall, stimulates maturation of oocyte, stimulates development of female sex characteristics, inhibits FSH secretion, increases LH secretion.	Ovarian follicle, corpus luteum (controlled by FSH)
FSH (follicle-stimulating hormone)	Causes immature oocyte and follicle to develop increases estrogen secretion, stimulates new gamete formation and development of uterine wall after menstruation.	Pituitary gland (controlled by hypothalamus)
GnRH (Gonadotrophin Releasing Hormone)	Controls pituitary secretion.	Hypothalamus
HCG (Human Chorionic Gonadotropin)	Prevents corpus luteum from disintegrating, stimulates estrogen and progesterone secretion from corpus luteum.	Embryonic membranes placenta
LH (luteinizing hormone)	Stimulates further development of oocyte and follicle, stimulates ovulation, increases progesterone secretion, aids development of corpus luteum.	Pituitary gland (controlled by hypothalamus)
Progesterone	Stimulates thickening of uterine wall, stimulates formation of mammary ducts.	Corpus luteum (controlled by LH)
Prolactine	Promotes milk production by mammary glands after child birth	Pituitary gland (controlled by hypothalamus)
MALE FSH	Increases testosterone production, aids sperm maturation	Pituitary gland (controlled by hypothalamus)
GnRH	Controls pituitary secretion.	Hypothalamus
Inhibin	Inhibits FSH secretior	Sustentacular (sertoli) cell of Testes
LH	Stimulates testosterone secretion	Pituitary gland (controlled by hypothalamus)
Testosterone	Increases sperm production, stimulates development of male primary and secondary sex characteristics, and inhibits LH secretion.	Interstitial endocrinocytes (Leydig cells) in testes (controlled by LH)

GLOSSARY قائمة كلمات

Absorption passage of a substance through a membrane such as skin or mucosa, into blood
Accessory sex ducts series of ducts leading sperm from testis out.
Accessory sex glands glands that add secretion to sperm.
Acetylcholine chemical neurotransmitter
Actin contractile protein found in the thin myofilaments of skeletal muscle
Action potential nerve impulse
Active movement passage of substance across cell membrane using energy
Afferent carrying or conveying toward the center (for example, an afferent neuron carries nerve impulses toward the central nervous system)
Alveolus one of the millions of tiny air sacs in the lungs through which gases is exchanged between the outside air and the blood.
Antagonist muscle those having opposing actions; for example, muscles that flex the upper arm are antagonists to muscles that extend it
Anastomosis communication between two structures such as blood vessels
Anatomy Study of body structure
Anatomical position a starting positional reference

Antidiuretic hormone (ADH) Hormone released from the posterior pituitary gland that increases the reabsorption of water in the kidneys, thus decreasing the volume of urine excreted.
Aorta large artery that carries blood out of the left ventricle of the heart
Appendicular skeleton Part of the skeleton that includes the bones of the upper extremities, lower extremities, shoulder girdle, and hips.
Arteriole vessel between a small artery and a capillary
Artery vessel that carry blood away from the heart
Articulations formation of joints
Atrium one of the two upper chambers of the heart
Autonomic nervous system division of the human nervous system that regulates involuntary actions
Axial towards the midline of the body
Axon nerve cell process that transmits impulses away from the cell body
Buffer system a weak base or acid in the body that serves as neutralizing agent
Bile substance that reduces large fat globules into smaller droplets of fat that is more easily broken down
Body cavities spaces in the body holding internal organs
Body plans imaginary flat lines dividing the body in to sections.
Bone cell cellular part of a bone
Brain stem portion of the brain that connects the cerebrum with the spinal cord; contains the midbrain, pons, and medulla oblongata.
Bronchiole one of the small subdivisions of the bronchi that branch through the lung
Bronchus one of the large air tubes in the lung
Cancer cells growing with out normal body control mechanism
Capillary microscopic vessel through exchanges take place between the blood and the tissues
Cartilage a firm but delicate connective tissue
Cell the basic structural and functional unit of the body
Cell inclusions divers group of substances produced and stored inside the cell
Cerebellum the second largest part of the human brain that plays an essential role in the production of normal movements
Cerebral cortex a thin layer of gray matter made up of neuron dendrites and cell bodies that compose the surface of the cerebrum
Cerebrum the largest and upper part of the human brain that controls consciousness, memory, sensations, emotions, and voluntary movements

Chemoreceptor receptor that detects chemical changes
Coagulation clotting, as of blood
Connective tissue tissues specialized for connecting and supporting the body
Coronary referring to the heart or to the arteries supplying blood to the heart
Corpus luteum Yellow body formed from ovarian follicle after ovulation; produces progesterone.
Chyme partially digested food mixture leaving the stomach
Chyle milky-appearing fluid absorbed into the lymphatic system from the small intestine. It consists of lymph and droplets of digested fat.
Colon intestine
Conception the process of fertilization and subsequent establishment of pregnancy
Contraception protection against pregnancy
Cortex, renal the outer, red part of the kidney
Cortical nephron the nephrone locating inside the renal cortex
Cranium a bony cavity holding the brain
Cytoplasm substances surrounding organelle out of the nucleus
Defecation act of eliminating undigested waste from the digestive tract
Deglutition swallowing
Dendrite branching or tree like; a nerve cell process that transmits impulse towards the body
Dermis part of the skin next beneath to epidermis
Diaphragm dome-shaped muscle under the lungs that flattens during inhalation; membrane or structure that serves to separate
Diaphysis the shaft of long bone
Diastole relaxation phase of the cardiac cycle
Diencephalons "between" brain; parts of the brain between the cerebral hemispheres and the mesencephalon or midbrain
Diffusion Movement of molecules from a region where they are in higher concentration to a region where they are in lower concentration.
Digestion the break down of food materials either mechanically (that is chewing) or chemically (that is digestive enzymes)
Directional terms terms used in anatomy to state direction of body parts
Duodenum the first subdivision of the small intestine where most chemical digestion occurs
Effector responding organ; for example, voluntary and involuntary muscle, the heart, and glands
Efferent carrying from, as neurons that transmit impulses from the central nervous system to the periphery; opposite of afferent
Electrolytes compounds that dissociate in to ions when in solution
Emulsify in digestion, when bile breaks up fat
Endocardium lining of the heart
Endocrine referring to a gland that releases its secretion directly and lymphatic vessels into the bloodstream
Endothelium epithelium that lines the heart, blood vessels,
Endosteum connective tissue layer covering the inner hallow of bone
Erythrocyte red blood cells
Enzyme biochemical catalyst allowing chemical reaction to take place Epidermis the outer layer of skin
Epiglottis leaf-shaped cartilage that covers the larynx during swallowing
Epimysium sheath of fibrous connective tissue that encloses muscle
Epinephrine adrenaline; secretion of the adrenal medulla
Epithelial tissue tissues covering body surface and lining cavities
Esophagus the muscular, mucus-lined tube that connects the pharynx with the stomach; also known as the food pipe
Essential organs reproductive organs that must be present for reproduction to occur and are known as gonads

Endosteum Epithelium that lines the heart, blood vessels, and lymphatic vessels.
Epiphysis End of a long bone.
Epiphyseal plate a plate between diaphysis and epiphysis
Eponychyem thin flap of skin covering the proximal end of the nail
Extra cellular fluid all fluid existing out of a cell
Filtrate the primary filtered substance in the renal tubule
Fibrin blood protein that forms a blood clot
Follicle-Stimulating Hormone (FSH) Hormone produced by the anterior pituitary that stimulates development of ova in the ovary and spermatozoa in the testes
Fontanel Area in the infant skull where bone formation has not yet occurred; "soft spor"
Foreskin a loose-fitting, retractable casing located over the glans of the penis; also known as the prepuce
FSH See Follicle-stimulating hormone
Gametogenesis the process of formation of sex cell
Glandular tissue forming the secretary portion of a gland
Gamete Reproductive cell; ovum or spermatozoon.
Ganglion collection of nerve cell bodies located outside the central nervous system
Hemoglobin iron-containing protein in red blood cells that functions mainly to transport oxygen
Hematocrit test that measures the volume percentage of red blood cells in whole blood
Hemolysis rupture of red blood cells
Hilus Area where vessels and nerves enter or leave an organ
Hormone secretion of an endocrine gland
Hydrolysis splitting of large molecules by addition of water, as in digestion
Hydrostatic pressure a pressure exerted by a fluid on the side of the surface
Hypodermis the layer of connective tissue between dermis and the deeper tissue
Hyponychym thin flap of skin covering the under surface of nail
Hypothalamus portion of the floor lateral wall of the third ventricle of the brain
Hypoxia reduced oxygen supply to the tissues
Insertion attachment of a muscle to the bone that it moves when contraction occurs (as distinguished from its origin)
Intracellular fluid a fluid inside the cell
Intervertebral discs cartilaginous structure between the vertebral bones
Juxta medullary nephrone nephrones whose tubule extend to the medulla of the kidney
Lacteal a lymphatic vessel located in each villus of the intestine; serves to absorb fat materials from the chyme passing through the small intestine
Larynx voice box; structure between the pharynx and the trachea that contains the vocal cords
Leukemia malignant blood disease
Leukocyte white blood cells
Lung one of the two organs of respiration
Luteinizing hormone Hormone produced by the anterior pituitary that induces ovulation and formation of the corpus luteum in females; in males it stimulates cells in the testes to produce testosterone and is called interstitial cell-stimulating hormone (ICSH).
Lunula the whitish proximal part of the nail
Mammary gland referring to breast
Mastication chewing
Mediastinum region between the lungs and the organs and vessels it contains
Medulla oblongata the lowest part of the brain stem; an enlarged extension of the spinal cord; the vitals centers are located within this area
Medulla, renal the inner part of kidney
Meiosis nuclear division in which the numbers of chromosomes are reduced to half their original number; produce gametes
Menarche beginnings of the menstrual function

Meninges fluid containing membranes surrounding the brain and spinal cord
Menstruation a monthly bleeding form the female genitalia
Metaphysis part of the bone between diaphysis and epiphysis
Midbrain one of the three parts of the brain stem
Mitochondria Cell organelles that manufacture ATP with the energy released from the oxidation of nutrients
Mitosis indirect cell division involving complex changes in the nucleus
Mons pubis skin covered pad of fat over the symphysis pubis in the female
Mucus membrane a membrane that line body cavities opening to the exterior
Muscle tissue a tissue forming the muscle
Murmur abnormal heart sound
Myeloid tissue a fatty tissue inside spongy bone
Myocardium middle layer of the heart; heart muscle
Myometrium muscle layer in the uterus
Myosin contractile protein found in the thick filaments of skeletal muscle
Nephron the functional part of the kidney
Nerve collection of nerve fibers
Nerve impulse signals that carry information along the nerves
Nervous tissue a tissue specialized for nerve impulse conduction
Neurotransmitter chemicals by which neurons communicate
Oocyte immature stage of the female sex cell
Orbit the socket of the eye
Origin the attachment of a muscle to the bone, which does not move when contraction occurs, as distinguished from the insertion
Organelle Specialized subdivision within a cell
Osmotic pressure a pressure exerted by pure water on a surface
Osmosis Movement of water through a semi-permeable membrane
Osseous tissue a bone tissue
Ossicles bones of the inner ear
Ossification Process of bone formation.
Ovaries female gonads that produce ova (sex cells)
Ovulation the process of releasing ovum from ovary to the fallopian tube
Oxygen debt continued increased metabolism that occurs in a cell to remove excess lactic acid that resulted from exercise
Passive movement a movement across cell membrane with out the use of energy
Physiology study of the function of living organisms
Penis forms part of the male genitalia; when sexually aroused, becomes stiff to enable it to enter and deposit sperm in the vagina
Pericardium fibrous sac lined with serous membrane that surrounds the heart
Perichondrium a connective tissue layer surrounding a cartilage
Perineum pelvic floor; external region between the anus and genital organs
Periosteum connective tissue membrane covering a bone
Peristalsis wavelike, rhythmic contractions of the stomach and intestines that move food material along the digestive tract
Peritoneum large, moist, slippery sheet of serous membrane that lines the abdominopelvic cavity (parietal layer) and its organs (visceral layer)
Pharynx throat; passageway between the mouth and the esophagus
Pituitary (hypophysis) endocrine gland located under and controlled by the hypothalamus; releases hormones that control other glands
Plasma liquid portion of the blood
Plasma membrane a membrane bounding the entire cell
Platelet cell fragment that forms a plug to stop bleeding and acts in blood clotting

Pleura serous membrane that lines the pleural cavity and covers the lungs
Plexus network of vessels or nerves
Pons the part of the brain stem between the medulla oblongata and the midbrain
Prime mover the muscle responsible for producing a particular movement
Prostaglandin group of hormones, produced by many cells of the body, that has a variety of effects
Pulse wave of increased pressure in the vessels produced by contraction of the heart
Receptor peripheral beginning of a sensory neuron's dendrites
Reflex involuntary action
Renal pelvis a large space in the kidney formed by the extension of the expanded portion of the ureter
Renal tubule a series of ducts conducting urine from the point of formation to minor calyx
Respiration exchange of oxygen and carbon dioxide between the outside air and body cells
Saliva secretion of the salivary glands; moistens food and contains an enzyme that digests starch
Sarcomere contractile unit of muscle; length of a myofibril between two Z bands
Semen male reproductive fluid
Septum dividing wall, as between the chambers of the heart
Serosa Serous memberane; epithelial membrane that secretes a thin, artery fluid
Serum liquid portion of blood without clotting factors; liquid that remains after blood has clotted
Sinusoid Enlarged capillary that serves as a blood channel
Skull a structure formed of cranial and facial bones
Sperm the male spermatozoon; sex cell
Spermatozoon male reproductive cell or gamete; pl., spermatozoa
Sphincter ring-shaped muscle
Steroid category of lipids that includes the hormones of the sex glands and the adrenal cortex
Subacious glands are oil producing glands
Sudoriferous glands are sweat glands
Surfactant substance in the alveoli that prevents their collapse by reducing surface tension of the fluids within
Suture immovable joints between cranial bones
Synapse junction between adjacent neurons
Synergist muscles that assists a prime mover
Systole contraction phase of the cardiac cycle
Sphincter Muscular ring that regulates the size of an opening
Sudoriferous Producing sweat; referring to the sweat glands
Synovial Pertaining to a thick lubricating fluid found in joints, bursa, and tendon sheaths; pertaining to a freely movable (diarthrotic) joint
Target tissue tissue that is capable of responding to a specific hormone
Tendon Cord of fibrous connective tissue that attaches a muscle to a bone
Testis Male reproductive gland; pl., testes (TES-teze)
Testosterone Male sex hormone produced in the testes; promotes the development teristics
Thalamus located just above the hypothalamus; its functions are to help produce sensations, associate sensations with emotions, and play a part in the arousal mechanism
Thorax Chest; adj., thoracic (tho-RAS-ik)
Tissue Group of similar cells that performs a specialized function
Tone partially contracted state of muscle
Trachea windpipe; tube that extends from the larynx to the bronchi
Tract bundle of neuron fibers within the central nervous system
Trigon a triangularly arranged area inside the urinary bladder
Urine Liquid waste excreted by the kidneys
Urination the process of voiding urine
Uterus Muscular, pear-shaped organ in the female pelvis within which the fetus develops during pregnancy.

Uterine tube/fallopian tube a tube leading from ovary to uterus
Vagina Lower part of the birth canal that opens to the outside of the body; female organ of sexual intercourse
Valve structure that keeps blood
Vascular tissue a fluid tissue constituting blood
Vasectomy Surgical removal of part or all of the ductus (vas) deferens; usually done on both sides to produce sterility
Vein vessel that carries blood toward the heart
Vena cava one of the two large veins that carry blood into the right atrium of the heart
Ventilation movement of the air into and out of the lungs
Ventricle cavity or chamber; one of the two lower chambers of the heart
Venules vessel between a capillary and a small vein
Vertebra A bone of the spinal column; pl., vertebrae (VER the- bre)
Villi finger like folds covering the plicae of the small intestine
Vulva external genitals of the female
Zygote Fertilized ovum; cell formed by the union of a sperm and an egg